# Strong-to-Weak Spontaneous Symmetry Breaking

**From Fundamentals to Research Frontiers**

Ze-Yu Xing

October 22, 2026

# Table of contents


**Welcome** 7
- The question that organizes the book . . . 7
- What you will learn . . . 7
- Reading routes . . . 8
- Conventions that prevent common confusions . . . 8
- Scope and source method . . . 8

**I I · Foundations** 10

**1 Quantum states, ensembles, and locality** 11
- 1.1 What a state must predict . . . 11
- 1.2 Why density matrices are necessary . . . 12
- 1.3 Composite systems and reduced states . . . 13
- 1.4 Matrix functions and purification . . . 13
- 1.5 Local observables and thermodynamic statements . . . 14
- 1.6 Connected correlations and clustering . . . 14
- 1.7 Exercises and worked solutions . . . 15

**2 Statistical mechanics and spontaneous symmetry breaking** 17
- 2.1 The puzzle of a symmetric magnet . . . 17
- 2.2 Deriving the Gibbs ensemble . . . 17
- 2.3 Conventional symmetry and selection rules . . . 18
- 2.4 The order of limits . . . 18
- 2.5 Exact one-dimensional Ising model . . . 19
- 2.6 Landau theory and its limits . . . 19
- 2.7 Quantum ground-state order . . . 20
- 2.8 What must survive in a new notion of SSB? . . . 20
- 2.9 Exercises and worked solutions . . . 21

**3 Quantum channels and open-system dynamics** 23
- 3.1 From inaccessible degrees of freedom to a map . . . 23
- 3.2 Measurement, selection, and discarded records . . . 23
- 3.3 Three elementary channels . . . 24
- 3.4 The Choi representation . . . 24
- 3.5 Continuous time and the Lindblad equation . . . 25
- 3.6 Heisenberg evolution and conservation laws . . . 25
- 3.7 Stationary states, gaps, and mixing . . . 26
- 3.8 Local channels and causal structure . . . 26
- 3.9 Exercises and worked solutions . . . 26

**4 Information, distinguishability, and recovery** 28
- 4.1 Why ordinary expectation values are not enough . . . 28
- 4.2 Entropy and correlations . . . 28
- 4.3 Relative entropy and data processing . . . 29
- 4.4 Trace distance and fidelity . . . 29
- 4.5 A fidelity average is not a linear expectation value . . . 30
- 4.6 Conditional mutual information . . . 31
- 4.7 Recovery maps and the Markov property . . . 32
- 4.8 Information lengths and geometry . . . 33
- 4.9 Exercises and worked solutions . . . 33

**II II · Strong-to-Weak Order** **35**

**5 Symmetry of states and dynamics** **36**
5.1 Two symmetry conditions . . . 36
5.2 The support theorem . . . 36
5.3 A three-state laboratory . . . 37
5.4 Canonical and grand-canonical ensembles . . . 38
5.5 Symmetry of a quantum channel . . . 38
5.6 Lindblad generators and conserved sectors . . . 39
5.7 Non-Abelian caution . . . 39
5.8 Historical precursor: relative time translations . . . 40
5.9 Exercises and worked solutions . . . 40

**6 Nonlinear order parameters** **42**
6.1 Charge transfer as a state-comparison experiment . . . 42
6.2 Rényi-1: a diagnostic equivalent to fidelity . . . 43
6.3 Rényi-2 and spectral reweighting . . . 44
6.4 Generalized Rényi correlators . . . 45
6.5 Nonunitary operators and normalization . . . 45
6.6 Distinguishability and measurement cost . . . 46
6.7 Channel response and fidelity susceptibility . . . 46
6.8 Exercises and worked solutions . . . 49

**7 Purification and doubled-space order** **50**
7.1 Vectorization with an explicit convention . . . 50
7.2 Canonical purification . . . 50
7.3 Why the doubled symmetry breaks to the diagonal subgroup . . . 51
7.4 Worked example: a cat state of local Bell pairs . . . 52
7.5 The Choi double is a different state . . . 52
7.6 Uhlmann optimization and the role of the environment . . . 53
7.7 Thermal example with a complete calculation . . . 54
7.8 A finite-temperature lower bound beyond the solvable example . . . 54
7.9 Exercises and worked solutions . . . 55

**8 Stability, locality, and recoverability** **57**
8.1 What locality and depth mean . . . 57
8.2 Stability theorem and its proof . . . 57
8.3 A complete irrecoverability example . . . 59
8.4 General local irrecoverability . . . 59
8.5 Conditional mutual information . . . 60
8.6 What is a mixed-state phase? . . . 60
8.7 The 2026 local formulation . . . 61
8.8 Infinite-volume states and a genuine interpretive disagreement . . . 62
8.9 Exercises and worked solutions . . . 62

**9 Solvable models and decoherence-driven transitions** **64**
9.1 The pair-error channel on a graph . . . 64
9.2 Exact fidelity as a classical overlap . . . 65
9.3 One-dimensional open chain: full solution . . . 65
9.4 Two dimensions: cycles become dual spins . . . 67
9.5 Charge transfer as a disorder defect . . . 68
9.6 Rényi-2 maps to a clean Ising model . . . 69
9.7 Recoverability and the physical phase boundary . . . 69
9.8 Beyond the Pauli model . . . 70
9.9 Exercises and worked solutions . . . 70

**III III · Connections and Frontiers** 72

**10 Error correction, inference, and topological order** 73
10.1 Stabilizers as constraints . . . 73
10.2 Toric-code geometry before statistical mechanics . . . 75
10.3 Deriving the Nishimori condition . . . 76
10.4 Quantum memory requires a reference . . . 78
10.5 From charge inference to gauging . . . 79
10.6 What doubling does and does not establish . . . 79
10.7 Limits of the correspondence . . . 79
10.8 Exercises and worked solutions . . . 80

**11 Mixed-state SPT order, boundaries, and anomalies** 82
11.1 A pure SPT state from commuting operators . . . 82
11.2 Strings expose hidden bulk order . . . 83
11.3 Exact sublattice dephasing . . . 83
11.4 From the cluster ensemble to SW-SSB . . . 84
11.5 Canonical purification and Choi SPT data . . . 85
11.6 Strange correlators and information access . . . 86
11.7 What an anomaly actually constrains . . . 86
11.8 State robustness versus steady-state robustness . . . 87
11.9 Developments through the cutoff . . . 87
11.10 Exercises and worked solutions . . . 87

**12 Higher-form symmetry, intrinsic mixed order, and non-Abelian fusion** 89
12.1 What makes a symmetry higher form? . . . 89
12.2 The toric code's loop algebra . . . 90
12.3 A classical loop ensemble calculated exactly . . . 90
12.4 Nonlinear loop diagnostics . . . 91
12.5 Incoherent anyon proliferation versus coherent condensation . . . 91
12.6 Anomalies as entanglement obstructions . . . 92
12.7 Noninvertible higher-form operations . . . 93
12.8 Non-Abelian ordinary symmetry needs fusion records . . . 93
12.9 A maximally mixed state can retain quantum entanglement . . . 94
12.10 Center ordering and a decoding hierarchy . . . 95
12.11 Connecting the frameworks without collapsing them . . . 96
12.12 Exercises and worked solutions . . . 96

**13 Hydrodynamics and the broken strong symmetry** 98
13.1 Conservation does not yet mean diffusion . . . 98
13.2 An exactly controlled example: the exclusion process . . . 98
13.3 A doubled ferromagnet and a diffusive Goldstone mode . . . 100
13.4 Deriving fluctuating diffusion . . . 100
13.5 The response field as the broken-symmetry phase . . . 101
13.6 A microscopic slow-mode bound and its limits . . . 101
13.7 Finite-time information spreading is faster than particle diffusion . . . 102
13.8 Worked exercises . . . 103

**14 Charge scrambling, fluctuations, and learning** 105
14.1 What does a charge-transfer correlator compare? . . . 105
14.2 The subsystem characteristic function . . . 105
14.3 Separating classical uncertainty from coherence . . . 106
14.4 Why sufficiently fast nonlinear order forces fluctuations . . . 107
14.5 Three examples that prevent false equivalences . . . 108
14.6 Records turn charge scrambling into inference . . . 108
14.7 Worked exercises . . . 109

**15 Information-critical phases and frontier transitions** 110
15.1 A memory with more than two charge values . . . 110
15.2 Deriving coherent information from posterior uncertainty . . . 111

15.3 Why a discrete clock can behave like a continuous angle . . . 111
15.4 Scale-free information without ordinary critical correlations . . . 112
15.5 Charge-informed decoding: a different BKT problem . . . 113
15.6 Dephased fermions and an exact counting-statistics mapping . . . 113
15.7 Long-range channels change preparation constraints . . . 114
15.8 Absorbing states, stationary transitions, and postselection . . . 114
15.9 All-to-all fermions and approximate quantum codes . . . 115
15.10 Worked exercises . . . 116

**16 Detecting mixed-state order in the laboratory** **117**
16.1 Why one expectation value is insufficient . . . 117
16.2 Randomized measurements and their statistical cost . . . 117
16.3 Canonical purification is a measurement resource . . . 118
16.4 What the hardness theorem does and does not say . . . 118
16.5 Dephased Fermi gas: turning snapshots into nonlinear correlations . . . 119
16.6 A five-qubit cluster array: the role of discarded records . . . 120
16.7 Processor transitions using a known-channel kernel . . . 120
16.8 Reading finite-size evidence responsibly . . . 121
16.9 Worked exercises . . . 121

**IV IV · Working with the Theory** **123**

**17 Computational laboratory** **124**
17.1 What a small calculation can establish . . . 124
17.2 A global parity bit under weak-only noise . . . 124
17.3 Exact pair-error enumeration on a chain . . . 125
17.4 What changes on a square lattice? . . . 126
17.5 A noncommuting example: the correlators differ . . . 127
17.6 Diffusion: an exact conservation check . . . 127
17.7 Canonical block fluctuations . . . 128
17.8 Exercises and worked solutions . . . 129

**18 Synthesis and research problems** **131**
18.1 A connected picture . . . 131
18.2 A ledger of what has been established . . . 131
18.3 How to formulate a precise project . . . 132
18.4 Project 1: local versus global information . . . 132
18.5 Project 2: when a nonlinear transition is intrinsic . . . 133
18.6 Project 3: information length versus transport length . . . 133
18.7 Project 4: what non-Abelian noise leaves behind . . . 133
18.8 Project 5: an experimentally controlled estimator . . . 133
18.9 Several questions that remain connected . . . 134
18.10 Exercises and worked solutions . . . 134

**19 A conceptual source atlas** **136**
19.1 From open-system symmetry to spontaneous order . . . 136
19.2 Which diagnostic and which phase relation? . . . 137
19.3 Noise, inference, and quantum memory . . . 138
19.4 Average symmetry and protected mixed-state structure . . . 139
19.5 Generalized symmetry and non-Abelian information . . . 139
19.6 Hydrodynamics, charge scrambling, and learning . . . 140
19.7 Detection and the experimental timeline . . . 141
19.8 Established core and open boundaries . . . 142
19.9 Exercises in interpreting a frontier claim . . . 142

**Appendices** **143**

**A Mathematical toolkit** **143**
A.1 Spectra, singular values, and matrix functions . . . 143
A.2 Three norms and their roles . . . 143
A.3 Vectorization without ambiguity . . . 144
A.4 Cumulants and small twists . . . 144
A.5 Fourier transforms and zero modes . . . 145
A.6 Errors, loops, and boundaries over binary arithmetic . . . 145
A.7 Turning a posterior into a Boltzmann weight . . . 145
A.8 Scaling statements and their quantifiers . . . 146
A.9 Exercises and worked solutions . . . 146

**B Glossary and notation** **147**
B.1 States and limits . . . 147
B.2 Symmetry and operations . . . 147
B.3 Diagnostics and information . . . 148
B.4 Topology, dynamics, and evidence . . . 149

**References** **150**

# Welcome

## The question that organizes the book

Consider $N$ spins with global spin flip $U = \prod_i X_i$. The mixed state

$$\rho_+ = \frac{I + U}{2^N}$$

has a definite global symmetry charge. It has no ordinary spin order: $\mathrm{Tr}(\rho_+ Z_i Z_j) = 0$ for distinct sites. Yet applying $Z_i Z_j$ at arbitrarily distant points leaves it unchanged. Erasing even one spin loses a global bit that cannot be restored while leaving a distant region untouched. The state is separable, but its information is organized across the entire system.

Strong-to-weak spontaneous symmetry breaking gives a language for this phenomenon and for much less elementary states. Its central question is not only whether a symmetry is present, but how its charge is encoded, what local operations can reveal or erase, and which information structures persist under dynamics.

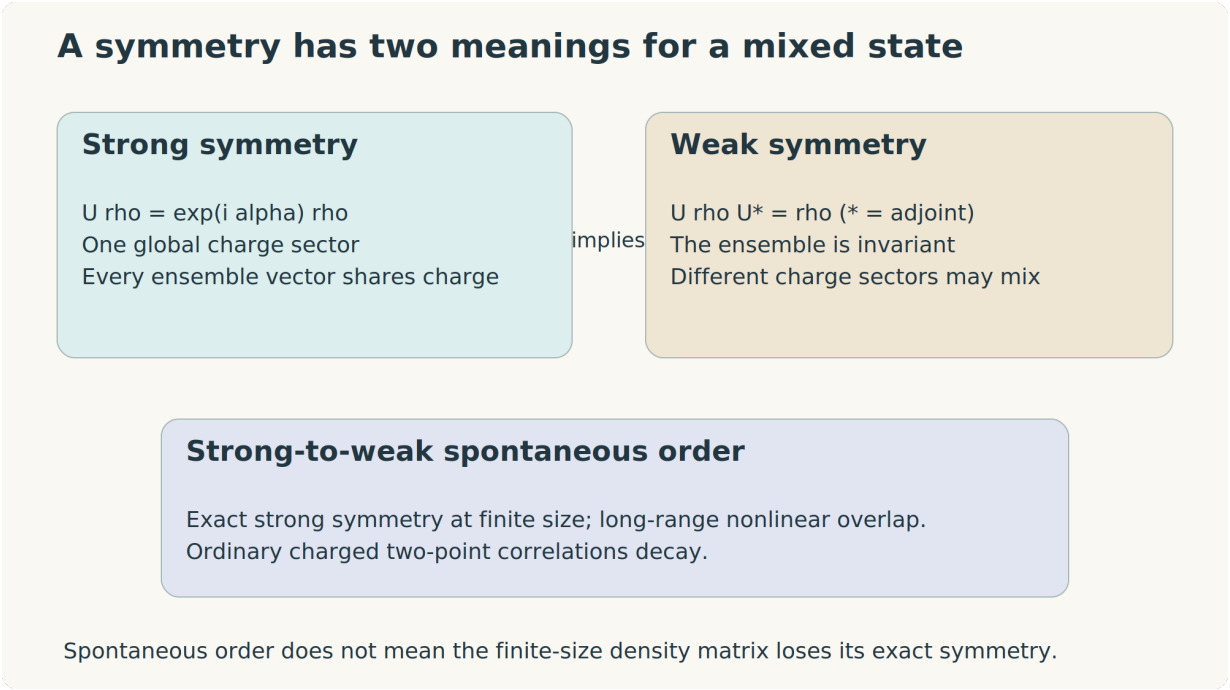



Figure 1: The two notions of mixed-state symmetry and the spontaneous-order question. Precise definitions and the finite-size/infinite-volume distinction are developed in Part II.

## What you will learn

The first part develops the tools rather than assuming them: density matrices, statistical ensembles, conventional SSB, quantum channels, Lindblad dynamics, fidelity, entropy, conditional mutual information, and recovery. The second part derives the nonlinear diagnostics, purification constructions, stability arguments, and solvable decoherence models. The third connects these ideas to error correction, topology, SPT phases, generalized symmetries,

diffusion, charge scrambling, information-critical behavior, and experiments. The final part provides reproducible Julia calculations, research projects, and an annotated source map.

Each chapter contains worked examples and exercises with solutions. Derivations distinguish exact finite-size identities, asymptotic arguments, model-dependent numerical findings, and conjectures. The numerical laboratory uses exact enumeration and standard-library Julia; no package installation is needed for the calculations.

## Reading routes

| Your starting point | A useful route |
|---|---|
| First course in quantum mechanics and statistical mechanics | Read Chapters 1–9 in order; work the exercises before entering the research chapters. |
| Comfortable with open systems and quantum information | Review the conventions below, then begin with symmetry in Chapter 5. |
| Interested in decoherence and quantum memories | Chapters 5–10, then information-critical phases and the computational laboratory. |
| Interested in transport and thermalization | Chapters 5–8, then hydrodynamics and charge scrambling. |
| Interested in mixed-state topology | Chapters 5–8 and 10–12; pay particular attention to the chosen phase equivalence. |
| Planning a project | Read the synthesis and source map after the relevant derivation chapters; use their assumptions as a checklist for a precise question. |

The appendices supply matrix inequalities, vectorization rules, Fourier conventions, a notation guide, and a glossary. Cross-references connect prerequisites to their later use. The sidebar search includes the chapter text and bibliography. A combined PDF is available from the download control.

## Conventions that prevent common confusions

We set $\hbar = k_B = 1$ and use natural logarithms unless bits are specified. $F(\rho, \sigma)$ is **root fidelity**, not its square. The canonical purification is $|\sqrt{\rho}\rangle\!\rangle$; the normalized vectorization $|\rho\rangle\!\rangle / \sqrt{\mathrm{Tr}\,\rho^2}$ is a different doubled state. Rényi-1 and Rényi-2 correlators refer to these different constructions.

"Strong symmetry" of a state means support in a single symmetry-charge sector, when that finite-volume description applies. "Strong symmetry" of a channel means that its local operations preserve that charge. Neither phrase follows from covariance alone. Global and local SW-SSB formulations are distinguished throughout: an exact finite-size charge constraint can disappear from the algebra of all finite-region observables in the infinite-volume limit.

A phase statement always depends on the allowed operations. Strongly symmetric shallow channels, two-way channel connectivity, and locally reversible channels impose different conditions. Thresholds extracted from nonlinear replicas need not be intrinsic phase boundaries.

## Scope and source method

The literature search begins with Chong Wang's June 2026 review, *Strong-to-Weak Spontaneous Symmetry Breaking*, and traces its primary sources backward and newer work forward. Coverage is frozen at **22 October 2026**. The review is a guide to the field, while the derivations here are organized by physical concepts. Essential calculations are developed in the chapters rather than delegated to references. The source map records which papers support each connection and where the distinctions remain unsettled.

The newest experiments and theoretical frameworks are treated at their demonstrated scope. A finite processor can show signatures and finite-size scaling; an exact thermodynamic theorem is a different kind of evidence. A

field this active has competing definitions and useful counterexamples. Understanding those boundaries is part of mastering the subject.

# Part I

# I · Foundations

# 1 Quantum states, ensembles, and locality

## 1.1 What a state must predict

A theory of a noisy quantum magnet must answer two kinds of questions. What is the probability of a measurement outcome on one spin? What information is shared by distant regions, possibly only after conditioning on the rest? A wavefunction can answer both questions for an isolated pure system. A density matrix can answer them when the system is entangled with an inaccessible environment or prepared with classical randomness. SW-SSB concerns properties of that density matrix which ordinary expectation values can miss.

We begin with finite-dimensional systems and set $\hbar = k_B = 1$. Sites form a lattice of linear size $L$, containing $N$ sites. Each site has a finite-dimensional Hilbert space. Thermodynamic limits always refer to a sequence of such finite systems; they are not an instruction to manipulate an infinite matrix.

### 1.1.1 Vectors and operators

A pure state is a normalized vector $|\psi\rangle$ in a complex vector space $\mathcal{H}$, with vectors differing by an overall phase representing the same physical state. In an orthonormal basis,

$$|\psi\rangle = \sum_a \psi_a |a\rangle, \qquad \sum_a |\psi_a|^2 = 1.$$

A linear operator $A$ maps vectors to vectors. Its adjoint satisfies $(A^\dagger)_{ab} = A^*_{ba}$. A Hermitian operator $A = A^\dagger$ represents an observable, with real eigenvalues. A unitary operator $U^\dagger U = I$ preserves inner products. The trace is $\mathrm{Tr}\, A = \sum_a A_{aa}$, independent of basis, and obeys $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. Cyclicity permits cyclic permutations of factors, not arbitrary reorderings.

For a spin one-half we use

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy $X^2 = Y^2 = Z^2 = I$, $XY = iZ$, and $\{X, Z\} = 0$. The $Z$ eigenvectors are $|0\rangle, |1\rangle$; the $X$ eigenvectors are $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Subscripts indicate sites: $X_i$ is the identity everywhere except site $i$.

An important identity is

$$\mathrm{Tr}(|\psi\rangle\langle\phi|\, A) = \langle\phi|A|\psi\rangle.$$

It converts familiar wavefunction expressions into operator expressions.

## 1.2 Why density matrices are necessary

Suppose preparation $a$ occurs with probability $p_a$ and produces $|\psi_a\rangle$. The probability of a projective measurement outcome with projector $P$ is

$$\sum_a p_a \langle\psi_a|P|\psi_a\rangle = \mathrm{Tr}(\rho P), \qquad \rho = \sum_a p_a |\psi_a\rangle\langle\psi_a|. \tag{1.1}$$

Every density matrix obeys three conditions:

1. $\rho = \rho^\dagger$;
2. $\rho \geq 0$, meaning $\langle v|\rho|v\rangle \geq 0$ for every vector;
3. $\mathrm{Tr}\,\rho = 1$.

Conversely, a matrix with these properties has a spectral decomposition

$$\rho = \sum_\alpha \lambda_\alpha |\alpha\rangle\langle\alpha|, \qquad \lambda_\alpha \geq 0, \quad \sum_\alpha \lambda_\alpha = 1,$$

and therefore describes a possible statistical preparation. The support of $\rho$ is the span of eigenvectors with positive eigenvalue. A projector onto the support is denoted $P_\rho$.

For any observable,

$$\langle A\rangle_\rho = \mathrm{Tr}(\rho A). \tag{1.2}$$

A state is pure precisely when $\rho^2 = \rho$, or equivalently $\mathrm{Tr}\,\rho^2 = 1$. Otherwise $0 < \mathrm{Tr}\,\rho^2 < 1$. The quantity $\mathrm{Tr}\,\rho^2$ is its purity. For dimension $d$, the minimum purity is $1/d$, attained by $I/d$.

### 1.2.1 Worked example: the Bloch ball

Any qubit density matrix can be written

$$\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z).$$

Because $(\mathbf{r}\cdot\sigma)^2 = |\mathbf{r}|^2 I$, the eigenvalues are $(1 \pm |\mathbf{r}|)/2$. Positivity is equivalent to $|\mathbf{r}| \leq 1$. The boundary sphere contains pure states; its interior contains mixtures. Direct multiplication gives

$$\mathrm{Tr}\,\rho^2 = \frac{1 + |\mathbf{r}|^2}{2}.$$

The maximally mixed state $I/2$ admits both decompositions

$$\frac{I}{2} = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} = \frac{|+\rangle\langle +| + |-\rangle\langle -|}{2}.$$

Thus an ensemble is not determined by its density matrix. A property of a chosen preparation record is not automatically a property of the state. This distinction becomes decisive when comparing trajectory order with intrinsic SW-SSB.

## 1.3 Composite systems and reduced states

For subsystems $A, B$, the joint space is $\mathcal{H}_A \otimes \mathcal{H}_B$. If $A$ has basis $|a\rangle$ and $B$ has basis $|b\rangle$, a general vector is

$$|\Psi\rangle = \sum_{a,b} \Psi_{ab} |a\rangle \otimes |b\rangle.$$

A product vector factorizes as $\psi_a \phi_b$. A vector that cannot factorize is entangled. A mixed state is separable if it is a convex combination of product density matrices; otherwise it is entangled. Classical correlations need not be entanglement.

If measurements act only on $A$, the appropriate state is the partial trace

$$\rho_A = \mathrm{Tr}_B \, \rho_{AB}, \qquad (\rho_A)_{aa'} = \sum_b (\rho_{AB})_{ab,a'b}. \tag{1.3}$$

This definition is forced by $\mathrm{Tr}_{AB}[\rho_{AB}(O_A \otimes I_B)] = \mathrm{Tr}_A(\rho_A O_A)$ for every $O_A$. It discards correlations accessible only by joint measurements. It does not describe measuring $B$ and retaining its outcome.

### 1.3.1 Worked example: a Bell pair and a classical coin

For $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, write the projector as four outer products. The cross terms vanish on tracing $B$, because $\mathrm{Tr}(|0\rangle\langle 1|) = 0$. Consequently $\rho_A = I/2$.

The classical state

$$\rho_{\text{coin}} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$$

has exactly the same reduced state. The two joint states are distinguished by $\langle X_A X_B \rangle$: it equals one for the Bell pair and zero for the coin. A subsystem's mixedness alone does not tell us whether its uncertainty originates in entanglement or classical preparation.

For the $N$-spin cat state

$$|\text{cat}\rangle = \frac{|0\rangle^{\otimes N} + |1\rangle^{\otimes N}}{\sqrt{2}},$$

tracing out even one spin removes the relative-phase coherence. Every proper reduced state equals that of the corresponding classical coin. Distinguishing these states globally requires access to a nonlocal observable such as $\prod_i X_i$.

## 1.4 Matrix functions and purification

For a positive matrix $\rho$, define $f(\rho) = \sum_\alpha f(\lambda_\alpha)|\alpha\rangle\langle\alpha|$. In particular $\sqrt{\rho}$ is the unique positive square root. Matrix square roots generally do not distribute over products: $\sqrt{AB} \neq \sqrt{A}\sqrt{B}$ unless suitable commutativity conditions hold.

Every density matrix can be obtained by discarding an auxiliary system. Starting from its spectral decomposition, define

$$|\Psi_\rho\rangle_{SE} = \sum_\alpha \sqrt{\lambda_\alpha} |\alpha\rangle_S |\alpha\rangle_E. \tag{1.4}$$

The norm is one. The orthogonality of the environment basis gives $\mathrm{Tr}_E |\Psi_\rho\rangle\langle\Psi_\rho| = \rho$. This is a purification. Applying a unitary on $E$ changes the purification without changing $\rho$.

More generally any bipartite pure vector has a Schmidt decomposition

$$|\Psi\rangle = \sum_{\alpha=1}^{r} \sqrt{\lambda_\alpha} |u_\alpha\rangle_A |v_\alpha\rangle_B.$$

It follows from the singular-value decomposition of the coefficient matrix $\Psi_{ab}$. The two reduced states have identical nonzero spectra. The Schmidt rank $r$ is one for a product vector; a large $r$ allows entanglement but its distribution of weights determines how much.

In a fixed basis, one particularly useful purification is

$$|\sqrt{\rho}\rangle\!\rangle = \sum_{a,b} (\sqrt{\rho})_{ab} |a\rangle_L |b\rangle_R. \tag{1.5}$$

Its norm is $\mathrm{Tr}\,\rho = 1$. We introduce it now to make its ingredients familiar; its relation to doubled symmetry and Rényi-1 order is developed in Chapter 7. The right factor transforms in the conjugate representation under a change of basis. Forgetting this conjugation produces incorrect symmetry generators.

## 1.5 Local observables and thermodynamic statements

An operator is local if its support lies in a region of bounded diameter as $L \to \infty$. A sum of local terms, such as a Hamiltonian $H = \sum_i h_i$, is an extensive operator; it is not itself local. A global symmetry $U = \bigotimes_i u_i$ has support across the entire system even when each factor is on-site.

Two sequences of states can be globally orthogonal while becoming indistinguishable by every fixed local measurement. For example, the two parity-sector states

$$\rho_\pm = \frac{I \pm \prod_i X_i}{2^N} \tag{1.6}$$

have orthogonal supports. Yet every proper subsystem has reduced state $I/2^{|A|}$ in both sectors, since tracing out an omitted factor gives $\mathrm{Tr}\, X = 0$. Global charge information is entirely invisible on a proper region. These states will provide the simplest exact SW-SSB model.

This example explains why the order of limits matters. A global fidelity computed before tracing is different from a fidelity of reduced states after tracing. Both are meaningful; they answer different operational questions.

## 1.6 Connected correlations and clustering

For operators supported near $i$ and $j$, define

$$\langle A_i B_j \rangle_c = \mathrm{Tr}(\rho A_i B_j) - \mathrm{Tr}(\rho A_i)\,\mathrm{Tr}(\rho B_j).$$

Exponential clustering means that the magnitude is bounded by $c\|A_i\|\|B_j\| e^{-r/\xi}$, with $r = |i-j|$ and a correlation length $\xi$ independent of system size. Algebraic decay $r^{-\eta}$ has no finite exponential correlation length. A nonzero limit is long-range order.

For the parity state in Equation 1.6, all fixed-support connected correlators vanish once their total support omits a site. Nonetheless a global constraint survives. SW-SSB is not an exception to the Born rule: it uses nonlinear functions of the density matrix to expose a kind of organization that fixed-order linear observables do not detect.

## 1.7 Exercises and worked solutions

### 1.7.1 1. Positivity is stronger than positive diagonal entries

Consider $M = \frac{1}{2}\begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$ with real $a$. For which values is $M$ a state?

> **Solution**
>
> Its eigenvectors are $|\pm\rangle$ with eigenvalues $(1 \pm a)/2$. Hermiticity and trace one hold for every real $a$, but positivity requires $-1 \leq a \leq 1$. A matrix with apparently reasonable occupation probabilities can fail to define a physical state because its off-diagonal coherence is too large.

### 1.7.2 2. A mixed state has many ensembles

Show that the uniform mixture of the four pure states $|\pm x\rangle, |\pm y\rangle$ is $I/2$. Does the preparation record follow from the density matrix?

> **Solution**
>
> The two $x$ projectors sum to $I$ and the two $y$ projectors also sum to $I$. Dividing their sum by four gives $I/2$. All measurements on the qubit alone reproduce the same probabilities as a $z$-basis coin. The record distinguishing these preparations is additional physical information, typically stored in an environment or a classical register.

### 1.7.3 3. Entanglement versus correlation

Compute $\langle Z_A Z_B\rangle$, $\langle Z_A\rangle$, and $\langle X_A X_B\rangle$ for the Bell state and the classical coin. Which measurement distinguishes them?

> **Solution**
>
> Both have $\langle Z_A Z_B\rangle = 1$ and zero one-point $Z$ expectations. For the Bell state, $X_A X_B$ exchanges the two terms of the vector and hence has expectation one. For the coin, each diagonal basis vector has zero expectation of $X_A X_B$, giving zero. Thus correlated $Z$ outcomes alone do not certify entanglement; the complementary-basis correlation distinguishes these two examples.

### 1.7.4 4. The invisible parity bit

Prove normalization and positivity of $\rho_\pm$ and compute its purity and every proper reduced state.

> **Solution**
>
> Let $P_\pm = (I \pm U)/2$, where $U = \prod_i X_i$. Since $U^2 = I$, these are orthogonal projectors of rank $2^{N-1}$. Therefore $\rho_\pm = P_\pm/2^{N-1}$ is positive and normalized. Its purity is $2^{1-N}$. If at least one site is traced out, the $U$ term vanishes. The identity term gives $\rho_A = I_A/2^{|A|}$. The sectors are globally distinguishable by measuring $U$ but locally indistinguishable on every proper subset.

### 1.7.5 5. Basis conventions in a doubled space

For the unnormalized vector $|I\rangle\!\rangle = \sum_a |a\rangle_L |a\rangle_R$, show $(A \otimes I)|I\rangle\!\rangle = (I \otimes A^T)|I\rangle\!\rangle$.

**Solution**

The first side is $\sum_{ab} A_{ba}|b\rangle_L|a\rangle_R$. The second is $\sum_{ab}(A^T)_{ba}|a\rangle_L|b\rangle_R = \sum_{ab} A_{ab}|a\rangle_L|b\rangle_R$, identical after exchanging dummy indices. A transpose, rather than an adjoint, arises because coefficients of kets have not been complex conjugated. This identity fixes all later vectorization rules.

# 2 Statistical mechanics and spontaneous symmetry breaking

## 2.1 The puzzle of a symmetric magnet

A finite ferromagnet with no magnetic field has a spin-flip symmetric Hamiltonian and a spin-flip symmetric Gibbs state. Its magnetization is exactly zero. Nevertheless a macroscopic magnet points in a definite direction. Resolving this apparent contradiction introduces the thermodynamic limit, selection of a state by an arbitrarily small perturbation, and long-range correlations. SW-SSB keeps these lessons but changes which correlations carry the order.

An extensive thermodynamic quantity scales as $N$. An intensive quantity stays finite as $N \to \infty$. Energy density, entropy density, and magnetization per site are intensive. A phase is a stable family of macroscopic behavior, rather than an isolated density matrix with an unusual identity.

## 2.2 Deriving the Gibbs ensemble

For probabilities $p_a$ over energy eigenstates with energies $E_a$, maximize the Shannon entropy $-\sum_a p_a \ln p_a$ subject to normalization and a specified mean energy. With multipliers $\alpha, \beta$, stationarity gives

$$-\ln p_a - 1 - \alpha - \beta E_a = 0, \qquad p_a = \frac{e^{-\beta E_a}}{Z}, \quad Z = \sum_a e^{-\beta E_a}.$$

In quantum mechanics, maximizing $S(\rho) = -\operatorname{Tr}(\rho \ln \rho)$ yields the same result in operator form,

$$\rho_\beta = \frac{e^{-\beta H}}{Z}, \qquad F_{\text{th}} = -T \ln Z. \tag{2.1}$$

Here $F_{\text{th}}$ denotes thermodynamic free energy, not the root fidelity $F$ used later. Differentiation gives

$$\langle H \rangle = -\partial_\beta \ln Z, \qquad \partial_\beta^2 \ln Z = \operatorname{Var}(H) \geq 0.$$

The entropy is $S = \beta \langle H \rangle + \ln Z$. These identities illustrate a recurring principle: derivatives of a generating function measure fluctuations.

### 2.2.1 Fixed charge and fluctuating charge

Suppose $[H, Q] = 0$ and $P_q$ projects onto a charge eigenspace. A fixed-charge, or canonical-with-respect-to-$Q$, ensemble is

$$\rho_{\beta,q} = \frac{P_q e^{-\beta H}}{Z_q}, \qquad Z_q = \operatorname{Tr}(P_q e^{-\beta H}). \tag{2.2}$$

The grand-canonical ensemble is

$$\rho_{\beta,\mu} = \frac{e^{-\beta(H-\mu Q)}}{Z_{\text{gc}}} = \sum_q \frac{e^{\beta \mu q} Z_q}{Z_{\text{gc}}} \rho_{\beta,q}. \tag{2.3}$$

The distinction is physical: an environment may exchange energy while preserving charge, or may exchange both. Both states commute with $Q$. Only the first has support in one charge sector. This is the seed of strong versus weak symmetry.

Ensemble equivalence means that local observables agree in suitable thermodynamic limits, typically away from pathologies such as phase coexistence, long-range nonadditive interactions, or anomalous constraints. It does not mean that the global density matrices approach one another in trace distance. Measuring total charge still distinguishes an exactly fixed sector from a distribution of sectors.

### 2.2.2 Worked example: a fixed number of occupied sites

Take $N$ two-level sites with occupation $n_i = 0, 1$. The infinite-temperature state at fixed $Q = \sum_i n_i = M$ is the uniform distribution over $\binom{N}{M}$ configurations. By symmetry,

$$\langle n_i \rangle = \nu = \frac{M}{N}, \qquad \langle n_i n_j \rangle = \frac{M(M-1)}{N(N-1)} \quad (i \neq j).$$

Thus

$$\langle n_i n_j \rangle_c = -\frac{\nu(1-\nu)}{N-1}.$$

Pairwise correlations vanish at fixed separation as $N \to \infty$, yet summing all of them cancels the on-site variances exactly so that $\mathrm{Var}(Q) = 0$. A block of $\ell$ sites has a hypergeometric occupation distribution and

$$\mathrm{Var}(Q_A) = \ell\nu(1-\nu)\frac{N-\ell}{N-1}. \tag{2.4}$$

For $\ell \ll N$, this approaches the grand-canonical binomial result. For $\ell = N$, it vanishes. Subsystem fluctuations and a fixed total charge are fully compatible. Charge scrambling later concerns additional information beyond this variance alone.

## 2.3 Conventional symmetry and selection rules

Let $U$ be a unitary symmetry, $UHU^\dagger = H$. For a finite Gibbs state, $U\rho U^\dagger = \rho$. If an observable is charged, $UOU^\dagger = e^{i\theta}O$ with $e^{i\theta} \neq 1$, then cyclicity gives

$$\mathrm{Tr}(\rho O) = \mathrm{Tr}(U\rho U^\dagger O) = e^{-i\theta}\,\mathrm{Tr}(\rho O) = 0. \tag{2.5}$$

For the Ising symmetry $U = \prod_i X_i$, $Z_i$ is odd. A symmetric state must have $\langle Z_i \rangle = 0$. Its two-point function $\langle Z_i Z_j \rangle$ is neutral and can remain nonzero. A symmetric representative of a broken phase can therefore display order through correlations even when its one-point function vanishes.

## 2.4 The order of limits

Consider the classical Ising Hamiltonian

$$H_h = -J\sum_{\langle ij \rangle} Z_i Z_j - h\sum_i Z_i, \qquad J > 0.$$

At finite $N$, the partition function is a finite sum of analytic functions, and $m_N(0) = N^{-1}\langle \sum_i Z_i \rangle = 0$. Spontaneous magnetization is defined by

$$m_* = \lim_{h\to 0^+} \lim_{N\to\infty} m_N(h). \tag{2.6}$$

Reversing the limits gives zero. A useful two-state approximation deep in an ordered phase retains only the two uniformly magnetized configurations. Their weights are proportional to $e^{\pm\beta hNm_0}$, giving

$$m_N(h) \simeq m_0 \tanh(\beta h N m_0).$$

A field of order $1/N$ can select a branch. The approximation neglects domain walls and is justified only when their effects are sufficiently suppressed. Its purpose is to expose the nonuniformity of the limit, not to calculate a critical temperature.

In the symmetric mixture of the two branches,

$$\rho_{\text{coin}} = \tfrac{1}{2}|0\cdots0\rangle\langle 0\cdots0| + \tfrac{1}{2}|1\cdots1\rangle\langle 1\cdots1|,$$

we have $\langle Z_i\rangle = 0$ but $\langle Z_i Z_j\rangle = 1$. In either extremal branch the one-point magnetization is nonzero and the connected two-point function clusters. Confusing a symmetric mixture with a selected branch obscures what a nondecaying connected correlator means.

## 2.5 Exact one-dimensional Ising model

The zero-field nearest-neighbor chain provides a complete example of how fluctuations destroy finite-temperature order. For a periodic chain, introduce the transfer matrix

$$T_{s,s'} = e^{Kss'}, \qquad T = \begin{pmatrix} e^{K} & e^{-K} \\ e^{-K} & e^{K} \end{pmatrix}, \quad K = \beta J.$$

Its normalized eigenvectors are the symmetric and antisymmetric combinations, with eigenvalues $\lambda_+ = 2\cosh K$ and $\lambda_- = 2\sinh K$. The partition function is $Z_N = \lambda_+^N + \lambda_-^N$. Insertions of $Z$ exchange the two transfer-matrix eigenvectors. In the infinite chain this gives

$$\langle Z_i Z_{i+r}\rangle = \left(\frac{\lambda_-}{\lambda_+}\right)^r = (\tanh K)^r = e^{-r/\xi}, \qquad \xi^{-1} = -\ln\tanh K. \tag{2.7}$$

For every finite $K$, $\xi$ is finite. Only at $T = 0$ does it diverge. At low temperature $\xi \simeq e^{2\beta J}/2$.

The domain-wall explanation gives the same scale. Flipping all spins to one side of a bond costs $2J$. At nonzero temperature a domain wall has nonzero density of order $e^{-2\beta J}$. Distant spins are separated by many independently fluctuating walls, so their relative orientation becomes random. A finite chain shorter than $\xi$ appears ordered; increasing $L$ reveals that it is a crossover.

For diagonal classical states, nonlinear overlaps also reduce to statistical-mechanical sums. The transfer matrix will reappear in the decoherence model, but its effective coupling and its boundary insertions must be derived afresh. A resemblance to an Ising partition function does not determine the physical transition automatically.

## 2.6 Landau theory and its limits

For a uniform Ising order parameter $m$, symmetry permits an even free-energy density

$$f(m) = f_0 + \frac{a}{2}m^2 + \frac{b}{4}m^4 - hm, \qquad b > 0.$$

At $h = 0$, stationarity requires $m(a + bm^2) = 0$. For $a > 0$, $m = 0$ minimizes $f$. For $a < 0$, the minima are $m = \pm\sqrt{-a/b}$. The susceptibility in the symmetric phase is $\chi = \partial m/\partial h = 1/a$.

Allow spatial variation by adding $\kappa(\nabla m)^2/2$. In the Gaussian symmetric regime,

$$\langle |m_{\mathbf{k}}|^2 \rangle \propto \frac{T}{a + \kappa k^2}, \qquad \xi = \sqrt{\kappa/a}.$$

Landau theory assumes a local analytic functional in a suitable order parameter and neglects sufficiently strong critical fluctuations. It predicts mean-field exponents, not universal exact exponents in arbitrary dimension. Disorder, topology, conservation laws, and nonlinear replica structures can require different theories. In particular, a purity-normalized observable may change singularly because it reweights the spectrum of $\rho$, even when the intrinsic mixed-state phase does not change.

### 2.6.1 Continuous symmetry and the infrared

For a continuous order parameter with phase $\theta(\mathbf{x})$, the low-energy energy cost is approximately $\frac{\rho_s}{2} \int d^d x \, (\nabla\theta)^2$. Thermal fluctuations give

$$\langle [\theta(\mathbf{r}) - \theta(0)]^2 \rangle \propto \frac{T}{\rho_s} \int \frac{d^d k}{(2\pi)^d} \frac{1 - \cos(\mathbf{k} \cdot \mathbf{r})}{k^2}. \tag{2.8}$$

For $d = 1$ this grows linearly in $r$; for $d = 2$ logarithmically. Gaussian averaging of $e^{i[\theta(\mathbf{r}) - \theta(0)]}$ then gives exponential or algebraic decay. The rigorous Mermin-Wagner theorem concerns continuous symmetries, locality assumptions, and equilibrium states, not all possible nonlinear observables [1]. In two dimensions, vortex binding can produce a Berezinskii-Kosterlitz-Thouless phase with power-law correlations. Later SW-SSB analogues must specify which doubled-space state and which finite-time limit inherit such infrared physics.

## 2.7 Quantum ground-state order

For the transverse-field Ising chain,

$$H = -J \sum_i Z_i Z_{i+1} - g \sum_i X_i,$$

the two terms compete. At $g \gg J$, the ground state approaches $|+\rangle^{\otimes N}$. At $J \gg g$, its low-energy space is spanned approximately by $|0\rangle^{\otimes N}$ and $|1\rangle^{\otimes N}$. At finite size, tunneling combines these into even and odd cats. Connecting the branches requires acting on all spins, so the splitting is exponentially small in system size in the ordered regime. A local perturbation can select a branch before that tunneling matters.

This pure-state order differs from the state $\rho_+ = (I + \prod_i X_i)/2^N$. Both have an exact even global charge, but the cat has $\langle Z_i Z_j \rangle = 1$ while $\rho_+$ has zero ordinary two-point correlations. The latter is not an ordinary ferromagnet. Understanding how it can still carry a robust form of order is the task of Part II.

## 2.8 What must survive in a new notion of SSB?

A useful extension should have a charged diagnostic, a thermodynamic long-distance limit, a response to suitable symmetry-breaking perturbations, and stability under physically appropriate local deformations. These conditions need not take the same mathematical form as equilibrium ground-state SSB. A dissipative channel can destroy information, so channel reachability has a direction. Exact global constraints can be locally invisible. Mixed states have no unique ensemble decomposition. Each fact forces a refinement of the conventional picture.

## 2.9 Exercises and worked solutions

### 2.9.1 1. Susceptibility and fluctuations

For a classical Gibbs distribution with energy $H_h = H_0 - hM$, derive $\partial_h \langle M \rangle = \beta \operatorname{Var}(M)$. What changes for a quantum $M$ not commuting with $H_0$?

**Solution**

Differentiate the numerator and denominator of $\langle M \rangle = Z^{-1} \sum_s M(s) e^{-\beta H_h(s)}$. The two terms give $\beta \langle M^2 \rangle - \beta \langle M \rangle^2$. For noncommuting quantum operators, differentiating an exponential requires the Duhamel integral: $\partial_h e^{-\beta H_h} = \int_0^\beta d\tau \, e^{-(\beta-\tau)H_h} M e^{-\tau H_h}$. The susceptibility is an imaginary-time integrated connected correlator. Replacing it by $\beta$ times the equal-time variance is generally incorrect.

### 2.9.2 2. Finite-size symmetry selection

Using the two-state expression for $m_N(h)$, take $h = c/N$, then $h = c/N^2$, then fixed positive $h$, as $N \to \infty$.

**Solution**

The arguments of the hyperbolic tangent tend to $\beta c m_0$, zero, and infinity respectively. The magnetizations tend to $m_0 \tanh(\beta c m_0)$, zero, and $m_0$. The thermodynamic limit depends on the scaling of the perturbation. A finite-size numerical calculation must specify that scaling before it is interpreted as spontaneous selection.

### 2.9.3 3. A constraint hidden in weak correlations

Derive Equation 2.4 from the on-site variance and the off-site covariance.

**Solution**

There are $\ell$ diagonal terms, each $\nu(1-\nu)$, and $\ell(\ell-1)$ ordered off-diagonal terms, each $-\nu(1-\nu)/(N-1)$. Their sum is $\ell\nu(1-\nu)[1-(\ell-1)/(N-1)]$. Every individual covariance is $O(1/N)$, but there are $O(N^2)$ such terms for the whole system. Dropping them before summing would falsely predict a fluctuating total charge.

### 2.9.4 4. Correlation length and apparent order

At $\beta J = 3$, estimate the one-dimensional correlation length. Would a chain of length 30 reliably distinguish this finite-temperature state from a perfectly ordered state through short-distance correlations?

**Solution**

The low-temperature estimate gives $\xi \simeq e^6/2 \simeq 202$. At separation 30, $e^{-30/202} \simeq 0.862$, still large. The finite chain could look strongly ordered. Establishing long-range order requires varying both size and separation beyond the inferred scale, rather than interpreting a large finite-size correlator as a nonzero asymptote.

### 2.9.5 5. Canonical and grand-canonical are globally different

Show that $\rho_{\beta,q}$ and $\rho_{\beta,\mu}$ can agree locally while a measurement of $P_q$ distinguishes them with probability approaching one.

**Solution**

The fixed-sector state returns outcome $q$ with probability one. In the grand-canonical state it returns $w_q = e^{\beta\mu q} Z_q / Z_{\mathrm{gc}}$. When a central-limit distribution spans $O(\sqrt{N})$ sectors, an individual typical sector has weight $O(N^{-1/2})$. Thus the two-outcome test $\{P_q, I - P_q\}$ distinguishes them increasingly well even when every bounded local observable agrees. Local ensemble equivalence is not a statement of global trace-norm convergence.

# 3 Quantum channels and open-system dynamics

## 3.1 From inaccessible degrees of freedom to a map

A system need not be thermally equilibrated to be mixed. An unobserved photon can carry away phase information; a random unitary can produce classical uncertainty; a measurement can leave a record that is discarded. The common description is a quantum channel, a map taking input density matrices to output density matrices.

Assume initially that system and environment are uncorrelated, $\rho \otimes |0\rangle\langle 0|_E$. Let a joint unitary $V$ act, and discard $E$:

$$\mathcal{E}(\rho) = \mathrm{Tr}_E \left[ V(\rho \otimes |0\rangle\langle 0|_E) V^\dagger \right]. \tag{3.1}$$

The initial-product assumption matters. If the input is correlated with $E$, a single channel on arbitrary system states need not describe the reduced dynamics. The environment can always be enlarged to purify its initially mixed state.

Choose an orthonormal basis $|a\rangle_E$ and define $K_a = {}_E\langle a|V|0\rangle_E$. Inserting this basis into the partial trace gives

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \qquad \sum_a K_a^\dagger K_a = I. \tag{3.2}$$

The second identity follows from $V^\dagger V = I$ and guarantees trace preservation. Each term is positive on positive inputs. The same construction remains positive when an arbitrary spectator system is included. This is complete positivity. A completely positive trace-preserving map is abbreviated CPTP.

Ordinary positivity is insufficient because the input system may be entangled with a spectator. Matrix transposition preserves positivity on a single system but is not completely positive. Applied to one half of a Bell pair, it produces a matrix with a negative eigenvalue. It cannot be a deterministic physical channel.

Kraus operators are not unique. Replacing $K_a$ by $\widetilde{K}_b = \sum_a w_{ba} K_a$ with an isometry $w$ gives the same channel. Thus a particular list of Kraus outcomes is additional structure until a physical measurement of the environment selects that list.

## 3.2 Measurement, selection, and discarded records

A measurement instrument consists of completely positive maps $\mathcal{E}_a$ with $\sum_a \mathcal{E}_a$ trace preserving. Outcome $a$ has probability $p_a = \mathrm{Tr}\, \mathcal{E}_a(\rho)$, and the conditioned state is $\rho_a = \mathcal{E}_a(\rho)/p_a$. The map $\rho \mapsto \rho_a$ is nonlinear because of this normalization. The unconditioned state is the linear average $\sum_a p_a \rho_a$.

For a projective $Z$ measurement on a qubit,

$$\mathcal{M}_Z(\rho) = P_0 \rho P_0 + P_1 \rho P_1 = \frac{1}{2}(\rho + Z\rho Z).$$

If the input is $|+\rangle$, either conditioned state is pure, while the discarded-record output is $I/2$. A monitored trajectory and its record-averaged density matrix are different physical objects. They may undergo different phase transitions.

To keep the record explicitly, introduce a classical register:

$$\rho_{SC} = \sum_a p_a \rho_a \otimes |a\rangle\langle a|_C.$$

Tracing $C$ produces the unconditioned state. Reading $C$ can reveal correlations absent from the density matrix of $S$ alone. Later comparisons with charge sharpening, spin-glass-like trajectory order, and decoding will always specify which records are available.

## 3.3 Three elementary channels

### 3.3.1 Dephasing

With probability $p$ apply $Z$:

$$\mathcal{D}_p(\rho) = (1-p)\rho + pZ\rho Z. \tag{3.3}$$

In the $Z$ basis, diagonal entries remain fixed and off-diagonal entries are multiplied by $1-2p$. The Bloch vector transforms as $(r_x, r_y, r_z) \mapsto ((1-2p)r_x, (1-2p)r_y, r_z)$. At $p = 1/2$ all coherence in this basis disappears. At $p = 1$ the map is a reversible unitary, so a noise parameter need not be globally monotone unless its domain is specified. We usually take $0 \le p \le 1/2$.

Composition gives

$$\mathcal{D}_p\mathcal{D}_q = \mathcal{D}_{p+q-2pq}.$$

The attenuation factors multiply. This simple identity lets us translate finite-time dephasing into a noise probability.

### 3.3.2 Amplitude damping

Spontaneous relaxation from $|1\rangle$ to $|0\rangle$ is described by

$$K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|, \qquad K_1 = \sqrt{\gamma}|0\rangle\langle 1|.$$

Then $\rho_{11} \mapsto (1-\gamma)\rho_{11}$ and $\rho_{01} \mapsto \sqrt{1-\gamma}\rho_{01}$. The maximally mixed state does not remain maximally mixed. A channel satisfying $\mathcal{E}(I) = I$ is called unital; amplitude damping is not unital. Unitality and trace preservation are different conditions.

### 3.3.3 Erasure and replacement

The map $\mathcal{R}(\rho) = \tau \operatorname{Tr}\rho$ replaces every input by a fixed state $\tau$. It is CPTP and destroys all input information. If arbitrary one-way local channels defined phase equivalence, every state could be sent to a trivial product state. Any useful mixed-state phase definition therefore needs additional conditions, such as a reverse channel, local reversibility, or symmetry constraints.

## 3.4 The Choi representation

Let $d = \dim \mathcal{H}$ and $|\Phi\rangle = d^{-1/2}\sum_i |i\rangle|i\rangle$. Define

$$J_{\mathcal{E}} = (\mathcal{E} \otimes \mathrm{id})(|\Phi\rangle\langle\Phi|).$$

Complete positivity is equivalent to $J_{\mathcal{E}} \ge 0$, and trace preservation is equivalent to $\operatorname{Tr}_{\text{out}} J_{\mathcal{E}} = I_{\text{in}}/d$. Conversely,

$$\mathcal{E}(A) = d \operatorname{Tr}_{\text{in}}\left[J_{\mathcal{E}}(I \otimes A^T)\right].$$

This follows by expanding $|\Phi\rangle\langle\Phi| = d^{-1}\sum_{ij}|i\rangle\langle j|\otimes|i\rangle\langle j|$. It provides a finite matrix test for physicality and makes clear that a channel has more structure than its action on one selected state.

The Choi matrix of a channel should not be confused with vectorizing a density matrix as $|\rho\rangle\rangle$. Both use doubled spaces, but one represents a map and the other a state. The canonical vector $|\sqrt{\rho}\rangle\rangle$ is a third construction, with yet another normalization.

## 3.5 Continuous time and the Lindblad equation

Markovian time-homogeneous dynamics assumes $\mathcal{E}_{t+s} = \mathcal{E}_t\mathcal{E}_s$ and $\mathcal{E}_0 = \mathrm{id}$. Under the usual continuity assumptions in finite dimension, $\mathcal{E}_t = e^{t\mathcal{L}}$ with a generator of the Gorini-Kossakowski-Sudarshan-Lindblad form [2,3]:

$$\partial_t\rho = \mathcal{L}(\rho) = -i[H,\rho] + \sum_a \left(L_a\rho L_a^\dagger - \frac{1}{2}\{L_a^\dagger L_a,\rho\}\right). \tag{3.4}$$

The $L_a$ are jump operators; rates may be absorbed into their normalization. The anticommutator terms compensate the probability lost from no-jump evolution.

To see the structure, take a small step $dt$ with

$$K_0 = I - dt\left(iH + \frac{1}{2}\sum_a L_a^\dagger L_a\right), \qquad K_a = \sqrt{dt}\,L_a.$$

Then $\sum_\alpha K_\alpha^\dagger K_\alpha = I + O(dt^2)$. Expanding the Kraus sum to first order gives Equation 3.4. The exact exponential is CPTP. A first-order Euler step is only an approximation and need not stay positive for a large step.

Trace preservation follows directly by cyclicity. Hermiticity is also preserved. The nontrivial positivity guarantee comes from the generator's structure, not from these two easier checks.

### 3.5.1 Worked example: continuous dephasing

For $L = \sqrt{\kappa}Z$, the master equation is $\dot\rho = \kappa(Z\rho Z - \rho)$. The off-diagonal equation is $\dot\rho_{01} = -2\kappa\rho_{01}$, so

$$\rho_{01}(t) = e^{-2\kappa t}\rho_{01}(0), \qquad p(t) = \frac{1-e^{-2\kappa t}}{2}.$$

The diagonal entries remain fixed. Complete dephasing is reached only at infinite time. A collective or many-body transition may nevertheless occur at a finite $p_c < 1/2$, hence at a finite time. Conversely, in some one-dimensional models only the infinite-time limit has long-range nonlinear order.

## 3.6 Heisenberg evolution and conservation laws

The adjoint generator is defined by $\mathrm{Tr}[O\mathcal{L}(\rho)] = \mathrm{Tr}[\mathcal{L}^\dagger(O)\rho]$. It is

$$\mathcal{L}^\dagger(O) = i[H,O] + \sum_a \left(L_a^\dagger O L_a - \frac{1}{2}\{L_a^\dagger L_a, O\}\right).$$

An observable $Q$ is conserved in mean for all states if $\mathcal{L}^\dagger(Q) = 0$. The sufficient conditions $[H,Q] = [L_a,Q] = 0$ are stronger: each jump preserves every charge sector. That is the dynamical strong-symmetry condition relevant to this book. Covariance under conjugation by a symmetry is weaker and can allow jumps that exchange charge with the environment.

For example, local raising and lowering jumps can preserve covariance under $e^{i\theta Q}$ because their phases cancel between $L_a$ and $L_a^\dagger$, yet they do not preserve fixed-$Q$ sectors. These distinctions are developed precisely in Chapter 5.

## 3.7 Stationary states, gaps, and mixing

A stationary state satisfies $\mathcal{L}(\rho_*) = 0$. A stationary manifold may contain multiple states, especially if sectors cannot communicate. Uniqueness must be stated either globally or within a selected sector.

The spectrum of $\mathcal{L}$ lies in the closed left half-plane for a finite CPTP semigroup. A positive spectral gap is the smallest nonzero decay rate $-\operatorname{Re}\lambda$ after stationary and other nondecaying modes are treated appropriately. It is tempting to equate its inverse with equilibration time. That can fail.

Define a trace-distance mixing time within an invariant sector by

$$t_{\text{mix}}(\epsilon) = \inf\left\{ t : \sup_{\rho \text{ in sector}} \frac{1}{2}\|e^{t\mathcal{L}}(\rho) - \rho_*\|_1 \le \epsilon \right\}.$$

Even when eigenvalues are known, coefficients, nonorthogonal eigenvectors, Jordan blocks, and the size of the state space can introduce size-dependent factors. A uniform spectral gap alone does not establish size-independent mixing. This is relevant to proposed links between steady-state SW-SSB and rapid mixing.

## 3.8 Local channels and causal structure

A finite-depth local channel circuit has a bounded number of layers; each layer contains disjoint gates supported on bounded regions. Information can propagate only a bounded distance through a fixed-depth circuit. A local Lindbladian with bounded, short-range terms has an approximate causal cone, with exponentially small tails outside a Lieb-Robinson-type cone. Finite-time dynamics is therefore related to shallow channels, but the desired accuracy may introduce logarithmic corrections in system size.

Depth, range, norm bounds, and error convention must all be specified before claiming that a phase cannot be transformed locally. A circuit of depth proportional to $L$ can communicate across the sample and is outside fixed-depth stability statements.

The environment also matters. A collection of fresh, initially uncorrelated local ancillas is different from preloading a globally correlated environment with the state one wants to prepare. Allowing the latter would insert the order by assumption.

## 3.9 Exercises and worked solutions

### 3.9.1 1. Complete positivity of transposition

Apply transposition to the second qubit of $|\Phi^+\rangle\langle\Phi^+|$ and find the spectrum.

> **Solution**
>
> The partially transposed matrix is half the swap operator: $\frac{1}{2}\sum_{ij}|i\rangle\langle j| \otimes |j\rangle\langle i|$. Swap has eigenvalue $+1$ on the three-dimensional symmetric subspace and $-1$ on the singlet. The spectrum is $1/2, 1/2, 1/2, -1/2$. Hence transposition cannot act as a deterministic channel on an arbitrary subsystem entangled with a spectator.

### 3.9.2 2. An environment realization of dephasing

Construct an isometry producing Equation 3.3, and identify the record basis.

**Solution**

Take $W|\psi\rangle = \sqrt{1-p}|\psi\rangle|0\rangle_E + \sqrt{p}\,Z|\psi\rangle|1\rangle_E$. Orthogonality of the environment states gives $W^\dagger W = I$. Tracing $E$ yields the Kraus sum. Measuring $E$ in this basis reveals which random unitary occurred. Measuring a different basis gives a different unraveling of the same reduced channel.

### 3.9.3 3. Strong symmetry of pair noise

Let $U = \prod_i X_i$ and let each bond jump be $L_{ij} = \sqrt{\kappa} Z_i Z_j$. Show that the sector of $U$ is conserved. Compare a single-site $Z_i$ jump.

**Solution**

Conjugating $Z_i Z_j$ by $U$ produces two minus signs and leaves it unchanged. Thus every pair jump commutes with $U$, preserving each eigenspace in every trajectory. A single $Z_i$ anticommutes with $U$ and flips its eigenvalue. The single-site dissipator is still covariant under $U$ because the two signs in $Z_i \rho Z_i$ cancel. It preserves weak symmetry of the averaged state but exchanges strong-charge sectors.

### 3.9.4 4. What does a gap prove?

Suppose a bound has the form $\|\rho(t) - \rho_*\|_1 \leq C_N e^{-\Delta t}$, with $C_N = e^{aN}$. Estimate the time needed to reach error $\epsilon$ when $\Delta$ is independent of $N$.

**Solution**

It is sufficient to take $t \geq (aN + \ln(1/\epsilon))/\Delta$. Thus this estimate allows a mixing time linear in $N$ despite a constant gap. Better estimates may exist, but the eigenvalue statement alone does not supply them. One must control the prefactor or an appropriate stronger mixing inequality.

### 3.9.5 5. Selection versus averaging

Starting from a two-spin cat, projectively measure $Z_1$. Give the conditioned and unconditioned states and identify which have a definite value of $X_1 X_2$.

**Solution**

The outcomes give $|00\rangle$ or $|11\rangle$ with equal probability. Discarding the outcome gives their classical coin. The input cat is an eigenvector of $X_1 X_2$ with eigenvalue $+1$. Neither selected branch nor the unconditioned coin is supported in that eigenspace. The coin remains invariant under conjugation by $X_1 X_2$, whereas the selected branches do not. This is an explicit channel-induced loss of strong symmetry, distinct from spontaneous breaking diagnosed in a strongly symmetric state.

# 4 Information, distinguishability, and recovery

## 4.1 Why ordinary expectation values are not enough

The density matrix contains all measurable probabilities, but different functions of it ask different questions. An expectation value is linear in $\rho$. Entropy, fidelity, and conditional mutual information are nonlinear. Measuring such quantities usually requires repeated preparations, multiple copies, controlled preparation access, or structural assumptions. Nonlinearity is not a violation of quantum mechanics; it is a statement about how data from many preparations are combined.

This chapter develops the information quantities used to define mixed-state order. All logarithms are natural unless explicitly marked $\log_2$. Entropies are therefore in nats; dividing by $\ln 2$ converts to bits. Fidelity throughout the book is the unsquared, or root, fidelity.

## 4.2 Entropy and correlations

For eigenvalues $\lambda_a$ of a density matrix,

$$S(\rho) = -\operatorname{Tr}(\rho \ln \rho) = -\sum_a \lambda_a \ln \lambda_a,$$

with $0 \ln 0 = 0$. It vanishes for pure states and is bounded above by $\ln d$. The Rényi entropies are

$$S_\alpha(\rho) = \frac{1}{1-\alpha} \ln \operatorname{Tr} \rho^\alpha, \qquad \alpha > 0,\ \alpha \neq 1.$$

Taking $\alpha \to 1$ gives $S$. In particular $S_2 = -\ln \operatorname{Tr} \rho^2$. The label "Rényi-1 correlator" used later refers to a canonical-purification construction, not to substituting $\alpha = 1$ into a purity denominator.

For a classical joint distribution, the conditional entropy $H(A|B) = H(AB) - H(B)$ measures residual uncertainty in $A$ after $B$ is known. Quantum conditional entropy $S(A|B) = S(AB) - S(B)$ can be negative: a Bell pair has $S(AB) = 0$ and $S(B) = \ln 2$. This is one reason not to interpret every quantum entropy difference as an ordinary classical ignorance.

The mutual information is

$$I(A:C) = S(A) + S(C) - S(AC). \tag{4.1}$$

It is nonnegative and vanishes exactly when $\rho_{AC} = \rho_A \otimes \rho_C$. It captures both quantum and classical correlations. A classical coin shared by two sites has $I = \ln 2$ without entanglement. A Bell pair has $I = 2 \ln 2$.

## 4.3 Relative entropy and data processing

Quantum relative entropy is

$$D(\rho\|\sigma) = \operatorname{Tr}\rho(\ln\rho - \ln\sigma),$$

provided $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$; otherwise it is infinite. It is nonnegative and zero exactly for identical states. It is not symmetric and is not a metric.

An important identity is

$$I(A:C) = D(\rho_{AC}\|\rho_A \otimes \rho_C).$$

It follows by using $\ln(\rho_A \otimes \rho_C) = \ln\rho_A \otimes I + I \otimes \ln\rho_C$ and taking partial traces.

Relative entropy decreases under every channel:

$$D(\mathcal{E}\rho\|\mathcal{E}\sigma) \le D(\rho\|\sigma). \tag{4.2}$$

This data-processing inequality says that forgetting or processing information cannot improve optimal statistical discrimination between two hypotheses. The inequality is a nontrivial matrix theorem; its operational direction is worth fixing before we use analogous statements for fidelity.

For density matrices, quantum Pinsker's inequality in natural-log convention is $D(\rho\|\sigma) \ge \frac{1}{2}\|\rho - \sigma\|_1^2$. Since $|\operatorname{Tr}(XA)| \le \|X\|_1\|A\|_\infty$, it implies for unit-norm local operators

$$|\langle A_A C_C\rangle_c| \le \sqrt{2I(A:C)}.$$

Thus a nondecaying ordinary connected correlation requires nonzero distant mutual information. The converse does not identify a particular order parameter, since MI combines all correlations.

## 4.4 Trace distance and fidelity

The trace norm is $\|A\|_1 = \operatorname{Tr}\sqrt{A^\dagger A}$. The trace distance is

$$T(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1.$$

For equal prior probabilities, optimal one-shot discrimination succeeds with probability $(1+T)/2$. Root fidelity is

$$F(\rho,\sigma) = \operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}} = \|\sqrt{\rho}\sqrt{\sigma}\|_1. \tag{4.3}$$

For normalized states, $0 \le F \le 1$. It equals one exactly when the states coincide and zero exactly when their supports are orthogonal. For two pure states it is $|\langle\psi|\phi\rangle|$. Some papers use its square as "fidelity"; their exponents and bounds must be translated before comparing formulas.

If $\rho$ and $\sigma$ commute, diagonalize them together. Then

$$F(\rho,\sigma) = \sum_a \sqrt{p_a q_a}. \tag{4.4}$$

This is the classical Bhattacharyya overlap. It is the central computational shortcut in several solvable SW-SSB models.

Fidelity and trace distance obey

$$1 - F(\rho,\sigma) \le T(\rho,\sigma) \le \sqrt{1 - F(\rho,\sigma)^2}. \tag{4.5}$$

Fidelity increases under channels: $F(\mathcal{E}\rho, \mathcal{E}\sigma) \ge F(\rho,\sigma)$. The direction is opposite to distinguishability: two states become more similar when information is discarded.

### 4.4.1 Purification derivation of the fidelity formula

All purifications of $\rho$ in a sufficiently large environment are related by an environment isometry. Using the vectorization identity from Chapter 1, their overlaps with a purification of $\sigma$ can be written as $\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma}\,V)$ for an appropriate unitary $V$ on the support. The polar decomposition $A = W|A|$ gives

$$\max_{V\ munitary} |\,\mathrm{Tr}(AV)| = \mathrm{Tr}\,|A|.$$

Indeed the choice $V = W^\dagger$ attains the trace norm, and the upper bound follows from $|\,\mathrm{Tr}(AV)| \le \|A\|_1 \|V\|_\infty$. Setting $A = \sqrt{\rho}\sqrt{\sigma}$ yields Uhlmann's characterization: fidelity is the largest overlap of purifications. A common channel dilation followed by discarding part of the environment cannot reduce that available maximum, explaining fidelity monotonicity.

This does not mean that an arbitrary physical purification has the maximizing overlap. Optimizing over the environment is precisely what makes fidelity a property of $\rho, \sigma$, independent of a particular preparation.

### 4.4.2 Worked example: flipping a biased classical bit

For $\rho = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$, let $\sigma = X\rho X$. Then

$$F(\rho, X\rho X) = 2\sqrt{p(1-p)}, \qquad T(\rho, X\rho X) = |2p-1|.$$

A pure classical bit is orthogonal to its flip. A maximally uncertain bit is unchanged by flipping. Large overlap after a charge transfer can therefore signal delocalized or erased charge information without any ordinary ordered expectation value.

## 4.5 A fidelity average is not a linear expectation value

For a unitary $A$, define $F_A = F(\rho, A\rho A^\dagger)$. The identity

$$F_A = \|\sqrt{\rho}A\sqrt{\rho}\|_1 \ge |\,\mathrm{Tr}(\rho A)| \tag{4.6}$$

follows from the trace-norm bound. For a pure state equality holds. For a mixed state it can be strict: $\rho = I/2$ and $A = Z$ give $F_A = 1$ but $\mathrm{Tr}(\rho Z) = 0$.

For a nonunitary $A$, $A\rho A^\dagger$ may have trace different from one. There are two distinct choices: use the same algebraic homogeneous overlap on positive operators, or normalize the postselected output and separately track its success probability. They need not agree. Unless explicitly stated, worked fidelity comparisons in this book use unitary charge-transfer operators. Ladder-operator examples state their normalization explicitly.

Fidelity alone does not establish SW-SSB. The maximally mixed state has $F_A = 1$ for every unitary but has no fixed global strong charge. The symmetry assumption, absence of conventional order, and long-distance limit are all part of the definition.

## 4.6 Conditional mutual information

Partition the complete system into a region $A$, a buffer $B$, and a distant remainder $C$. The conditional mutual information is

$$I(A:C|B) = S(AB) + S(BC) - S(B) - S(ABC). \tag{4.7}$$

Equivalently it is $I(A:BC) - I(A:B)$. Strong subadditivity states $I(A:C|B) \geq 0$. Unlike ordinary mutual information, CMI asks whether knowing the buffer exhausts the correlation with the rest.

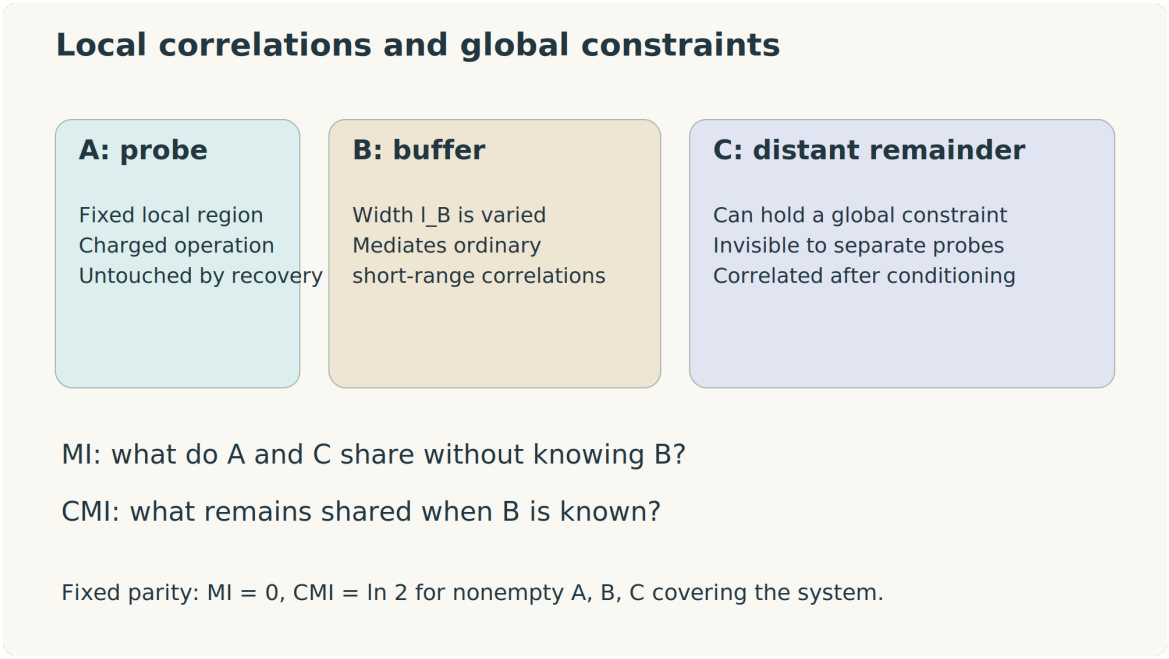



Figure 4.1: Ordinary and conditional correlations ask different questions. The buffer separates A from C; its width is increased before deciding whether an information length is finite.

For a pure state of all $ABC$, $S(AB) = S(C)$, $S(BC) = S(A)$, and $S(B) = S(AC)$, so $I(A:C|B) = I(A:C)$. Their distinction is intrinsically mixed-state physics.

### 4.6.1 Exact comparison: three kinds of Ising order

Take nonempty $A, B, C$ covering $N$ qubits.

For the classical coin of all-up and all-down configurations, every nonempty reduced state has entropy $\ln 2$. Thus $I(A:C) = \ln 2$ but $I(A:C|B) = 0$: the bit in $B$ already tells us the orientation.

For the pure cat, every proper nonempty reduced state has entropy $\ln 2$, while the full entropy vanishes. Thus both MI and CMI equal $\ln 2$.

For the parity state $\rho_+ = (I + \prod_i X_i)/2^N$, every proper region $R$ has entropy $|R| \ln 2$, while the full entropy is $(N-1)\ln 2$. Hence

$$I(A:C) = 0, \qquad I(A:C|B) = \ln 2. \tag{4.8}$$

Every ordinary distant correlation is absent, yet conditioning on the buffer exposes one global constraint. In the $X$ basis this is simply a uniform collection of bits with even total parity. Given all the other bits, the parity of $A$ determines the parity of $C$.

| State | Distant MI | Buffered CMI | Information carried |
|---|---|---|---|
| Product state | 0 | 0 | No shared bit |
| Classical Ising coin | $\ln 2$ | 0 | A locally readable common orientation |
| Pure Ising cat | $\ln 2$ | $\ln 2$ | Orientation and global coherence |
| Fixed-parity maximally mixed state | 0 | $\ln 2$ | A global charge constraint |

These are representative examples, not an exhaustive classification of every state by two numbers.

## 4.7 Recovery maps and the Markov property

Suppose $C$ is erased, leaving $\rho_{AB}$. Can a map acting only on $B$ append a new $C$ and reconstruct the original correlations with untouched $A$? Exact quantum Markov states have $I(A:C|B) = 0$ and admit such a recovery map.

For a compatible state the Petz candidate is

$$\mathcal{R}^{\mathrm{P}}_{B\to BC}(X_B) = \rho_{BC}^{1/2}\left(\rho_B^{-1/2} X_B \rho_B^{-1/2} \otimes I_C\right)\rho_{BC}^{1/2}. \tag{4.9}$$

Inverse square roots are taken on the support. The map can be extended trace preservingly outside that support. It satisfies $\mathcal{R}^{\mathrm{P}}(\rho_B) = \rho_{BC}$, but that identity alone does not establish recovery of correlations with $A$. Exact recovery holds under the Markov condition. For approximate recovery, an appropriately optimized or rotated recovery map is generally needed; the plain Petz map should not automatically be assigned every sharp bound.

The Fawzi-Renner theorem establishes the existence of a CPTP map with [4]

$$F\left(\rho_{ABC}, (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB})\right) \geq e^{-I(A:C|B)/2}. \tag{4.10}$$

Thus small CMI guarantees good recovery. A nonzero CMI is not, by itself, a quantitative proof that every recovery has small fidelity. To establish an obstruction, one may separately upper-bound all possible recovery fidelities; combining that bound with Equation 4.10 yields a lower bound on CMI.

### 4.7.1 Worked example: why parity cannot be recovered locally

Erase nonempty $C$ from $\rho_+$. The input $\rho_{AB} = I_A/2^{|A|} \otimes I_B/2^{|B|}$ is a product across $A|B$. Any channel on $B$ produces $I_A/2^{|A|} \otimes \tau_{BC}$. It cannot correlate the parity of $A$ with $BC$.

Measure full parity $U_A U_B U_C$. Since $\langle U_A \rangle = 0$ in the unchanged maximally mixed $A$, its two outcomes are equiprobable in every recovered state. The target has even parity with probability one. Monotonicity of fidelity under this measurement gives

$$F(\rho_+, \rho_{\mathrm{rec}}) \leq \frac{1}{\sqrt{2}}.$$

The bound is attained by the fully maximally mixed recovered state. Combining it with the recovery theorem requires $I \geq \ln 2$, saturated by Equation 4.8. The obstruction concerns correlation with untouched $A$, not an inability to prepare a parity constraint if allowed to act globally.

## 4.8 Information lengths and geometry

If $I(A:C)$ decays as $e^{-\ell_B/\xi_c}$, $\xi_c$ is an information correlation length. If $I(A:C|B)$ decays as $e^{-\ell_B/\xi_M}$ in a specified family of partitions, $\xi_M$ is a Markov length. The geometry and prefactors matter. A bound proportional to $|C|e^{-\ell_B/\xi_M}$ does not remain small when $|C|$ is taken to infinity before the buffer grows.

Long-range CMI can arise from global charge constraints, topological structure, or other forms of conditional organization. It is not a unique signature of SW-SSB. Conversely, the implication from SW-SSB to long-range CMI uses the appropriate global or local formulation and its assumptions; it is not merely a resemblance between the parity example and an arbitrary state.

## 4.9 Exercises and worked solutions

### 4.9.1 1. Fidelity of commuting distributions

For two independent classical bits with distributions $p \otimes p$ and $q \otimes q$, prove that their root fidelity is the square of the single-bit fidelity. Generalize to $N$ bits.

**Solution**

The sum factorizes: $\sum_{ab} \sqrt{p_a p_b q_a q_b} = (\sum_a \sqrt{p_a q_a})^2$. For $N$ bits it is $F(p,q)^N$. Two locally very similar distributions with single-bit fidelity below one can become globally nearly orthogonal. This elementary extensivity is a warning against equating global overlap with local similarity.

### 4.9.2 2. Root fidelity versus squared fidelity

If another convention defines $\mathfrak{F} = F^2$, rewrite the recovery bound and the trace-distance upper bound.

**Solution**

They become $\mathfrak{F}(\rho, \rho_{\text{rec}}) \geq e^{-I}$ and $T \leq \sqrt{1-\mathfrak{F}}$. A factor of two in an exponent may reflect convention rather than a scientific disagreement. Check definitions before comparing threshold scaling or susceptibility formulas.

### 4.9.3 3. A classical conditional correlation

Let $a, b$ be independent uniform bits and $c = a \oplus b$. Compute $I(a:c)$ and $I(a:c|b)$ in bits.

**Solution**

Each pair of variables is uniform on four outcomes, so $H(a) = H(c) = 1$ and $H(ac) = 2$, giving $I(a:c) = 0$. The full distribution has four outcomes, hence $H(abc) = 2$. With $H(ab) = H(bc) = 2$ and $H(b) = 1$, CMI is one bit. This is the smallest parity model of hidden conditional information.

### 4.9.4 4. Pure-state MI equals CMI

Prove the equality for a pure state on $ABC$, and explain why it fails for the parity ensemble.

**Solution**

Schmidt decomposition across each bipartition gives $S(AB) = S(C)$, $S(BC) = S(A)$, $S(B) = S(AC)$, and $S(ABC) = 0$. Substitution gives equality. The parity state has nonzero full entropy and no such

complement-entropy identities. Its vanishing MI and nonzero CMI demonstrate that mixedness creates an independent kind of information structure.

#### 4.9.5 5. Recovery directions

Does a state admitting a good recovery map necessarily have small CMI independent of dimension? What can safely be concluded from Equation 4.10 alone?

**Solution**

The displayed theorem states a sufficient condition: small CMI implies existence of good recovery. Its logical converse does not follow. A lower bound on the best fidelity also does not upper-bound CMI. To bound CMI from below, establish that every allowed recovery has fidelity at most $f_* < 1$; then $e^{-I/2} \leq f_*$ gives $I \geq -2 \ln f_*$. Converse estimates require additional results and may involve dimensions or other assumptions.

# Part II

# II · Strong-to-Weak Order

# 5 Symmetry of states and dynamics

Symmetry can constrain a probability distribution without constraining every sample drawn from it. A fair coin respects interchange of heads and tails; a particular toss does not. The corresponding distinction for a quantum ensemble is the starting point of strong-to-weak spontaneous symmetry breaking. We will first formulate it precisely, then distinguish symmetry of a *state* from symmetry of its *dynamics*. Confusing these two questions is a common source of incorrect conclusions.

The prerequisite is the density-matrix language: a state is a positive operator $\rho$ of unit trace, and its support is the span of eigenvectors with nonzero eigenvalues. A symmetry group $G$ acts through unitaries $U_g$ satisfying $U_g U_h = U_{gh}$. Unless stated otherwise, the symmetry is unitary and internal, and the lattice has finite-dimensional local Hilbert spaces. Antiunitary, spatial, fermionic, and higher-form symmetries need additional care; some appear later.

## 5.1 Two symmetry conditions

A state has **weak symmetry** when

$$U_g \rho U_g^\dagger = \rho \qquad (g \in G). \tag{5.1}$$

Equivalently, $[U_g, \rho] = 0$. All expectation values transform as those of a symmetric statistical ensemble. In particular, if $O$ has nontrivial charge, meaning

$$U_g O U_g^\dagger = \chi_O(g) O, \tag{5.2}$$

where $\chi_O(g)$ is a one-dimensional character and differs from one for some $g$, then $\mathrm{Tr}(\rho O) = 0$. To see this, insert Equation 5.1 in the trace and cycle $U_g$ through it. The expectation equals $\chi_O(g)^{-1}$ times itself.

A stronger condition is

$$U_g \rho = \chi_\rho(g) \rho \qquad (g \in G), \tag{5.3}$$

where $\chi_\rho(g)$ is a phase. This is **strong symmetry** of the state. Taking the adjoint gives $\rho U_g^\dagger = \chi_\rho(g)^* \rho$, so strong symmetry implies weak symmetry. In an Abelian charge basis, strong symmetry fixes a single total charge; weak symmetry allows statistical weight in several charge sectors. This terminology extends earlier work on symmetries of Lindblad generators [5,6] and underlies the modern mixed-state theory [7,8].

The phase in Equation 5.3 is not arbitrary. Applying $U_g U_h$ to a nonzero $\rho$ gives

$$\chi_\rho(gh) = \chi_\rho(g)\chi_\rho(h).$$

Thus it is a character of the group. For $U(1)$ with $U_\theta = e^{i\theta Q}$, this condition reads $Q\rho = q\rho$. For an Ising symmetry $U = \prod_i X_i$, it reads $U\rho = \pm\rho$.

## 5.2 The support theorem

Statements about "every state in the ensemble" must be independent of how we choose a convex decomposition. The following elementary result ensures that they are.

**Support theorem**

For a positive density operator, $U\rho = \lambda\rho$ if and only if $U$ equals $\lambda I$ on the support of $\rho$. Consequently, every vector occurring with nonzero weight in *any* convex decomposition of $\rho$ is a $U$ eigenvector with the same eigenvalue $\lambda$.

Write the spectral decomposition as $\rho = \sum_{a:p_a>0} p_a|a\rangle\langle a|$. Acting with $U\rho = \lambda\rho$ on $|a\rangle$ gives $p_a U|a\rangle = \lambda p_a|a\rangle$. Therefore $U|a\rangle = \lambda|a\rangle$ throughout the support. The converse follows by substituting the spectral decomposition.

Now let $\rho = \sum_b p_b|\psi_b\rangle\langle\psi_b|$ be an arbitrary decomposition. If $|v\rangle$ belongs to the kernel, then

$$0 = \langle v|\rho|v\rangle = \sum_b p_b|\langle v|\psi_b\rangle|^2.$$

Every summand is nonnegative, so each nonzero-weight $|\psi_b\rangle$ is orthogonal to the kernel. It lies in the support and consequently has eigenvalue $\lambda$. The words "every convex decomposition" follow from positivity, not from a preferred preparation history.

Weak symmetry has a different implication. Commuting normal operators can be simultaneously diagonalized, so for an Abelian symmetry one *can* choose an ensemble of definite-charge states. Their charges need not agree. Another allowed decomposition may consist entirely of states without definite charge. For example,

$$\frac{I}{2} = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} = \frac{|+\rangle\langle +| + |-\rangle\langle -|}{2}.$$

For the symmetry $X$, the first ensemble has nonsymmetric constituents, while the second has charges $+1$ and $-1$. Neither representation makes $I/2$ strongly symmetric.

## 5.3 A three-state laboratory

Take $N > 2$ qubits and $U = \prod_i X_i$. Three states will recur throughout the book.

The paramagnet

$$\rho_{\text{pm}} = |+\rangle\langle +|^{\otimes N}$$

has strong symmetry with charge $+1$. Its charged $Z$ correlations vanish between distinct sites. The pure cat state

$$|\text{GHZ}\rangle = \frac{|0\cdots 0\rangle + |1\cdots 1\rangle}{\sqrt{2}}$$

has the same strong charge but $\langle Z_i Z_j\rangle = 1$. The state is symmetric at finite $N$; its long-range correlations diagnose conventional SSB in the thermodynamic sequence. Finally,

$$\rho_+ = \frac{I+U}{2^N} = \frac{P_+}{2^{N-1}}, \qquad P_+ = \frac{I+U}{2}, \tag{5.4}$$

is maximally mixed within the even-parity sector. It also has strong charge $+1$, but $\langle Z_i Z_j\rangle = 0$ for distinct sites. Nevertheless, $Z_i Z_j \rho_+ Z_i Z_j = \rho_+$. Transferring a pair of charges leaves the state unchanged. This last property will become nonlinear long-range order.

The classical ferromagnetic mixture

$$\rho_{\text{fm}} = \frac{|0\cdots 0\rangle\langle 0\cdots 0| + |1\cdots 1\rangle\langle 1\cdots 1|}{2}$$

has only weak $U$ symmetry. Its support contains both even and odd cat states. Its ordinary correlations are long-ranged. These examples separate two axes: the exact algebraic symmetry of a finite state, and the long-distance order of a thermodynamic family. SSB never requires us to falsely assert that a symmetric finite-volume density matrix violates its own defining equation.

## 5.4 Canonical and grand-canonical ensembles

Let $[H, Q] = 0$. The fixed-charge thermal state is

$$\rho_{\beta,q} = \frac{P_q e^{-\beta H}}{Z_q}, \qquad Z_q = \mathrm{Tr}(P_q e^{-\beta H}). \tag{5.5}$$

It is strongly $U(1)$ symmetric because $Q\rho_{\beta,q} = q\rho_{\beta,q}$. The grand-canonical state

$$\rho_{\beta,\mu} = \frac{e^{-\beta(H-\mu Q)}}{Z(\beta,\mu)}$$

is generally only weakly symmetric: it includes several charges. In suitable thermodynamic regimes, the two ensembles have the same reduced states on small regions after matching charge density. They retain different *global* charge statistics. Ensemble equivalence therefore means local operational equivalence, not equality in global trace norm.

This observation gives physical intuition for SW-SSB. A state can have an exact global charge while local regions behave as though they exchange charge with an effective reservoir formed by the rest of the system. Information about total charge then resides in a constraint shared by the whole sample. Whether this intuition produces a particular nonlinear order parameter requires assumptions about the state; ensemble equivalence alone is not a general proof of every proposed SW-SSB criterion.

## 5.5 Symmetry of a quantum channel

A channel $\mathcal{E}$ is **weakly symmetric**, or covariant, if

$$\mathcal{E}(U_g \rho U_g^\dagger) = U_g \mathcal{E}(\rho) U_g^\dagger \tag{5.6}$$

for every input $\rho$. This property preserves weak symmetry of inputs. It does not require a fixed charge to remain fixed.

A charge-preserving **strongly symmetric channel**, in the convention used here, has Kraus operators satisfying

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \qquad [K_a, U_g] = 0 \quad \text{for every } a, g. \tag{5.7}$$

Then $U_g \mathcal{E}(\rho) = \chi_\rho(g)\mathcal{E}(\rho)$ whenever the input has charge $\chi_\rho$. It preserves each charge sector, not merely the set of symmetric ensembles. The property that the Kraus *span* lies in the commutant is unchanged by isometric mixing of Kraus labels. A different convention sometimes calls Kraus operators with a common nontrivial character "strong"; such channels can shift charge deterministically. We use the sector-preserving version because it is the one in the stability theorem below.

In a Stinespring dilation, the distinction has a simple meaning. If $V$ is the system-environment unitary, strong symmetry requires

$$[V, U_g \otimes I_E] = 0.$$

The environment cannot absorb the system's symmetry charge. Conservation only of a *combined* charge, $[V, U_g \otimes U_{g,E}] = 0$, permits exchange between system and bath and generally leads to weak covariance instead. "The universe conserves charge" is therefore insufficient to establish strong symmetry of an open subsystem.

### 5.5.1 Worked example: single-site and pair dephasing

For $U = \prod_i X_i$, single-site dephasing

$$\mathcal{D}_i(\rho) = (1-p)\rho + pZ_i\rho Z_i$$

is covariant because conjugation by $U$ changes the sign of $Z_i$ twice. Its Kraus operator $Z_i$ anticommutes with $U$, so it is not strongly symmetric. For Equation 5.4,

$$\mathcal{D}_i(\rho_+) = \frac{I + (1-2p)U}{2^N}.$$

For $0 < p < 1$, both parity sectors acquire weight. By contrast,

$$\mathcal{D}_{ij}(\rho) = (1-p)\rho + pZ_iZ_j\rho Z_jZ_i$$

is strongly symmetric. Each error creates or transports charges in pairs and leaves total parity unchanged. Applying such channels on a connected lattice drives the paramagnet toward Equation 5.4 at maximal dephasing. Strong symmetry of a channel consequently does not imply absence of SW-SSB in its output.

## 5.6 Lindblad generators and conserved sectors

For

$$\mathcal{L}(\rho) = -i[H,\rho] + \sum_a \left( L_a \rho L_a^\dagger - \frac{1}{2}\{L_a^\dagger L_a, \rho\} \right),$$

the sufficient strong-symmetry conditions are $[H, U_g] = [L_a, U_g] = 0$. Every finite-time channel $e^{t\mathcal{L}}$ then preserves charge sectors. Its adjoint annihilates their projectors: $\mathcal{L}^\dagger(P_q) = 0$. Thus sector weights are constants of motion. At least one stationary density matrix exists within each invariant finite-dimensional sector, although it need not be unique there. Strong symmetry therefore produces a stationary-state structure richer than a single globally mixing fixed point.

Weak symmetry of a generator means that $\mathcal{L}$ commutes with conjugation by $U_g$. The jumps can mix under that conjugation; individual commutators need not vanish. Also, Lindblad representations are not unique: unitary mixing of jumps and certain shifts accompanied by Hamiltonian changes leave the generator unchanged. Whenever a claim is made directly about the jumps, specify the representation or formulate the intrinsic statement using invariant sectors or the superoperator [5,6].

These observations do not prove that every strongly symmetric steady state has SW-SSB. Ergodicity within sectors, extra conserved quantities, dark states, and the order of long-time and thermodynamic limits matter. Later chapters examine conditions under which the connection to equilibration and diffusion becomes sharp.

## 5.7 Non-Abelian caution

For a non-Abelian group, a multiplet with definite total spin is not automatically a strongly symmetric state. The condition $U_g\rho = \chi_\rho(g)\rho$ for *every* $g$ restricts the support to copies of a one-dimensional representation. For $SU(2)$ this is the singlet representation. A rotationally invariant mixed state supported on a spin-$j > 0$ multiplet satisfies $[U_g, \rho] = 0$ but generally not the strong condition.

Weak invariance has block form

$$\mathcal{H} = \bigoplus_\lambda V_\lambda \otimes M_\lambda, \qquad \rho = \bigoplus_\lambda I_{V_\lambda} \otimes r_\lambda,$$

with positive multiplicity-space operators $r_\lambda$ and normalization $\sum_\lambda \dim(V_\lambda) \operatorname{Tr} r_\lambda = 1$. This is Schur's lemma. A strong singlet state can still be highly mixed in its multiplicity space and highly entangled in real space. The simplicity of parity ensembles must not be assumed for non-Abelian charge constraints.

## 5.8 Historical precursor: relative time translations

The idea of using a doubled symmetry to organize equilibration predates the present fixed-charge density-matrix formulation. [9] studies random Floquet spin chains through the disorder-averaged spectral form factor

$$K(t) = \mathbb{E}\left[\operatorname{Tr}(U_F^t)\operatorname{Tr}(U_F^{\dagger t})\right].$$

The two traces generate forward and backward histories. Each history has a cyclic choice of its time origin, giving independent discrete time-shift symmetries in the doubled description. A pairing between the two histories selects a relative time offset. Simultaneously shifting both histories preserves that pairing; shifting one alone changes it. This is a relative-to-diagonal symmetry-breaking structure closely related to the later strong-to-weak language.

The authors express $K(t)$ through a transfer matrix acting along the spatial direction. In their localized regime its dominant eigenvalue is unique. In the late-time ergodic regime the leading eigenvalues become asymptotically degenerate, and the associated order correlations become long-ranged. The symmetry-breaking interpretation organizes the emergence of ergodic spectral behavior. In suitable random-matrix regimes, different relative cyclic offsets also explain the familiar multiplicity behind a spectral ramp.

This is a conceptual precursor, not an assertion that every thermalizing pure state has global mixed-state fidelity order. The object being averaged is a spectral quantity with doubled histories; the symmetry acts on those histories. The fixed-charge state definition in this chapter concerns the support of a density matrix. Comparing the two viewpoints is fruitful precisely because their mathematical objects and limits are specified. Later hydrodynamic formulations similarly use two copies, but for density evolution and fluctuations rather than a spectral form factor.

## 5.9 Exercises and worked solutions

### 5.9.1 Exercise: a symmetry witness

Let $\rho$ be any density matrix and $U$ unitary. Show that $|\operatorname{Tr}(\rho U)| = 1$ if and only if $\rho$ has strong symmetry under $U$.

Solution

Diagonalize $\rho = \sum_a p_a |a\rangle\langle a|$. The bound $|\sum_a p_a \langle a|U|a\rangle| \le \sum_a p_a |\langle a|U|a\rangle| \le 1$ saturates only if each nonzero-weight expectation has modulus one and the same phase. Equality in Cauchy-Schwarz then forces $U|a\rangle = e^{i\alpha}|a\rangle$. The support theorem gives $U\rho = e^{i\alpha}\rho$. The reverse direction follows immediately by taking the trace. For $U = e^{i\theta Q}$, one angle can have accidental degeneracies; strong symmetry of the whole $U(1)$ requires the condition for all angles.

### 5.9.2 Exercise: covariance permits charge exchange

For a qubit with charge $Q = |1\rangle\langle 1|$, analyze amplitude damping with $K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|$ and $K_1 = \sqrt{\gamma}|0\rangle\langle 1|$.

Solution

Under $U_\theta = e^{i\theta Q}$, $K_0$ is unchanged and $K_1$ acquires phase $e^{-i\theta}$. The phases cancel between $K_1$ and $K_1^\dagger$ in the channel, so it is covariant. It is not strongly symmetric because $[Q, K_1] = -K_1$. The input $|1\rangle\langle 1|$ becomes $(1-\gamma)|1\rangle\langle 1| + \gamma|0\rangle\langle 0|$. It is weakly symmetric but has no fixed charge for $0 < \gamma < 1$.

### 5.9.3 Exercise: singlets and multiplets

Two spin-$1/2$ particles have singlet projector $P_0$ and triplet projector $P_1$. Classify $P_0$, $P_1/3$, and $I/4$ under collective $SU(2)$.

Solution

All three commute with collective rotations and are weakly symmetric. The singlet is invariant under left multiplication by every rotation, hence strongly symmetric with trivial character. The triplet density matrix is proportional to identity *within a nontrivial three-dimensional representation*, so left multiplication rotates its support and does not equal a scalar times the density matrix. It is only weakly symmetric, as is $I/4$.

# 6 Nonlinear order parameters

The even-parity state in Chapter 5 has no ordinary $Z$ correlations, yet it is unchanged when $Z_i Z_j$ transports charge between arbitrarily distant sites. An order parameter should detect that insensitivity. For pure states, expectation values already measure overlap with an altered state. For mixed states, expectation and overlap separate, and there are several inequivalent ways to generalize them.

This chapter develops the distinctions before introducing statistical-mechanical mappings. All fidelities use the **root convention**

$$F(\rho,\sigma) = \operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}. \tag{6.1}$$

Some papers call $F^2$ the fidelity. Squaring a nonnegative correlator changes a critical exponent by a factor of two, although it does not change whether its thermodynamic limit vanishes. Checking conventions is therefore essential when comparing numerical work.

## 6.1 Charge transfer as a state-comparison experiment

Let $O_x$ have nonzero symmetry charge. The product $A_{xy} = O_x O_y^\dagger$ is neutral and transfers charge from one neighborhood to another. Initially take $O_x$ and $O_y$ to be unitary, so $\sigma_{xy} = A_{xy}\rho A_{xy}^\dagger$ is normalized. Define

$$C_O(x,y) = \operatorname{Tr}(\rho A_{xy}), \qquad F_O(x,y) = F(\rho,\sigma_{xy}). \tag{6.2}$$

The first number asks for a coherent expectation value. The second asks how distinguishable the original state is from the state after charge transfer. They agree in magnitude for a pure state:

$$F_O(x,y) = |\langle\psi|A_{xy}|\psi\rangle| \quad \text{if } \rho = |\psi\rangle\langle\psi|.$$

For a general state, put $X = \sqrt{\rho}A_{xy}\sqrt{\rho}$. Then $XX^\dagger = \sqrt{\rho}\sigma_{xy}\sqrt{\rho}$, so

$$F_O(x,y) = \|\sqrt{\rho}A_{xy}\sqrt{\rho}\|_1 \geq |\operatorname{Tr}(\rho A_{xy})|. \tag{6.3}$$

The inequality is $|\operatorname{Tr} X| \leq \|X\|_1$. Long-range ordinary order necessarily gives long-range fidelity order, but the converse fails.

For a single charged unitary $O_x$, strong symmetry implies

$$F(\rho, O_x\rho O_x^\dagger) = 0.$$

Indeed, $O_x$ moves the support into a different charge sector, orthogonal to the original one. Two separated insertions can have nonzero fidelity because their total charge is zero. This is directly analogous to a vanishing one-point function and nonvanishing two-point function in a symmetric finite-volume magnet, with fidelity replacing expectation.

> **Global SW-SSB: the convention in this book**
>
> A thermodynamic family with strong symmetry has global SW-SSB if some bounded, fixed-support charged operators have $F_O(x,y) \to c > 0$ at long distance, while ordinary correlations of all charged local order parameters decay. The thermodynamic limit and the subsequent long-distance limit must be specified; sup-

ports do not grow with separation. This is the fidelity-based definition of [7]. The local definition introduced in 2026 is discussed in Chapter 8 and is broader.

For $\rho_+ = (I + \prod_i X_i)/2^N$, $F_Z(i,j) = 1$ whereas $C_Z(i,j) = 0$ for distinct sites. For the paramagnet, both vanish. For a GHZ state, both have unit magnitude. A pure state cannot satisfy this *global* definition of SW-SSB, because its fidelity and ordinary correlators coincide in magnitude.

## 6.2 Rényi-1: a diagnostic equivalent to fidelity

For unitary $A = A_{xy}$ define

$$R_1(A) = \mathrm{Tr}(\sqrt{\rho}\, A \sqrt{\rho}\, A^\dagger). \tag{6.4}$$

This nonnegative quantity is sometimes called a Wightman correlator or quantum affinity. Since $\sqrt{A\rho A^\dagger} = A\sqrt{\rho}A^\dagger$, it is $\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma})$ for $\sigma = A\rho A^\dagger$. It is experimentally useful when the canonical purification of $\rho$ is preparable [10]. Independent work develops its relation to fidelity and dynamical diagnostics [11].

The decisive inequalities are

$$F(\rho,\sigma)^2 \le \mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma}) \le F(\rho,\sigma). \tag{6.5}$$

Thus

$$R_1(A) \le F(A) \le \sqrt{R_1(A)}. \tag{6.6}$$

These are volume-independent bounds. Fidelity long-range order exists exactly when Rényi-1 long-range order does. They need not have identical amplitudes or identical critical exponents in a noncommuting model.

### 6.2.1 Derivation of the bounds

The upper bound follows from

$$\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma}) \le \|\sqrt{\rho}\sqrt{\sigma}\|_1 = F(\rho,\sigma).$$

For the other direction choose a unitary $V$ from the polar decomposition such that $F = |\mathrm{Tr}(V\sqrt{\rho}\sqrt{\sigma})|$. Write this trace as $\mathrm{Tr}(BC)$, with

$$B = \sigma^{1/4} V \rho^{1/4}, \qquad C = \rho^{1/4}\sigma^{1/4}.$$

The Hilbert-Schmidt Cauchy-Schwarz inequality gives

$$F^2 \le \mathrm{Tr}(B^\dagger B)\,\mathrm{Tr}(C^\dagger C).$$

The second factor is $\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma})$. The first is $\mathrm{Tr}(\sqrt{\rho}V^\dagger\sqrt{\sigma}V) \le \sqrt{\mathrm{Tr}\,\rho\,\mathrm{Tr}\,\sigma} = 1$, again by Cauchy-Schwarz. This proves Equation 6.5 without a replica limit or a thermodynamic assumption.

### 6.2.2 Commuting and pure limits

If $\rho$ and $\sigma$ commute with eigenvalues $p_a, q_a$ in the same basis, then

$$F = R_1 = \sum_a \sqrt{p_a q_a}. \tag{6.7}$$

This is the classical Bhattacharyya overlap. Many exactly solvable Pauli-noise models have this property. At the other extreme, for pure $\rho$, $R_1(A) = |\langle A\rangle|^2 = F(A)^2$. Both bounds can therefore be saturated.

## 6.3 Rényi-2 and spectral reweighting

A more easily expressed polynomial diagnostic is

$$R_2(A) = \frac{\mathrm{Tr}(\rho A \rho A^\dagger)}{\mathrm{Tr}\,\rho^2}. \tag{6.8}$$

It equals a two-point function after vectorizing $\rho$ in a doubled Hilbert space [7,12]. For unitary $A$, $0 \le R_2(A) \le 1$: positivity follows from $\mathrm{Tr}(\rho\sigma) \ge 0$, while the upper bound is Hilbert-Schmidt Cauchy-Schwarz and $\mathrm{Tr}\,\sigma^2 = \mathrm{Tr}\,\rho^2$.

There is no analogue of Equation 6.6 with bounds uniform in system size. The denominator is often exponentially small. Normalizing by it changes the relative importance of different parts of the spectrum. If $\rho = \sum_a p_a |a\rangle\langle a|$, then

$$R_1(A) = \sum_{a,b} \sqrt{p_a p_b}|A_{ab}|^2, \qquad R_2(A) = \frac{\sum_{a,b} p_a p_b |A_{ab}|^2}{\sum_a p_a^2}. \tag{6.9}$$

Rényi-2 favors the largest eigenvalues more strongly. It is an informative observable with its own statistical-mechanical transitions; it is not a universally interchangeable substitute for fidelity.

### 6.3.1 Exact counterexample: fidelity order without Rényi-2 order

Let $d = 2^{N-1}$ be the dimension of the even-parity sector and set

$$\rho = w|+\cdots+\rangle\langle+\cdots+| + (1-w)\frac{P_+}{d}, \qquad 0 < w < 1. \tag{6.10}$$

For $A = Z_i Z_j$, the state and its conjugate commute. In the $X$ basis, $A$ swaps the distinguished all-plus vector with the vector carrying minus signs at $i, j$. All remaining vectors have common weight $t = (1-w)/d$. Consequently,

$$F = R_1 = (d-2)t + 2\sqrt{t(w+t)} \longrightarrow 1 - w,$$

but

$$R_2 = \frac{1-w^2}{dw^2 + 1 - w^2} \longrightarrow 0. \tag{6.11}$$

The purity is dominated by the single large eigenvalue $w$, whose eigenvector has no charged long-range order. The broad parity ensemble has finite total probability but contributes vanishing relative purity. Ordinary $Z_i Z_j$ expectation values are zero throughout. This is a genuine distinction between nonlinear order criteria, not a matter of squaring a correlator [7].

### 6.3.2 Exact counterexample in the opposite direction

The reverse discrepancy can be constructed by giving an increasingly rare component disproportionately large purity. This construction is deliberately artificial, which helps isolate the mechanism.

Attach three neutral flag qubits to each of $N$ system qubits. Let $|f_0\rangle$ be the all-zero flag vector and $\tau_F$ the maximally mixed state on its orthogonal complement, of rank $D = 2^{3N} - 1$. Define

$$\rho_N = \epsilon_N |f_0\rangle\langle f_0| \otimes \frac{P_+}{d} + (1-\epsilon_N)\tau_F \otimes |+\cdots+\rangle\langle+\cdots+|, \quad \epsilon_N = 2^{-N/4}. \tag{6.12}$$

The symmetry acts only on system qubits, and both blocks have strong even parity. For $A = Z_i Z_j$ on those qubits, orthogonality of the flags gives

$$F = \epsilon_N \longrightarrow 0, \qquad R_2 = \frac{\epsilon_N^2/d}{\epsilon_N^2/d + (1-\epsilon_N)^2/D} \longrightarrow 1.$$

The second limit follows because $\epsilon_N^2/d$ scales as $2^{-3N/2}$, while $1/D$ scales as $2^{-3N}$. The rare parity block wins the purity competition while its actual probability vanishes. In trace distance the state approaches $\tau_F \otimes |+\cdots+\rangle\langle+\cdots+|$. This limiting family has zero fidelity between charge-transfer states for separated bounded-support charged operators, because the system factor is a symmetric product state. Continuity of fidelity excludes hidden global fidelity order in the vanishing-weight admixture. Related spectral counterexamples are given in [7]; this flag construction makes the reweighting transparent.

Neither example proves that Rényi-2 is unhelpful in physical models. The important physical example is the decohered square-lattice paramagnet: its fidelity transition occurs near $p = 0.109$, whereas its Rényi-2 transition occurs near $p = 0.178$. Both are meaningful singularities of their stated quantities. Only the first diagnoses the intrinsic fidelity order and associated recoverability boundary in that model.

## 6.4 Generalized Rényi correlators

A useful family organizing the doubled-state picture is

$$R_\alpha(A) = \frac{\mathrm{Tr}(\rho^{\alpha/2} A \rho^{\alpha/2} A^\dagger)}{\mathrm{Tr}\,\rho^\alpha}, \qquad \alpha > 0. \tag{6.13}$$

At $\alpha = 1$ this is Equation 6.4; at $\alpha = 2$ it is Equation 6.8. It is a correlator in the normalized vector $|\rho^{\alpha/2}\rangle\rangle$. Changing $\alpha$ changes the effective statistical ensemble. A transition in that family need not occur at the same channel parameter for every $\alpha$. Analytic continuation from integer replicas can be powerful, but it assumes control over the continuation and the order of replica, volume, and distance limits. Integer-replica calculations alone do not prove a result at $\alpha = 1$.

For a Gibbs state, $\rho^\alpha / \mathrm{Tr}\,\rho^\alpha$ corresponds to inverse temperature $\alpha\beta$, with the same projector if a charge sector is fixed. Rényi reweighting can therefore move a state across an ordinary thermal transition. This is a concrete reason to be cautious about interpreting a Rényi singularity as a phase transition of the original probability weights.

## 6.5 Nonunitary operators and normalization

For a local annihilation operator or spin ladder operator, $A\rho A^\dagger$ need not have unit trace. Define the success probability $s_A = \mathrm{Tr}(\rho A^\dagger A)$ and, for $s_A > 0$, the conditioned state $\hat{\sigma}_A = A\rho A^\dagger / s_A$. The homogeneous fidelity expression obeys

$$\|\sqrt{\rho} A \sqrt{\rho}\|_1 = \sqrt{s_A}\, F(\rho, \hat{\sigma}_A). \tag{6.14}$$

Thus an unnormalized insertion combines an overlap with a postselection amplitude. Scaling $A \mapsto cA$ rescales it by $|c|$. In contrast, the polynomial expressions $\mathrm{Tr}(\sqrt{\rho}A\sqrt{\rho}A^\dagger)$ and $\mathrm{Tr}(\rho A \rho A^\dagger)$ scale by $|c|^2$. Moreover, $\sqrt{A\rho A^\dagger} =$

$A\sqrt{\rho}A^\dagger$ is generally false. The unitary proof of Equation 6.5 therefore cannot simply be reused by substituting a ladder operator. A theorem stated for more general insertions must specify its norm and normalization factors. In applications, keep local norms bounded independently of volume and report whether insertions are normalized or conditioned.

## 6.6 Distinguishability and measurement cost

For trace distance $T(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1$, the Fuchs-van de Graaf inequalities give

$$1 - F \le T \le \sqrt{1-F^2}.$$

Hence vanishing fidelity means asymptotically perfect distinguishability, while fidelity bounded below prevents perfect discrimination. Fidelity near one is a stronger statement than merely nonzero long-range fidelity. SW-SSB requires the latter.

It is tempting to declare that all information-theoretic measures must detect the same boundary. This needs qualification. Relative entropy can diverge because of support zeros even when fidelity is nonzero; a trace-distance criterion must specify whether it tests convergence to one or to a smaller value. One should compare precise order definitions, rather than the names of distance measures.

Another distinction is between an algebraically simple expression and a statistically efficient experiment. A two-copy SWAP measurement can access $\mathrm{Tr}\,\rho^2$, but if this purity is exponentially small, estimating a normalized ratio to fixed accuracy can require exponentially many repetitions. Randomized measurement protocols help in structured families and finite systems; their existence does not establish universal polynomial complexity. [13] proves an information-theoretic obstruction to detecting generic global SW-SSB from polynomially many copies without suitable prior structure. Canonical purification, Gaussianity, known noise channels, and local formulations supply different ways to change the available resources or the precise question.

## 6.7 Channel response and fidelity susceptibility

Conventional susceptibility relates a magnet's response to an applied field to its equilibrium fluctuations. SW-SSB has an analogous relation, but the perturbation is a channel and the responding quantity is nonlinear. The fidelity convention is crucial: [14] uses the **squared** Uhlmann fidelity. Translating its formulas into the root convention of this book requires squaring the one-point response, rather than differentiating root fidelity directly.

We derive the result for an Ising symmetry, where every charged probe $Z_i$ is a Hermitian unitary. This avoids the normalization and finite-spectrum issues of a nonunitary particle-creation operator. Let the initial $\rho$ have fixed global parity and apply weak independent single-site dephasing,

$$\rho_p = \mathcal{D}_p(\rho), \qquad \mathcal{D}_p = \prod_j \left[(1-p)\,\mathrm{id} + p\,\mathrm{Ad}_{Z_j}\right]. \tag{6.15}$$

This channel explicitly breaks strong symmetry while preserving weak covariance. Define the squared-fidelity magnetization and its response by

$$M_2(p) = \frac{1}{N}\sum_i F(\rho_p, Z_i\rho_p Z_i)^2, \qquad \chi_N = \lim_{p\to 0^+}\frac{M_2(p)-M_2(0)}{p}. \tag{6.16}$$

Here the subscript in $M_2$ denotes squaring root fidelity; this is not the Rényi-2 correlator $R_2$. Strong symmetry gives $M_2(0) = 0$. The limit is taken at fixed finite $N$ before studying its size dependence; the perturbative regime requires $Np \ll 1$.

### 6.7.1 Deriving the response from charge-sector blocks

Set $S = \sum_j Z_j \rho Z_j$. To first order,

$$\rho_p = (1 - Np)\rho + pS + O(p^2).$$

The support of $S$ has parity opposite to that of $\rho$. Meanwhile

$$Z_i \rho_p Z_i = (1 - Np) Z_i \rho Z_i + p Z_i S Z_i + O(p^2).$$

Root fidelity adds over orthogonal blocks. Its homogeneity is $F(a\rho, b\sigma) = \sqrt{ab}\, F(\rho, \sigma)$ for positive scalars. The two parity blocks therefore contribute, to leading order,

$$F(\rho_p, Z_i \rho_p Z_i) = \sqrt{p}\, [F(\rho, Z_i S Z_i) + F(S, Z_i \rho Z_i)] + o(\sqrt{p}).$$

By symmetry of fidelity and conjugation by $Z_i$, the two bracketed terms are equal. Define the normalized mixture

$$\overline{\sigma}_i = \frac{1}{N} \sum_j Z_i Z_j \rho Z_j Z_i.$$

Since $Z_i S Z_i = N \overline{\sigma}_i$, this gives

$$F(\rho_p, Z_i \rho_p Z_i) = 2\sqrt{Np}\, F(\rho, \overline{\sigma}_i) + o(\sqrt{p}). \tag{6.17}$$

For a translation-invariant family the coefficient is independent of $i$, so

$$\frac{\chi_N}{4} = N F(\rho, \overline{\sigma}_i)^2. \tag{6.18}$$

The factor four comes from adding the two opposite-parity block overlaps *before* squaring. For other charge groups and oriented charge-insertion channels, the number of matching blocks can differ. We do not import that factor into unrelated $U(1)$ protocols. The identity in Equation 6.18 describes the explicitly specified Ising response.

### 6.7.2 Bounding susceptibility by the spatial fidelity correlator

Write $f_{ij} = F(\rho, Z_i Z_j \rho Z_j Z_i)$, including $f_{ii} = 1$. Separate concavity of squared fidelity gives

$$F(\rho, \overline{\sigma}_i)^2 \geq \frac{1}{N} \sum_j f_{ij}^2.$$

One can see this directly from Uhlmann's theorem. For each $j$, choose purifications with overlap $f_{ij}$, and attach an orthogonal flag $j$. A purification of the mixture uses flag amplitudes $1/\sqrt{N}$. A purification of the unchanged $\rho$ may use any normalized flag amplitudes $\sqrt{q_j}$, since every flagged branch reduces to $\rho$. Optimizing their overlap $\sum_j \sqrt{q_j/N} f_{ij}$ over $q_j$ gives $\sqrt{\sum_j f_{ij}^2/N}$.

For the upper bound set $X_j = \sqrt{\rho} Z_i Z_j \sqrt{\rho}$. Then

$$\sqrt{N} F(\rho, \overline{\sigma}_i) = \mathrm{Tr} \sqrt{\sum_j X_j X_j^\dagger} \leq \sum_j \|X_j\|_1 = \sum_j f_{ij}.$$

The inequality is the trace-norm triangle inequality applied to the block row $[X_1\ X_2\ \cdots\ X_N]$, decomposed into rows with one nonzero block each. Combining these two inequalities with normalized fidelity at most one yields

$$\sum_j f_{ij}^2 \le \frac{\chi_N}{4} \le \min\left\{N, \left(\sum_j f_{ij}\right)^2\right\}. \tag{6.19}$$

These are the fidelity fluctuation-response bounds, with every root/squared conversion and the Ising prefactor explicit. For an inhomogeneous system, average the corresponding row bounds over $i$ rather than invoking translation invariance.

If $f_{ij}$ approaches a nonzero constant at large separation, the lower bound is proportional to $N$, while the normalized-fidelity bound caps the response by $4N$. Thus $\chi_N = \Theta(N)$ for fidelity long-range order. If $f_{ij}$ decays exponentially with a size-independent length on a fixed-dimensional lattice, both spatial sums remain finite and $\chi_N = O(1)$. This includes conventional strong-to-none order as well as SW-SSB; the ordinary correlator is still needed to distinguish them.

At a critical point with root fidelity $f(r) \sim r^{-\eta}$ in $d$ dimensions, the lower sum scales as $L^{d-2\eta}$, $\log L$, or a constant according as $2\eta < d$, $2\eta = d$, or $2\eta > d$. The upper bound is the smaller of $L^d$ and the square of the sum of $r^{-\eta}$. These bounds do not generally determine a unique critical response exponent. In particular, for a sufficiently slow critical tail they alone do not exclude an extensive response.

### 6.7.3 Two exact checks and a convention trap

For the product paramagnet, each dephased qubit has $X$-basis probabilities $(1-p, p)$. Its root one-point fidelity is $2\sqrt{p(1-p)}$, so $M_2(p) = 4p(1-p)$ and $\chi_N = 4$, independently of size. Its initial spatial correlator is $f_{ij} = \delta_{ij}$, and both sides of Equation 6.19 equal one.

For the even-parity ensemble, the dephased state is $[I + (1-2p)^N U]/2^N$. Its two parity weights give

$$M_2(p) = 1 - (1-2p)^{2N}, \qquad \chi_N = 4N. \tag{6.20}$$

All initial $f_{ij} = 1$, so the lower bound and the normalization cap are saturated. Taking the thermodynamic limit first instead gives $M_2(p) \to 1$ at every fixed $p > 0$: the linear-response window shrinks with size, as expected for spontaneous order.

If we had differentiated the **root** one-point fidelity with respect to $p$, even the unordered product paramagnet would have a divergent derivative, proportional to $p^{-1/2}$. That singularity reflects entering a previously empty charge sector, not long-range order. A legitimate alternative is to parameterize noise by an amplitude $p = \sin^2\theta$; then root fidelity has a finite one-sided linear response in $|\theta|$. Either convention can be useful, but they define different susceptibilities. The volume dependence in Equation 6.19 belongs to the squared-fidelity definition Equation 6.16.

### 6.7.4 Information geometry of the perturbation

The amplitude variables also clarify the Bures geometry analyzed in [14]. Let $p_j = \sin^2\theta_j$, take small nonnegative $\theta_j$, and write $\rho_\theta = \mathcal{D}_\theta(\rho)$. The squared Bures distance is $D_B^2(\rho,\sigma) = 2 - 2F(\rho,\sigma)$. To second order,

$$D_B^2(\rho_\theta, \rho) = \sum_j \theta_j^2 + o(\|\theta\|^2),$$

because every one-error state is orthogonal to the initial charge sector. Distance from the unperturbed state alone therefore misses spatial ordering. Compare two perturbed states instead through the polarization $\frac{1}{2}[D_B^2(\rho_\theta,\rho) + D_B^2(\rho,\rho_\phi) - D_B^2(\rho_\theta,\rho_\phi)]$. Its leading term is

$$g(\theta,\phi) = F\left(\sum_j \theta_j^2 Z_j \rho Z_j, \sum_j \phi_j^2 Z_j \rho Z_j\right). \tag{6.21}$$

For the pure paramagnet, the single-error states are mutually orthogonal, giving $g = \sum_j \theta_j\phi_j$ in the nonnegative orthant. For the parity ensemble they are all the same opposite-parity state, giving $g = \|\theta\|\|\phi\|$. The second

expression is not bilinear: many distinct spatial perturbations become indistinguishable once only the total charge change matters. At the rank-changing boundary this construction need not be a smooth Riemannian metric. This is a geometric manifestation of the same charge delocalization diagnosed by the fidelity correlator, rather than an equilibrium thermodynamic fluctuation-dissipation theorem assumed for arbitrary open dynamics.

## 6.8 Exercises and worked solutions

### 6.8.1 Exercise: a qubit overlap

Let $\rho = (I + rZ)/2$, $|r| < 1$, and $A = X$. Compute $F$, $R_1$, and $R_2$.

**Solution**

The conjugate state has eigenvalues interchanged. With $p_\pm = (1 \pm r)/2$, $F = R_1 = 2\sqrt{p_+ p_-} = \sqrt{1 - r^2}$. The numerator of $R_2$ is $2p_+p_- = (1 - r^2)/2$, while the purity is $(1 + r^2)/2$. Thus $R_2 = (1 - r^2)/(1 + r^2)$. All vanish as $|r| \to 1$ and become one at $r = 0$, but their quantitative responses differ.

### 6.8.2 Exercise: exponent bounds

At a critical point, suppose $F(r) \sim r^{-\eta_F}$ and $R_1(r) \sim r^{-\eta_1}$ with positive nonzero prefactors. What follows from Equation 6.5?

**Solution**

The inequality $F^2 \leq R_1$ forbids $\eta_1 > 2\eta_F$, while $R_1 \leq F$ forbids $\eta_1 < \eta_F$. Thus $\eta_F \leq \eta_1 \leq 2\eta_F$. If the altered and original states commute, $F = R_1$ and $\eta_1 = \eta_F$. The mere equivalence of ordered phases does not fix the critical exponents in the general case.

### 6.8.3 Exercise: purity bias

In Equation 6.10, take $w = 1/2$. Find the limiting fidelity and the leading size dependence of $R_2$.

**Solution**

The fidelity approaches $1/2$. The exact ratio becomes $R_2 = 3/(d + 3)$, with $d = 2^{N-1}$, so $R_2 \sim 3\,2^{1-N}$. Finite probability in the parity sector does not prevent its purity contribution from becoming exponentially subdominant.

# 7 Purification and doubled-space order

A mixed state can be understood as part of a larger pure state. This elementary fact becomes especially useful for SW-SSB: a nonlinear mixed-state correlator becomes an ordinary expectation value in a suitable doubled system. The qualification "suitable" matters. An arbitrary environmental purification, the canonical purification, and the vectorized density matrix encode different objects.

We will derive the relevant identities directly. Besides clarifying the symmetry-breaking pattern, they explain both the promise and the resource requirements of purification-based experiments [10].

## 7.1 Vectorization with an explicit convention

Choose a basis $\{|a\rangle\}$ and write a matrix $M = \sum_{ab} M_{ab}|a\rangle\langle b|$. Introduce a conjugate copy of the Hilbert space and define

$$|M\rangle\rangle = \sum_{ab} M_{ab}|a\rangle_L|\bar{b}\rangle_R. \tag{7.1}$$

The bar indicates the conjugate representation. It keeps track of complex conjugation under a basis change. Once this convention is fixed,

$$\langle\langle M|N\rangle\rangle = \mathrm{Tr}(M^\dagger N), \qquad (A \otimes B^*)|M\rangle\rangle = |AMB^\dagger\rangle\rangle. \tag{7.2}$$

Check the second identity by multiplying matrix elements: the coefficient of $|a\rangle_L|\bar{b}\rangle_R$ is $\sum_{cd} A_{ac}M_{cd}B^*_{bd} = (AMB^\dagger)_{ab}$. This prevents a common transpose error. The right-copy operator is $B^*$ in this notation, not an unqualified $B$ for a complex matrix.

The partial trace is equally useful:

$$\mathrm{Tr}_R\,|M\rangle\rangle\langle\langle M| = MM^\dagger. \tag{7.3}$$

Indeed, tracing over $b$ leaves $\sum_{abc} M_{ab}M^*_{cb}|a\rangle\langle c|$. All later distinctions follow from this identity.

## 7.2 Canonical purification

The **canonical purification** is

$$|\sqrt{\rho}\rangle\rangle. \tag{7.4}$$

It has norm one because $\langle\langle\sqrt{\rho}|\sqrt{\rho}\rangle\rangle = \mathrm{Tr}\,\rho = 1$, and its left marginal is $\rho$. In the eigenbasis of $\rho$,

$$|\sqrt{\rho}\rangle\rangle = \sum_a \sqrt{p_a}|a\rangle_L|\bar{a}\rangle_R.$$

The Schmidt coefficients are $\sqrt{p_a}$, exactly those needed to reproduce the original probabilities. This is a particular purification selected by positivity of the matrix square root and the pairing between the two copies. An arbitrary purification has the form $(I \otimes V)|\sqrt{\rho}\rangle\rangle$ with an isometry on an adequately large auxiliary space. That isometry need not be local.

For any unitary insertion $A$, Equation 7.2 gives

$$\langle\langle\sqrt{\rho}|A\otimes A^{*}|\sqrt{\rho}\rangle\rangle = \mathrm{Tr}(\sqrt{\rho}A\sqrt{\rho}A^{\dagger}) = R_{1}(A). \tag{7.5}$$

If $A = O_{x}O_{y}^{\dagger}$, define a local doubled operator

$$B_{x} = O_{x}\otimes O_{x}^{*}.$$

Then $R_{1}(x,y) = \langle B_{x}B_{y}^{\dagger}\rangle_{\mathrm{CP}}$. A nonlinear correlation in a mixed state has become a conventional two-point correlation in a pure state.

The locality assignment is part of the construction: the left and right degrees of freedom associated with site $x$ are treated as occupying the same doubled site. Without it, "local doubled operator" would have no fixed spatial meaning.

## 7.3 Why the doubled symmetry breaks to the diagonal subgroup

Suppose $U_{g}\rho = \chi_{\rho}(g)\rho$. The support theorem implies the same relation for $\sqrt{\rho}$. Hence

$$(U_{g}\otimes I)|\sqrt{\rho}\rangle\rangle = \chi_{\rho}(g)|\sqrt{\rho}\rangle\rangle,$$

and similarly $(I\otimes U_{g}^{*})|\sqrt{\rho}\rangle\rangle = \chi_{\rho}(g)^{*}|\sqrt{\rho}\rangle\rangle$. The canonical state has separate left and right symmetries, denoted $G_{L}\times G_{R}$ for the Abelian examples of interest.

For a weakly symmetric $\rho$, its square root commutes with $U_{g}$, giving only

$$(U_{g}\otimes U_{g}^{*})|\sqrt{\rho}\rangle\rangle = |\sqrt{\rho}\rangle\rangle. \tag{7.6}$$

This is the diagonal subgroup. If $O_{x}$ carries character $\chi_{O}$, then $B_{x}$ carries charge $\chi_{O}$ under left transformations and the conjugate charge under right transformations. It is neutral under their diagonal combination. Long-range $\langle B_{x}B_{y}^{\dagger}\rangle$ therefore signals the breaking pattern

$$G_{L}\times G_{R}\longrightarrow G_{\mathrm{diag}}. \tag{7.7}$$

At finite volume, separate symmetries remain exact and $\langle B_{x}\rangle = 0$ for a charged $B_{x}$; the long-range two-point function carries the spontaneous-breaking information. This is a faithful ordinary-SSB interpretation because Rényi-1 and fidelity order are equivalent by volume-independent inequalities. The same drawing applied to a different doubled vector need not diagnose the same mixed-state phase.

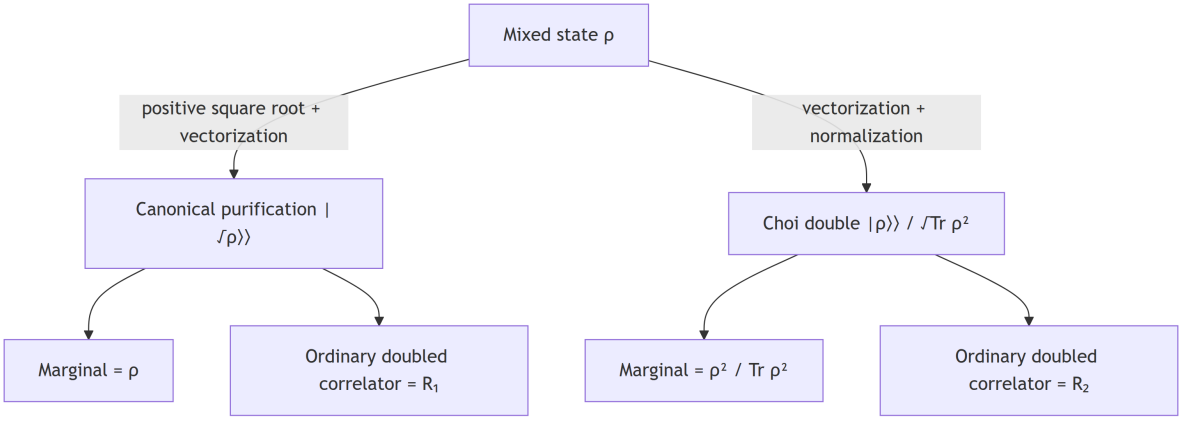



Figure 7.1: The square-root and density-matrix doubles use different spectral weights.

## 7.4 Worked example: a cat state of local Bell pairs

For the parity state $\rho_+ = P_+/2^{N-1}$, positivity and $P_+^2 = P_+$ give

$$\sqrt{\rho_+} = \frac{I+U}{2^{(N+1)/2}}.$$

Let $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and $|\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$ on each left-right pair. Since $|I\rangle\rangle = 2^{N/2}|\Phi^+\rangle^{\otimes N}$ and $|U\rangle\rangle = 2^{N/2}|\Psi^+\rangle^{\otimes N}$,

$$|\sqrt{\rho_+}\rangle\rangle = \frac{|\Phi^+\rangle^{\otimes N} + |\Psi^+\rangle^{\otimes N}}{\sqrt{2}}. \tag{7.8}$$

The local doubled operator $B_i = Z_{i,L}Z_{i,R}$ has eigenvalue $+1$ on $|\Phi^+\rangle$ and $-1$ on $|\Psi^+\rangle$. Its one-point expectation in the cat vanishes, but $\langle B_i B_j\rangle = 1$. This is precisely the mixed-state value $R_1(Z_iZ_j) = 1$.

The original $\rho_+$ is separable: in the $X$ basis it is the classical uniform distribution over configurations with even parity. Its canonical purification has long-range entanglement of a cat type. Canonical-purification entanglement is therefore not, by itself, entanglement between physical sites in the original mixed state. It is an auxiliary diagnostic carrying both classical and quantum correlations in a useful pure-state representation.

This example also exposes a complexity issue. The mixed state can be prepared by parallel local pair-dephasing channels applied to $|+\rangle^{\otimes N}$. Its canonical cat purification cannot be created from uncorrelated doubled sites by a bounded-depth geometrically local unitary circuit: such a circuit has disjoint causal cones at sufficiently large separation and cannot establish the displayed long-range connected correlations. A short-depth *mixed-state preparation* does not guarantee a short-depth canonical-purification preparation.

## 7.5 The Choi double is a different state

The normalized vectorized density matrix is

$$|\rho\rangle\rangle_{\text{norm}} = \frac{|\rho\rangle\rangle}{\sqrt{\operatorname{Tr}\rho^2}}. \tag{7.9}$$

Its left marginal is

$$\rho_L^{\text{Choi}} = \frac{\rho^2}{\operatorname{Tr}\rho^2}, \tag{7.10}$$

not $\rho$ in general. Its ordinary doubled correlator is $R_2(A)$, obtained by putting $M = \rho$ in Equation 7.2. It is often called a Choi state or Choi double in the mixed-state literature. This use of vectorization should not be confused with the Choi matrix of a channel, although both constructions use the same matrix-to-vector technology.

The two doubles coincide for a state with a flat nonzero spectrum, because then $\rho$ is proportional to a projector and $\rho/\sqrt{\operatorname{Tr}\rho^2} = \sqrt{\rho}$. They also coincide for pure states. For a generic thermal or noisy state they differ, sometimes enough to put their correlation functions in different phases.

The practical advantage of the Choi double is linear evolution. Vectorizing a Lindblad equation gives

$$\partial_t|\rho\rangle\rangle = \mathbb{L}|\rho\rangle\rangle,$$

where

$$\mathbb{L} = -i(H\otimes I - I\otimes H^T) + \sum_a \left[L_a\otimes L_a^* - \frac{1}{2}L_a^\dagger L_a\otimes I - \frac{1}{2}I\otimes (L_a^\dagger L_a)^T\right]. \tag{7.11}$$

The generator is generally non-Hermitian. This representation allows tensor-network, transfer-matrix, and field-theory methods to work directly with the density operator. No comparably simple linear equation holds for $\sqrt{\rho}$. A tractable evolution in doubled space consequently does not automatically compute the fidelity or Rényi-1 correlator.

## 7.6 Uhlmann optimization and the role of the environment

For normalized states, Uhlmann's theorem says

$$F(\rho,\sigma) = \max_V \left|\langle\langle\sqrt{\rho}|(I\otimes V)|\sqrt{\sigma}\rangle\rangle\right|. \tag{7.12}$$

The maximization is over auxiliary unitaries, with isometric extensions if needed. An intuitive derivation uses $\max_V |\operatorname{Tr}(VX)| = \|X\|_1$, proved by choosing the polar unitary of $X$; apply it to $X = \sqrt{\rho}\sqrt{\sigma}$. Fidelity chooses the most favorable matching of purifications. It does not depend on an accidentally chosen environmental basis.

If $\sigma = A\rho A^\dagger$ for unitary $A$, then $|\sqrt{\sigma}\rangle\rangle = (A\otimes A^*)|\sqrt{\rho}\rangle\rangle$. The Rényi-1 correlator is the overlap for one prescribed canonical matching. Uhlmann optimization can increase it, but Equation 6.5 shows that it cannot change a nonzero thermodynamic order parameter into a vanishing one or vice versa.

A physical bath purification obtained from a noisy preparation may look completely different. Measuring the bath in different bases produces different pure-state ensembles of the same reduced $\rho$. A trajectory-resolved quantity must specify the record and the unraveling; an intrinsic density-matrix diagnostic cannot depend on that choice.

### 7.6.1 Hidden Edwards-Anderson order

Write the parity state as an equal mixture of $Z$-basis cats

$$|c_s\rangle = \frac{|s\rangle + |\bar{s}\rangle}{\sqrt{2}},$$

where one representative is chosen from each complementary pair. In each cat, $\langle Z_i Z_j\rangle = s_i s_j = \pm 1$. The absolute value remains one after averaging, while the signed average vanishes. This resembles an Edwards-Anderson diagnostic of glassy order.

The same $\rho_+$ is a mixture of $X$-basis even-parity product states, each having zero separated $Z$ correlation. Thus the ensemble average $\sum_a p_a|\langle\psi_a|A|\psi_a\rangle|$ is not intrinsic to $\rho$. Joint concavity of fidelity nevertheless gives the useful bound

$$\sum_a p_a|\langle\psi_a|A|\psi_a\rangle| \le F(\rho, A\rho A^\dagger) \tag{7.13}$$

for any chosen decomposition and unitary $A$. The cat decomposition saturates it here. The connection to decompositions with hidden long-range order is developed in [15] and discussed in [8]. The safe physical message is that fidelity detects a property of the ensemble *as a density matrix*, not the ordering of every possible trajectory representation.

## 7.7 Thermal example with a complete calculation

Consider $H = -h\sum_i X_i$, $h > 0$, projected into even parity. At any fixed finite temperature and large $N$, the parity constraint changes finite-site probabilities only by corrections proportional to powers of $\tanh(\beta h)$. The local $X_i = \pm 1$ variables therefore have probabilities

$$P(x_i) = \frac{e^{\beta h x_i}}{2\cosh(\beta h)}.$$

Acting with $A = Z_i Z_j$ flips $x_i, x_j$. The energy change is $\Delta H = 2h(x_i + x_j)$. The states commute, and the fidelity is

$$F_Z(i,j) = \sum_x P(x) e^{-\beta \Delta H(x)/2} \longrightarrow \left[\sum_{x_i=\pm 1} \frac{e^{\beta h x_i} e^{-\beta h x_i}}{2\cosh(\beta h)}\right]^2 = \operatorname{sech}^2(\beta h). \tag{7.14}$$

It is positive for every finite $\beta$, independent of separation, although $\langle Z_i Z_j \rangle = 0$. At zero temperature it tends to zero and the state becomes the pure paramagnet. This is an exact model illustrating why thermal charge fluctuations can generate SW-SSB without ordinary order.

The Rényi-2 calculation weights configurations at inverse temperature $2\beta$ and gives $R_2 \to \operatorname{sech}^2(2\beta h)$. Its amplitude differs, but in this model both criteria are ordered for every nonzero temperature. Interacting systems require control over correlations and imaginary-time locality; this simple result is an example, not a proof for every conceivable Hamiltonian or charge sector.

The canonical thermal purification is the thermofield double

$$|\mathrm{TFD}_\beta\rangle = \frac{1}{\sqrt{Z}} \sum_n e^{-\beta E_n/2} |n\rangle_L |\bar{n}\rangle_R.$$

For the charge-projected ensemble the sum is restricted to that charge. The unprojected grand-canonical sum instead entangles different left-copy charges with corresponding right-copy charges. Preparation by the formal expression $e^{-\beta H/2}$ is imaginary-time filtering; it is not automatically an efficient deterministic unitary protocol.

## 7.8 A finite-temperature lower bound beyond the solvable example

The thermal paramagnet suggests a much broader principle: a finite-energy charge transfer cannot make a finite-temperature state perfectly distinguishable from itself. Under useful boundedness assumptions this can be proved without solving the Hamiltonian.

Let $H = \sum_X h_X$ be a finite-range lattice Hamiltonian with uniformly bounded interactions and bounded coordination. It commutes with an Abelian symmetry, and $P_q$ projects onto a fixed charge sector. Assume that each endpoint admits a bounded-support **unitary** charged operator $O_x$, and that $A = O_x O_y^\dagger$ preserves this sector. Discrete on-site Ising symmetry has exactly this property. In the sector,

$$\rho = \frac{P_q e^{-\beta H}}{Z_q}, \qquad \sigma = A\rho A^\dagger = \frac{P_q e^{-\beta A H A^\dagger}}{Z_q}.$$

The two states have the same support and partition function. Their quantum relative entropy is

$$S(\rho\|\sigma) = \operatorname{Tr}\rho(\ln\rho - \ln\sigma) = \beta \operatorname{Tr}[\rho(AHA^\dagger - H)] \le \beta \|AHA^\dagger - H\|_\infty. \tag{7.15}$$

Terms in $H$ disjoint from both endpoints commute with $A$ and cancel. The remaining terms occupy a bounded number of neighborhoods of fixed size. Consequently

$$\|AHA^\dagger - H\|_\infty \le 2 \sum_{X:\, X \text{ meets an endpoint support}} \|h_X\|_\infty \le C,$$

with $C$ independent of $|x-y|$ and system volume.

For clarity, the needed entropy-overlap inequality can also be derived directly. Diagonalize $\rho = \sum_i p_i |i\rangle\langle i|$ and $\sigma = \sum_j q_j |j'\rangle\langle j'|$ on their common support. The weights $w_{ij} = p_i |\langle i|j'\rangle|^2$ form a probability distribution. Then

$$\mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma}) = \sum_{ij} w_{ij} \exp\left[\frac{1}{2}\ln\frac{q_j}{p_i}\right] \ge \exp[-S(\rho\|\sigma)/2]$$

by scalar Jensen's inequality. Since $F \ge \mathrm{Tr}(\sqrt{\rho}\sqrt{\sigma})$, we obtain

$$F(\rho, A\rho A^\dagger) \ge e^{-\beta C/2} > 0 \qquad (\beta < \infty). \tag{7.16}$$

This lower bound is uniform in separation. It proves fidelity long-range order for this class of fixed-sector thermal states. If ordinary charged correlations also decay, the order is specifically strong-to-weak; if they remain nonzero, strong symmetry is broken further in the conventional sense. The bound can be extremely small at low temperature without being zero, and gives no nonzero guarantee at $\beta = \infty$.

The unitary-insertion assumption is substantive. In a finite-dimensional system with genuine $U(1)$ charge, a nonzero-charge unitary cannot simply raise every charge eigenvalue: a finite spectrum has an upper endpoint. Particle annihilation and creation operators are nonunitary, and a vacuum or completely filled charge sector may lack the necessary fluctuations. Unbounded rotor or bosonic Hilbert spaces introduce norm and domain issues as well. Extending the broad thermal expectation to such cases requires further assumptions about charge density, compressibility, finite-energy insertions, and normalization. The thermal arguments in [7] and the synthesis in [8] should be read with those qualifications, rather than as a theorem that every fixed-charge state at any positive temperature must exhibit every version of SW-SSB.

## 7.9 Exercises and worked solutions

### 7.9.1 Exercise: when is the Choi double a purification?

Determine all $\rho$ satisfying $\rho^2/\operatorname{Tr}\rho^2 = \rho$.

**Solution**

For each nonzero eigenvalue $p_a$, $p_a^2 = p_a \operatorname{Tr}\rho^2$, so all nonzero eigenvalues are equal. If the rank is $d$, normalization gives $p_a = 1/d$ and $\rho = P/d$ for a projector $P$. Conversely, every such state satisfies the relation. The condition includes pure states and maximally mixed sector states, but not a generic Gibbs state.

### 7.9.2 Exercise: trace out a Bell cat

Trace the right copy out of Equation 7.8. Why do the cross terms survive?

**Solution**

On one site, $\mathrm{Tr}_R\,|\Phi^+\rangle\langle\Phi^+| = I/2$ and $\mathrm{Tr}_R\,|\Psi^+\rangle\langle\Psi^+| = I/2$, while $\mathrm{Tr}_R\,|\Phi^+\rangle\langle\Psi^+| = X/2$. Tensoring gives diagonal terms $I/2^N$ and cross terms $U/2^N$. Including the cat's factor $1/2$ yields $(I+U)/2^N$. Orthogonal global Bell strings can have a nonzero partial trace of their off-diagonal outer product; orthogonality only guarantees that its full trace vanishes.

### 7.9.3 Exercise: preparation and observation

A device prepares an arbitrary purification of $\rho$. Does measuring the simple doubled operator $A \otimes A^*$ necessarily return $R_1(A)$?

Solution

No. If the state is $(I \otimes V)|\sqrt{\rho}\rangle\rangle$, that measurement returns the canonical expectation with right operator $V^\dagger A^* V$. It equals $R_1$ only if the relevant transformation is accounted for, or if special commutation relations make it irrelevant. A known local $V$ can be compensated in the observable; an unknown or highly nonlocal one defeats the simple two-point protocol. Canonical-purification availability is a physical resource assumption.

# 8 Stability, locality, and recoverability

An order parameter is more useful when it distinguishes a phase rather than an isolated algebraic curiosity. Fidelity order has two fundamental properties: strongly symmetric shallow local channels cannot remove it, and some local symmetry-breaking noise cannot be repaired locally. These statements involve different operations and different quantifiers. Neither follows from a slogan that noise always increases fidelity.

We first prove the stability mechanism, then connect it to conditional mutual information. Finally, we examine a major 2026 development: the difference between global finite-volume definitions and intrinsic infinite-volume local definitions. Keeping that distinction explicit is necessary for understanding current disagreements about what should count as a mixed-state phase.

## 8.1 What locality and depth mean

Consider a lattice of bounded-degree geometry with finite-dimensional on-site Hilbert spaces. A depth-$D$ local circuit has $D$ layers of gates of bounded spatial range. Its light-cone radius is $O(D)$. A local channel circuit is represented by adding product-state ancillas, applying a local unitary circuit $W$, and tracing out those ancillas. For strong symmetry, every gate of $W$ commutes with the physical symmetry while the ancillas transform trivially. No charge leaks into a discarded bath.

For **finite depth**, $D$ is bounded independently of total system size $L$. For **polylogarithmic depth**, $D$ may grow as a power of $\log L$. These cases must not be silently identified: the support and number of terms in a dressed operator are constant in the first case and grow in the second.

## 8.2 Stability theorem and its proof

Under the preceding assumptions, fidelity long-range order survives a strongly symmetric finite-depth local channel, possibly with a different charged order parameter of somewhat larger but still bounded support [7]. If ordinary charged correlations were short-ranged initially, they remain without long-range order. Thus global SW-SSB is preserved.

Here is the proof architecture with the step involving discarded ancillas made explicit. Suppose

$$F(\rho, O_x O_y^\dagger \rho O_y O_x^\dagger) \geq c > 0$$

at large separation, and let $|\Psi\rangle$ purify $\rho$ on a reference $R$. The trace-norm variational principle gives an auxiliary unitary $V_R$ such that

$$\left|\langle \Psi | O_x O_y^\dagger \otimes V_R | \Psi \rangle\right| \geq c.$$

Apply $W$ to the system and fresh local ancillas, obtaining a purification $|\Phi\rangle$ of $\sigma = \mathcal{E}(\rho)$. Define the dressed operators

$$\widetilde{O}_x = W(O_x \otimes I_E)W^\dagger, \qquad \widetilde{O}_y = W(O_y \otimes I_E)W^\dagger.$$

Then

$$\left|\langle \Phi | \widetilde{O}_x \widetilde{O}_y^\dagger \otimes V_R | \Phi \rangle\right| \geq c. \tag{8.1}$$

Finite depth keeps the two supports disjoint at large separation. Expand each dressed operator in a physical-ancilla operator basis on its finite light cone:

$$\widetilde{O}_x = \sum_{\mu=1}^{M_x} A_{x,\mu} \otimes B_{x,\mu}, \qquad \widetilde{O}_y = \sum_{\nu=1}^{M_y} A_{y,\nu} \otimes B_{y,\nu}. \tag{8.2}$$

There are finitely many terms, with $M_x, M_y$ and all local operator norms bounded independently of $L$ and $|x-y|$. Since the ancillas are symmetry neutral and $W$ is strongly symmetric, each physical coefficient $A_{x,\mu}$ can be chosen with the same charge as $O_x$. The corresponding statement holds at $y$.

Expanding Equation 8.1, the triangle inequality implies that at least one pair of physical coefficients has an overlap bounded below by a positive constant, up to these fixed expansion factors. To turn that statement into a fidelity correlator, use

$$|\langle \Phi | A \otimes B | \Phi \rangle| \leq \|B\|_\infty \|\sqrt{\sigma} A \sqrt{\sigma}\|_1. \tag{8.3}$$

For completeness, this follows by writing $|\Phi\rangle$ in its Schmidt basis. The left side is a trace pairing of $\sqrt{\sigma}A\sqrt{\sigma}$ with a transpose of an auxiliary contraction. Hölder's inequality bounds that pairing by the trace norm times $\|B\|_\infty$. The reference and discarded ancillas together constitute the auxiliary system; no locality of $B$ is required at this last step.

Therefore some charged local physical insertions have

$$F(\sigma, A_{x,\mu} A^\dagger_{y,\nu} \sigma A_{y,\nu} A^\dagger_{x,\mu}) \geq c' > 0,$$

where $c'$ is independent of separation and size. More invariantly, a finite matrix of correlators of charged local operators retains a nonzero long-distance norm. In a homogeneous system one can select fixed operator types; in an inhomogeneous system the local basis may vary with position. This is why the theorem asserts existence of order after the channel, not preservation of the original microscopic $O_x$.

To complete the SW-SSB claim, ordinary order must remain absent. In the adjoint channel, a charged local observable is mapped to a charged observable on its bounded backward cone. At sufficiently large separation the backward cones do not meet, and the product-state ancillas imply factorization of their pullbacks. Long-range ordinary charged correlations at the output would then imply such correlations for bounded-support charged observables at the input, contradicting the assumption. This finishes the finite-depth argument.

### 8.2.1 What the theorem does and does not say

Monotonicity alone would only control $F(\mathcal{E}\rho, \mathcal{E}\sigma)$. In general $\mathcal{E}(A\rho A^\dagger)$ is not $A\mathcal{E}(\rho)A^\dagger$. The dressing and finite ancilla expansion bridge that gap.

The result is one-way. Strong local channels can create fidelity order from an unordered state: the pair-dephasing model in the next chapter does exactly this. They cannot take an ordered state back to a state with exponentially decaying fidelity correlations. Consequently, preparation by one strongly symmetric shallow channel does not establish phase equivalence; one must examine an inverse process.

For growing depth, the number of terms in Equation 8.2 grows exponentially in the light-cone volume. The quantitative extension yields a bound of the form $F \gtrsim \exp[-\mathrm{poly}(D)]$ for appropriately dressed operators [7]. For polylogarithmic $D$, that can vanish as $L \to \infty$. It still prevents certain exponentially small behaviors on system-size separations, but is not automatically a nonzero-order theorem for fixed-support operators. Definitions of phase equivalence using polylogarithmic circuits must include their allowed errors and localization scales.

## 8.3 A complete irrecoverability example

For the parity state $\rho_+ = (I + U)/2^N$, dephase one site $i$ completely:

$$\mathcal{T}_i(\rho) = \tfrac{1}{2}(\rho + Z_i \rho Z_i), \qquad \mathcal{T}_i(\rho_+) = I/2^N. \tag{8.4}$$

Let a recovery channel act on a region $R$ that does not cover the entire system. Its output from the maximally mixed input is $\omega_R \otimes I_{\bar{R}}/2^{|\bar{R}|}$. Since $\bar{R}$ contains at least one site, the output has $\langle U \rangle = 0$, whereas $\rho_+$ has $\langle U \rangle = 1$. Measuring global parity therefore gives

$$T(\rho_+, \mathcal{R}_R \mathcal{T}_i(\rho_+)) \geq \frac{1}{2}. \tag{8.5}$$

The bound is tight: doing nothing leaves $I/2^N$, whose trace distance from $\rho_+$ is exactly $1/2$. Repair requires access to the global parity constraint. Resetting or manipulating a bounded neighborhood cannot reestablish it.

This is why the breaking is called spontaneous even though the original finite-volume state remains strongly symmetric. Arbitrarily weak strong-symmetry-breaking noise distributed through a large sample destroys the constraint, and its effect cannot be undone near a single damaged region. For independent single-site dephasing of strength $p$,

$$\mathcal{D}_p^{\otimes N}(\rho_+) = \frac{I + (1-2p)^N U}{2^N}.$$

For every fixed $p > 0$, the large-$N$ state approaches the unrestricted maximally mixed state. Taking $N \to \infty$ before $p \to 0$ differs from reversing those limits, as in conventional spontaneous ordering.

## 8.4 General local irrecoverability

The parity example is representative. For a simple quantitative derivation assume the charged order operators are Hermitian unitaries, such as Pauli $Z$, and $F_{ij} \geq c$. Apply $\mathcal{T}_i$ and attempt any recovery $\mathcal{R}$ whose support excludes a distant site $j$. Put $\sigma = \mathcal{R}\mathcal{T}_i(\rho)$. Joint concavity and data processing of fidelity give the conservative lower bound

$$F(\sigma, Z_j \sigma Z_j) \geq c/2. \tag{8.6}$$

To verify it, expand both states as equal mixtures of the recovered original and locally flipped states, pair an original branch with the branch carrying $Z_i Z_j$, and use that $\mathcal{R}$ commutes with conjugation at $j$. The paired fidelity is at least $F_{ij}$ by data processing; discarding the other nonnegative term gives Equation 8.6.

Strong symmetry of the target implies $T(\rho, Z_j \rho Z_j) = 1$, using the trace distance $T = \frac{1}{2}\|\rho - \sigma\|_1$ from Chapter 4. By the triangle inequality and unitary invariance,

$$1 \leq 2T(\rho, \sigma) + T(\sigma, Z_j \sigma Z_j) \leq 2T(\rho, \sigma) + \sqrt{1 - c^2/4}.$$

Thus every such local recovery has

$$T(\rho, \sigma) \geq \frac{1 - \sqrt{1 - c^2/4}}{2} > 0. \tag{8.7}$$

The full theorem allows bounded nonunitary charged operators with appropriate Kraus completions and norm factors [7]. The unitary proof already displays all the physics: a local attempted repair cannot eliminate a residual one-point fidelity at a remote site, whereas the strongly symmetric target requires that fidelity to vanish exactly.

## 8.5 Conditional mutual information

Partition a *finite complete system* into $A$, a buffer $B$, and the remaining region $C$. Use natural logarithms and

$$I(A:C|B) = S_{AB} + S_{BC} - S_B - S_{ABC}.$$

For the parity state, every proper subsystem is maximally mixed, but $S_{ABC} = (N-1)\ln 2$. If $A, B, C$ are all nonempty,

$$I(A:C) = 0, \qquad I(A:C|B) = \ln 2. \tag{8.8}$$

Knowing the buffer does not remove the global parity relation between the two remaining regions. The classical ferromagnetic mixture instead has $I(A:C) = \ln 2$ and $I(A:C|B) = 0$: once the buffer reveals which magnetization branch occurred, no additional correlation remains. A GHZ pure state has both quantities equal to $\ln 2$ in this geometry.

The recovery theorem states that a suitable channel $\mathcal{R}_{B\to BC}$ achieves

$$F(\rho_{ABC}, \mathcal{R}_{B\to BC}(\rho_{AB})) \geq e^{-I(A:C|B)/2}. \tag{8.9}$$

The channel can be chosen using rotated or averaged Petz recovery constructions; the plain unrotated Petz map does not satisfy every general strengthened bound automatically [4]. Small CMI guarantees a good recovery. Conversely, a proof that *all* such local recoveries have nonzero error forces CMI to remain nonzero. Large CMI by itself should not be used as the reverse of a one-sided fidelity bound.

To apply this to local noise in $A$, interchange $A$ and $C$ in the recovery theorem. Small CMI then supplies a map $B \to AB$ recovering the erased region $A$ from $\rho_{BC}$. After any channel supported in $A$, discard $A$ entirely: the marginal on $BC$ is still $\rho_{BC}$ by no-signaling. The same recovery map would consequently repair that local channel using only $AB$. This contradicts a uniform lower bound such as Equation 8.7. The local irrecoverability theorem therefore implies long-range CMI for strongly symmetric finite-volume SW-SSB families in this complete-system geometry. The conclusion concerns a uniform large-system limit retaining the global charge constraint. It is not a theorem about every possible infinite-volume CMI construction.

## 8.6 What is a mixed-state phase?

For pure ground states, a gapped path and a quasi-local unitary often provide equivalent pictures of phase connectivity. General channels discard information, so the analogous mixed-state classification is not unique.

**Two-way channel connectivity** asks for shallow local channels $\mathcal{E}$ and $\mathcal{F}$ such that $\mathcal{E}(\rho) \approx \sigma$ and $\mathcal{F}(\sigma) \approx \rho$. Symmetry restrictions may be imposed. This is a state-dependent notion; $\mathcal{F}\mathcal{E}$ need not equal identity on arbitrary inputs. Without symmetry it can identify states with different global conditional information.

For example, the parity state and unrestricted maximally mixed state are two-way connected if symmetry restrictions are removed. One-site dephasing gives the forward direction. For the reverse, reset every site to $|+\rangle$ and apply complete pair dephasing on a connected nearest-neighbor graph. Edge coloring implements this in bounded depth. It prepares the parity ensemble anew. This preparation does not repair the lost global constraint locally; it discards and rebuilds the state.

**Local reversibility** adds a stronger condition: restricting an evolution to a finite region must admit a recovery in a slightly enlarged region, with controlled tails and errors. The recovery must follow the local deformation, rather than merely prepare an equivalent final state from scratch. [16] develops this refinement and shows that it preserves more mixed-state topological data than unrestricted two-way connectivity.

**Finite Markov length** is a useful path criterion. If CMI decays exponentially with buffer width, its decay scale is a Markov length. Under locality and uniformity assumptions, maintaining a finite such scale along local Lindblad evolution permits quasi-local recovery of the evolution using Petz constructions [17]. A diverging Markov length obstructs this argument. When discussing a "phase transition," specify the equivalence relation, the path, whether symmetry is imposed, and which finite-size or infinite-volume information is retained. None of these choices is supplied just by writing $\rho(p)$.

### 8.6.1 Reachability, coarse graining, and stability of dynamics

The distinction between reachability and reversibility has an earlier operational formulation. Fast dissipative reachability compares states using a time-independent local Lindbladian, locality-preserving ancillas, and a global approximation error vanishing with size. In [18], "fast" permits polylogarithmic time and suitable polylogarithmic locality scales. Mutual reachability defines equivalence classes, while one-way reachability orders their dissipative complexity. These are broader asymptotic resources than the strictly bounded-depth, bounded-range circuits in the stability proof above. A conclusion in one resource class should therefore not be transferred to the other without controlling its growing scales and errors.

Real-space renormalization makes two-way connectivity constructive. A coarse-graining channel can discard microscopic variables while preserving their relevant correlations with the complementary region; when those correlations are preserved exactly, recovery reverses its action on the state. Iterating suitably controlled steps can then supply both directions of phase connectivity. This approach maps the two-dimensional thermal toric code at nonzero temperature toward infinite temperature, while decoder-based local channels establish a nontrivial regime for a toric-code state subjected to sufficiently weak local dephasing [19]. Thermal noise and a specified finite-depth dephasing preparation are consequently different state families, even when both create anyons. This construction uses its stated circuit ranges and convergence bounds; it does not make a practical decoder's threshold universally identical to every phase boundary.

A third question concerns the dynamics itself: does a small local perturbation of a generator leave its stationary behavior in a stable phase? The **uniformity** criterion asks, roughly, for nearby perturbed and unperturbed steady states to be related by controlled finite-time dissipative evolution [20]. It addresses a neighborhood of generators rather than only two endpoint density matrices. It supplies stability consequences beyond an isolated spectral-gap assertion; proposed generic relations between the gap and relaxation within uniform phases retain conjectural components. Thus endpoint connectivity, correlation-preserving coarse graining, locally reversible paths, and dynamical uniformity organize related but different questions. None is established as a universal replacement for all the others.

## 8.7 The 2026 local formulation

Global fidelity depends on the entire density matrix. For a finite region $R$ containing $x$, define instead

$$f_R(x) = F(\rho_R, O_x \rho_R O_x^\dagger). \tag{8.10}$$

For unitary $O_x$, data processing under partial trace implies $f_R(x) \geq f_{R'}(x)$ whenever $R \subset R'$. The values are bounded below by zero, so they have a limit as nested regions exhaust the lattice. Local SW-SSB, or local charge incoherence, is diagnosed by a positive limit [21–23]. On a large finite sample, the region should grow while remaining small compared with the whole system.

Global fidelity order implies this local order. Place $y$ outside $R$. Tracing out its complement gives

$$F(\rho, O_x O_y^\dagger \rho O_y O_x^\dagger) \leq F(\rho_R, O_x \rho_R O_x^\dagger),$$

so a positive global lower bound survives. The converse fails. A highly excited thermalizing pure eigenstate may have thermal reduced states on every sufficiently small region. Their local charged fidelity is positive at finite temperature, while the *global* pure-state fidelity correlator equals the decaying ordinary correlator. ETH is an assumption about these eigenstates, not a theorem for all Hamiltonians; under that assumption it provides a clear distinction between local and global definitions.

Local tomography costs exponentially in $|R|$ rather than total volume. With polynomial resources in system size, one can in principle inspect regions with $|R| = O(\log N)$. Extrapolating their fidelity to an infinite-region limit still requires control over convergence. A finite positive $f_R$ alone is only an upper bound on the limit and does not prove order.

## 8.8 Infinite-volume states and a genuine interpretive disagreement

An infinite spin-system state is specified by a compatible family of finite-region density matrices, or equivalently by expectation values of all local observables. The sequence $\rho_+ = (I+U)/2^N$ and the sequence $I/2^N$ have exactly the same reduced state on every fixed proper region once the volume is large enough. They therefore define the same infinite-volume local state. Yet one finite-volume sequence has fixed total parity and global SW-SSB, while the other has no strong finite-volume symmetry at all.

This is a change in which information is retained, not a contradiction. The global definition characterizes a sequence together with its total-charge constraint. An intrinsic infinite-volume definition cannot distinguish two sequences with identical limiting local data.

[23] defines strong symmetry of an infinite state by requiring that the symmetry act trivially on that subsystem in *every extension* of the state to an auxiliary system. In finite dimensions the canonical purification shows that this agrees with $U\rho \propto \rho$. They prove, for compact symmetry groups in their operator-algebraic framework, that strong symmetry is equivalent to zero fidelity with every state obtained by a charged local insertion. In their terminology a positive limiting local fidelity is absence of infinite-volume strong symmetry. They argue that some globally distinct finite-size families should be regarded as the same infinite-volume phase, and explicitly question conclusions drawn solely from their global distinguishability. This is a substantive conceptual perspective, not a proof that global charge diagnostics are experimentally meaningless.

The same caution applies to CMI. For the infinite maximally mixed product state, ordinary CMI of finite regions is zero. The $\ln 2$ in Equation 8.8 survives only when the finite-volume complete-system constraint is retained. [21] states this distinction explicitly and obtains an infinite-volume counterpart using **symmetry-averaged CMI**: first resolve the charge of a large region, compute CMI within each charge sector, then average. Its long-range lower bound requires an additional local-indistinguishability assumption on those charge sectors. It is not ordinary CMI of the unconstrained state. [23] further shows that for fixed bounded $A$ and an exhausting buffer, intrinsic infinite-volume CMI must eventually vanish, without asserting any universal decay rate.

These developments refine the conceptual target of SW-SSB. Questions about a quantum memory with known total charge naturally retain global data. Questions about an infinite stationary local state may prefer the extension-based definition. A careful calculation states which question it answers; a comprehensive theory has to explain how the answers connect.

## 8.9 Exercises and worked solutions

### 8.9.1 Exercise: CMI of three representative states

Compute $I(A:C)$ and $I(A:C|B)$ for a GHZ state, the classical ferromagnetic mixture, and $\rho_+$, with nonempty $A, B, C$ covering all sites.

**Solution**

Every nonempty proper reduction of GHZ has entropy $\ln 2$, while the full state has entropy zero. Thus MI and CMI are both $\ln 2$. Every nonempty reduction of the classical mixture, including the full state, has entropy $\ln 2$, giving MI $\ln 2$ and CMI zero. For $\rho_+$, every proper reduction has entropy equal to its number of sites times $\ln 2$, while the full entropy is $(N-1)\ln 2$. The MI cancels to zero and the CMI leaves $\ln 2$.

### 8.9.2 Exercise: why covariance is insufficient

Apply single-site complete $Z$ dephasing to $\rho_+$. Does this contradict the stability theorem?

**Solution**

The output is $I/2^N$, which is not strongly symmetric under $U$. The channel is weakly covariant but its $Z_i$ Kraus operator anticommutes with $U$. It violates the theorem's strong channel assumption, so there is no

contradiction. Symmetry of averaged probabilities is insufficient to protect a fixed charge sector.

### 8.9.3 Exercise: local evidence and extrapolation

Suppose numerical data show $f_R \approx 0.3$ for regions of linear sizes $4, 6, 8$. What can be concluded without further assumptions?

Solution

Monotonicity gives an upper bound of approximately $0.3$ on the infinite-region local order parameter, not a lower bound. A later crossover could still lead to zero. To establish positive limiting order one needs a rigorous lower bound, a controlled saturation estimate, or finite-size scaling with a justified model. This is particularly relevant near critical points, where the convergence scale can be much larger than accessible regions.

# 9 Solvable models and decoherence-driven transitions

We now build a mixed-state transition from a preparation protocol simple enough to solve. Start with an unentangled paramagnet, apply random nearest-neighbor pair errors, and discard the error record. Total symmetry charge stays fixed, ordinary order never develops, but nonlinear correlations can become long-ranged. The model connects SW-SSB to classical disorder, decoding thresholds, and the distinction between intrinsic and Rényi-weighted transitions [7,12,15].

The derivation has three complementary representations. The error representation gives exact probabilities on a graph. On a chain it produces closed-form correlators. In two dimensions, its cycle redundancy becomes a random-bond Ising model. A second calculation in the $Z$ basis gives the clean Ising model governing Rényi-2. Deriving both makes the inequivalence of the diagnostics concrete.

## 9.1 The pair-error channel on a graph

Let a connected graph have $N$ vertices and $E$ edges, with a qubit at each vertex. Initialize

$$\rho_0 = |+\rangle\langle+|^{\otimes N}.$$

On each edge $e = (ij)$ apply independently

$$\mathcal{E}_{p,e}(\rho) = (1-p)\rho + pZ_iZ_j\rho Z_iZ_j, \qquad 0 \le p \le \tfrac{1}{2}. \tag{9.1}$$

All these superoperators commute. The edge gates can be scheduled in a bounded number of layers on a bounded-degree lattice. The output $\rho_p = \prod_e \mathcal{E}_{p,e}(\rho_0)$ is therefore generated by a strongly symmetric finite-depth channel: each Kraus operator commutes with $U = \prod_i X_i$.

Let $s_e \in \{0,1\}$ indicate whether the error on edge $e$ occurred. Its probability is

$$w_p(s) = \prod_e p^{s_e}(1-p)^{1-s_e}.$$

The error product flips the $X$ eigenvalue at each endpoint of an occupied edge. Define a vertex charge bit

$$n_i = (\partial s)_i = \sum_{e \ni i} s_e \pmod 2. \tag{9.2}$$

Here $\partial$ is the graph incidence map over the two-element field. Every edge has two endpoints, so $\sum_i n_i = 0$ modulo two. The output is diagonal in the $X$ basis:

$$\rho_p = \sum_{n:\sum_i n_i = 0} P_p(n)|n\rangle_X\langle n|, \quad P_p(n) = \sum_{s:\partial s = n} w_p(s). \tag{9.3}$$

The word *syndrome* anticipates error correction: $n$ records error endpoints but forgets which path connected them. Different edge strings with the same boundary differ by cycles.

For $i \neq j$, $\mathrm{Tr}(\rho_p Z_iZ_j) = 0$, because $Z_iZ_j$ has no diagonal matrix elements in this basis. In fact, no conventional long-range charged order is generated. All states have exact even total parity. The interesting structure is entirely in the nonlinear comparison between the probability distribution and its charge-shifted version.

At $p = 0$, only $n = 0$ occurs and the state is pure. At $p = 1/2$, all edge strings are equally likely. The incidence map of a connected graph has rank $N-1$, so each even $n$ has $2^{E-N+1}$ preimages. Therefore

$$P_{1/2}(n) = 2^{-(N-1)}, \qquad \rho_{1/2} = \frac{I+U}{2^N}. \tag{9.4}$$

The protocol interpolates between the pure paramagnet and the canonical parity ensemble without changing its total charge.

## 9.2 Exact fidelity as a classical overlap

Let $\delta^{ij}$ denote the vertex bit string with ones at $i, j$ only. Conjugation by $A = Z_i Z_j$ permutes charge configurations:

$$A\rho_p A = \sum_n P_p(n \oplus \delta^{ij}) |n\rangle_X \langle n|.$$

The two matrices commute. Hence

$$F_Z(i,j) = R_1(i,j) = \sum_n \sqrt{P_p(n) P_p(n \oplus \delta^{ij})}, \tag{9.5}$$

and

$$R_2(i,j) = \frac{\sum_n P_p(n) P_p(n \oplus \delta^{ij})}{\sum_n P_p(n)^2}. \tag{9.6}$$

These expressions contain no replica continuation. They are exact finite-size formulas and provide a direct numerical route: enumerate edge strings, accumulate probabilities by their boundaries, and shift the two endpoint bits.

One should sum probabilities for all indistinguishable error strings *before* taking square roots. Averaging the overlap for each error trajectory instead would keep an error record that the physical mixed state has discarded and would compute a different quantity.

## 9.3 One-dimensional open chain: full solution

On a tree, the boundary $n$ uniquely determines the edge string $s$ subject to even total charge. There are no nontrivial cycles. On an open chain, toggling endpoint charges at $i, j$ therefore toggles exactly the $r = |i - j|$ bonds on the unique connecting path $\gamma$:

$$s \longmapsto s \oplus \gamma.$$

Use Equation 9.5 and factor the sum edge by edge. Off the path, the contribution is $p + (1-p) = 1$. On each path edge,

$$\sum_{s_e=0,1} \sqrt{w_p(s_e) w_p(1-s_e)} = 2\sqrt{p(1-p)}.$$

Thus

$$F_Z(r) = R_1(r) = \left[2\sqrt{p(1-p)}\right]^r. \tag{9.7}$$

The same factorization gives the purity and Rényi-2 numerator:

$$\operatorname{Tr} \rho_p^2 = [p^2 + (1-p)^2]^{N-1},$$

$$R_2(r) = \left[\frac{2p(1-p)}{p^2+(1-p)^2}\right]^r. \tag{9.8}$$

Both have a finite correlation length for every $p < 1/2$, with

$$\xi_F^{-1} = -\ln[2\sqrt{p(1-p)}], \qquad \xi_2^{-1} = -\ln\frac{2p(1-p)}{p^2+(1-p)^2}. \tag{9.9}$$

Writing $p = 1/2 - \epsilon$ gives $\xi_F^{-1} = 2\epsilon^2 + O(\epsilon^4)$ and $\xi_2^{-1} = 8\epsilon^2 + O(\epsilon^4)$. Their common endpoint transition has different approach scales. At $p = 1/2$ both correlators equal one at every separation. There is no finite-noise-strength global SW-SSB transition on this nearest-neighbor open chain; order is reached only at complete dephasing.

The chain result also checks every normalization. At $p = 0$ the state is pure and both correlators vanish for $r > 0$. At $p = 1/2$ the purity is $2^{-(N-1)}$ and the state is uniform on the even sector. For $r = 0$ the insertion is identity and the correlators are one.

### 9.3.1 A controlled local-fidelity calculation

The same chain also lets us test the 2026 local definition without guessing whether a finite-region value has saturated. Take an interval $R = [a, b]$ deep inside a much longer chain and a site $x \in R$. A reduced charge configuration on $R$ depends on the $m + 1$ independent error bits $s_{a-1}, \ldots, s_b$, where $m = b - a + 1$. The two boundary-crossing bonds matter because charge can enter the interval from its complement.

For given charges $n_a, \ldots, n_b$, these $m + 1$ bits have two possible values: a string $s$ and its bitwise complement $\bar{s}$. Their probabilities add:

$$P_R(n) = w_p(s) + w_p(\bar{s}).$$

Flipping only $n_x$ is represented by toggling the left path of $\ell_L = x - a + 1$ bonds from $x$ to the exterior. Toggling the right path of $\ell_R = b - x + 1$ bonds gives the same reduced charge change. The two paths are complementary within these $m + 1$ bits. Put $b_p = 2\sqrt{p(1-p)}$. The reduced-state one-point fidelity obeys the useful exact bounds

$$\max\{b_p^{\ell_L}, b_p^{\ell_R}\} \le F(\rho_R, Z_x\rho_R Z_x) \le \min\{1, b_p^{\ell_L} + b_p^{\ell_R}\}. \tag{9.10}$$

To obtain the lower bound, retain the individual error bits as a classical auxiliary variable. Their distribution and the distribution shifted by the left path have overlap $b_p^{\ell_L}$, by the same edgewise calculation as Equation 9.7. Forgetting the error bits and retaining only the reduced charges is a stochastic channel, so fidelity cannot decrease. Repeat with the right path and take the better bound.

For the upper bound, sum over one representative from each pair $s, \bar{s}$ and use

$$\sqrt{(u+v)(u'+v')} \le \sqrt{uu'} + \sqrt{uv'} + \sqrt{vu'} + \sqrt{vv'}.$$

Here $u = w(s)$, $v = w(\bar{s})$, and the primed weights use the left-path shift. Two terms reconstruct the full edge-string overlap for that left shift; the other two reconstruct the overlap for the complementary right shift. Their sums are respectively $b_p^{\ell_L}$ and $b_p^{\ell_R}$. Normalized fidelity is also at most one, completing Equation 9.10.

For a central probe with $\ell_L = \ell_R = \ell$, the local fidelity lies between $b_p^\ell$ and $2b_p^\ell$. It consequently vanishes as the probe moves far from the boundary for every $p < 1/2$. At $p = 1/2$, the reduced state is maximally mixed and its local fidelity equals one. Thus both the local and global definitions locate the same endpoint in this particular nearest-neighbor chain, despite their inequivalence in general.

This calculation illustrates a productive numerical strategy. A positive small-region fidelity is not sufficient evidence for order, but an analytic upper bound decaying to zero certifies absence of local order. Conversely, a volume-independent positive lower bound certifies its presence. Here both bounds are available from the two alternative paths by which a local charge can communicate with the boundary. The boundary distance, rather than only the number of sites, is the physically relevant scale.

### 9.3.2 Time and the order of limits

The Lindblad generator $\mathcal{L}_e(\rho) = \gamma(Z_iZ_j\rho Z_iZ_j - \rho)$ produces

$$p(t) = \frac{1 - e^{-2\gamma t}}{2}.$$

Inserting this into Equation 9.7 gives $F(r,t) = [\sqrt{1 - e^{-4\gamma t}}]^r$. At large time, $\xi_F \simeq 2e^{4\gamma t}$. For every finite time, the infinite-distance correlator still vanishes; at fixed distance it approaches one as $t \to \infty$. The limits do not commute. This discrete-symmetry model should not be confused with a $U(1)$ diffusive model, whose finite-time length scales can grow very differently.

### 9.3.3 What changes on a ring?

On a ring, $s$ and its complement $s \oplus \mathbf{1}$ have the same boundary. Therefore $P(n)$ sums their two probabilities. A two-point charge shift can use either arc of the ring. The open-chain formula is recovered in the thermodynamic bulk limit with $r$ fixed or much smaller than the ring circumference, but it is not the exact finite-ring formula at arbitrary $r$. This is a useful warning: global cycles and boundary conditions matter for nonlinear state overlaps.

## 9.4 Two dimensions: cycles become dual spins

On a planar square lattice, choose one error string $s^0$ with boundary $n$. Every other solution is $s = s^0 \oplus \ell$, where $\ell$ is a closed loop configuration. Represent loops as domain walls of Ising variables $t_p = \pm 1$ on dual plaquettes, including an exterior plaquette with fixed spin for an open planar sample. If edge $e$ separates dual plaquettes $p, q$,

$$(-1)^{s_e} = \eta_e t_p t_q, \qquad \eta_e = (-1)^{s^0_e}.$$

An individual edge probability can be written as

$$p^{s_e}(1-p)^{1-s_e} = \sqrt{p(1-p)}\exp[K(-1)^{s_e}], \quad e^{-2K} = \frac{p}{1-p}.$$

Substitute this in Equation 9.3:

$$P_p(n) = [p(1-p)]^{E/2} \sum_{\{t_p\}} \exp\left[K \sum_{e=(pq)} \eta_e t_p t_q\right]. \tag{9.11}$$

The sum is the partition function of a random-bond Ising model on the dual lattice. The disorder signs encode the frustration required by the charge endpoints. Changing the representative $s^0$ by a loop is a dual gauge transformation and leaves this partition function unchanged. On a torus, one must additionally sum over noncontractible loop sectors; choosing only plaquette domain walls would omit them.

The disorder strength and coupling obey $\Pr(\eta_e = -1) = p$ and $e^{-2K} = p/(1-p)$ when the representative is sampled from the actual error distribution. This relation is the **Nishimori line** [24,25]. It ties the inference model's assumed noise probabilities to those that generated the data.

To see the disorder average precisely, note that for any function $g$ of a syndrome,

$$\sum_n P_p(n) g(n) = \sum_s w_p(s) g(\partial s).$$

The right side permits choosing the actual sampled string as representative, giving independent random bonds. There is no need to guess a probability distribution for frustrations separately.

## 9.5 Charge transfer as a disorder defect

Choose a path $\gamma$ from $i$ to $j$. The representative $s^0 \oplus \gamma$ has boundary $n \oplus \delta^{ij}$. In the dual Ising partition function this reverses the signs of the bonds crossing the path. The ratio

$$\frac{P_p(n \oplus \delta^{ij})}{P_p(n)} = \frac{Z_{\eta \oplus \gamma}}{Z_\eta} = e^{-\Delta \mathcal{F}_{ij}(\eta)} \tag{9.12}$$

defines a dimensionless defect free-energy cost. Its path dependence is removable by dual gauge transformations, up to global topology. The endpoints are disorder-operator insertions. Consequently,

$$F_Z(i,j) = R_1(i,j) = \mathbb{E}_{\eta,\text{Nishimori}}\left[e^{-\Delta \mathcal{F}_{ij}(\eta)/2}\right]. \tag{9.13}$$

This is a half-moment of a partition-function ratio, not the ordinary disorder-averaged spin correlation and not simply the exponential of a typical free energy.

At weak noise the dual RBIM is ferromagnetically ordered. A long defect line costs free energy proportional to its length, and the fidelity decays. At stronger noise the dual model is disordered, defect endpoints have a finite asymptotic cost, and fidelity becomes long-ranged. Thus *disorder* of the dual Ising model corresponds to SW-SSB *order* of the original density matrix.

### 9.5.1 Why a typical-defect argument is insufficient

Jensen's inequality gives

$$F_Z(i,j) \geq \exp[-\mathbb{E}\Delta \mathcal{F}_{ij}/2].$$

A bounded mean defect cost is therefore sufficient for a nonzero lower bound. An extensive positive mean does not, by itself, prove exponential decay: rare disorder samples can dominate an exponential moment. In fact, normalization gives the exact identity

$$\mathbb{E}[e^{-\Delta \mathcal{F}_{ij}}] = \sum_n P_p(n) \frac{P_p(n \oplus \delta^{ij})}{P_p(n)} = 1 \tag{9.14}$$

for $0 < p < 1/2$. The first moment stays one even where the half-moment decays. This explicitly demonstrates the importance of the full defect-cost distribution. The identification of the critical point and exponents uses the established Nishimori statistical mechanics and direct numerical work, beyond the elementary typical-domain-wall argument [7,8].

For the nearest-neighbor square-lattice model the intrinsic transition occurs at approximately

$$p_c \simeq 0.109. \tag{9.15}$$

This is a numerical estimate, not a closed-form exact result. At criticality the root-fidelity/Rényi-1 correlator decays algebraically; the review reports numerical estimates around $\eta \simeq 0.16$–$0.17$ for this half-moment and correlation-length exponents around $\nu \simeq 1.3$–$1.5$. These ranges reflect numerical determinations and conventions, not rigorous universal bounds. The equality $F = R_1$ is exact in this commuting model.

## 9.6 Rényi-2 maps to a clean Ising model

The clean mapping is most transparent in the $Z$ basis. Label configurations by $z_i = \pm 1$. The initial density matrix has every entry equal to $2^{-N}$. Conjugation by an edge error multiplies the entry $(z, z')$ by $(z_i z_j)(z'_i z'_j)$. Thus

$$\rho_p(z, z') = 2^{-N} \prod_{e=(ij)} \left[(1-p) + p\tau_i\tau_j\right], \qquad \tau_i = z_i z'_i. \tag{9.16}$$

Put $a = 1 - 2p > 0$. The bracket equals one for $\tau_i\tau_j = +1$ and $a$ for $-1$, so it is $\sqrt{a}\, e^{K_0\tau_i\tau_j}$ with $e^{-2K_0} = a$. Squaring the density-matrix entries and summing gives

$$\operatorname{Tr}\rho_p^2 = 2^{-N} a^E \sum_{\{\tau\}} e^{K_2 \sum_{\langle ij\rangle}\tau_i\tau_j}, \qquad K_2 = 2K_0 = -\ln(1-2p). \tag{9.17}$$

The factor $2^{-N}$ follows because each relative configuration $\tau$ has $2^N$ pairs $(z, z')$. For the numerator with $A = Z_x Z_y$, an extra factor $z_x z_y z'_x z'_y = \tau_x\tau_y$ appears. Therefore

$$R_2(x, y) = \langle \tau_x\tau_y \rangle_{\text{clean Ising}, K_2}. \tag{9.18}$$

On the infinite square lattice, the clean Ising critical coupling is $K_c = \frac{1}{2}\ln(1+\sqrt{2})$. Solving $K_2(p_c^{(2)}) = K_c$ gives

$$p_c^{(2)} = \frac{1-\sqrt{\sqrt{2}-1}}{2} \simeq 0.178203. \tag{9.19}$$

This threshold follows exactly from the clean model's critical point. At it $R_2(r) \sim r^{-1/4}$ and $\xi_2 \sim |p - p_c^{(2)}|^{-1}$. It lies well inside the fidelity-ordered regime $p > p_c$. Thus the two-copy correlator has an additional critical singularity without defining an additional fidelity-order boundary.

| Channel parameter | Fidelity and Rényi-1 | Rényi-2 | Ordinary charged correlator |
|---|---|---|---|
| $0 \le p < p_c$ | Short-ranged | Short-ranged | Zero for separated $Z$ insertions |
| $p_c < p < p_c^{(2)}$ | Long-ranged | Short-ranged | Zero |
| $p_c^{(2)} < p \le 1/2$ | Long-ranged | Long-ranged | Zero |

This table excludes the critical points themselves, where the relevant correlator is algebraic rather than a nonzero asymptotic constant.

## 9.7 Recoverability and the physical phase boundary

If $p_1 < p_2 < 1/2$, applying additional pair dephasing of strength

$$q = \frac{p_2 - p_1}{1 - 2p_1} \tag{9.20}$$

takes $\rho_{p_1}$ to $\rho_{p_2}$. This follows from $1 - 2p_2 = (1-2q)(1-2p_1)$. The nontrivial question is whether a strongly symmetric local channel can reverse that change on this family.

[7] constructs recovery using local Petz maps and the finite correlation scale of the associated RBIM away from its transition. States on the same side of the Nishimori transition can be connected in both directions with controlled shallow, logarithmic-scale constructions. Crossing the intrinsic boundary in the restoring direction is obstructed

by fidelity stability. Crossing $p_c^{(2)}$ while remaining above $p_c$ does not obstruct that recovery. The Rényi-2 order can therefore disappear under a suitable strongly symmetric recovery channel even though fidelity order remains.

There is a subtle information-theoretic point. The SW-SSB phase can have a nonzero CMI plateau due to fixed global parity. Recovery of a strongly symmetric local deformation need not erase that plateau. The relevant sufficient condition is exponential smallness of the **CMI loss**

$$\Delta I = I(A:C|B)_{\rho_{\text{initial}}} - I(A:C|B)_{\rho_{\text{after local channel}}}, \tag{9.21}$$

as the buffer grows, rather than exponential smallness of CMI itself. A charge-preserving deformation leaves the global constraint in place, and the common plateau cancels in $\Delta I$. Strong-symmetry-breaking erasure of one site does destroy it and cannot be repaired locally. These are different operations, fully consistent with the irrecoverability theorem.

The recovery derivation uses the statistical model's finite correlation length and control of boundary free energies, including intermediate spatially inhomogeneous states. It should not be promoted into a theorem that every arbitrary noisy family with one nonzero fidelity correlator has exactly two phases. The solvable model provides a well-supported instance of the more general framework.

## 9.8 Beyond the Pauli model

Several deformations are scientifically distinct. Changing the initial state introduces correlations and possibly SPT order. Changing graph geometry or adding long-range pair channels changes the effective statistical mechanics. Replacing $Z_iZ_j$ by a continuous phase rotation changes the noise while preserving total parity. Adding single-site $Z_i$ errors explicitly removes strong symmetry and rounds or changes the problem.

For example, consider $(1-p)\rho + pe^{i\theta Z_iZ_j}\rho e^{-i\theta Z_iZ_j}$. Its off-diagonal weights involve $b = |1-p+pe^{2i\theta}|$. Replica calculations show the same relative-domain-wall structure as the Pauli channel, with a modified tension. [7] argues for Nishimori-type universality under suitable continuation assumptions. The Pauli mapping above is exact; the general non-Pauli universality conclusion uses replica reasoning and is accordingly less rigorous. Preserving that distinction prevents a useful physical conjecture from being mistaken for an unrestricted theorem.

The next chapters explain why the same error-endpoint probabilities govern quantum memory, how gauging promotes charge transport to string operators, and why different logical sectors can survive different amounts of decoherence.

## 9.9 Exercises and worked solutions

### 9.9.1 Exercise: the tree generalization

Derive the fidelity correlator on an arbitrary finite tree with nonuniform edge error probabilities $p_e$.

Solution

The incidence map on a tree is a bijection from edge strings to even vertex syndromes. The unique path $\gamma_{ij}$ toggles the charge endpoints. Edgewise summation therefore yields $F_Z(i,j) = R_1(i,j) = \prod_{e\in\gamma_{ij}} 2\sqrt{p_e(1-p_e)}$. Nonuniformity changes the accumulated tension, while branching alone does not create the cycle redundancy responsible for the two-dimensional planar mapping.

### 9.9.2 Exercise: derive the clean threshold

Starting from Equation 9.16, explain why the coupling for Rényi-2 is $-\ln(1-2p)$ rather than $-\frac{1}{2}\ln(1-2p)$, and solve for the square-lattice critical $p$.

Solution

An individual matrix element contains $e^{K_0 \sum \tau_i \tau_j}$ with $K_0 = -\frac{1}{2}\ln(1-2p)$. The purity multiplies two matrix elements, doubling the coupling. Thus $K_2 = 2K_0$. Setting it equal to $K_c = \frac{1}{2}\ln(1+\sqrt{2})$ gives $1-2p = e^{-K_c} = \sqrt{\sqrt{2}-1}$ and Equation 9.19. Forgetting the square would give the wrong threshold.

### 9.9.3 Exercise: a misleading disorder average

Why does Equation 9.14 not force the half-moment fidelity to stay nonzero at large distance?

Solution

Let $X = e^{-\Delta\mathcal{F}} \geq 0$. Knowing $\mathbb{E}X = 1$ gives only $\mathbb{E}\sqrt{X} \leq 1$ by concavity. For a simple example, let $X = M$ with probability $1/M$ and zero otherwise. Then $\mathbb{E}X = 1$ but $\mathbb{E}\sqrt{X} = M^{-1/2} \to 0$. Rare large likelihood ratios can preserve the first moment while the overlap measured by the half-moment vanishes.

### 9.9.4 Exercise: composition is not inversion

Can the additional-noise formula Equation 9.20 be used with $p_2 < p_1$ to construct a physical inverse dephasing channel?

Solution

It would require $q < 0$, which is not a valid probability and does not define this CPTP channel. Recovery within a phase is state-dependent and uses information in neighboring degrees of freedom. It is not obtained by assigning negative noise to every edge or by inverting the channel on all possible input states.

# Part III

# III · Connections and Frontiers

# 10 Error correction, inference, and topological order

A symmetry charge can remain exactly fixed while its location becomes impossible to infer. The preceding chapters described this as strong-to-weak order. Error correction asks a closely related question: after local errors have occurred, can an observer infer enough about them to recover a global quantum state? The useful connection is between **recovering information** and **distinguishing physically different error classes**. It is more precise than the loose statement that noise destroys order.

We will construct the connection from stabilizers, derive its statistical mechanics, and identify exactly where a quantum-memory interpretation enters. Throughout this chapter syndrome measurements are perfect unless stated otherwise, the noise is an unconditioned channel, and logarithms are natural. A decoder may use the complete syndrome record. Its classical computational cost is a separate issue from whether recovery is possible in principle.

## 10.1 Stabilizers as constraints

A Pauli string is a tensor product of single-qubit operators chosen from $I, X, Y, Z$, with an overall phase. Two Hermitian Pauli strings either commute or anticommute. Let $S_1, \dots, S_r$ be independent, commuting Hermitian Pauli strings on $n$ qubits, with no product equal to $-I$. Their common $+1$ eigenspace is a stabilizer code. Its projector is

$$P_{\mathcal{C}} = \prod_{a=1}^{r} \frac{I + S_a}{2}. \tag{10.1}$$

Each independent constraint halves the dimension, so $\dim \mathcal{C} = 2^{n-r} = 2^k$. The code encodes $k$ logical qubits. The maximally mixed code state is $\rho_{\mathcal{C}} = P_{\mathcal{C}}/2^k$. Expansion of the product gives an especially useful form,

$$\rho_{\mathcal{C}} = 2^{-n} \sum_{S \in \mathcal{S}} S, \qquad S(\rho_{\mathcal{C}}) = k \ln 2, \tag{10.2}$$

where $\mathcal{S}$ is the group generated by the stabilizers. The trace of a nonidentity Pauli string vanishes; this checks normalization directly.

A Pauli error $E$ changes the measured value of $S_a$ precisely when it anticommutes with it. Write the syndrome as bits $s_a$:

$$S_a E = (-1)^{s_a} E S_a.$$

Errors $E$ and $ES$ with $S \in \mathcal{S}$ act identically on every codeword. Errors with the same syndrome can nevertheless act differently: their ratio may be a logical operator that preserves the code space but changes its encoded state. Correcting the syndrome alone does not determine whether a logical error remains.

For a stochastic Pauli channel $\mathcal{N}(\rho) = \sum_E P(E) E \rho E^\dagger$, the decoder should infer an equivalence class, not the exact microscopic history. Fix a reference error for each syndrome, and label the remaining logical Pauli action by $\ell$. Then

$$P(s, \ell) = \sum_{E \in (s,\ell)} P(E), \qquad P(\ell|s) = \frac{P(s,\ell)}{\sum_{\ell'} P(s,\ell')}. \tag{10.3}$$

The optimal syndrome-dependent Pauli correction picks the most probable $\ell$. Its success probability is

$$P_{\text{succ}}^{\text{ML}} = \sum_{s} \max_{\ell} P(s, \ell). \tag{10.4}$$

"Maximum likelihood" here sums every history in a class. A minimum-weight decoder chooses a single low-weight representative; entropy can make that choice suboptimal.

### 10.1.1 Why correcting a subspace is possible

The exact recovery theorem originates in the general theory of quantum error-correcting codes [26].

There is an exact criterion behind the syndrome construction. Let $V$ encode a logical Hilbert space into the physical one, so $V^\dagger V = I_L$. A collection of errors $E_a$ is exactly correctable on that entire space when

$$V^\dagger E_a^\dagger E_b V = c_{ab} I_L. \tag{10.5}$$

This is the Knill-Laflamme condition. It says that overlaps between error images carry no information about which logical vector was encoded. To see why that matters, use a dilation $|\psi_L\rangle \mapsto \sum_a E_a V|\psi_L\rangle \otimes |a\rangle_E$, with the error weights absorbed into $E_a$. The environment matrix has elements $\langle\psi_L|V^\dagger E_b^\dagger E_a V|\psi_L\rangle$. If it depends on the logical input, the environment has learned information about a superposition and the system alone cannot reverse the interaction on every input. Independence for every vector forces the matrix to be scalar, by the polarization identity. Correcting arbitrary entangled inputs requires this full operator condition, rather than agreement only on a chosen basis.

The sufficiency is constructive. Diagonalize the positive matrix $c$ by mixing the error operators. In that error basis, $V^\dagger F_a^\dagger F_b V = d_a \delta_{ab} I_L$. For $d_a > 0$, each $F_a V/\sqrt{d_a}$ is an isometry, and different $a$ have orthogonal image spaces. A recovery can identify that image space without learning the logical vector, then invert its isometry. An error label can thus be measured harmlessly when it contains no logical information. This is exactly what an ideal syndrome does.

For Pauli stabilizer codes, distinct syndromes give orthogonal spaces. Errors differing by a stabilizer act identically on the logical input. Errors with the same syndrome but different logical actions violate the scalar condition when included together. Decoding is therefore an inference problem about which of those conflicting alternatives occurred.

At any finite size, a stochastic channel assigning positive probability to every Pauli error generally cannot be corrected exactly: its rare logical errors remain possible. A threshold asserts **approximate** recovery with failure tending to zero as size grows. It does not assert the exact condition for the complete error set at every nonzero noise rate. This distinction explains why a positive-noise threshold and the existence of arbitrarily unlikely undetectable logical errors are compatible.

### 10.1.2 Worked example: a three-qubit repetition code

Take $S_1 = Z_1 Z_2$, $S_2 = Z_2 Z_3$, with codewords $|000\rangle$ and $|111\rangle$. For independent bit flips of probability $p < 1/2$, the trivial syndrome can arise from $I$ or $X_1 X_2 X_3$. Their probabilities are $(1-p)^3$ and $p^3$. A nontrivial syndrome likewise has a one-flip representative and its complementary two-flip representative. Choosing the lower-weight member gives

$$P_{\text{fail}} = 3p^2(1-p) + p^3.$$

The logical bit is more reliable than one physical bit for small $p$. This code does not protect an arbitrary qubit against arbitrary noise: a single $Z_i$ already acts as a logical phase error. The example illustrates both the value and the limitation of inference from a restricted noise model.

## 10.2 Toric-code geometry before statistical mechanics

Put a qubit on every edge of a square lattice with periodic boundaries. Define

$$A_v = \prod_{e \ni v} X_e, \qquad B_f = \prod_{e \in \partial f} Z_e, \qquad H_{\mathrm{TC}} = -\sum_v A_v - \sum_f B_f. \tag{10.6}$$

A star and a face share zero or two edges and therefore commute. On a torus, $\prod_v A_v = \prod_f B_f = I$. Euler's formula $V - E + F = 0$ then gives $k = E - (V-1) - (F-1) = 2$: four ground states.

A string of $Z$ operators along an open direct-lattice path anticommutes with $A_v$ at its two endpoints. The endpoint violations are called electric anyons, or $e$ particles. An $X$ string on the dual lattice creates magnetic $m$ particles at its endpoints, where $B_f = -1$. Contractible closed strings are stabilizers. Noncontractible strings wind around a handle and act on logical qubits.

The crucial distinction is topological. A syndrome specifies the endpoints of an error chain. It does not specify whether that chain wound around the torus. Local stabilizer measurements cannot see this difference, because a noncontractible closed string has no endpoints. This is why local constraints can protect nonlocal information. The foundational error-correction interpretation was developed in [25] and extended to faulty syndrome measurements in [27].

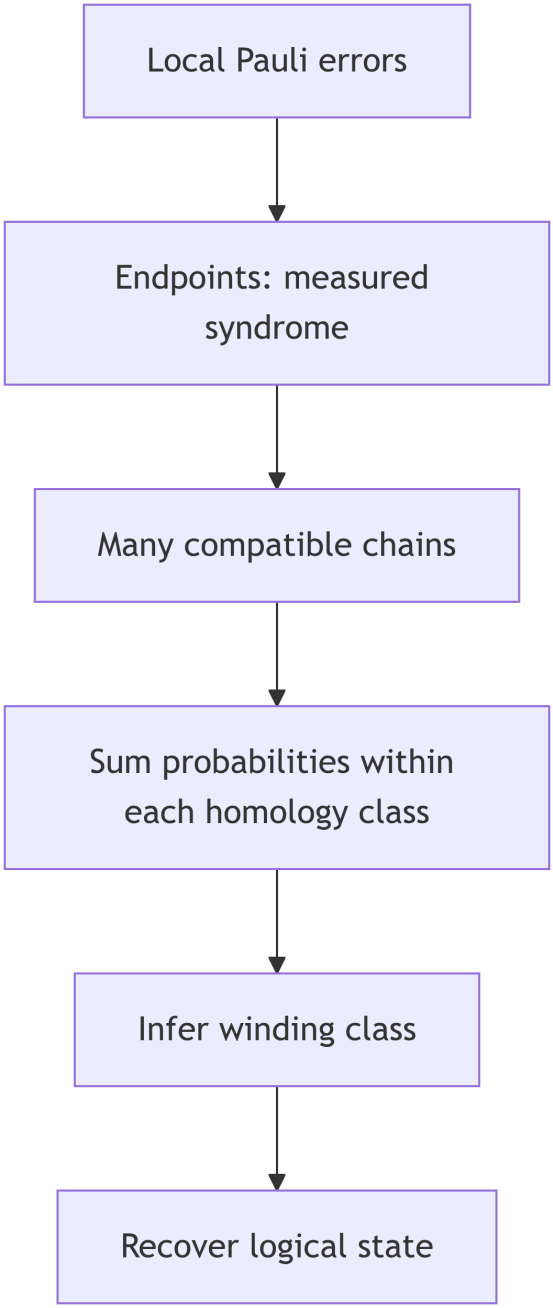



Figure 10.1: Decoding infers a logical class by summing compatible microscopic histories.

For the moment consider only $Z$ errors. Represent an error by binary edge variables $e_a \in \{0, 1\}$, with probability

$$P(e) = p^{|e|}(1-p)^{N_e-|e|}.$$

The syndrome is the boundary $s = \partial e$, computed modulo two. Adding the boundary of a face changes the error by $B_f$ and leaves its effect on the code unchanged. On a torus, two chains with the same endpoints differ by face boundaries and one of four homology classes $h \in \mathbb{Z}_2^2$.

## 10.3 Deriving the Nishimori condition

Choose one representative $e$ consistent with a measured syndrome. Introduce signs $\tau_a = (-1)^{e_a}$. Multiplying by a face stabilizer can be represented by an Ising variable $\sigma_f = \pm 1$ on each face: an edge between faces $f$ and $f'$

is toggled when $\sigma_f \sigma_{f'} = -1$. A fixed sign pattern $\eta_a(h)$ inserts the required noncontractible twist. Thus every chain in class $h$ has signs

$$(-1)^{e'_a} = \tau_a \eta_a(h) \sigma_f \sigma_{f'}.$$

For one binary error variable,

$$p^{e'_a}(1-p)^{1-e'_a} = \frac{e^{K(-1)^{e'_a}}}{2\cosh K}, \qquad e^{-2K} = \frac{p}{1-p}. \tag{10.7}$$

This identity fixes the effective inverse temperature $K$. It is not an independently adjustable laboratory temperature. Summing the probabilities in an error class therefore gives, up to an $h$-independent redundancy factor,

$$Z_h(\tau) = \sum_{\{\sigma_f\}} \exp\left[K \sum_a \tau_a \eta_a(h) \sigma_f \sigma_{f'}\right], \qquad P(h|s) = \frac{Z_h(\tau)}{\sum_{h'} Z_{h'}(\tau)}. \tag{10.8}$$

This is a random-bond Ising model. For disorder averages, we may use the actually drawn physical error as the reference chain: then $\tau_a = -1$ independently with probability $p$, tied to $K$ by Equation 10.7. This matched disorder-temperature relation is the Nishimori condition. It is a device for computing ensemble-averaged performance, not a prescription requiring the decoder to know the error. A representative chosen from the syndrome alone need not have independent bond signs; changing that representative changes variables and relabels the four sectors without changing the inference problem.

To see why a transition is relevant, compare free energies of twisted and untwisted sectors,

$$\Delta F_h = -\ln \frac{Z_h}{Z_0}.$$

In a phase with a macroscopic domain-wall cost, incorrect winding sectors are suppressed as the system grows. Their posterior probabilities vanish, and recovery becomes asymptotically reliable. When a spanning domain wall costs only order one, distinct windings can have comparable posterior probabilities. Global logical information is no longer recoverable with probability approaching one.

The familiar threshold near $p \simeq 0.109$ belongs to the square-lattice toric code with independent errors of one Pauli type, perfect syndrome information, and optimal decoding. It should not be transplanted to depolarizing noise, correlated channels, different lattices, finite measurement rounds, or a particular practical decoder. The spin-model mapping explains the meaning of the threshold; the numerical value is a property of that model, obtained by solving its disordered statistical mechanics. Information-theoretic mixed-state diagnostics reproduce this threshold in the specified toric-code setting [28].

### 10.3.1 The origin of quenched rather than annealed averages

In an experiment, a syndrome is drawn once and then used for inference. The relevant calculation first normalizes the posterior for that syndrome and only afterward averages the outcome over syndromes. An expression such as $\mathbb{E}_\tau[\ln Z_\tau]$ is consequently natural. Replacing it by $\ln \mathbb{E}_\tau[Z_\tau]$ changes the inference problem. Replica calculations sometimes make this distinction easy to overlook: the replica number required for an entropy or fidelity limit need not be the integer replica number describing a purity.

## 10.4 Quantum memory requires a reference

Recovering one known state is a weak test. A channel that discards the input and prepares that state passes it. To test a code as a quantum memory, entangle its logical space $L$ with an inaccessible reference $R$:

$$|\Phi\rangle_{RL} = 2^{-k/2} \sum_{a=1}^{2^k} |a\rangle_R |a\rangle_L,$$

encode $L$ into the physical qubits $B$, and apply the noise only to $B$. Define coherent information by

$$I_c(R\rangle B) = S(B) - S(RB) = -S(R|B).$$

For a noiseless code, $RB$ is pure and $I_c = k \ln 2$. For the stochastic Pauli channel above, distinct pairs $(s, \ell)$ produce orthogonal syndrome-resolved logical Bell states. Hence $S(RB) = H(s, \ell)$. Tracing out $R$ makes the logical state maximally mixed in every syndrome sector, so

$$S(B) = H(s) + k \ln 2, \qquad I_c = k \ln 2 - H(\ell|s). \tag{10.9}$$

This derivation uses a maximally entangled logical input, Pauli noise, and orthogonal perfect syndromes. It is not a general formula for arbitrary non-Pauli channels. It nevertheless exposes the physics particularly clearly: inaccessible logical uncertainty subtracts directly from coherent information.

At vanishing posterior logical entropy, correction returns every logical state and its entanglement with $R$. At extensive uncertainty among logical Paulis, it cannot. For the toric code, $k = 2$ is fixed, so "extensive" here means uncertainty over the full logical algebra rather than a quantity proportional to physical volume. Coherent information is a useful diagnostic for this family, not automatically the capacity of a general channel after arbitrary block coding or multiple uses.

### 10.4.1 Worked example: quantum information can become classical

Consider one encoded qubit with an effective logical dephasing channel

$$\mathcal{D}_q(\rho_L) = (1-q)\rho_L + q\overline{Z}\rho_L\overline{Z}.$$

Applied to half a Bell pair, it creates a mixture of two orthogonal Bell states, giving $I_c = \ln 2 - h(q)$, where $h(q) = -q \ln q - (1-q)\ln(1-q)$. At $q = 1/2$, $I_c = 0$. Yet $|0_L\rangle$ and $|1_L\rangle$ remain perfectly distinguishable. The memory retains a classical bit and loses the phase needed for an arbitrary qubit. This is the simplest reason that "topological memory survives" must specify whether classical or quantum information is meant.

For independent $X$ and $Z$ noise, the syndrome/Pauli-class bookkeeping includes both types of logical uncertainty. When only one type proliferates, a classical logical algebra can remain. When both complementary logical observables are randomized, even that remnant can disappear. Correlations between $X$ and $Z$ errors require their joint posterior; treating them as independent would discard useful information.

The distinction between one codeword and its logical family also enters phase classification. For the toric code under local noise, correlation-preserving renormalization and truncated decoding channels establish a low-noise mixed topological phase; within that framework, destroying logical information requires leaving the toric-code phase [19]. This is a model-specific connection between recoverability and phase structure, with locality and approximation hypotheses on the channels.

## 10.5 From charge inference to gauging

Why do global symmetry problems and topological decoding problems repeatedly produce the same spin models? A useful bridge is gauging. In a spin system, a global $\mathbb{Z}_2$ operation flips every spin simultaneously. Gauging introduces link variables and makes a local flip into a redundancy, with a Gauss constraint at each vertex. In a pure gauge description, a fixed-point toric-code wavefunction can be constructed explicitly:

$$|\psi_{\mathrm{TC}}\rangle \propto \prod_v (I + A_v)|0\cdots 0\rangle_Z. \tag{10.10}$$

The starting state has $B_f = +1$ everywhere. Applying all products of stars preserves those flux constraints and sums over gauge-related configurations. A global redundancy $\prod_v A_v = I$ accounts for one repeated term. Additional noncontractible choices produce the four torus sectors.

The state is a **coherent** sum. Replacing it by a classical probability distribution over the same configurations removes off-diagonal matrix elements between configurations. The resulting loop ensemble can retain global winding constraints and classical sector labels without preserving all quantum logical superpositions. Gauging thus turns questions about delocalized charge and domain walls into questions about flux, strings, and logical sectors. It does not assert that every ungauged SW-SSB phase is equivalent to every mixed topological phase.

The correspondence is especially transparent for Abelian noise models, where errors and charges form additive groups. A non-Abelian theory requires fusion channels as well as charge labels; Chapter 12 develops that extra structure.

## 10.6 What doubling does and does not establish

Vectorizing a density matrix turns a channel into an operator on ket and bra copies. In a topological theory, those copies carry conjugate topological data. A noise event can create correlated excitations across them. This motivates interpreting decoherence as a defect coupling the two copies, and a noise-driven transition as condensation at that defect. The errorfield-double construction develops this idea for Abelian topological order [29].

For the argument's physical content, imagine an error string on the ket copy together with its partner on the bra copy. Their endpoint pairs can proliferate under averaging. Operators that braid nontrivially with those proliferated pairs no longer distinguish coherent sectors in the same way. The surviving algebra depends on which paired excitations proliferate, which is why different error channels can leave different logical information.

A vectorized state $|\rho\rangle\rangle/\sqrt{\mathrm{Tr}\,\rho^2}$ weights eigenstates by their probabilities rather than their square roots. Its entanglement and ordinary correlators describe purity-normalized quantities. An integer-replica transition in this doubled state can therefore differ from a fidelity or optimal-recovery transition. To connect a doubled calculation to a physical memory, identify the channel, input/reference, retained record, replica limit, and recovery criterion explicitly.

Measurements provide another instructive comparison. A measurement-prepared state with its outcomes retained may be correctable by classical feedforward. Discarding the record produces a different density matrix. The relation between measurement-prepared order, classical statistical models, and decoding protocols was made explicit in [30]. The distinction between conditioning and averaging is indispensable when comparing such constructions with decoherence.

## 10.7 Limits of the correspondence

There are at least four independent axes. First, noise may be Pauli or coherent. Coherent amplitudes can interfere, so a positive classical probability model may no longer exist in the same form. Second, syndrome measurements may themselves fail; repeated extraction then produces a spacetime inference problem. Third, a practical decoder can fail below the information-theoretic threshold because its algorithm ignores correlations. Fourth, a mixed-state phase definition can preserve more data than a particular decoder tests.

Strong-to-weak order concerns nonlinear distinguishability of charge-modified states. Error correction concerns distinguishability and reversibility of an entire family of encoded states. The overlap is powerful because both ask

what global information remains after local noise, but neither phrase replaces the other's hypotheses. A density matrix with nonzero topological negativity need not encode a recoverable qubit, and a classical loop ensemble can contain global conditional information without ordinary bipartite entanglement.

## 10.8 Exercises and worked solutions

### 10.8.1 Exercise 1: entropy of a stabilizer marginal

Let $\rho = 2^{-n}\sum_{S\in\mathcal{S}} S$, and let $A$ contain $n_A$ qubits. Suppose the subgroup $\mathcal{S}_A$ supported entirely inside $A$ has $r_A$ independent generators. Find $S(A)$.

**Worked solution**

Partial trace annihilates every Pauli string acting nontrivially outside $A$. Thus $\rho_A = 2^{-n_A}\sum_{S\in\mathcal{S}_A} S_A$. This is the normalized projector onto the common $+1$ eigenspace of $r_A$ constraints, of rank $2^{n_A-r_A}$. Its nonzero eigenvalues are all $2^{-(n_A-r_A)}$, and $S(A) = (n_A - r_A)\ln 2$. The formula applies to this flat stabilizer density matrix; a general weighted mixture of stabilizer sectors need not have a flat spectrum.

### 10.8.2 Exercise 2: posterior probability, not representative weight

A measured syndrome has two logical classes. Class 0 contains one history of probability $0.10$ and class 1 contains four histories of probability $0.04$ each. Which class should a maximum-likelihood decoder select? What is its conditional success probability?

**Worked solution**

The total syndrome probability is $0.10+4(0.04) = 0.26$. The class posteriors are $5/13$ and $8/13$. Select class 1, with success probability $8/13$. Picking the most probable individual history selects class 0 and succeeds only with probability $5/13$. The example isolates the configurational entropy that becomes a partition sum in Equation 10.8.

### 10.8.3 Exercise 3: logical dephasing with a classical flag

For the one-qubit channel $\mathcal{D}_q$, suppose the environment supplies the error bit $a \in \{0,1\}$ to the decoder. Calculate coherent information when that classical flag is included in the output.

**Worked solution**

The joint state is $\sum_a p_a|a\rangle\langle a|_C \otimes |\Phi_a\rangle\langle\Phi_a|_{RB}$. Its entropy is $H(a) = h(q)$. The $BC$ marginal has entropy $\ln 2 + h(q)$ because each conditional $B$ state is maximally mixed. Therefore $I_c(R\rangle BC) = \ln 2$ for all $q$. Applying $\overline{Z}^a$ recovers the Bell state exactly. The same physical interaction is reversible or irreversible depending on whether its record is available.

### 10.8.4 Exercise 4: a twist costs information

Suppose one syndrome has two possible logical classes with partition functions $Z_0$ and $Z_1 = e^{-\Delta}Z_0$, with $\Delta \geq 0$. Compute its maximum-likelihood success probability and its contribution to the conditional logical entropy.

Worked solution

The posterior is $P_0 = 1/(1+e^{-\Delta})$, $P_1 = 1/(1+e^{\Delta})$. Success is $P_0$. Its entropy is $h(P_1)$. If $\Delta = cL$ with $c > 0$, $P_1 \sim e^{-cL}$ and $h(P_1) \sim (cL+1)e^{-cL}$, so this syndrome becomes reliably correctable. If $\Delta$ stays finite, its uncertainty remains nonzero. A threshold statement requires averaging over the syndrome distribution and controlling rare syndromes as well as typical ones.

### 10.8.5 Exercise 5: torus versus sphere

Count toric-code logical qubits on a sphere using $V - E + F = 2$.

Worked solution

There are again two redundant stabilizer constraints. Thus $k = E - V - F + 2 = 0$. The ground state is unique on a sphere. Local anyon excitations and their braiding still exist, but there are no noncontractible cycles carrying ground-space logical qubits. Topological order is more than ground-state degeneracy on one chosen manifold.

# 11 Mixed-state SPT order, boundaries, and anomalies

An ordinary paramagnet and a one-dimensional cluster state have short-ranged bulk correlations. Nevertheless, if an appropriate symmetry must be preserved, they cannot be smoothly connected while keeping a unique gapped ground state. A boundary reveals the difference: it carries degrees of freedom on which the symmetry acts projectively. These are symmetry-protected topological, or SPT, phases.

The mixed-state question is subtler. Part of a protecting symmetry may remain strong while another part becomes weak. Some string correlations survive, some information becomes classical, and doubled-space diagnostics can retain structure after ordinary correlations have disappeared. To understand these statements, we first derive a pure cluster state's protection and then perform a dephasing channel exactly. This also gives an example of why nonzero string order, quantum entanglement, SW-SSB, and robustness of a Lindblad steady state must be distinguished.

## 11.1 A pure SPT state from commuting operators

On a chain of qubits define

$$K_j = Z_{j-1} X_j Z_{j+1}.$$

Neighboring $K_j$ share two anticommuting Pauli pairs; the minus signs cancel, so all $K_j$ commute. On a ring their common $+1$ eigenstate is the cluster state $|C\rangle$, the unique ground state of $H_C = -\sum_j K_j$.

For an even-length ring, split the sites into two sublattices and define

$$G_A = \prod_{j\in A} X_j, \qquad G_B = \prod_{j\in B} X_j.$$

The neighboring $Z$ factors cancel in a product over one sublattice, giving $G_A = \prod_{j\in A} K_j$ and similarly for $G_B$. Thus $G_A|C\rangle = G_B|C\rangle = |C\rangle$. Both symmetries are strong.

There is a finite-depth unitary preparation. Begin with $|+\rangle$ on every site and apply a controlled-$Z$ gate on every bond. The gate is diagonal, with entries $(1, 1, 1, -1)$ in the computational basis, and conjugates $X_j$ to $K_j$. Two bond layers suffice. Hence the cluster state is short-range entangled when symmetry is ignored. Its nontriviality concerns symmetric deformations, not an impossibility of preparing it with any shallow circuit.

### 11.1.1 Deriving the edge projective representation

Open a chain of even length $N$, retain $K_2, \dots, K_{N-1}$, and omit boundary stabilizers. There are two unconstrained qubits. Operators for the left one can be chosen as

$$\overline{Z}_L = Z_1, \qquad \overline{X}_L = X_1 Z_2;$$

the right ones are $\overline{Z}_R = Z_N$, $\overline{X}_R = Z_{N-1} X_N$. These commute with all retained stabilizers, while the two operators on the same edge anticommute. Within the ground space, multiply each global symmetry by suitable bulk stabilizers to obtain

$$G_A \simeq \overline{X}_L \overline{Z}_R, \qquad G_B \simeq \overline{Z}_L \overline{X}_R, \tag{11.1}$$

where $A$ denotes odd sites in this formula. The two global symmetries commute, but their left-edge factors anticommute. This is a projective representation: the group multiplication law holds up to phases at a single edge, and the opposite edge cancels the phase for the entire system.

A local Hamiltonian on the left edge that respects both independent edge Pauli actions must be proportional to identity. Indeed, an operator $aI + bX + cY + dZ$ commuting with both $X$ and $Z$ has $b = c = d = 0$. A symmetry-preserving local perturbation therefore cannot lift that edge degeneracy on its own. Coupling the two distant edges can split it, but a local gapped bulk makes that coupling exponentially small with length.

This edge algebra is a concrete version of a cohomological SPT invariant. For a general group $G$, edge operators obey $V_g V_h = \omega(g,h)V_{gh}$. Associativity requires the two-cocycle equation $\omega(g,h)\omega(gh,k) = \omega(h,k)\omega(g,hk)$. Rephasing $V_g$ changes $\omega$ by a coboundary. The equivalence class, rather than a particular phase convention, labels the projective action. For the cluster state's $\mathbb{Z}_2 \times \mathbb{Z}_2$, the anticommutator above is the nontrivial invariant.

## 11.2 Strings expose hidden bulk order

Relabel the ring as alternating sites $a_i, b_i$, $i = 1, \dots, L$, with

$$K_i^A = Z_{i-1}^B X_i^A Z_i^B, \qquad K_i^B = Z_i^A X_i^B Z_{i+1}^A. \tag{11.2}$$

Products telescope:

$$\mathcal{S}_A(i,j) = \prod_{k=i}^{j} K_k^A = Z_{i-1}^B \left( \prod_{k=i}^{j} X_k^A \right) Z_j^B. \tag{11.3}$$

Its expectation is one in $|C\rangle$, independent of length. A similar $B$ string has $Z^A$ endpoints. Ordinary $Z$ two-point functions vanish at long distance. The string compensates for a fluctuating symmetry charge in the interior, leaving endpoint information.

The endpoint charge is the important feature: a truncated $A$ symmetry operation leaves $B$-charged endpoint operators. This is a decorated-defect picture of SPT order. It provides intuition for why the two symmetries work together. A nonzero string expectation without its symmetry and endpoint transformation properties would not by itself classify an SPT phase.

## 11.3 Exact sublattice dephasing

Apply independent $Z_i^B$ dephasing to the $B$ sites:

$$\mathcal{E}_p = \prod_i \mathcal{E}_{p,i}, \qquad \mathcal{E}_{p,i}(\rho) = (1-p)\rho + p Z_i^B \rho Z_i^B, \qquad 0 \le p \le \tfrac{1}{2}. \tag{11.4}$$

Every Kraus operator commutes with $G_A = \prod_i X_i^A$, so that symmetry is preserved strongly. A single $Z_i^B$ anticommutes with $G_B = \prod_i X_i^B$, so the channel exchanges $B$ charge with its environment. It is still covariant under $G_B$: the minus sign cancels between the two sides of each Kraus term. Thus $\rho_p = \mathcal{E}_p(|C\rangle\langle C|)$ has strong $G_A$ and, for generic $p > 0$, only weak $G_B$.

The channel acts diagonally on Pauli strings. A Pauli string containing $X$ or $Y$ at $m$ dephased sites is multiplied by $(1-2p)^m$. Therefore

$$\mathrm{Tr}(\rho_p \mathcal{S}_A(i,j)) = 1, \qquad \mathrm{Tr}(\rho_p \mathcal{S}_B(i,j)) = (1-2p)^{j-i+1}. \tag{11.5}$$

The two strings no longer play interchangeable roles. One remains an ordinary linear correlator; the other decays for every $p > 0$. This is an exact calculation, not evidence of a finite-noise thermodynamic transition in one dimension.

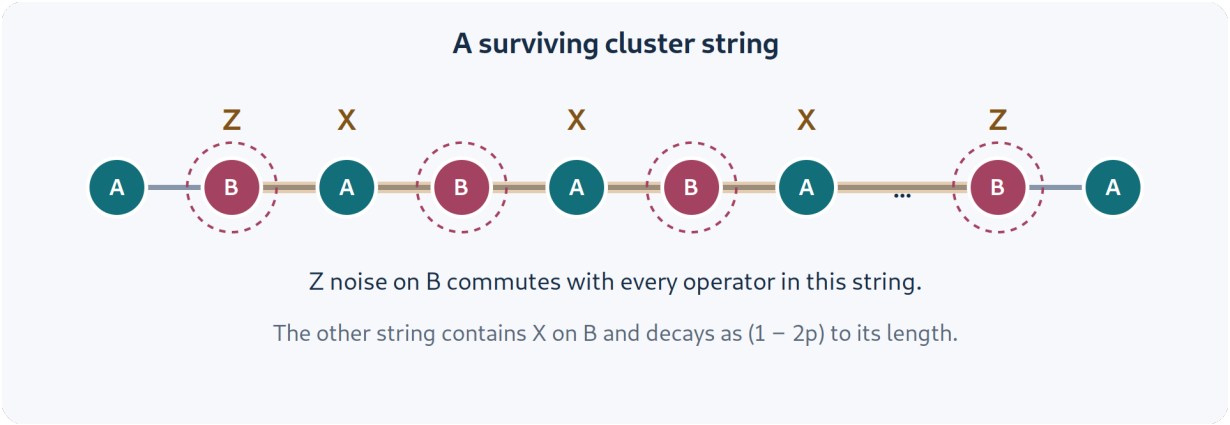



Figure 11.1: The endpoint decoration determines which string is insensitive to the chosen dephasing channel. Dashed circles mark the noisy sublattice.

At maximal dephasing, the full stabilizer density matrix simplifies. Terms containing any $K_i^B$ have $X$ or $Y$ on at least one $B$ site and vanish. The surviving state is

$$\rho_* = 2^{-2L} \prod_{i=1}^{L} (I + K_i^A). \tag{11.6}$$

The $L$ independent surviving constraints give rank $2^L$ and entropy $L \ln 2$. Since their product equals $G_A$, we have $G_A \rho_* = \rho_*$. Conjugation by $G_B$ leaves each $K_i^A$ invariant, but $G_B \rho_*$ is not $\rho_*$. This checks the strong/weak statements directly, without inferring them from terminology.

### 11.3.1 Worked example: separability and symmetry protection

Diagonalize the $B$ spins in the $Z$ basis, with values $s_i = \pm 1$. The constraints require $X_i^A = s_{i-1} s_i$. Consequently

$$\rho_* = 2^{-L} \sum_{\{s_i\}} \bigotimes_i |Z_i^B = s_i\rangle\langle Z_i^B = s_i| \otimes \bigotimes_i |X_i^A = s_{i-1} s_i\rangle\langle X_i^A = s_{i-1} s_i|. \tag{11.7}$$

This is a convex mixture of product states, so $\rho_*$ is fully separable. Yet its string relation is exact, and the strong/weak symmetry constrains admissible preparations and reversals. There is no contradiction: a phase distinction under symmetry-restricted operations need not be ordinary entanglement across a spatial cut. Also, this decomposition does not make each component $G_B$ symmetric: $G_B$ sends $s_i \mapsto -s_i$, pairing different members of the ensemble.

This example warns against interpreting every mixed SPT invariant as distillable entanglement. Average-symmetry phases, quantum information retained from a pure SPT, and the operational class of allowed channels must be specified. The systematic distinction between disorder and decoherence was developed in [31] and [32]; doubled-space treatments provide complementary classifications [33,34].

## 11.4 From the cluster ensemble to SW-SSB

The state $\rho_*$ is not already a featureless parity ensemble on $A$. Its local constraints remember where the $A$ charges sit relative to $B$ spins. A local $Z_i^A$ flips $K_i^A$ and moves the support to an orthogonal constraint sector. Acting with $Z_i^A Z_j^A$ at distinct sites still changes two locally readable constraints, so its fidelity correlator is zero. A strong $G_A$ symmetry alone does not force SW-SSB.

Now maximally dephase every neighboring $A$ pair with the operator $Z_i^A Z_{i+1}^A$. These pair errors commute with $G_A$. A product $\prod_i (K_i^A)^{t_i}$ survives every pair dephasing only if $t_i + t_{i+1} = 0$ modulo two on each bond. On a connected ring all $t_i$ are equal. Thus only identity and $\prod_i K_i^A = G_A$ remain:

$$\rho_{\text{end}} = 2^{-2L}(I + G_A). \tag{11.8}$$

The $B$ sites are maximally mixed, and the $A$ sites form the fixed-parity ensemble. For $i \neq j$, its ordinary $\langle Z_i^A Z_j^A \rangle$ vanishes, while $Z_i^A Z_j^A \rho_{\text{end}} Z_i^A Z_j^A = \rho_{\text{end}}$, so $F = 1$. We have explicitly converted constrained cluster-string structure into SW-SSB under a strongly $A$-symmetric channel.

At this endpoint, the old linear string with two $B$ endpoints vanishes: its expectation contains a nontrivial Pauli operator on the maximally mixed $B$ sector. Thus the persistence of a strong symmetry does not guarantee the persistence of the earlier SPT diagnostic. This one-way conversion does not prove equivalence of the initial and final phases under locally reversible symmetric channels.

The endpoint calculation is also consistent with the one-dimensional finite-depth obstruction: a transition can occur at maximal dephasing, where local channel invertibility is lost. It does not demonstrate a generic finite-noise transition under a finite-depth strictly local channel. Long-range channels can change this conclusion, as the later frontier chapters explain.

## 11.5 Canonical purification and Choi SPT data

Two different doubled states must be kept distinct. If $\rho = \sum_a \lambda_a |a\rangle\langle a|$, the canonical purification is $|\sqrt{\rho}\rangle\rangle = \sum_a \sqrt{\lambda_a}|a\rangle|a^*\rangle$. Vectorization instead gives $|\rho\rangle\rangle = \sum_a \lambda_a |a\rangle|a^*\rangle$, which must be normalized by $\sqrt{\operatorname{Tr}\rho^2}$.

A strong symmetry acts on ket and bra copies separately, with conjugate characters. A weak symmetry acts diagonally. This makes mixed SPT invariants resemble ordinary SPT invariants for an enlarged group. But a physical density matrix obeys Hermiticity, positivity, and unit trace. An arbitrary pure SPT state in the doubled space need not satisfy those conditions when reshaped into a matrix.

For short-range-correlated, suitably injective tensor-network states with direct-product unitary strong group $G$ and weak group $K$, a useful one-dimensional classification contains the ordinary strong-group projective term and a mixed term,

$$H^2(G, U(1)) \oplus H^1(K, H^1(G, U(1))). \tag{11.9}$$

The second term records a weak-symmetry defect decorated by a strong-symmetry charge. This is a classification within the stated tensor-network assumptions, not a statement that arbitrary mixed density matrices with arbitrary long-range conditional information obey this formula [34].

Choi-space long-range order naturally involves $R_2$ quantities. A claimed transition derived only there should remain identified as such. Canonical-purification or fidelity statements require their own argument. Even when the fixed-point examples agree, stability under physical channels is an additional question.

### 11.5.1 A positive tensor-network representation

Positivity can be built into a numerical representation. Write a purification amplitude as a matrix $T_{s,a}$, where $s$ collects system indices and $a$ collects environment indices. Then

$$\tilde{\rho}_{s,s'} = \sum_a T_{s,a} T^*_{s',a}, \qquad \rho = TT^\dagger / \operatorname{Tr}(TT^\dagger). \tag{11.10}$$

This remains positive for every choice of $T$. If the amplitudes of $T$ are expressed as a matrix product,

$$T_{\{s_i\},\{a_i\}} = \operatorname{Tr}\left[A_1^{s_1,a_1} A_2^{s_2,a_2} \cdots A_L^{s_L,a_L}\right],$$

the auxiliary matrix indices describe bonds between neighboring sites, while $a_i$ is a local purification index. Tracing those local indices gives a locally purified density operator, or LPDO. This is a practical way to avoid unphysical negative eigenvalues that unconstrained truncation of a density-matrix representation can introduce.

A strong symmetry restricts the system support itself; a weak symmetry can instead be implemented by a compensating transformation on the purification indices. Symmetry relations on the virtual bonds can then reveal the projective data. To turn that observation into a phase classification, one needs injectivity conditions: after blocking a bounded number of sites, the tensors must map their virtual boundary data faithfully into physical block data in the specified sense. These hypotheses exclude arbitrary long-range correlations or unbounded hidden information. LPDO classifications and fixed-point constructions establish average-SPT examples for more general symmetry extensions than the direct-product illustration above [35]. The tensor ansatz supplies both a positivity guarantee and a domain of applicability; it is not a theorem that every mixed phase has an efficient fixed-bond-dimension description.

## 11.6 Strange correlators and information access

A pure-state strange correlator compares an SPT wavefunction $|\psi\rangle$ with a trivial reference $|\Omega\rangle$:

$$C_{\text{str}}(i,j) = \frac{\langle\Omega|O_i O_j^\dagger|\psi\rangle}{\langle\Omega|\psi\rangle}.$$

It resembles a correlation function at an interface between the two states. That interface can retain boundary-like long-range or critical structure even when ordinary bulk correlators are short-ranged. Its usefulness depends on the operator, the reference, and a nonzero denominator; it is not a universal correlator for every state without those choices.

Mixed-state extensions can use linear overlaps, replicated overlaps, or fidelity. These differ in which information is accessible and in whether normalization amplifies rare components. Type-I and type-II strange correlators were used to separate operationally accessible SPT information from a fuller doubled-state topological response [36]. Fidelity strange correlators give another bulk route to average-SPT diagnosis [37].

A purification offers physical intuition: symmetry defects in the environment can carry information about charge or domain walls in the system. Conditioning on an environment record may reveal a pure ordered state; averaging can hide that order from linear observables. The resulting correspondence depends on a specified local purification or preparation, rather than on a unique environment intrinsic to $\rho$ [38].

## 11.7 What an anomaly actually constrains

An anomaly is an obstruction to realizing or gauging a symmetry in a strictly lower-dimensional, local, short-range-trivial manner. The edge anticommutation in Equation 11.1 is the simplest example. In field theory, a bulk response can cancel the boundary's anomalous variation, called anomaly inflow. On a lattice, one can often diagnose it through the algebra of symmetry operators truncated to intersecting regions.

Mixed-state constraints depend critically on whether the anomalous symmetry is strong. A strong condition restricts every vector in the support, whereas a weak condition only constrains the ensemble. For group-cohomology anomalies of a strong symmetry in $d$ dimensions, [39] connects the anomaly to long-range multipartite nonseparability: its rigorous arguments cover $d \leq 1$, with higher-dimensional extensions supported by plausibility arguments. A mixed strong/weak anomaly can instead constrain recoverability and long-range correlations without forcing quantum entanglement. Our separable cluster ensemble makes that possibility less surprising.

"The boundary cannot be trivial" is therefore incomplete until triviality is defined. It can mean a unique gapped symmetric pure ground state, a short-range-entangled symmetric mixed state, a state with finite correlation and Markov lengths, or a state preparable by a specified symmetric circuit. These are different restrictions. The anomaly must be matched by the degrees of freedom and the operational structure that actually remain.

## 11.8 State robustness versus steady-state robustness

Suppose a density matrix has a protected invariant under a class of locally reversible symmetric deformations. It does not follow that any local Lindbladian with that state as a stationary state has a stable stationary phase. A small perturbation to the generator acts for arbitrarily long times before the steady state is reached. Finite-time stability theorems do not automatically commute with this infinite-time limit.

For decohered cluster states, a parent Lindbladian can be mapped to reaction-diffusion dynamics. Typical allowed perturbations that create the relevant strong-symmetry defects produce SW-SSB at arbitrarily small strength and destabilize its steady-state mixed SPT order. Perturbations creating only weak-symmetry defects can behave differently [40]. The result concerns that family of generators and perturbations; it is not a theorem that every conceivable mixed SPT state is unstable under every symmetric channel.

The distinction is physically instructive. A stationary constraint can rely on defects being absent forever. A tiny creation rate accumulated over infinite time can populate those defects throughout the sample. Whether they retain enough correlations to preserve a string or anomaly diagnostic is then a dynamical question, even if the microscopic perturbation respects the stated covariance.

## 11.9 Developments through the cutoff

The decorated-defect mechanism also runs in the opposite direction: dephasing the charge attached to a defect can induce SW-SSB of the defect's symmetry. Extensions to higher-form, modulated, and gapless SPT settings were developed in [41]. This gives a channel-dependent topological response rather than a single linear-response coefficient.

Extended cluster states with multiple subsystem symmetries can generate a hierarchy of mixed SPT states under successive subsystem decoherence, diagnosed by surviving $R_2$ strings [42]. Choi-space analyses also distinguish trace-preserving decoherence from no-click postselection: filtering and renormalizing a state can alter the relevant stabilizer support and drive different limiting orders [43]. These are model-resolved developments; a unified classification of all such hierarchies is not established.

Double-ASPT constructions combine strong-to-weak order with a remaining strong/weak SPT structure [44]. SymTFT constructions organize mixed SPTs using folded higher-dimensional topological theories [45,46]. They provide useful conceptual maps, while positivity and the particular phase equivalence remain essential constraints.

Experiments on a five-qubit superconducting array probe cluster-string order, selective dephasing, and the difference between conditioned and discarded measurement records [47]. Such finite-size signatures connect directly to the algebra derived here. Establishing thermodynamic phases additionally requires a controlled family of sizes and an understanding of noise and reconstruction assumptions.

## 11.10 Exercises and worked solutions

### 11.10.1 Exercise 1: check the cluster symmetry

Show directly that $K_j$ commutes with each of $G_A, G_B$, and explain why a one-site $Z$ dephasing channel may preserve covariance while violating strong symmetry.

> **i** Worked solution
>
> If $j$ lies on $A$, $G_A$ overlaps the central $X_j$ with $X_j$ and commutes, while $G_B$ overlaps both neighboring $Z$ factors with $X$ and gives two minus signs. Reverse the sublattices for $j \in B$. For a jump $Z_j$ on $B$, $G_B Z_j G_B^\dagger = -Z_j$, so the jump fails to commute with $G_B$. Yet $(-Z_j)\rho(-Z_j) = Z_j \rho Z_j$, making the channel covariant. A state initially in one $G_B$ sector becomes a mixture of the two sectors.

### 11.10.2 Exercise 2: trace out the dephased sublattice

Compute $\rho_A = \mathrm{Tr}_B\, \rho_*$ from Equation 11.7 and find its nonlinear $Z_i^A Z_j^A$ correlators.

> **Worked solution**
>
> Set $x_i = s_{i-1}s_i$. Periodicity enforces $\prod_i x_i = 1$. Every allowed $x$ pattern has two preimages $s$, related by an overall sign. Thus $\rho_A = 2^{-(L-1)} \sum_{\prod x_i = 1} |\{x_i\}\rangle_X \langle\{x_i\}| = (I + G_A)/2^L$. For separated distinct sites, $\langle Z_i^A Z_j^A \rangle = 0$ for $L > 2$, while conjugation by the pair preserves $\rho_A$. Hence $F = R_1 = R_2 = 1$. The full state $\rho_*$ had locally readable charge constraints; discarding $B$ hides them and produces SW-SSB on $A$. This is a concrete effect of changing which information is accessible.

### 11.10.3 Exercise 3: symmetry allowed edge terms

For the pure open cluster state, suppose only $G_A$ is required. Can a local boundary term lift the left-edge degeneracy?

> **Worked solution**
>
> With our odd-$A$ convention, $G_A$ acts as $\overline{X}_L$ on the left edge. A term $-h\overline{X}_L$ commutes with that symmetry and selects an eigenstate, lifting the degeneracy. It is forbidden if $G_B$, acting as $\overline{Z}_L$, is also imposed. Both protecting factors are necessary for this projective edge protection. For mixed states one must additionally specify strong versus weak implementations of each factor.

### 11.10.4 Exercise 4: dephasing is not postselection

Compare $\mathcal{M}(\rho) = P_+\rho P_+ + P_-\rho P_-$ with the conditioned state $\rho_+ = P_+\rho P_+ / \mathrm{Tr}(P_+\rho)$, where $P_\pm = (I \pm Z)/2$.

> **Worked solution**
>
> The first map is linear, completely positive and trace preserving; it retains both outcomes and discards the record. The conditioned expression is nonlinear because its success probability depends on $\rho$. It can change expectations by selecting a rare component. One can represent the unnormalized branch $P_+\rho P_+$ as a completely positive trace-nonincreasing map, but normalization and postselection cannot be treated as deterministic local noise. Stability theorems for CPTP channels do not automatically apply to that normalized branch.

### 11.10.5 Exercise 5: two cancelling projective phases

Use Equation 11.1 to show that the full global symmetries commute although their factors on each edge anticommute.

> **Worked solution**
>
> Reversing the order of $\overline{X}_L\overline{Z}_R$ and $\overline{Z}_L\overline{X}_R$ introduces one minus sign at $L$ and one at $R$. Operators on different edges commute. The product of the signs is $+1$, so the global representations commute. The obstruction belongs to realizing the edge alone with a nondegenerate symmetric state, not to representing the full group on the whole chain.

# 12 Higher-form symmetry, intrinsic mixed order, and non-Abelian fusion

The examples so far involved a global operation acting on all sites and a charged operator localized near a point. Topological phases invite a broader language: the charged objects can be strings, and the symmetry operations can be deformable loops. Mixed states then support strong and weak versions of those generalized symmetries. A second generalization replaces Abelian charges by non-Abelian representations or anyons, whose combination requires fusion information beyond local labels.

These two generalizations are related but distinct. A non-Abelian ordinary symmetry such as $SU(2)$ is a group acting globally. A noninvertible higher-form symmetry has extended operators whose multiplication can be a sum of operators rather than another group element. We will develop both using explicit algebras, and retain the different assumptions behind the available classification proposals.

## 12.1 What makes a symmetry higher form?

An ordinary, or zero-form, symmetry acts on point-like charged operators. A one-form symmetry acts on line-like charged operators, such as Wilson loops. In $d$ spatial dimensions, its symmetry operator is supported on a closed surface of dimension $d-1$. Linking that surface with the charged line multiplies the line by a phase. In two spatial dimensions, both can be represented by loops, and their equal-time crossing algebra is especially easy to draw.

For a $\mathbb{Z}_2$ example, write a symmetry loop $U(\gamma)$ and a charged loop or string $W(\gamma')$. If they cross an odd number of times,

$$U(\gamma)W(\gamma') = -W(\gamma')U(\gamma). \tag{12.1}$$

An even number gives a plus sign. “Topological” means that smoothly deforming the symmetry operator without crossing a charged insertion does not change its action on the relevant low-energy or constrained-state space. On a lattice this usually follows because two deformations differ by a product of local constraints.

This definition is more informative than simply calling a nonlocal conserved operator a higher-form symmetry. Its deformability, support dimension, action on extended charged objects, and allowed defects all matter. In a generic microscopic model these properties can be emergent at long distances rather than exact at every energy.

### 12.1.1 Wilson loops and symmetry breaking

A conventional broken symmetry has a nondecaying correlator of two distant charged endpoints. For a one-form symmetry, the analogous order parameter is an extended charged loop. In a confining phase, a large Wilson loop often decays as $e^{-a\,\mathrm{Area}}$; in a deconfined phase, its leading local suppression is a perimeter term $e^{-b\,\mathrm{Perimeter}}$. After removing this short-distance boundary contribution, the loop can retain order. At an exactly solvable fixed point it can equal one.

These area/perimeter statements assume a setting where Wilson loops are appropriate diagnostics and dynamical screening is controlled. With fundamental matter, non-Abelian charges, or mixed-state channels, one must specify the operator and the symmetry being tested. A zero ordinary loop expectation may reflect noise in a microscopic representative rather than the disappearance of all dressed loop structure.

## 12.2 The toric code's loop algebra

Return to Equation 10.6. A closed direct-lattice $Z$ string $W_e(\gamma) = \prod_{a\in\gamma} Z_a$ and a closed dual-lattice $X$ string $W_m(\gamma^*) = \prod_{a\perp\gamma^*} X_a$ obey

$$W_e(\gamma)W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)}W_m(\gamma^*)W_e(\gamma), \tag{12.2}$$

where $I$ counts intersections modulo two. A contractible $Z$ loop is a product of enclosed $B_f$; a contractible dual $X$ loop is a product of enclosed $A_v$. Deforming a path across a face or star therefore leaves its action unchanged in the code space.

The sign is the mutual braiding phase of $e$ and $m$. Transport an $e$ around an $m$, compare the order of their string operations, and the amplitude changes by $-1$. On a torus, crossing noncontractible loops act as conjugate logical Pauli operators.

A useful consistency check follows immediately. No nonzero density matrix can satisfy both $W_e\rho = \rho$ and $W_m\rho = \rho$ for two anticommuting loops. Applying their product in the two orders would imply $\rho = -\rho$. Thus a statement that a toric-code state has strong loop symmetries must identify which compatible loops and sectors are fixed; it cannot assign simultaneous sharp eigenvalues to every noncontractible logical loop.

The incompatibility is a manifestation of a mutual higher-form anomaly. It is encoded in an algebra of operators and survives many changes of microscopic Hamiltonian. Which part remains meaningful for mixed states depends on whether the operations are strong, weak, or dressed into channel symmetries.

## 12.3 A classical loop ensemble calculated exactly

Write a link configuration as signs $z_a = \pm 1$ in the $Z$ basis. The flatness condition is $\prod_{a\in\partial f} z_a = 1$ on every face. On a torus, flat configurations have two winding holonomies $w_x, w_y = \pm 1$ that cannot be changed by local star flips.

For fixed $w = (w_x, w_y)$, define

$$\rho_w = \frac{1}{N_w} \sum_{z\,\text{flat},\,\text{hol}(z)=w} |z\rangle\langle z|. \tag{12.3}$$

This is a classical probability distribution written as a density matrix. It can be obtained by fully dephasing a corresponding toric-code wavefunction in the $Z$ basis. Since $V-1$ independent star flips generate every configuration in a fixed flat sector, $N_w = 2^{V-1}$ on a connected square torus. Thus $S(\rho_w) = (V-1)\ln 2$.

Every contractible $Z$ loop is strongly fixed at $+1$, and the two noncontractible $Z$ loops have sharp values $w_x, w_y$. Star operators permute configurations inside the sum, giving $A_v\rho_w A_v = \rho_w$. However, $A_v\rho_w \neq \rho_w$: left multiplication creates off-diagonal matrix elements absent from the diagonal state. The corresponding symmetry has become weak.

A noncontractible dual $X$ loop changes one holonomy and maps $\rho_w$ into a different sector. It is not a weak symmetry of a single fixed-$w$ state. The uniform mixture $\overline{\rho} = \frac{1}{4}\sum_w \rho_w$ is invariant under those permutations, but it no longer fixes the individual noncontractible $Z$ holonomies strongly. This explicit example prevents an inconsistent assignment of all strong and weak loop conditions at once.

### 12.3.1 Locally invisible, globally readable information

The four distributions $\rho_w$ have identical marginals on a small contractible region that does not wind around the torus. To see this, changing a winding sector can be accomplished by a dual $X$ loop placed outside that region; conjugation then changes the sector while leaving its reduced state unchanged. Nevertheless a noncontractible $Z$ measurement distinguishes the four sectors perfectly.

The convex family $\rho = \sum_w p_w \rho_w$ therefore stores two classical bits. It does not store arbitrary superpositions of the four original toric-code ground states, because dephasing removed the off-diagonal coherences. All spatial entanglement measures vanish for each $\rho_w$: the displayed decomposition is separable. Global conditional structure and local indistinguishability can survive without quantum entanglement.

This classical loop ensemble is an instructive separator of phase definitions. Under two-way channel connectivity it can be trivial in a sense that allows global recoverability without preserving every local operator relation. Under locally reversible channel circuits, its topological degeneracy and strong/weak loop structure are protected [16]. The change is in the equivalence relation, not in the elementary probability distribution.

## 12.4 Nonlinear loop diagnostics

For an extended operator $W$ that is unitary, define

$$F_W = F(\rho, W\rho W^\dagger), \quad R_{1,W} = \mathrm{Tr}(\sqrt{\rho}\, W \sqrt{\rho}\, W^\dagger), \quad R_{2,W} = \frac{\mathrm{Tr}(\rho W \rho W^\dagger)}{\mathrm{Tr}\,\rho^2}. \tag{12.4}$$

These measure indistinguishability under an insertion, rather than the ordinary expectation $\mathrm{Tr}(\rho W)$. To formulate higher-form SW-SSB, choose the appropriate charged extended insertion and compare the nonlinear and ordinary large-distance or large-loop behavior. The geometry and perimeter renormalization must be included; replacing a point operator by any arbitrary long string does not automatically define an order parameter.

For a particularly transparent classical example, let $W = A_v$. In $\rho_w$, its ordinary expectation vanishes, while $A_v \rho_w A_v = \rho_w$, so all three quantities above equal one. This shows the algebraic strong-to-weak distinction for that operation. Because $A_v$ is local, it is not alone a thermodynamic one-form order parameter; an extended family and its deformation properties establish the generalized-symmetry interpretation.

In noisy topological states, different replica indices can again produce different transitions. The updated analysis of disordered toric-code ensembles in [48] explicitly uses a Rényi-2 finite-Markov-length equivalence when calling certain phases intrinsically mixed. A fidelity-based statement or a theorem under locally reversible physical channels requires the corresponding additional argument.

## 12.5 Incoherent anyon proliferation versus coherent condensation

For Abelian anyons, let $M_{ab}$ be the phase from fully braiding $a$ around $b$. A boson can sometimes condense coherently in a pure state, identifying certain sectors with vacuum and confining sectors that braid nontrivially with it. A mixed-state channel can instead create a random distribution of anyon pairs. This **incoherent proliferation** need not satisfy the same rules as a coherent condensate.

Suppose noise proliferates an anyon set $\mathcal{A}$. A probe $b$ that braids nontrivially with that random background acquires fluctuating phases. Its coherent distinguishability can be lost. A natural surviving set is the commutant

$$\mathcal{A}' = \{b : M_{ab} = 1 \text{ for every } a \in \mathcal{A}\}. \tag{12.5}$$

This mechanism is realized in concrete decohered topological states and subsystem-code constructions [49,50]. It is an organizing rule within those frameworks, not a complete classification of arbitrary noise-driven phases.

### 12.5.1 Worked example: proliferating a toric-code fermion

The toric-code anyons are $1, e, m, f = e \times m$. Their spins are $\theta_e = \theta_m = 1, \theta_f = -1$. Mutual braiding gives $M_{e,m} = -1$, and hence $M_{e,f} = M_{m,f} = -1$, while $M_{f,f} = 1$. Incoherent $f$ proliferation can therefore leave the set $\{1, f\}$.

The surviving $f$ braids trivially with everything in that remaining set but has fermionic self-statistics. Such a transparent fermion makes the remaining braiding theory nonmodular: braiding alone does not distinguish all nontrivial sectors from vacuum. It cannot be treated as the complete modular anyon content of a conventional bosonic two-dimensional gapped pure ground state.

Mixed-state examples realizing this structure retain topological entanglement negativity while losing the original topological quantum memory [51]. This shows why residual quantum entanglement and recoverable logical qubits are distinct. It also explains why calling the process ordinary condensation of a bosonic order parameter would miss the point: the proliferated excitation is fermionic, and the channel averages histories rather than preparing a coherent fermion condensate.

A **premodular** theory allows such transparent sectors; a modular theory has nondegenerate braiding. Proposed classifications of two-dimensional mixed topological orders by premodular data are powerful but partial or conjectural. Incoherent proliferation and classical gauging are mechanisms known to generate examples, not a proof that no additional mechanisms exist [50].

### 12.5.2 An ensemble criterion for long-range quantum structure

Another diagnostic asks whether $\rho$ admits any decomposition $\rho = \sum_a p_a |\psi_a\rangle\langle\psi_a|$ in which every component is short-range entangled. This is weaker than full separability into on-site product states: a short-range-entangled component can contain local Bell pairs or other entanglement removable by a bounded local unitary. It is also an existence question over all decompositions, rather than a statement about the experimentally selected trajectory ensemble. For locally decohered toric-code and X-cube models, explicit decompositions and statistical-mechanical mappings give evidence that such separability transitions coincide with optimal error-correction thresholds in the studied settings [52]. That coincidence is not a universal equivalence between lost quantum memory and absence of all mixed-state entanglement. In particular, the fermion-proliferated examples above show why residual entanglement and the original logical memory need separate tests.

## 12.6 Anomalies as entanglement obstructions

The preceding classical loop example had global structure without entanglement. Strong anomalous higher-form symmetries can impose a stronger obstruction. In particular, nontrivial braiding or self-statistics can forbid preparing a state from short-range or separable resources by the relevant local channels. The intuition is locality: distant pieces of a would-be product preparation cannot reproduce a topologically invariant phase attached to intersecting extended operations unless the state already contains nonlocal quantum structure.

One proof strategy is symmetry pullback. Let a local channel map $\rho$ to $\sigma$, and suppose an extended operator is a strong symmetry of $\sigma$. Through a Stinespring dilation, pull its action backward to an operator on the input and local environment. Locality bounds the support thickening. If the channel class permits the required pullback, relations between intersecting symmetry operators, including their anomalous phase, are inherited by the input. A purported trivial input can then contradict that algebra.

The theorem's channel class matters. Stochastic local channels that mod out classical correlations support an entanglement-oriented phase definition; strong higher-form anomalies are stable under that definition. In $2 + 1$ dimensions they imply long-range bipartite entanglement and lower bounds on a topological entanglement-of-formation quantity [53]. Higher-dimensional multipartite extensions are not all established by the same proof.

Independent higher-form anomaly arguments bound overlap with short-range-entangled states and construct a $3 + 1$-dimensional mixed order that violates the pure-state expectation of remote detectability [54]. Remote detectability says, roughly, that a nontrivial excitation can be distinguished by a distant braiding experiment. If some detecting sectors have been erased by decoherence, an excitation may survive without that complete detector set. The pure-state consistency conditions must then be modified rather than silently assumed.

Weak symmetries need extra care. Their action can be stable as a **channel symmetry** after a locally reversible deformation even when no simple dressed unitary represents it. Local reversibility preserves more structure than two-way reachability of the global density matrix [16]. Consequently, entanglement classifications that intentionally discard classical information and phase classifications that preserve classical topological memories should not be presented as competing answers to one identical question.

## 12.7 Noninvertible higher-form operations

A group symmetry operator has an inverse. Anyon line operators in a non-Abelian topological theory instead obey fusion rules

$$W_a W_b = \sum_c N_{ab}^c W_c. \tag{12.6}$$

The nonnegative integers $N_{ab}^c$ count fusion channels. When several $c$ appear, the operation is not invertible in the group sense. Calling it a symmetry requires its topological action and fusion algebra, not unitarity of every $W_a$.

For finite-group quantum doubles $D(G)$, excitations include flux and representation data. Decoherence can convert quantum locally indistinguishable sectors into a convex set of classical sectors. In the quantum-double constructions of [55], the number of extremal density matrices in the resulting information convex equals the pure ground-state degeneracy. A convex combination of $D$ independent extremal points has affine dimension $D-1$, so "number of extrema" and "dimension of the convex set" must not be interchanged.

An **information convex** is the set of density matrices compatible with specified local reduced-state constraints. Its shape tells us what global information remains hidden from those local measurements. A simplex signals classical alternatives; a full matrix-state space permits coherent quantum alternatives. More complicated non-Abelian structures can impose recovery and secret-sharing constraints not visible in a count of sectors alone. Axiomatic approaches based on local recoverability, short-range ordinary correlations, and spatial uniformity develop these distinctions systematically [56].

## 12.8 Non-Abelian ordinary symmetry needs fusion records

For an ordinary non-Abelian group, the Hilbert space decomposes as

$$\mathcal{H} = \bigoplus_\lambda V_\lambda \otimes M_\lambda, \qquad U_g = \bigoplus_\lambda D_\lambda(g) \otimes I_{M_\lambda}. \tag{12.7}$$

$V_\lambda$ carries an irreducible representation and $M_\lambda$ is its multiplicity space. A weakly invariant density matrix is proportional to identity on each $V_\lambda$ factor, with arbitrary positive operators on multiplicity spaces. The strict strong condition $U_g\rho = \chi(g)\rho$ for every $g$ requires support in a common one-dimensional representation. For $SU(2)$ this means the singlet sector. One should not replace that condition by "fixed spin $j$" for $j > 0$.

Local regions can each carry nontrivial irreps even when the whole system is a singlet. Combining those irreps requires a fusion tree. For $SU(2)$,

$$j_1 \otimes j_2 = \bigoplus_{J=|j_1-j_2|}^{j_1+j_2} J.$$

Knowing $j_1, j_2$ does not determine $J$. That ambiguity has operational consequences for a decoder.

### 12.8.1 Worked example: four spin halves

Each site has fixed local spin $1/2$. Four sites contain two independent total singlets. In one basis they are

$$|\alpha\rangle = |s_{12}\rangle|s_{34}\rangle,$$

and

$$|\beta\rangle = \frac{1}{\sqrt{3}}\left(|t_+\rangle_{12}|t_-\rangle_{34} - |t_0\rangle_{12}|t_0\rangle_{34} + |t_-\rangle_{12}|t_+\rangle_{34}\right). \tag{12.8}$$

Here $s = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$, $t_\pm = |\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$, and $t_0 = (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$. Every one-site Casimir has the same value $3/4$ in both states. The total Casimir also vanishes in both. Yet measuring the two-site Casimir $(\mathrm{S}_1 + \mathrm{S}_2)^2$ distinguishes them: $J_{12} = 0$ for $\alpha$ and $J_{12} = 1$ for $\beta$.

Thus neither the local irrep list nor the total charge specifies how charges fused. A decoder retaining nested-region fusion outcomes can know more than one retaining only the smallest-region labels. Measuring incompatible fusion trees can disturb the state, so a hierarchy must specify a compatible measurement protocol rather than assume all fusion data are simultaneously available.

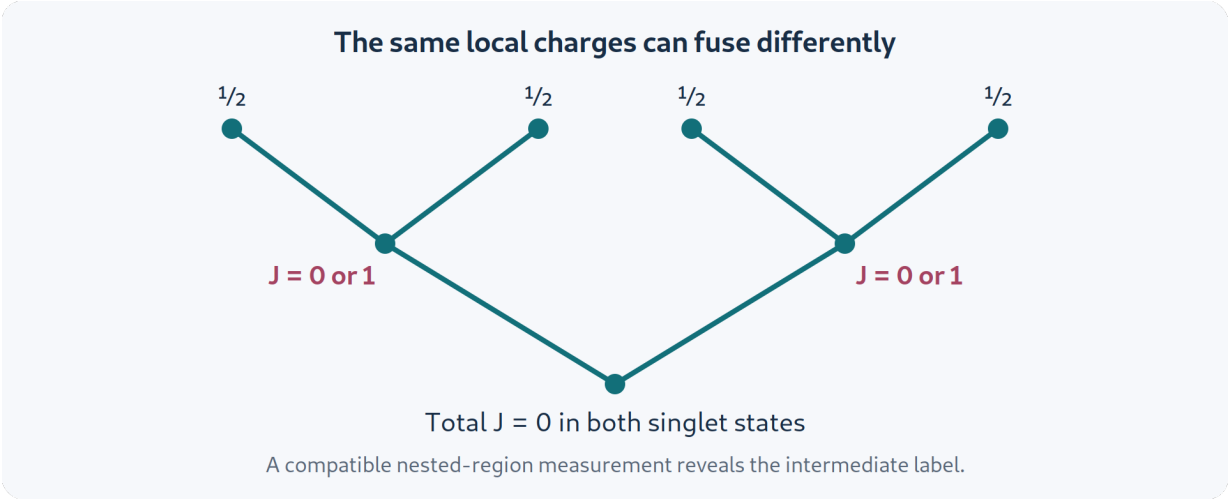



Figure 12.1: Intermediate fusion labels distinguish two singlets with identical one-site charge labels and the same total charge.

## 12.9 A maximally mixed state can retain quantum entanglement

The four-spin example generalizes to an instructive stationary ensemble. Let $P_0$ project $N$ spin halves onto total spin zero, and take $\rho_0 = P_0/\operatorname{Tr} P_0$. A unital strongly $SU(2)$-symmetric channel preserves this state; if it mixes within the singlet sector with no further invariant subspaces, this is the sector's unique stationary state. Unitality and strong symmetry alone do not guarantee that mixing hypothesis.

Partition the spins into two macroscopic regions. Write $m_A(j) = \dim M_j^A$ and $m_B(j) = \dim M_j^B$. A total singlet requires equal spins $j$ on the two sides. There is one singlet $|S_j\rangle$ in $V_j \otimes V_j$, maximally entangled with Schmidt rank $d_j = 2j + 1$. Multiplicity spaces need no entanglement. Hence

$$\rho_0 = \bigoplus_j p_j\, |S_j\rangle\langle S_j| \otimes \frac{I_{M_j^A}}{m_A(j)} \otimes \frac{I_{M_j^B}}{m_B(j)}, \quad p_j = \frac{m_A(j)m_B(j)}{\sum_l m_A(l)m_B(l)}. \tag{12.9}$$

The label $j$ is readable independently on each side by a local total-spin measurement. Conditional on it, all quantum entanglement resides in $|S_j\rangle$. The multiplicity factors are product mixed states.

Entanglement of formation $E_F$ is the minimum average pure-state entanglement over convex decompositions. Distillable entanglement $E_D$ is the asymptotic singlet yield under local operations and classical communication. For this state, measuring the locally accessible $j$ flag and retaining $|S_j\rangle$ gives a yield $\sum_j p_j \ln d_j$. Conversely, the displayed block decomposition, with product-basis decompositions of its multiplicity factors, attains that same average formation cost. Since $E_D \leq E_F$, the two bounds meet:

$$E_D(\rho_0) = E_F(\rho_0) = \sum_j p_j \ln(2j+1). \tag{12.10}$$

These measures use ordinary unrestricted bipartite operations; imposing additional local symmetry or reference-frame restrictions would define a different operational resource theory. The logarithmic negativity, $E_{\mathcal{N}} = \ln \|\rho_0^{T_B}\|_1$, follows because the partially transposed rank-$d_j$ singlet has trace norm $d_j$, while the flags occupy orthogonal blocks:

$$E_{\mathcal{N}}(\rho_0) = \ln \sum_j p_j (2j+1). \tag{12.11}$$

The four-spin equal mixture of $\alpha$ and $\beta$ has $p_0 = p_1 = 1/2$, giving $E_D = E_F = \frac{1}{2}\ln 3$ and $E_{\mathcal{N}} = \ln 2$. These are measures of the mixed state itself, not averages over a chosen trajectory ensemble.

For an even block of $n$ spin halves, the multiplicity is

$$m_n(j) = \binom{n}{n/2-j} - \binom{n}{n/2-j-1}.$$

This follows by subtracting the number of states with magnetization $j+1$ from those with magnetization $j$: every spin-$J \geq j+1$ irrep contributes to both, while spin $j$ contributes only to the latter. Stirling's approximation for two blocks each containing a fixed fraction of $N$ gives typical $j \sim \sqrt{N}$. Thus both formulas above grow as $\frac{1}{2}\ln N + O(1)$ for this spin-half singlet family. A fixed-size block cannot show that growth because its accessible $j$ is bounded.

The representation-theoretic entanglement mechanism is established for broad invariant-sector ensembles [57], and exact negativity results extend to non-Abelian symmetries and more general commutant structures [58]. In contrast, an Abelian on-site charge constraint admits a product charge-basis decomposition of its maximally mixed sector state. Strong non-Abelian symmetry can therefore enforce quantum entanglement even when the ensemble is otherwise maximally mixed. This is a concrete limit of interpreting all charge thermalization as classical diffusion.

## 12.10 Center ordering and a decoding hierarchy

The October 2026 work [59] formulates non-Abelian SW-SSB as distinguishability between strongly symmetric states differing by a pair of inserted irreps after symmetric decoherence. For finite $G$, it relates the task to decoding a quantum double $D(G)$ under pure-charge noise. Nested-region fusion records produce a hierarchy of statistical-mechanics decoding problems with Nishimori-like disorder matching.

The **center** $Z(G)$ consists of group elements commuting with every $g \in G$. By Schur's lemma, a center element acts within an irrep as a scalar. It therefore supplies an Abelian label that survives some ambiguity of non-Abelian fusion. For $SU(2)$, the center is $\{I, -I\}$ and acts as $(-1)^{2j}$: integer and half-integer spins have opposite center charge. For $S_3$, the center is trivial.

At finite levels of that decoding hierarchy, under its stated group assumptions, noncentral representation information can hide in noise-generated fusion channels. The surviving nonzero-noise ordering transition is associated with the center. The paper establishes specific finite-level statements for nonnilpotent groups and tests $S_3$ and $SU(2)$ with tensor-network calculations. In the $SU(2)$ example the center transition is consistent with Nishimori-Ising universality, and a second-level decoder raises the threshold by about 16 percent relative to the first. These finite-level results are not a proof of the threshold of an unrestricted optimal decoder with unlimited fusion information.

The monotonic direction has a simple information-theoretic proof. If a higher-level record contains the lower-level record, a decoder can ignore the extra information and reproduce its earlier performance. Optimal success therefore cannot decrease. That statement is rigorous independently of the numerical threshold improvement. It does not determine whether the hierarchy converges quickly, or whether a finite level captures all thermodynamic information.

## 12.11 Connecting the frameworks without collapsing them

Higher-form order generalizes point-charge order to strings and surfaces. Anomalies encode their intersection phases. Topological error correction asks which globally indistinguishable sectors and coherent logical algebras survive noise. Non-Abelian fusion enriches what a syndrome must record. These connections explain why statistical mechanics, symmetry, and quantum information recur together.

Several unresolved questions remain precise. Which invariants are complete under locally reversible circuits? How do stochastic-channel entanglement classes map onto classifications retaining classical memories? When can a Rényi-2 transition be promoted to a stable fidelity-based statement? What is the optimal fusion decoder beyond finite hierarchy levels? How much anomaly data is experimentally accessible through local regions? Each asks about a specified operation or diagnostic; none is answered by naming a single universal "mixed-state topology."

## 12.12 Exercises and worked solutions

### 12.12.1 Exercise 1: crossing loops and sharp sectors

Let Hermitian unitaries $A, B$ anticommute. Show that a state with $A\rho = \rho$ has $\mathrm{Tr}(\rho B) = 0$ and $F(\rho, B\rho B) = 0$.

> **Worked solution**
>
> The support of $\rho$ lies in the $A = +1$ subspace. Anticommutation sends that support into $A = -1$ after applying $B$. The two supports are orthogonal, so the fidelity vanishes. Also $\mathrm{Tr}(\rho B) = \mathrm{Tr}(A\rho AB) = -\mathrm{Tr}(\rho B)$ and hence is zero. This is the same support argument used for charged local operators, now applied to a crossing logical loop.

### 12.12.2 Exercise 2: classical-sector entropy

For $\rho = \sum_w p_w \rho_w$ in Equation 12.3, compute the entropy and the Holevo information available to an observer receiving the whole lattice.

> **Worked solution**
>
> Different $w$ sectors have disjoint support. Therefore $S(\rho) = H(p) + \sum_w p_w S(\rho_w) = H(p) + (V-1)\ln 2$. The Holevo information of the ensemble is $S(\rho) - \sum_w p_w S(\rho_w) = H(p)$. Measuring both winding holonomies attains it. A contractible local observer obtains identical marginals for every $w$, so its Holevo information is zero. Global classical information is present without spatial quantum entanglement.

### 12.12.3 Exercise 3: surviving anyons

In the toric code, find the commutant of $\{1, e\}$ and compare it with the commutant of $\{1, f\}$.

> **Worked solution**
>
> $e$ braids trivially with $1, e$ and nontrivially with $m, f$, so the first commutant is $\{1, e\}$. Similarly $f$ braids trivially with $1, f$ and nontrivially with $e, m$, so the second is $\{1, f\}$. The first has a transparent boson,

the second a transparent fermion. The common word "transparent" refers to mutual braiding within the surviving set; the self-statistics still distinguishes them.

### 12.12.4 Exercise 4: more records cannot worsen optimal decoding

Suppose $X$ is a coarse syndrome record, $Y$ an additional fusion record, and $H$ the hidden logical class. Prove $\sum_{x,y} \max_h P(h,x,y) \geq \sum_x \max_h P(h,x)$.

**Worked solution**

For each fixed $x$, let $h_x$ maximize $P(h,x) = \sum_y P(h,x,y)$. Then $\sum_y \max_h P(h,x,y) \geq \sum_y P(h_x,x,y) = \max_h P(h,x)$. Sum over $x$. Operationally, a decoder with $(X,Y)$ can ignore $Y$. The result requires actual inclusion of records; two incompatible measurement schemes are not automatically ordered this way.

### 12.12.5 Exercise 5: count four-spin singlets

Use angular-momentum addition to count total singlets of four spin halves and relate the count to Equation 12.8.

**Worked solution**

Group the spins into pairs. Each pair has $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$. The product $(0 \oplus 1) \otimes (0 \oplus 1)$ contains a singlet once in $0 \otimes 0$ and once in $1 \otimes 1 = 0 \oplus 1 \oplus 2$. Thus the singlet multiplicity is two, represented by $\alpha$ and $\beta$. Their distinction lies in an intermediate fusion label despite identical one-site and total irreps.

# 13 Hydrodynamics and the broken strong symmetry

An ink drop spreads through water even though its molecules never stop obeying microscopic mechanics. Hydrodynamics explains this loss of microscopic detail with a few slowly varying densities. SW-SSB asks a complementary question: when does a many-body probability distribution cease to remember where an individual conserved charge came from? The connection is powerful because the first question concerns ordinary response, while the second concerns nonlinear distinguishability. This chapter develops the connection without identifying the two questions by definition.

Our route is concrete. We derive diffusion on a lattice, solve its density modes, identify a stationary SW-SSB state, and then formulate the continuum theory. The final sections explain why finite-time order in a conserved system has different dimensional restrictions from stationary order.

## 13.1 Conservation does not yet mean diffusion

Let $n_i$ be a particle number at site $i$ and $Q = \sum_i n_i$. Strong conservation by a Lindblad generator means that the Hamiltonian and every jump operator commute with $Q$. Thus $\mathcal{L}^\dagger(Q) = 0$. For local dynamics one can assign currents to bonds so that

$$\frac{d}{dt}\langle n_i\rangle = \sum_{j\sim i}\langle j_{j\to i}\rangle. \tag{13.1}$$

This is a continuity equation. It says that particles leave a region through its boundary. It does not determine whether transport is ballistic, diffusive, subdiffusive, or absent. A translation-invariant noninteracting Hamiltonian can move particles ballistically; a constrained stochastic system can relax more slowly than diffusion.

Diffusion requires a constitutive relation. In a locally equilibrated, isotropic system with no persistent drift, the leading gradient term is

$$\mathbf{j} = -D\nabla n, \qquad \partial_t n = D\nabla^2 n. \tag{13.2}$$

The positive coefficient $D$ sets the transport timescale. A Fourier component evolves as $n_{\mathbf{k}}(t) = e^{-Dk^2 t} n_{\mathbf{k}}(0)$. A disturbance with size $\ell$ therefore relaxes in time of order $\ell^2/D$. Conservation supplies the vanishing rate at $k = 0$; locality and an ordinary constitutive expansion supply the $k^2$ form.

## 13.2 An exactly controlled example: the exclusion process

Consider a periodic chain of $L$ sites. Each occupation $n_i$ is either zero or one. A particle jumps across a nearest-neighbor bond at rate $\gamma$ if the destination is empty. This is the symmetric simple exclusion process. It is also the diagonal sector of a strongly number-conserving quantum Lindblad evolution with jumps

$$L_{i,+} = \sqrt{\gamma}\, b_{i+1}^\dagger b_i, \qquad L_{i,-} = \sqrt{\gamma}\, b_i^\dagger b_{i+1}, \tag{13.3}$$

where $b_i$ are hard-core boson lowering operators. Diagonal initial density matrices remain diagonal. We may therefore calculate with a classical distribution $p(n, t)$.

The incoming particle current is $\gamma n_{i-1}(1 - n_i)$ and the outgoing current is $\gamma n_i(1 - n_{i-1})$. Their difference is exactly $\gamma(n_{i-1} - n_i)$: the two-particle terms cancel. Repeating on the other bond gives

$$\partial_t \langle n_i \rangle = \gamma(\langle n_{i+1} \rangle + \langle n_{i-1} \rangle - 2\langle n_i \rangle). \tag{13.4}$$

No factorization of correlations was used. Define $n_k = L^{-1/2} \sum_j e^{-ikj}(n_j - \nu)$ with $k = 2\pi m/L$ and $\nu = Q/L$. Then

$$\mathcal{L}^\dagger(n_k) = -\lambda(k) n_k, \qquad \lambda(k) = 2\gamma(1 - \cos k) = 4\gamma \sin^2(k/2). \tag{13.5}$$

This is an exact density-sector dispersion, not a list of every many-body eigenvalue. Restoring lattice spacing $a$ gives $D = \gamma a^2$ at small wavevector. On a ring, the longest nonuniform density wave has relaxation time

$$\tau_{\text{dens}} = \frac{1}{4\gamma \sin^2(\pi/L)} \simeq \frac{L^2}{4\pi^2 \gamma}. \tag{13.6}$$

For example, at $L = 8$, the first density decay rate is $(2 - \sqrt{2})\gamma$; doubling $L$ approaches a fourfold increase in relaxation time. This exact prediction is a useful numerical check for any implementation of the model.

Detailed balance makes the uniform fixed-$Q$ distribution stationary:

$$\rho_Q = \binom{L}{Q}^{-1} \sum_{\sum_i n_i = Q} |n\rangle\langle n|. \tag{13.7}$$

Every allowed hop has a reverse hop of equal rate. On a connected ring, all configurations at fixed $Q$ communicate, so this is the unique stationary diagonal distribution in that sector. The word *sector* matters: the full generator retains one stationary distribution for every $Q$.

For $i \neq j$, take $A_{ij} = b_i^\dagger b_j$. Since $A_{ij}$ preserves total number, $\sqrt{\rho_Q}$ is a constant times the identity on its support. Consequently

$$R_1(i,j) = \text{Tr}(\sqrt{\rho_Q} A_{ij} \sqrt{\rho_Q} A_{ij}^\dagger) = \frac{\binom{L-2}{Q-1}}{\binom{L}{Q}} = \frac{Q(L-Q)}{L(L-1)}. \tag{13.8}$$

The combinatorial factor counts an occupied source and an empty destination. At fixed filling $0 < \nu < 1$, it tends to $\nu(1-\nu)$ independent of separation. The ordinary off-diagonal correlator vanishes because $\rho_Q$ is diagonal. The homogeneous root-fidelity insertion gives the same number here: $\sqrt{\rho_Q} A_{ij} \sqrt{\rho_Q} = A_{ij}/\binom{L}{Q}$ on this sector, and $A_{ij}$ is a partial isometry with $\binom{L-2}{Q-1}$ unit singular values. Thus its trace norm equals Equation 13.8, establishing fidelity-based U(1) order directly rather than invoking unitary-probe inequalities for a nonunitary hop. For a normalized postselected state the fidelity would instead be divided by the square root of the transfer success probability. The retained vacancy/occupancy factor has physical content.

The same counting gives

$$\text{Cov}(n_i, n_j) = \begin{cases} \nu(1-\nu), & i = j, \\ -\nu(1-\nu)/(L-1), & i \neq j. \end{cases} \tag{13.9}$$

The negative, system-wide background makes $\text{Var}(Q) = 0$. Nevertheless, every nonzero Fourier mode has structure factor $S(k) = \nu(1-\nu)L/(L-1)$. At stationarity, the regression of density fluctuations is exactly $\langle n_k(t) n_{-k}(0) \rangle = S(k) e^{-\lambda(k)t}$. This example exhibits both nonlinear order and diffusion, and lets us see precisely which static and dynamical ingredients produce each.

## 13.3 A doubled ferromagnet and a diffusive Goldstone mode

The classical generator has a useful spin representation. Let $P_{i,i+1}$ exchange two occupations. Acting on probability vectors,

$$-\mathcal{L} = H_{\text{st}} = \gamma \sum_i (1 - P_{i,i+1}) = \frac{\gamma}{2} \sum_i (1 - \sigma_i \cdot \sigma_{i+1}). \tag{13.10}$$

It is positive and frustration free: each term annihilates a fully symmetric state. The square root of the uniform distribution is the normalized Dicke wavefunction in the fixed-$Q$ sector. Thus a stationary probability distribution is related to a ferromagnetic ground state, and a long-wavelength density disturbance is related to a spin-wave excitation. Its energy is a decay rate rather than a real-time oscillation frequency.

More generally, vectorizing operators maps a Lindbladian to an operator on a doubled Hilbert space. Strong symmetry produces separate left and right actions. A state with SW-SSB has order that locks these actions while retaining the diagonal weak symmetry. Low-energy twists of the relative order can then supply slow modes. The microscopic constructions of [60] and [61] make this relation explicit for Brownian averaged dynamics, including constrained and long-range variants.

This correspondence also explains a potential trap. The doubled operator is not in general Hermitian. Even when a detailed-balance transformation makes it Hermitian, it is a zero-temperature problem with special frustration-free structure. Applying the finite-temperature Mermin–Wagner theorem to it as though it were an ordinary thermal magnet is unjustified. Quadratic, rather than linear, low-energy dispersion changes the infrared reasoning; frustration-free one-dimensional examples are analyzed in [62].

## 13.4 Deriving fluctuating diffusion

A continuum theory should describe fluctuations as well as the average density. Write $n = \bar{n} + \delta n$, and let $\mu(n)$ be a dimensionless thermodynamic chemical potential, with $k_B T = 1$. Suppose the static susceptibility is finite and positive,

$$\chi = \left.\frac{\partial n}{\partial \mu}\right|_{\bar{n}} > 0.$$

Near equilibrium $\delta\mu = \delta n/\chi$. A dissipative current with mobility $\sigma > 0$ is $\mathbf{j} = -\sigma\nabla\mu + \xi$. Detailed balance fixes the Gaussian current noise to

$$\langle \xi_a(\mathbf{x}, t)\xi_b(\mathbf{x}', t')\rangle = 2\sigma\delta_{ab}\delta^{(d)}(\mathbf{x} - \mathbf{x}')\delta(t - t'). \tag{13.11}$$

Combining with continuity gives

$$\partial_t \delta n = D\nabla^2 \delta n - \nabla \cdot \xi, \qquad D = \frac{\sigma}{\chi}. \tag{13.12}$$

This is the Einstein relation in our units. In physical units the temperature factors depend on whether $\mu$ means the energy-valued chemical potential or $\mu/(k_B T)$; mixing these conventions produces incorrect dimensions.

For each $k \neq 0$, this is an Ornstein–Uhlenbeck process. If the initial coarse-grained density is definite, its variance solves

$$\partial_t C(k, t) = -2Dk^2 C(k, t) + 2\sigma k^2, \qquad C(k, t) = \chi(1 - e^{-2Dk^2 t}). \tag{13.13}$$

The stationary variance is $\chi$ and the retarded density response to a chemical-potential source is

$$G^R_{nn}(\omega, k) = \frac{\chi D k^2}{Dk^2 - i\omega}. \tag{13.14}$$

The pole at $\omega = -iDk^2$ is the hydrodynamic mode. The numerator ensures a spatially uniform conserved charge cannot change under a source that merely shifts its energy in a closed sector. Taking $\omega \to 0$ at nonzero $k$, and then $k \to 0$, gives $\chi$; reversing these limits gives a different answer.

## 13.5 The response field as the broken-symmetry phase

Introduce a real response field $\varphi$ to impose the noisy continuity equation in a path integral. Integrating out the Gaussian noise yields the quadratic action

$$S = \int dt\, d^d x \left[-\varphi(\partial_t \delta n - D\nabla^2 \delta n) + i\sigma(\nabla\varphi)^2\right]. \tag{13.15}$$

The $i\sigma$ term ensures that $e^{iS}$ suppresses fluctuations of the response field with the correct noise weight. A convenient nonlinear completion is

$$\mathcal{L} = n\partial_t\varphi + i\sigma(n)\nabla\varphi\cdot\nabla[\varphi + i\mu(n)]. \tag{13.16}$$

Varying $\varphi$ and choosing the physical saddle $\varphi = 0$ gives $\partial_t n = \nabla\cdot[\sigma(n)\nabla\mu(n)]$. Expanding around $\bar{n}$ recovers Equation 13.15. This is the simplest realization of the SW-SSB hydrodynamic effective theory of [63]; related microscopic Liouvillian constructions appear in [64]. The doubled formalism organizes $\varphi$ as a phase associated with a broken strong symmetry, while the unbroken weak symmetry constrains the action. Hydrodynamic effective theories are developed further in [65].

Calling diffusion a Goldstone mode is an organizing statement with hypotheses. The steady state must carry the appropriate order; the charge fluctuations must support a nontrivial long-wavelength mode; locality and a controlled derivative expansion must apply. Detailed balance additionally ties fluctuations to response. An arbitrary mixed state does not determine a susceptibility or a transport coefficient until a dynamics and a perturbation protocol have been specified. The next chapter examines this limitation in detail.

## 13.6 A microscopic slow-mode bound and its limits

There is a precise way to isolate the static input. Consider a reversible classical Markov process, or a Lindbladian admitting the appropriate self-adjoint detailed-balance representation. We derive the classical version to avoid hiding the assumptions in notation. For configurations $\eta$, let $\pi_\eta$ be the stationary distribution and $w_{\eta\eta'}$ the transition rate. Detailed balance means $\pi_\eta w_{\eta\eta'} = \pi_{\eta'} w_{\eta'\eta}$. The stationary inner product is $\langle f, g\rangle_\pi = \sum_\eta \pi_\eta f_\eta^* g_\eta$.

The positive quadratic form of the generator is

$$\mathcal{D}(f) = -\langle f, \mathcal{L}f\rangle_\pi = \frac{1}{2}\sum_{\eta,\eta'} \pi_\eta w_{\eta\eta'} |f_{\eta'} - f_\eta|^2. \tag{13.17}$$

To derive it, expand the square. The two diagonal terms become identical after exchanging $\eta$ and $\eta'$ using detailed balance. Their difference with the cross terms is precisely the negative generator expectation. Positivity follows term by term. This is the stochastic counterpart of a variational energy expectation.

Choose $f = n_k$, normalized with a factor $V^{-1/2}$. A bounded-range number-conserving hop from $x$ to $x + a$ changes it by $V^{-1/2}e^{-ikx}(e^{-ika} - 1)$, of magnitude at most $|ka|/\sqrt{V}$. If local rates and local charge-transfer magnitudes are bounded, order $V$ possible events contribute a Dirichlet form no greater than $Ck^2$. Meanwhile $\langle |n_k|^2\rangle_\pi = S(k)$. The Rayleigh variational principle then bounds the lowest decay eigenvalue with density overlap in the corresponding sector:

$$\lambda_{\text{trial}}(k) = \frac{\mathcal{D}(n_k)}{S(k)} \le \frac{Ck^2}{S(k)}. \tag{13.18}$$

More precisely, some nonzero spectral weight of the trial vector lies at rates no greater than its Rayleigh quotient. If translation invariance is present, it remains in the wavevector-$k$ sector. For $S(k \to 0^+) > 0$, the bound forces

rates approaching zero at least as quickly as $k^2$. It does not establish that the full spectrum has a single diffusion pole, that $D$ is nonzero, or that slower modes are absent. Those require further dynamical information.

If $S(k) \sim |k|$, the bound still forces gaplessness, now through an upper bound proportional to $|k|$. If $S(k) \sim k^2$, it supplies no vanishing bound at all. Thus a finite nonzero structure factor is a sufficient static route to slow modes under these dynamical assumptions, rather than a necessary definition of SW-SSB. This distinction is central to the charge-scrambling analysis of [66]. The microscopic doubled-space constructions cited earlier use closely related single-mode reasoning; their conclusions about diffusion rely on more than the presence of a conserved symbol in the generator.

For a generic nonreversible or nonnormal Lindbladian, replacing $-\mathcal{L}$ by a Hermitian Hamiltonian without justification is invalid. Complex eigenvalues, left and right eigenvectors, and transient growth alter how decay modes must be bounded. A conservation law always yields an exact constraint, but the particular Rayleigh proof above belongs to its stated self-adjoint setting.

## 13.7 Finite-time information spreading is faster than particle diffusion

There is a revealing calculation entirely within Gaussian hydrodynamics. Let $p[n]$ be a zero-mean Gaussian distribution with covariance $C$, restricted to a fixed total charge. Compare it with the distribution obtained by moving a small charge $q$ from $y$ to $x$. The shift is $\delta n = q(\delta_x - \delta_y)$, which has zero total charge. Completing the square gives the Bhattacharyya overlap

$$\int Dn \sqrt{p[n]p[n-\delta n]} = \exp\left[-\frac{1}{8}\delta n^T C^{-1} \delta n\right]. \tag{13.19}$$

Using Equation 13.13, the logarithm contains

$$-\log R_1(r,t) = \frac{q^2}{4\chi} \int^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1-\cos(\mathbf{k}\cdot\mathbf{r})}{1-e^{-2Dk^2 t}}. \tag{13.20}$$

A microscopic cutoff $\Lambda$ regularizes the local cost. The infrared term uses $1 - e^{-2Dk^2 t} \simeq 2Dk^2 t$. In one dimension, $\int dk(1-\cos kr)/(2\pi k^2) = |r|/2$. Therefore, at large $r$,

$$R_1(r,t) \propto \exp\left[-\frac{q^2|r|}{16\chi D t}\right]. \tag{13.21}$$

The overlap length grows as $t$, whereas a density disturbance spreads only as $\sqrt{t}$. The information question asks whether two *entire distributions* differing by a distant transfer can be distinguished. It need not track the path of a typical particle. This separation is one of the central conclusions supported by microscopic models, field theory, and numerical diagnostics in [67].

In two dimensions the same low-$k$ integral grows as $\log r/t$, giving an algebraic overlap. In $d > 2$ its separation-dependent infrared part stays bounded. These statements describe a noncompact Gaussian continuum theory. They do not establish that every discrete microscopic system immediately has the same order. Integer particle worldlines and compact phase defects contain information omitted by the continuum approximation.

In microscopic U(1) models, two-dimensional field theory and simulations support a finite-time BKT-like onset of algebraic nonlinear order. Vortices are bound on one side and proliferate on the other. In one dimension there is no such finite-time long-range onset in the local models studied; the information length grows without becoming infinite at finite time. Infinite-time fixed-sector steady states can nevertheless display SW-SSB in any dimension, as Equation 13.8 demonstrates. The order of limits is therefore part of the physics:

$$\lim_{t\to\infty}\lim_{r\to\infty} R_1(r,t) \quad \text{need not equal} \quad \lim_{r\to\infty}\lim_{t\to\infty} R_1(r,t).$$

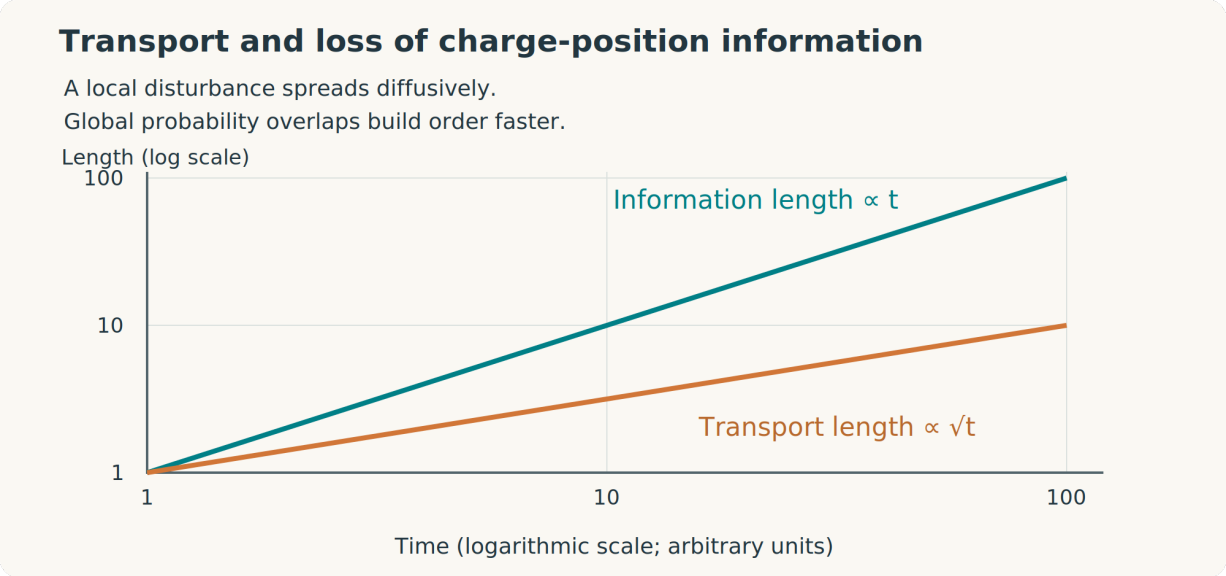



Figure 13.1: Two different lengths in a conserved open system. The curves illustrate the scaling laws derived above in dimensionless units, with freely chosen equal prefactors at time one. They are a mechanism diagram, not a fit to experimental or microscopic simulation data.

## 13.8 Worked exercises

### 13.8.1 A density wave on a ring

Prepare $\langle n_j(0)\rangle = \nu + \epsilon\cos(2\pi j/L)$ in the exclusion process. Find its amplitude at time $t$, and the time needed to reduce it by a factor ten.

**Solution**

The cosine is a sum of the two Fourier modes $k = \pm 2\pi/L$. Each decays with the same exact rate. Thus the amplitude is $\epsilon e^{-4\gamma\sin^2(\pi/L)t}$ and the required time is $\log 10/[4\gamma\sin^2(\pi/L)]$. This is an exact one-point prediction even though the evolving distribution contains correlations.

### 13.8.2 Fluctuations in a finite block

For Equation 13.7, calculate the variance of the charge in a block containing $\ell$ sites. Check the limits $\ell = 1$ and $\ell = L$.

**Solution**

Sum the covariance over all ordered pairs in the block:

$$\mathrm{Var}(Q_A) = \nu(1-\nu)\left[\ell - \frac{\ell(\ell-1)}{L-1}\right] = \nu(1-\nu)\frac{\ell(L-\ell)}{L-1}.$$

It equals the Bernoulli variance for one site and vanishes for the full system. For $1 \ll \ell \ll L$ it is extensive even though the total charge is exact.

### 13.8.3 A dynamical countercheck

Could the same stationary $\rho_Q$ be stationary under a generator with no diffusion?

**Solution**

Yes. The zero generator leaves every state stationary and has no spreading. Pure number dephasing also leaves $\rho_Q$ stationary while each $n_i$ is separately conserved. SW-SSB is a property of the state; diffusion requires sufficiently mixing transport dynamics. This simple counterexample prevents a state-only order parameter from determining a unique transport law.

# 14 Charge scrambling, fluctuations, and learning

Suppose a box contains exactly one thousand particles. The charge in the whole box is perfectly sharp, but a small subregion can fluctuate strongly. Does that fluctuation mean that the charge has scrambled? What if the distribution is a random mixture of only a few density patterns? What if it comes from dephasing a superfluid? These questions distinguish three notions that are often compressed into one phrase: nonlinear charge-transfer order, uncertainty of subsystem charge, and quantum coherence between subsystem-charge sectors.

The distinction has practical consequences. Number fluctuations are easy to measure; fidelity is harder. A theorem relating them could make SW-SSB experimentally accessible. It must, however, state additional conditions. This chapter derives the useful relation and the counterexamples that delimit it, following the static framework of [66].

## 14.1 What does a charge-transfer correlator compare?

Let $\rho$ have a fixed total U(1) charge. A transfer operator $T_{xy}$ removes one unit near $y$ and inserts it near $x$. It commutes with total $Q$, but for a region $A$ containing $x$ and excluding $y$,

$$[Q_A, T_{xy}] = T_{xy}. \tag{14.1}$$

The Rényi-1 correlator is

$$R_1(x,y) = \mathrm{Tr}(\sqrt{\rho}\, T_{xy} \sqrt{\rho}\, T_{xy}^\dagger). \tag{14.2}$$

If $\rho$ is diagonal in occupation configurations $\eta$, this becomes a sum of $\sqrt{p_\eta p_{T\eta}}$ over allowed transfer pairs. It detects whether configurations remain statistically connected under a charge move. Large variance of $Q_A$ alone merely says that many block charges occur; it says nothing about this connectivity.

For unitary charge shifts the expression is an ordinary overlap of two normalized distributions. Hard-core creation and annihilation operators are not unitary, so configurations with an empty source or a full destination do not contribute. Their occupancy prefactors should be retained rather than absorbed into an unexplained normalization.

## 14.2 The subsystem characteristic function

The distribution of $Q_A$ is encoded by

$$Z_A(\theta) = \mathrm{Tr}(\rho e^{i\theta Q_A}).$$

Expand around $\theta = 0$: $Z_A = 1 + i\theta\langle Q_A\rangle - \theta^2\langle Q_A^2\rangle/2 + \cdots$. Multiplying by its complex conjugate and taking the logarithm yields

$$\log|Z_A(\theta)| = -\frac{\theta^2}{2}\,\mathrm{Var}(Q_A) + O(\theta^4). \tag{14.3}$$

Odd cumulants affect the phase, not the modulus. An extensive variance makes a small truncated symmetry rotation highly sensitive to region size. This is a linear statistic of the density matrix: both $\langle Q_A\rangle$ and $\langle Q_A^2\rangle$ are ordinary expectation values. It cannot, without assumptions on the family of states, carry all information in the nonlinear quantity Equation 14.2.

For a translation-invariant system, let $\delta n_k = V^{-1/2}\sum_x e^{-ikx}(n_x - \bar{n})$ and $S(k) = \langle \delta n_k \delta n_{-k}\rangle$. Then

$$\mathrm{Var}(Q_A) = \frac{1}{V}\sum_k S(k)|f_A(k)|^2, \qquad f_A(k) = \sum_{x\in A} e^{ikx}. \tag{14.4}$$

The form factor weights wavevectors below approximately the inverse block size. If $S(k \to 0^+) = S_0 > 0$ and correlations are sufficiently regular, Parseval's identity gives a leading volume term $S_0|A|$. The $+$ reminds us that at fixed total charge $S(0) = 0$ exactly; a single zero mode is removed before taking the thermodynamic limit.

If instead $S(k) \sim |k|$, sharp-boundary blocks generally have an area times logarithm contribution, $\ell^{d-1}\log \ell$, together with boundary-dependent short-distance terms. Thus a state can possess long-range order while its subsystem-number variance is subextensive. The long-wavelength fluctuation spectrum, not the word "ordered," determines this scaling.

## 14.3 Separating classical uncertainty from coherence

Use the canonical purification $|\sqrt{\rho}\rangle\!\rangle$ and choose an occupation basis where $Q_A$ is diagonal and real. The two copies have observables $Q_{A,L}$ and $Q_{A,R} = Q_A^T$. Define

$$Q_A^+ = Q_{A,L} + Q_{A,R}, \qquad Q_A^- = Q_{A,L} - Q_{A,R}. \tag{14.5}$$

The weak action is generated by $Q^-$, since $U\rho U^\dagger$ vectorizes as $(U \otimes U^*)|\rho\rangle\!\rangle$. The positive sum $Q^+$ is useful for analyzing the broken direction; it should not be called the unique "strong charge," because the actual strong symmetry has two independent copy actions.

Let $m = \langle Q_A\rangle$, $V_A = \mathrm{Var}_\rho(Q_A)$ and

$$C_A = \mathrm{Tr}\left[\sqrt{\rho}(Q_A - m)\sqrt{\rho}(Q_A - m)\right].$$

Both copies have variance $V_A$, and their covariance is $C_A$. Direct expansion gives

$$\mathrm{Var}(Q_A^+) = 2V_A + 2C_A, \qquad \mathrm{Var}(Q_A^-) = 2V_A - 2C_A,$$

and hence

$$V_A = \frac{1}{4}\mathrm{Var}(Q_A^+) + \frac{1}{4}\mathrm{Var}(Q_A^-). \tag{14.6}$$

This is an identity, without a hydrodynamic approximation. To identify the coherence term, define the Wigner–Yanase skew information

$$I_{\mathrm{WY}}(\rho, Q_A) = -\frac{1}{2}\mathrm{Tr}[\sqrt{\rho}, Q_A]^2 = \mathrm{Tr}(\rho Q_A^2) - \mathrm{Tr}(\sqrt{\rho}Q_A\sqrt{\rho}Q_A). \tag{14.7}$$

The second equality follows by expanding the commutator and using trace cyclicity. It immediately gives

$$I_{\mathrm{WY}}(\rho, Q_A) = V_A - C_A = \frac{1}{2}\mathrm{Var}(Q_A^-). \tag{14.8}$$

In an eigenbasis of $\rho$,

$$I_{\mathrm{WY}}(\rho, Q_A) = \frac{1}{2}\sum_{a,b}(\sqrt{p_a} - \sqrt{p_b})^2|\langle a|Q_A|b\rangle|^2 \geq 0. \tag{14.9}$$

It vanishes if $[\rho, Q_A] = 0$. Thus a classical mixture of different block charges can have large variance and zero skew information. For a pure state $\sqrt{\rho} = \rho$, the second term in Equation 14.7 equals $m^2$, so $I_{\mathrm{WY}} = V_A$. The skew

information extracts a coherent contribution to fluctuations; it is not the same object as the fidelity susceptibility introduced for a symmetry-breaking noise channel.

One can also define twist overlaps valid for discrete symmetries. For $U_A = e^{i\theta Q_A}$, set

$$\chi_A^w(\theta) = \mathrm{Tr}(\sqrt{\rho}U_A\sqrt{\rho}U_A^\dagger), \qquad \chi_A^s(\theta) = \mathrm{Tr}(\sqrt{\rho}U_A\sqrt{\rho}U_A). \tag{14.10}$$

They are expectations of rotations generated by $Q_A^-$ and $Q_A^+$, respectively, in the canonical double. Differentiating at zero gives $-\partial_\theta^2\chi_A^w(0) = 2I_{\mathrm{WY}}$. The modulus of the centered strong twist has curvature $\mathrm{Var}(Q_A^+)$. For a classical diagonal distribution $\chi_A^w = 1$: all its uncertainty comes from statistical mixing, while the two copies agree configuration by configuration.

## 14.4 Why sufficiently fast nonlinear order forces fluctuations

Here is the mechanism behind the variance bound. Define a local doubled order parameter $O_x = b_{x,L}^\dagger b_{x,R}^T$. It has charge two under $Q^+$ and is neutral under $Q^-$. Its two-point function is $R_1(x,y)$. Nonzero long-distance $R_1$ means the canonical purification has ordinary order in this doubled operator.

For a block $A$, introduce

$$\Phi_A = \sum_{x\in A} O_x, \qquad X_A = \frac{\Phi_A + \Phi_A^\dagger}{2}, \qquad Y_A = \frac{\Phi_A - \Phi_A^\dagger}{2i}.$$

The commutator is $[Q_A^+, Y_A] = -2iX_A$. In a phase-selected extremal state with $\langle X_A\rangle \simeq m|A|$, Robertson's uncertainty inequality gives

$$\mathrm{Var}(Q_A^+)\,\mathrm{Var}(Y_A) \geq \frac{1}{4}|\langle [Q_A^+, Y_A]\rangle|^2 \simeq m^2|A|^2. \tag{14.11}$$

Using the symmetric finite-volume state directly would give $\langle X_A\rangle = 0$ and no bound. One must first use the ordered thermodynamic state's decomposition into phase-selected states. Averaging the inequality over phases and applying the law of total variance preserves a lower bound on the symmetric state's charge variance. This is the same care with order of limits needed in ordinary SSB.

Suppose the connected transverse correlator in a selected phase is absolutely summable. Then

$$\mathrm{Var}(Y_A) = \sum_{x,y\in A}\langle Y_x Y_y\rangle_c \leq K|A|.$$

Combining Equation 14.11 with Equation 14.6 gives

$$\mathrm{Var}_\rho(Q_A) \geq c|A|, \qquad c > 0. \tag{14.12}$$

A sufficient tail condition in $d$ dimensions is decay faster than $r^{-d}$, with enough uniformity to make the sum finite. More generally the bound is proportional to $|A|^2/\mathrm{Var}(Y_A)$. This explains both the theorem and its limitation: a slowly converging nonlinear correlator can make the denominator superextensive. The statement is a lower bound; proving a full volume law also requires an upper-bound or regularity assumption. A response susceptibility further requires dynamics or equilibrium fluctuation-dissipation input.

## 14.5 Three examples that prevent false equivalences

**Uniform fixed-charge mixing.** The exclusion-process ensemble gives $R_1 \to \nu(1-\nu)$ immediately at all distinct sites and $\mathrm{Var}(Q_A) \simeq \nu(1-\nu)|A|$ for a small fraction of the system. It is diagonal, so $I_{\mathrm{WY}} = 0$. Charge-transfer order can be entirely classical.

**Dephasing a superfluid.** Take a fixed-charge hard-core-boson wavefunction with nonnegative amplitudes, $|\psi\rangle = \sum_\eta \psi_\eta |\eta\rangle$. Occupation dephasing produces $\rho_{\mathrm{d}} = \sum_\eta \psi_\eta^2 |\eta\rangle\langle\eta|$. Then

$$R_1^{\mathrm{d}}(x,y) = \sum_{\eta:\, y\to x \text{ allowed}} \psi_\eta \psi_{T\eta} = \langle\psi| b_x^\dagger b_y |\psi\rangle. \tag{14.13}$$

If the original wavefunction has superfluid long-range order, the dephased state has nonlinear long-range order and vanishing ordinary one-body order. Dephasing preserves every occupation statistic, including $\mathrm{Var}(Q_A)$. A ground-state superfluid can have $S(k) \sim |k|$ and subextensive block variance. Its long-distance nonlinear tail is correspondingly too slow for the summability assumption. Nonnegative amplitudes are essential to the equality: phases would otherwise cancel in the original ordinary correlator.

**Sparse fixed-charge mixing.** Choose $r(V)$ random configurations from the exponentially large fixed-charge sector, with $V^2 \ll r(V)$ but $\log r(V) = o(V)$, and mix them uniformly. Sampling can reproduce the fixed-sector block-charge variance. A fixed finite-support transfer, however, almost never maps a selected configuration to another selected configuration: its partner is present with probability of order $r/\binom{V}{Q}$. Hence local-transfer $R_1$ vanishes in the large-system limit despite extensive variance. Suitable buffers can identify the finite ensemble label, and the construction also lacks the long-range CMI of the uniform ensemble in the scaling analyzed in [66]. The connectivity of support is the missing ingredient.

## 14.6 Records turn charge scrambling into inference

A monitored process is an instrument $\{\mathcal{M}_m\}$, not just its averaged channel. Its record distribution is $p(m) = \mathrm{Tr}\,\mathcal{M}_m(\rho)$ and its conditioned state is $\rho_m = \mathcal{M}_m(\rho)/p(m)$. Keeping the classical record gives $\sum_m p(m)|m\rangle\langle m| \otimes \rho_m$; discarding it gives $\sum_m p(m)\rho_m$. These are different information resources.

A charge-sharpening experiment asks whether the record reveals which total-charge sector was initially present. Start with a prior $p(q)$ and compute

$$p(q|m) = \frac{p(m|q)p(q)}{\sum_{q'} p(m|q')p(q')}. \tag{14.14}$$

If the posterior narrows rapidly, the observer has sharpened the charge. This need not coincide with an entanglement transition: learning one global classical label is weaker than reconstructing an extensive quantum state. Such distinct transitions are explicit in U(1)-symmetric monitored circuits [68] and monitored noisy dynamics [69]. With simultaneous charge and dipole conservation, the two global quantities can become difficult to infer at separate transitions [70].

A small example clarifies why local samples can fail. The uniform even-parity distribution on $N$ bits has every proper marginal equal to a uniform independent distribution. No fixed-size marginal distinguishes it from the unconstrained distribution. Yet conditioning on all but two disjoint nonempty groups leaves their parities locked; the CMI is $\log 2$. A learner trained only to match local statistics can therefore fit every accessible marginal while missing the global charge.

This does not mean that parity is computationally hard when its rule is supplied: one simply computes the product of all bits. Nor does it mean an arbitrary neural network is forbidden to learn it from any data. The restricted statistical-query theorem and neural-network evidence in [71] specify the access model under which learning global properties becomes hard. The observational hardness result in Chapter 16 is likewise a statement about resources and prior knowledge. "Unlearnable" is meaningful only after those are stated.

## 14.7 Worked exercises

### 14.7.1 A mixed qubit and coherent fluctuations

Let $\rho = (I + rX)/2$ and $Q = Z/2$, with $0 \le r \le 1$. Compute $\mathrm{Var}(Q)$ and $I_{\mathrm{WY}}$.

**Solution**

$\langle Q \rangle = 0$ and $Q^2 = I/4$, so the variance is $1/4$ for every $r$. In the $X$ basis, $\rho$ has eigenvalues $p_\pm = (1 \pm r)/2$, while $Q$ has only off-diagonal matrix elements of magnitude $1/2$. Substitution into Equation 14.9 gives

$$I_{\mathrm{WY}} = \frac{1}{4}(\sqrt{p_+} - \sqrt{p_-})^2 = \frac{1}{4}(1 - \sqrt{1 - r^2}).$$

The maximally mixed qubit has entirely statistical variance; the pure endpoint has entirely coherent variance.

### 14.7.2 The marginal tail

Suppose an ordered doubled state's connected transverse correlations behave as $r^{-\alpha}$. Estimate the uncertainty bound for a block of side $\ell$ when $\alpha < d$, $\alpha = d$, and $\alpha > d$.

**Solution**

Integrating the tail gives $\mathrm{Var}(Y_A) \sim |A|\ell^{d-\alpha}$ for $\alpha < d$, $|A| \log \ell$ for $\alpha = d$, and $|A|$ for $\alpha > d$. The corresponding lower bounds on block-charge variance scale as $\ell^\alpha$, $\ell^d / \log \ell$, and $\ell^d$. They are lower bounds and may be far from saturated. At the marginal power, a logarithm prevents an automatic extensive lower bound.

### 14.7.3 Retained and discarded information

A hidden bit $q$ is uniform, and the apparatus reports $m = q$ without error. Compare the charge uncertainty with and without its record.

**Solution**

With $m$, $p(q|m)$ is a delta function and $H(Q|M) = 0$. Without it, $p(q) = 1/2$ and $H(Q) = \log 2$. The averaged system state alone does not specify which information an external observer retained. The same distinction must be made for measurement trajectories in many-body experiments.

# 15 Information-critical phases and frontier transitions

A mixed-state phase transition need not make any ordinary two-point function singular. We have already seen how a nonlinear correlator or a recovery obstruction can reveal structure invisible to linear observables. A further possibility is an *extended* region where information correlations remain scale free. It resembles a critical phase of an ordinary magnet, but the diverging scale is the Markov length rather than the ordinary correlation length.

This chapter develops the statistical mechanics behind that possibility. We first derive the connection between logical error probabilities and disordered clock models. We then explain why a clock variable can acquire an emergent continuous symmetry, and how this produces partial logical memory. Finally we examine other routes to transitions: informed decoding, long-range channels, absorbing-state dynamics, and postselection. These routes test different assumptions and should not be forced into one universality class.

## 15.1 A memory with more than two charge values

On each oriented link of a square lattice, place an $N$-level system with

$$Z|a\rangle = \omega^a|a\rangle, \qquad X|a\rangle = |a+1 \pmod N\rangle, \qquad ZX = \omega XZ, \quad \omega = e^{2\pi i/N}.$$

The $\mathbb{Z}_N$ toric code generalizes the qubit code. Vertex and plaquette constraints measure oriented products of $Z$ and $X$; appropriate inverse powers on incoming links make all stabilizers commute. On a torus, the ground space stores two logical $N$-level systems. An $X^k$ error shifts a charge by $k$ modulo $N$, and an error string's endpoints produce a syndrome.

Let an independent link channel apply $X^k$ with probability $p_k$:

$$\mathcal{E}_e(\rho) = \sum_{k=0}^{N-1} p_k X_e^k \rho X_e^{-k}, \qquad \sum_k p_k = 1. \tag{15.1}$$

A full error is a flow $k$ on links with probability $P(k) = \prod_e p_{k_e}$. Its divergence is the charge syndrome $s = \partial k$. Two errors have the same syndrome when their difference is divergence free. They can still differ by a flow winding around a cycle of the torus. That winding is a logical error.

Fix a reference flow $k_s$ for each syndrome. Every compatible flow can be written as the reference plus a contractible loop flow and a representative $h$ of the two winding numbers. Define

$$Z_{s,h} = \sum_{\substack{\partial k = s \\ [k-k_s]=h}} \prod_e p_{k_e}, \qquad p(h|s) = \frac{Z_{s,h}}{\sum_{h'} Z_{s,h'}}. \tag{15.2}$$

The maximum-likelihood decoder chooses the largest *total sector weight*. Choosing the single most probable microscopic error can fail because another sector can contain many individually less probable errors. This is the difference between energy minimization and free-energy minimization.

In two dimensions, contractible divergence-free flows are lattice curls of a dual height $a_p \in \mathbb{Z}_N$. Thus $Z_{s,h}$ becomes

$$Z_{s,h} = \sum_{\{a_p\}} \prod_e p_{(k_s)_e + (\operatorname{curl} a)_e + h_e}. \tag{15.3}$$

Writing each probability as an exponential of a bond energy makes this a disordered clock model, with boundary twists labeled by $h$. The syndrome or reference error supplies the disorder. Its distribution is not arbitrary: it comes

from the same $p_k$ that defines the bond weights. This matched statistical-mechanical condition is the Nishimori condition. The quantum memory problem therefore selects a particular submanifold of a disordered-model phase diagram.

## 15.2 Deriving coherent information from posterior uncertainty

To test quantum information, entangle the two encoded qudits with an ideal reference $R$. Let $Q$ denote the noisy physical code. Its initial logical entropy is $\log(N^2) = 2\log N$. For independent $X$- and $Z$-type Pauli channels, syndrome sectors and reference-labeled logical error sectors provide an orthogonal basis in which the joint output is diagonal.

Let $(s_x, h_x)$ and $(s_z, h_z)$ label the two error types. The physical code state alone is maximally mixed in the logical subspace within each syndrome sector, so

$$S(Q) = 2\log N + H(s_x) + H(s_z).$$

The reference resolves the logical error labels, so

$$S(QR) = H(s_x, h_x) + H(s_z, h_z).$$

Subtracting gives the exact operational expression

$$I_c(R\rangle Q) = 2\log N - H(h_x|s_x) - H(h_z|s_z). \tag{15.4}$$

The conditional entropies include the average over syndromes. This derivation uses the orthogonality and factorization of the Pauli-noise code construction; arbitrary correlated or coherent errors need not yield the same formula. It avoids analytic continuation from integer Rényi indices. The direct spectrum and information calculation is central to [72].

When both posterior distributions concentrate in a unique winding sector, the memory retains $2\log N$ coherent information. If only $X$ errors occur and their posterior becomes uniform over $N^2$ windings, $I_c = 0$. If both conjugate error types fully randomize their windings, $I_c = -2\log N$. Negative coherent information is permitted: it is not an entropy and should not be interpreted as a negative number of stored qudits.

An intermediate posterior distribution can have a nonzero, nonmaximal entropy that persists as the lattice grows. This is the information-theoretic origin of fractional memory in the critical phase. It is not, by itself, a declaration that a particular fraction of a single qudit can be deterministically recovered with zero error. One must specify the coding task, asymptotic limit, and decoder performance. The work cited above also evaluates a maximum-likelihood decoder, whose logical error approaches a nontrivial fractional value in the information-critical region.

## 15.3 Why a discrete clock can behave like a continuous angle

The clean clock model provides the simplest derivation of the mechanism. At long distances write the spin as $e^{i\theta}$, with effective action

$$S_{\text{sw}} = \frac{K}{2}\int d^2x\,(\nabla\theta)^2, \tag{15.5}$$

where $K$ is a dimensionless stiffness. In this Gaussian theory $\langle[\theta(r) - \theta(0)]^2\rangle = (\pi K)^{-1}\log(r/a)$. Using the characteristic function of a Gaussian variable gives

$$\langle e^{i\theta(r)}e^{-i\theta(0)}\rangle \sim r^{-1/(2\pi K)}. \tag{15.6}$$

There are two distinct ways this scale-free theory can fail. The microscopic $N$-fold anisotropy contributes $u_N \cos(N\theta)$, whose scaling dimension is $\Delta_N = N^2/(4\pi K)$. It becomes relevant in two dimensions when $\Delta_N < 2$, or $K > N^2/(8\pi)$. At large stiffness the angle therefore locks to one of $N$ values.

The second perturbation is a vortex, around which $\theta$ winds by $2\pi$. Its energy grows as $\pi K \log(L/a)$, while the entropy for locating it grows as $2\log(L/a)$. Vortices proliferate when $\pi K < 2$. Equivalently, their scaling dimension is $\pi K$.

Both perturbations are irrelevant in the interval

$$\frac{2}{\pi} < K < \frac{N^2}{8\pi}. \tag{15.7}$$

This interval exists only for $N > 4$. There the discrete model forgets its anisotropy on long scales and behaves as an emergent U(1) theory with algebraic order. The two boundaries are BKT transitions rather than ordinary isolated critical points.

This clean derivation explains *why* $N > 4$ is a natural dividing point. It does not supply exact critical stiffnesses for a quenched disordered Nishimori model. Disorder, replica constraints, and the observable under consideration modify the continuum analysis. In the decohered code, the intermediate region is established through the exact information mapping together with the disordered-model analysis and numerical evidence in [72]; inserting the clean values into a noisy-code threshold would be unjustified.

## 15.4 Scale-free information without ordinary critical correlations

Let $A$ be an interior disk, $B$ a surrounding annulus, and $C$ the exterior. The conditional mutual information measures what the exterior reveals about $A$ after $B$ is known. A finite Markov length means that this residual information falls exponentially as the annulus thickens. An information-critical phase has scale-free, nonconstant dependence on the geometry even while ordinary correlations have a finite correlation length. This algebraic behavior must be distinguished from the constant CMI offset of an exact global constraint, such as the parity ensemble in Chapter 8. A nonzero constant alone does not establish a critical phase. The finite-system geometry and treatment of global or topological constraints belong to the diagnostic's definition.

In the code mapping, reduced density matrices retain boundary and enclosed-charge constraints. Their entropies become Shannon entropies of syndrome distributions with different boundary conditions. The CMI is therefore sensitive to whether charge information remains inferable across arbitrarily large annuli. In the critical clock regime the cost of a long-wavelength twist is scale invariant, producing nonexponential information correlations.

One can see why partial winding uncertainty survives from the Gaussian action. A square torus with total twist $\Theta$ across one direction has minimum-energy profile $\theta(x) = \Theta x/L$. Substituting into Equation 15.5 gives

$$S_{\text{twist}} = \frac{K}{2}\Theta^2, \tag{15.8}$$

independent of $L$. Twisted sectors can therefore retain finite relative weights as the system grows. In an illustrative clean compact theory, $w_h \propto \sum_{m\in\mathbb{Z}} \exp[-K(2\pi m + 2\pi h/N)^2/2]$. These normalized weights have neither zero entropy nor necessarily the uniform entropy. The actual noisy-code posterior also contains disorder and its averaging, but the size-independent twist cost explains the mechanism for persistent intermediate logical uncertainty.

Ungauging the code replaces the topological charge constraint with a global symmetry. The information-critical regime is then described by algebraic order in an off-diagonal, or relative-copy, symmetry with an emergent continuous angle. This makes it a critical relative of SW-SSB. It should not be described as a conventional physical superfluid with a nonzero $\langle b_i^\dagger b_j \rangle$ in the original density matrix. Its phase coherence belongs to the appropriate purified or enlarged description, and ordinary local correlators can remain short ranged.

The decomposition into Coulombic pure states found in the code construction provides another interpretation: individual components have gapless gauge-like fluctuations even though the mixture's ordinary correlation length need not diverge. As elsewhere, a convex decomposition is not unique. The invariant content is established by the

information diagnostics and channel/recovery structure, not by assigning a unique hidden wavefunction to every experimental run.

## 15.5 Charge-informed decoding: a different BKT problem

If anyons carry U(1) charge and the apparatus resolves their integer charges, the syndrome contains more information than an uncharged anyon position. A phenomenological model takes integer error currents $k_e \in \mathbb{Z}$ with $P(k) \propto \exp[-\sum_e k_e^2/(2\alpha)]$. The measured charge is their divergence. Given one representative error $k$, compatible errors are $k + J$ with $\partial J = 0$:

$$Z_k \propto \sum_{\partial J=0} \exp\left[-\frac{1}{2\alpha}\sum_e (k_e + J_e)^2\right]. \tag{15.9}$$

This is a disordered integer-loop model. A correct recovery must choose the winding sector, not merely annihilate the measured charges. The maximum-likelihood sum and a minimum-cost flow optimization are different decoders. [73] finds a BKT transition for the optimal matched decoder, with a modified winding-number jump, and numerical evidence for a disorder-dominated loop-glass region for some mismatched decoders. "Confidently incorrect" means that a decoder assigns high posterior confidence under its *assumed* noise model to a winding sector that is wrong under the actual model.

The global U(1) charge, the finite $\mathbb{Z}_N$ memory, and the relative-copy U(1) of the previous section are distinct symmetries. Their loop and angle descriptions are related tools, but matching notation does not identify their universality classes or thresholds.

## 15.6 Dephased fermions and an exact counting-statistics mapping

The clock and loop examples started with specially structured memories. A late-September development gives a complementary route from ordinary metallic and insulating parent states [74]. The key is an exact Fourier representation that we can derive directly for a diagonal fermionic or hard-core occupation distribution. The transfer has unit-magnitude matrix elements on allowed configurations; ordinary bosonic ladder operators would require additional occupation factors.

Let $p_n$ be occupation probabilities after complete density dephasing. For index $\alpha > 0$, define a normalized auxiliary distribution

$$q_\alpha(n) = \frac{p_n^{\alpha/2}}{\sum_m p_m^{\alpha/2}}, \qquad \chi_\alpha(\theta) = \sum_n q_\alpha(n) e^{in\cdot\theta}. \tag{15.10}$$

This auxiliary distribution is sometimes called an escort distribution. It is a mathematical construction, not an extra experimental ensemble. Integrate every angle over $[0, 2\pi)$ with normalized measure. Fourier orthogonality gives

$$\int D\theta\, |\chi_\alpha(\theta)|^2 = \sum_n q_\alpha(n)^2.$$

Insert $e^{i(\theta_i - \theta_j)}$ into the same integral. It selects pairs of configurations differing by one particle transferred between $i$ and $j$. Dividing by the previous integral gives

$$\frac{\int D\theta\, e^{i(\theta_i-\theta_j)}|\chi_\alpha(\theta)|^2}{\int D\theta\, |\chi_\alpha(\theta)|^2} = \frac{\sum_n p_n^{\alpha/2} p_{n'}^{\alpha/2}}{\sum_n p_n^\alpha}. \tag{15.11}$$

The right side is the diagonal-state Rényi-$\alpha$ transfer correlator. The left side is exactly a spin correlator of compact angles with positive weight $|\chi_\alpha|^2$. Thus one may define an effective XY action $S_\alpha = -\log|\chi_\alpha|^2$. Zeros of the weight are permitted; the action need not be a local nearest-neighbor XY Hamiltonian.

For $\alpha = 2$, $q_2 = p$: the weight is the squared full-counting-statistics characteristic function of the parent occupation distribution itself. For $\alpha = 1$, $q_1 \propto \sqrt{p}$, so substituting the parent's ordinary structure factor without further work would be incorrect. The paper supplies an additional expansion to analyze this case.

The second cumulant of $\log \chi_\alpha$ gives a quadratic angle action determined by the covariance of $q_\alpha$. In a metallic parent the $\alpha = 2$ kernel behaves as $S(k) \sim |k|$. A spin-wave estimate therefore involves $\int d^d k (1 - \cos kr)/|k|$, logarithmically divergent in one dimension but infrared finite in higher dimensions. This explains the tendency to algebraic order in one dimension and true nonlinear long-range order in $d \geq 2$. It does not violate the two-dimensional finite-range XY theorem: a $|k|$ kernel is spatially nonlocal.

An ordinary insulator instead has an analytic kernel starting as $k^2$, yielding the usual logarithmic spin-wave fluctuations in two dimensions. Compact vortices and short-distance couplings then determine whether a quasi-ordered regime survives. Insulating parents can therefore yield either nonlinear order or a trivial phase, depending on microscopic details; "dephased insulator" is not a universal phase label. The work also distinguishes quantum Hall cases. Its finite-dephasing treatment uses even-index replicas and additional regularization assumptions, which should not be promoted to an all-index fidelity theorem. The exact identity Equation 15.11 itself requires complete dephasing and remains valid independently of those continuum approximations.

## 15.7 Long-range channels change preparation constraints

For an initially short-range-entangled chain, local finite-time evolution has a bounded information light cone. In familiar finite-depth local dephasing constructions this obstructs a finite-noise SW-SSB transition in one dimension. A channel with pair operations at arbitrarily long separations changes the premise.

For example, apply commuting pair-dephasing channels with probabilities decreasing as a power of distance. Each individual $Z_i Z_j$ operation preserves $\prod_i X_i$, but the entire channel has long spatial tails. In its doubled statistical-mechanical representation, long-range ferromagnetic couplings can support order in one dimension. A domain wall with interactions $J(r) \propto r^{-\alpha}$ crosses of order $r$ bonds at range $r$, so its large-distance cost scales as

$$\Delta E \sim \sum_{r \lesssim L} r J(r) \sim \begin{cases} L^{2-\alpha}, & \alpha < 2, \\ \log L, & \alpha = 2, \\ \text{constant}, & \alpha > 2. \end{cases} \tag{15.12}$$

This energetic illustration explains how locality restrictions can be evaded; it is not an exact phase diagram for every channel, and nonsummable interactions require care with thermodynamic normalization. [75] implements this strategy for a family of matrix-product states interpolating between trivial and SPT states. Its phase diagram is explicitly formulated using Rényi-2 diagnostics and includes a mixed-SPT regime with logical information and nonlinear string order. The distinction from fidelity/Rényi-1 phase definitions remains essential. The model also permits relatively high-purity regimes useful for the processor experiment discussed next.

## 15.8 Absorbing states, stationary transitions, and postselection

An absorbing configuration cannot escape once reached. Consider classical defects whose reactions conserve parity, such as pair annihilation and branching by an even number of new defects. An inactive absorbing state can retain local charge information, while an active steady distribution can scramble it. Strong symmetry alone therefore does not guarantee that every stationary state of every local generator has SW-SSB.

[22] constructs local symmetric dynamics with strong-paramagnetic and active SW-SSB regimes. In one spatial dimension, the observed critical scaling is consistent with parity-conserving branching-annihilating random walks; a two-dimensional pair-flip Toom construction supplies another route. These are dynamical universality questions, different from a single application of an equilibrium-noise channel. Mixing time and the Liouvillian spectral gap must be distinguished, especially for nonnormal generators and special absorbing sectors.

Postselection provides yet another route. An unnormalized operation $\mathcal{M}$ followed by $\rho \mapsto \mathcal{M}(\rho)/\operatorname{Tr}\mathcal{M}(\rho)$ is nonlinear in its input, even when $\mathcal{M}$ is completely positive. It is not a deterministic CPTP channel. Consequently

a theorem about symmetric trace-preserving evolution does not automatically constrain a postselected phase diagram. [76] studies SW-SSB steady ensembles and weak-to-strong postselection transitions in Abelian and non-Abelian examples. The success probability is part of the physical resource cost, rather than an optional technical detail.

## 15.9 All-to-all fermions and approximate quantum codes

Spatial locality is not the only route to an information transition. The Sachdev–Ye–Kitaev model has random all-to-all interactions among Majorana fermions. With a convention $\chi_i^2 = 1$, a quartic Hamiltonian is $H = \sum_{i<j<k<l} J_{ijkl}\chi_i\chi_j\chi_k\chi_l$, with disorder variance scaled as $N^{-3}$ to obtain an extensive energy. Its large-$N$, low-temperature entropy motivates an approximate logical subspace whose size is exponential in $N$.

The order of limits is essential. One first takes $N$ large while keeping a low-energy window, then lowers its temperature. Taking zero temperature at a generic finite size can leave a single state rather than an extensive code. Approximate coding also requires a tolerated error criterion; residual entropy alone does not supply an efficient encoder or decoder.

Single-Majorana errors are odd under fermion parity and exchange parity with the bath. Bilinear errors $i\chi_i\chi_j$ are even. A parity-preserving pair-noise family with probability of order $q/N$ per pair has an extensive total effect, because there are order $N^2$ pairs. [77] studies these two noise classes with a reference-entangled low-energy state and finds a robust threshold structure for the parity-preserving family, connected to strong-to-weak parity order.

The saddle mechanism has a simple general form. In a doubled description, an inter-copy bilinear $B$ is odd under either copy's parity and even under their product. A pair-noise term can supply an attractive interaction proportional to $B^2/N$. The Hubbard–Stratonovich identity

$$e^{gB^2/(2N)} = \sqrt{\frac{N}{2\pi g}} \int d\phi\, e^{-N\phi^2/(2g)+\phi B} \tag{15.13}$$

turns it into an auxiliary field coupled linearly to $B$. After integrating fermions, the extensive saddle free energy has schematic form $f(\phi) = \phi^2/(2g) + f_0(\phi)$. Parity makes it even. If the quadratic coefficient $1/g + f_0''(0)$ changes sign, a nonzero $\phi$ develops: independent parities are reduced to their diagonal action. This explains why a parity-preserving noise can induce effective relative-copy parity order without explicitly exchanging parity with the environment.

The quantitative analysis uses large-$N$ saddle equations, low-temperature extrapolation, and Rényi information diagnostics. A Rényi-2 entropy difference should be labeled as such rather than substituted silently for operational von Neumann coherent information. The thresholds, approximate-code tolerances, and nonlocal interactions belong to the specified SYK families; they are not thresholds for geometrically local toric memories. The connection to thermofield-double teleportation is a theoretical information-theory link, not a claim that a laboratory wormhole has been observed.

These frontiers share a common question—where and how global charge information survives—but answer it under different operations and limits. A useful research habit is to state, before quoting a transition: the state family, the allowed operations, the information diagnostic, and the thermodynamic and time limits.

CMI can additionally constrain how critical behavior changes under coarse graining. [78] derives inequalities for crossover functions assuming ultraviolet-finite CMI and specified scaling hypotheses. Such results can restrict candidate renormalization-group flows, and bounds for convex mixtures constrain stability arguments. Their hypotheses matter: CMI is not automatically ultraviolet finite for every continuum geometry, and an unrestricted monotonicity rule for every nonequilibrium phase does not follow. This offers a promising way to test proposed mixed-state field theories without first solving every microscopic model.

## 15.10 Worked exercises

### 15.10.1 Why five is different from four

Using Equation 15.7, find the clean spin-wave interval for $N = 4$ and $N = 5$. Does this predict a noisy-code threshold?

> **Solution**
>
> For $N = 4$ both bounds equal $2/\pi$, so no open interval remains. For $N = 5$ the interval is $2/\pi < K < 25/(8\pi)$, numerically about $0.637 < K < 0.995$. This is a clean continuum relevance argument. It predicts the possibility of an intermediate regime, not its microscopic temperature or the threshold of a disordered decoder.

### 15.10.2 Sector entropy and memory

Suppose only one Pauli error type occurs in a $\mathbb{Z}_3$ toric memory. Its posterior over the nine winding sectors is $(1/2, 1/2, 0, \dots, 0)$ for every syndrome. Find $I_c$ and the maximum-likelihood success probability.

> **Solution**
>
> The conditional winding entropy is $\log 2$. Thus $I_c = 2\log 3 - \log 2 = \log(9/2)$. Any choice between the two equally likely sectors succeeds with probability $1/2$. Nonmaximal coherent information and a nonzero logical-error probability coexist. Neither value can be replaced by a claim of perfect storage of a particular integer number of qutrits in this single block.

### 15.10.3 Maximum probability versus maximum likelihood

Sector A contains one error of probability $0.3$; sector B contains four errors of probability $0.175$ each. Which sector should an optimal decoder choose?

> **Solution**
>
> The most probable individual error is in A. But B has total probability $0.7$, so maximum-likelihood decoding chooses B. Ignoring entropy reduces the success probability from $0.7$ to $0.3$ in this example. This is why the partition sum in Equation 15.2 cannot generally be replaced by one minimum-energy configuration.

# 16 Detecting mixed-state order in the laboratory

An order parameter is useful only when its relation to a measurement is understood. For ordinary SSB, one measures a spin or density correlation repeatedly and averages the outcomes. A nonlinear function of $\rho$ requires a different resource: multiple copies, a particular purification, a tractable model, or a sufficiently small subsystem for tomography. This chapter derives those options and explains how the first experimental platforms used them.

The evidence reviewed here is restricted to work publicly available by **22 October 2026**. The cold-atom and superconducting-device results are described with their finite-size and modeling assumptions. Their existence does not turn arbitrary mixed-state fidelity into an efficiently measurable observable.

## 16.1 Why one expectation value is insufficient

Every fixed measurement outcome probability has the form $\mathrm{Tr}(M\rho)$ and is linear in $\rho$. If two ensembles have the same density matrix, no measurement on one copy can determine which ensemble decomposition was used. Conversely, a nonlinear diagnostic is not the expectation of a fixed operator on one copy for all $\rho$.

This does not forbid estimating a nonlinear quantity from many repetitions. Tomography estimates $\rho$ and then computes the quantity classically. The difficulty is scaling: an arbitrary $N$-qubit density matrix has $4^N - 1$ real parameters. The purpose of specialized protocols is to exploit more structure than arbitrary tomography allows.

For a unitary probe $A$, the Rényi-2 numerator and purity can be measured using two copies. The swap identity is

$$\mathrm{Tr}[S(X \otimes Y)] = \mathrm{Tr}(XY), \tag{16.1}$$

where $S|a\rangle|b\rangle = |b\rangle|a\rangle$. Therefore

$$\mathrm{Tr}(\rho A \rho A^\dagger) = \mathrm{Tr}\left[S(\rho \otimes A\rho A^\dagger)\right], \qquad \mathrm{Tr}(\rho^2) = \mathrm{Tr}[S(\rho \otimes \rho)]. \tag{16.2}$$

One may use a controlled-swap circuit, or local interference/Bell-basis measurements whose product measures the global swap. For qubits, a single-site swap is $(I + X \otimes X + Y \otimes Y + Z \otimes Z)/2$. Joint-copy protocols trade circuit complexity for direct access to quadratic quantities.

The denominator cannot be ignored. If purity is exponentially small, an additive error that is tiny on a scale of one can still overwhelm the normalized ratio. For instance, a swap outcome with variance of order one needs order $\epsilon^{-2}$ shots for additive precision $\epsilon$. Resolving a quantity of order $2^{-N}$ by such a generic estimator can require exponentially many repetitions.

## 16.2 Randomized measurements and their statistical cost

Two physical copies need not coexist. One can measure independent preparations in randomly selected local Pauli bases. For one qubit, an unbiased density-matrix estimator from basis $U$ and outcome $b$ is

$$\hat{\rho} = 3U^\dagger|b\rangle\langle b|U - I.$$

Tensor products give an $N$-qubit estimator with $\mathbb{E}\hat{\rho} = \rho$. Independent snapshots then satisfy $\mathbb{E}\,\mathrm{Tr}(\hat{\rho}\hat{\sigma}) = \mathrm{Tr}(\rho\sigma)$. Take $\sigma = A\rho A^\dagger$ and estimate the numerator and denominator separately. The resulting finite-sample ratio is generally biased even though each quadratic estimator is unbiased; resampling or bias analysis is needed for quantitative error bars.

[79] develops this route specifically for SW-SSB diagnostics and demonstrates it numerically for decohered Ising chains. Randomization provides an implementable measurement scheme, but does not remove the growth of variance for global nonlinear observables. Restricting to a region can make tomography or randomized estimates feasible, at the cost of replacing a global diagnostic with a local one.

## 16.3 Canonical purification is a measurement resource

If a device can prepare

$$|\sqrt{\rho}\rangle\!\rangle = \sum_{a,b} (\sqrt{\rho})_{ab} |a\rangle_L |b\rangle_R,$$

the Rényi-1 correlator becomes an ordinary doubled-state expectation:

$$R_1(A) = \langle\!\langle \sqrt{\rho} | A \otimes A^* | \sqrt{\rho} \rangle\!\rangle. \tag{16.3}$$

For a two-point charged probe, this is a two-point function of local inter-copy operators. [10] identifies this route to scalable detection for families whose canonical purifications have efficient preparations.

Preparing *some* purification is not sufficient. If $|\Psi\rangle = (I \otimes V)|\sqrt{\rho}\rangle\!\rangle$, then the right-copy operator must be transformed by $V$ to obtain Equation 16.3. That operator may become nonlocal or unknown. The resource is a controlled canonical purification or an equivalent preparation with a known, tractable environment operator.

Local formulations offer another compromise. Estimate $F(\rho_A, O\rho_A O^\dagger)$ in regions of increasing radius around a charged insertion. The region size controls both the physical approximation and the measurement cost. If the required resources grow exponentially with region volume $v$, a budget polynomial in total system size can reach $v = O(\log N)$. [21] develops this strategy; [23] gives an infinite-volume local formulation. Global order implies suitable local fidelity order, but the converse needs additional hypotheses. Pure locally thermal states provide a warning against replacing every global definition by a finite-region test.

If a matrix-product density-operator representation is available, variational sequential circuits can give certified lower and upper fidelity bounds [80]. This is valuable for one-dimensional numerical studies. Its efficiency presumes an economical representation and successful optimization; it is not a promise that arbitrary laboratory states have such a representation.

## 16.4 What the hardness theorem does and does not say

[13] constructs efficiently preparable ensembles without SW-SSB that are computationally indistinguishable from ensembles with it, under the paper's pseudorandomness assumptions. This rules out a universally efficient, state-agnostic detection algorithm with the stated experimental access. It does not prohibit efficient diagnostics for thermal states, known channels, Gaussian states, canonical purifications, or other structured families.

The distinction can be phrased as a resource table:

| Access supplied | Quantity obtained | Principal limitation |
|---|---|---|
| Arbitrary copies and global swap | $R_2$ numerator and purity | Small purity and global-measurement variance |
| Random local measurements | Quadratic overlaps | Snapshot variance and ratio bias |
| Canonical purification | $R_1$ as an ordinary correlator | Preparation and copy-basis control |
| Local tomography | Reduced-state fidelity | Region-size cost; local/global distinction |
| Known pure input and dephasing kernel | Fidelity and $R_2$ for compatible probes | Input-purity and channel assumptions; shot cost |
| Samples plus a learned probability model | Diagonal-state $R_1, R_2$ estimators | Model bias and support errors |

A successful experiment should explain which row it uses. The following platforms use different rows.

## 16.5 Dephased Fermi gas: turning snapshots into nonlinear correlations

Occupation imaging completely dephases the measured state in the number basis. If a configuration is $n$, the resulting diagonal density matrix has weights $p_n$. For a transfer from $j$ to $i$, let $n'$ be the allowed displaced configuration. Then

$$R_1(i,j) = \sum_n \sqrt{p_n p_{n'}}, \qquad R_2(i,j) = \frac{\sum_n p_n p_{n'}}{\sum_n p_n^2}. \tag{16.4}$$

Disallowed transfers contribute zero. For a diagonal state and a compatible one-to-one charge move, the root-fidelity expression has the same classical square-root-overlap structure, with the appropriate subnormalized support for a nonunitary hopping probe. Fermionic hopping signs cancel between the two matrix elements in these nonlinear expressions.

Direct frequency estimation of all $p_n$ remains exponentially expensive. Suppose, however, a classical model supplies probabilities $q_n$. Experimental samples $n_m \sim p$ give

$$\widehat{R}_{1,\mathrm{QC}} = \frac{1}{M}\sum_{m=1}^{M} \sqrt{\frac{q_{n'_m}}{q_{n_m}}}, \qquad \widehat{R}_{2,\mathrm{QC}} = \frac{\sum_m q_{n'_m}}{\sum_m q_{n_m}}. \tag{16.5}$$

If $q = p$, taking the sampling expectation reproduces Equation 16.4. If $q \neq p$, more shots remove sampling error but leave model bias. This distinction is essential: a smooth model can assign probability to a transferred partner that is actually absent from the true support, artificially raising an overlap. Validation must therefore probe the model assumptions as well as statistical convergence.

The cold-atom experiment of [81] uses a learned Gaussian fermionic reference because its weakly interacting gas is close to a number-conserving free-fermion state. Such a state is determined by its one-body correlation matrix $G$. To see why this helps, diagonalize the single-particle density matrix; the normal modes are independent Bernoulli occupations. In a nonsingular mixed Gaussian state, the probability for exactly the sites $S$ to be occupied can be written

$$q(S) = \det(I - G)\det\left([G(I-G)^{-1}]_S\right). \tag{16.6}$$

This determinant formula follows from the Gaussian operator representation $\rho = \det(I-G)\exp[c^\dagger \log(G(I-G)^{-1})c]$. Singular pure-state limits are obtained continuously; a fixed-number Slater determinant instead gives configuration weights as squared orbital determinants. These polynomial calculations replace an exponentially large table of probabilities by a matrix model. The fitted $G$ minimizes the empirical negative log likelihood, equivalently the data-to-model KL divergence up to a data-only constant.

The experiment loads $^6$Li into a square optical lattice, images one spin component, and analyzes a dephased effectively single-component gas. The full homogeneous system has approximately $14 \times 14$ sites; central subsystems are used to assess size dependence. It finds nondecaying nonlinear correlations in the dephased metallic regime. A commensurate superlattice drives a metal-to-insulator change in the underlying state, accompanied by suppression of the nonlinear order after dephasing. The paper compares quantum-classical estimates with correlators evaluated entirely in the learned reference, and studies subsystem condensate densities.

A useful finite-size statistic is

$$\varrho_1(L) = \frac{1}{V_L^2}\sum_{i,j\in A_L} R_1(i,j). \tag{16.7}$$

If $R_1 \to r_\infty > 0$, the double sum has leading term $r_\infty V_L^2$. Short-range contributions are only of order $V_L$, so $\varrho_1 \to r_\infty$. If $R_1$ decays with a finite correlation length, $\varrho_1$ instead vanishes as $1/V_L$ away from boundary corrections. Algebraic order gives a different, size-dependent power. This diagnostic avoids choosing just one separation, although nearby-site contributions and fit windows still matter.

At nonzero temperature, fixed-sector thermal states can already have nonlinear order on both metallic and insulating sides. Thus the ideal zero-temperature transition, the dephasing enhancement, and a finite-temperature experimental crossover should be separated. A finite subsystem is also not itself an exactly fixed-charge system, because it exchanges particles with its complement. The probability-based correlators remain meaningful, but interpreting them as bulk phase diagnostics uses a thermodynamic argument in addition to the estimator.

## 16.6 A five-qubit cluster array: the role of discarded records

The superconducting-array experiment of [47] prepares a five-qubit cluster state and contrasts selective dephasing, conditioned measurements, and discarded records. Its simplest ideal reduced state follows directly from cluster stabilizers. Trace out even sites 2 and 4. All stabilizer products with support on those sites vanish under the trace, leaving

$$\rho_{1,3,5} = \frac{1}{8}(I + X_1X_3X_5). \tag{16.8}$$

It is the maximally mixed positive-parity state of three qubits. Its ordinary $Z_iZ_j$ expectations vanish, while $Z_iZ_j$ commutes with the parity and leaves $\rho$ unchanged. The root fidelity is therefore one. Conditioned on measurements of the even sublattice, the odd sites can instead form GHZ-like states with outcome-dependent correlation signs. Averaging discards the signs and removes the ordinary correlation while retaining the nonlinear overlap.

This gives an exact consistency check: measuring the discarded subsystem in a different complete basis cannot change the unconditioned reduced density matrix. It can change the conditioned decomposition and the amount of visible trajectory order. A basis-dependent averaged state would indicate retained postselection, incomplete averaging, or an experimental imperfection.

The reported experiment uses full few-qubit tomography and also symmetry resolution. The unprojected reduced state has appreciable leakage between parity sectors; projecting the tomographic estimate with $P_+ = (I + X_1X_3X_5)/2$ produces a symmetry-resolved state. Such a projection is an analysis operation with a specified selection probability. The observations are therefore finite-size demonstrations of the diagnostic structure and mixed-SPT mechanisms, not a thermodynamic extrapolation from a three-site system.

## 16.7 Processor transitions using a known-channel kernel

The work of [82], submitted on 19 October 2026, studies matrix-product initial states under long-range pair dephasing. It reports finite-size crossings and scaling collapses for chains with $N = 6, 9, 12$, together with decaying ordinary correlations. Its reconstruction protocol is especially instructive because it makes the extra prior information explicit.

Let the initial state be pure, $|\psi\rangle$, and let pair errors $Z_kZ_l$ occur independently with known probabilities $p_{kl}$. Their net action is $Z(s) = \prod_i Z_i^{s_i}$ with even parity. Instead of enumerating all error masks, form their characteristic kernel

$$\mathcal{K}(w) = \mathbb{E}_s(-1)^{s\cdot w} = \prod_{k<l:\, w_k\neq w_l} (1 - 2p_{kl}). \tag{16.9}$$

For each pair, the random sign is nontrivial only when $w_k \neq w_l$; independence turns the expectation into a product. This derives the kernel using $O(N^2)$ arithmetic, without computing the full distribution of masks.

Write the input amplitudes as $e^{i\phi_z}\sqrt{P_z}$. After dephasing,

$$\rho_{zz'} = e^{i(\phi_z-\phi_{z'})}\sqrt{P_zP_{z'}}\,\mathcal{K}(z \oplus z'). \tag{16.10}$$

For a probe $A = Z_iZ_j$, the diagonal input-phase unitary commutes with both channel and probe. Common-unitary invariance of fidelity removes it. Consequently computational-basis samples from the input contain all the unknown state information needed for this particular reconstruction. The conclusion fails for arbitrary mixed

inputs: two mixed density matrices can share $P_z$ but have different off-diagonal structure and different fidelity correlators.

The Rényi-2 identity follows immediately by multiplying matrix elements:

$$R_2(i,j) = \frac{\mathbb{E}_{z,z'}[d(z)d(z')\mathcal{K}(z\oplus z')^2]}{\mathbb{E}_{z,z'}[\mathcal{K}(z\oplus z')^2]}, \qquad d(z) = (-1)^{z_i+z_j}. \tag{16.11}$$

The denominator is purity. Independent sample pairs should exclude a shot paired with itself when estimating the quadratic averages.

Fidelity can be reconstructed with a Gram matrix. For $M$ measured inputs $z_a$, define

$$G_{ab} = \frac{\mathcal{K}(z_a \oplus z_b)}{M}, \qquad D_{ab} = \delta_{ab} d(z_a).$$

Factoring the channel density matrix into error-ensemble vectors shows that the nonzero eigenvalues of the empirical probe-weighted operator equal those of $G^{1/2}DG^{1/2}$. Taking the trace norm yields

$$\widehat{F} = \left\| G^{1/2} D G^{1/2} \right\|_1 . \tag{16.12}$$

This shifts a potentially exponential Hilbert-space calculation to an $M \times M$ matrix calculation. Its dense classical cost scales as $O(N^2M^2 + M^3)$. That statement is polynomial in the *number of supplied shots*; it does not prove a polynomial number of shots for fixed precision across all states and channels. The nonlinear estimator can also have finite-shot bias.

Computational-basis measurements reconstruct the ordinary, fidelity, and Rényi-2 correlators. A second Hadamard-test setting accesses off-diagonal string expectations. The protocol evaluates a known target channel by classical processing of input data, so channel correctness and input purity are substantive assumptions. It is a different task from blind tomography of an unknown noisy output. The experiment explicitly distinguishes fidelity and Rényi-2 transition estimates; their agreement is not imposed as a universal identity.

## 16.8 Reading finite-size evidence responsibly

A thermodynamic phase is defined by a limit, while every device is finite. A nonzero correlator at one separation establishes an overlap, not spontaneous symmetry breaking by itself. Increasing the separation with size, testing ordinary correlator decay, and comparing alternative finite-size fits make the interpretation stronger.

For a conventional continuous transition a common ansatz is $R(L,p) = L^{-2\Delta}\mathcal{F}((p-p_c)L^{1/\nu})$. Crossings of $L^{2\Delta}R$ can estimate $p_c$, but the scaling dimension and correlation-length exponent are fitted parameters and may be effective over a narrow size window. BKT transitions have exponentially diverging lengths, $\xi \sim \exp[b/\sqrt{|p-p_c|}]$, so a simple power-law collapse may misidentify the asymptotic behavior. Hardware errors, symmetry leakage, boundary conditions, and estimator bias can imitate rounding or shift apparent crossings.

The most convincing conclusion through the cutoff is that nonlinear mixed-state order has become experimentally accessible with well-specified resources. The next stage is to enlarge scaling windows, certify model assumptions, and design observables that remain economical as entropy and system size increase.

## 16.9 Worked exercises

### 16.9.1 An incorrect learned model

Let the true distribution be concentrated on $|01\rangle$, and consider the hop between sites 1 and 2. A model assigns equal probabilities to 01 and 10. Compare true $R_1$ and the infinite-data quantum-classical estimator.

Solution

The transferred partner 10 has true probability zero, so the true overlap is zero. Every observed sample is 01, but the model ratio is one, so Equation 16.5 returns one. Sampling more accurately cannot repair this support error. Prior knowledge and validation of the learned distribution are part of the measurement protocol.

### 16.9.2 The kernel at zero and maximal pair noise

For a connected graph of pair channels with $p_{kl} = 1/2$ on every graph edge, which masks $w$ have $\mathcal{K}(w) \neq 0$?

Solution

Every edge crossing between $w = 0$ and $w = 1$ contributes a zero factor. A nonzero kernel therefore requires $w$ constant on each connected component. For a connected graph only the all-zero and all-one masks survive, with kernel one. This retains the global parity constraint while removing other computational-basis coherences.

### 16.9.3 Purity and error bars

Suppose numerator and purity both equal $10^{-4}$, each with additive uncertainty $10^{-5}$. Estimate the uncertainty of their ratio if the estimates are independent.

Solution

Linear propagation gives relative variance $(10^{-5}/10^{-4})^2 + (10^{-5}/10^{-4})^2 = 0.02$, so the ratio has uncertainty about $0.14$ around one. A seemingly small absolute uncertainty does not give a precise normalized correlator. Correlated estimators can improve or worsen this estimate and require their covariance in the propagation.

# Part IV

# IV · Working with the Theory

# 17 Computational laboratory

## 17.1 What a small calculation can establish

A numerical demonstration should connect a definition to a calculation that a reader can reproduce. It should also make its limitations visible. Exact diagonalization of eight spins can establish an algebraic identity for those eight spins, test an implementation, or reveal a useful trend. It cannot establish a thermodynamic phase by itself.

This laboratory contains five connected demonstrations: an exact parity response, independent pair errors on a chain, small periodic square lattices, noncommuting mixed-state overlaps, and diffusion. The calculations use Julia's standard libraries. The source, CSV data, and original vector figures are available with the book.

Run from the repository root:

```
julia --project=numerics numerics/generate.jl
```

No external Julia packages are required. The script fixes the random seed for its single random-matrix example. All other results are deterministic enumeration, analytic formulas, or exact Fourier sums. Figure conversion is separate from the physics calculation:

```
python3 -m pip install cairosvg
python3 scripts/convert_figures.py
```

The generated tables are in `numerics/data/`. The root-fidelity convention is the one defined in Equation 4.3.

## 17.2 A global parity bit under weak-only noise

Begin with the exactly strongly symmetric state $\rho_+ = (I + U)/2^N$, $U = \prod_i X_i$. Apply independent single-site $Z_i$ errors with probability $p$. Since $Z_i U Z_i = -U$, each site attenuates the coefficient of $U$ by $1 - 2p$. Thus

$$\rho(p) = \frac{I + \lambda U}{2^N}, \qquad \lambda = (1 - 2p)^N. \tag{17.1}$$

For $p > 0$ in the chosen range, this output has both global parity sectors and is only weakly symmetric. We are studying the explicit symmetry-breaking response of an initially SW-ordered state, not declaring every point of this path strongly symmetric.

Conjugating by one $Z_i$ exchanges the two sectors. Their total probabilities are $(1 \pm \lambda)/2$, uniformly distributed within each sector. Applying the classical fidelity formula gives

$$F(\rho(p), Z_i \rho(p) Z_i) = \sqrt{1 - \lambda^2}. \tag{17.2}$$

For every proper region the state is maximally mixed. The full entropy is $(N - 1)\ln 2 + h((1 + \lambda)/2)$, where $h(x) = -x \ln x - (1 - x)\ln(1 - x)$. Therefore a nonempty tripartition covering the whole system has

$$I(A : C|B) = \ln 2 - h\left(\frac{1 + \lambda}{2}\right). \tag{17.3}$$

For fixed $N$, the one-point fidelity tends to zero as $p \to 0$. For every fixed $p > 0$, it tends to one as $N \to \infty$. This exact calculation exposes the noncommuting limits without a fit.

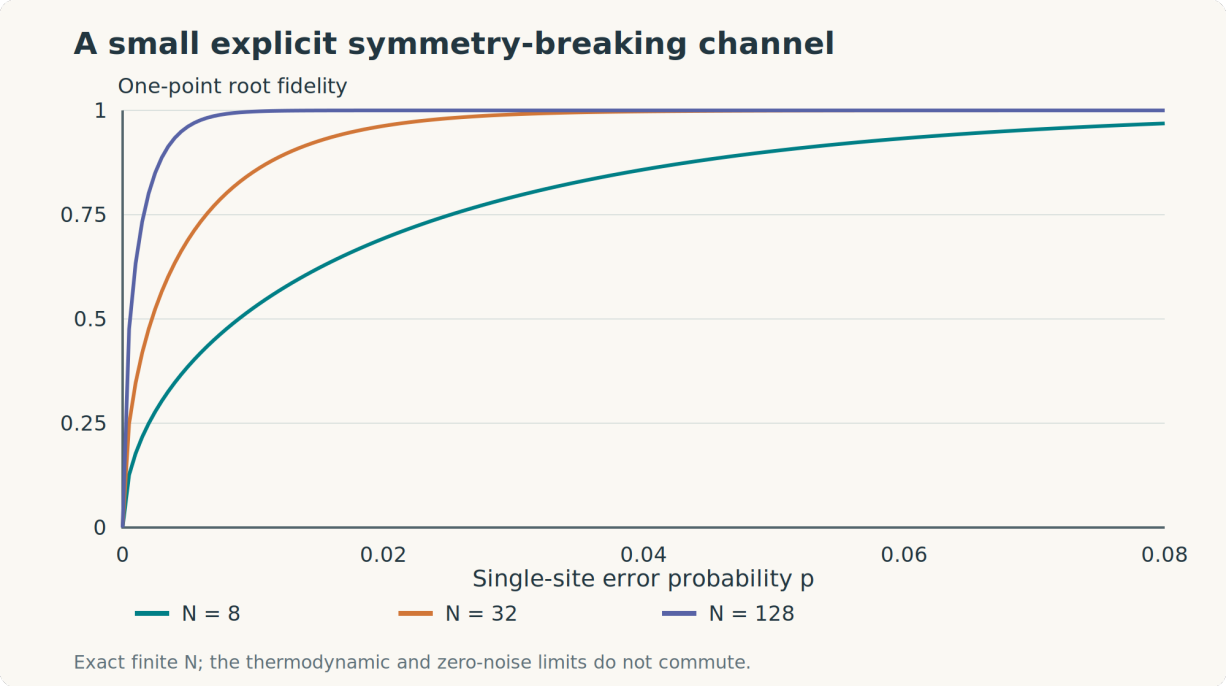



Figure 17.1: Exact response to single-site noise. Larger systems lose the strong parity constraint at smaller per-site error probabilities. The noise channel on this figure explicitly breaks strong symmetry.

The small-$p$ response is nonanalytic if expressed directly in root fidelity: for fixed finite $N$, $\lambda \simeq 1 - 2Np$ and $F \simeq 2\sqrt{Np}$. A finite linear susceptibility should therefore specify whether the response variable is $F$, $F^2$, or a reparameterized channel strength. Never differentiate a convention borrowed from a paper without checking that convention.

## 17.3 Exact pair-error enumeration on a chain

Now use a strongly symmetric channel. The input is $|+\rangle^{\otimes N}$, and each nearest-neighbor bond independently applies $Z_iZ_{i+1}$ with probability $p$. Label an $X$-basis spin configuration by an integer $s$, with bit one indicating $X_i = -1$. A bond error toggles two bits. Consequently every occupied output configuration has even parity.

The full output is diagonal in this basis. There is no need to store a $2^N \times 2^N$ matrix. Store its $2^N$ probabilities instead. Each bond updates them as

$$P_{\text{new}}(s) = (1-p)P(s) + pP(s \oplus m_{ij}),$$

where $m_{ij}$ has bits $i, j$ set and $\oplus$ denotes exclusive-or. The Julia implementation is short:

```
function bond_distribution(N, edges, p)
    P = zeros(2^N)
    P[1] = 1
    for (i, j) in edges
        mask = (1 << (i-1)) □ (1 << (j-1))
        P = [(1-p)*P[s+1] + p*P[(s□mask)+1]
             for s in 0:2^N-1]
    end
    P
end
```

The pair insertion $A = Z_iZ_j$ shifts the distribution by $m_{ij}$. For commuting states,

$$F_{ij} = R_{1,ij} = \sum_s \sqrt{P(s)P(s \oplus m_{ij})}, \qquad R_{2,ij} = \frac{\sum_s P(s)P(s \oplus m_{ij})}{\sum_s P(s)^2}. \tag{17.4}$$

An open chain has no bond cycles. Each allowed even spin configuration corresponds to exactly one bond-error pattern. Flipping the two endpoints of an interval toggles the $r$ bonds inside it. The classical overlap factorizes over those bonds:

$$F(r) = R_1(r) = [2\sqrt{p(1-p)}]^r, \qquad R_2(r) = \left[\frac{2p(1-p)}{p^2 + (1-p)^2}\right]^r. \tag{17.5}$$

The script verifies these formulas for $N = 4, 6, 8$, all allowed separations, and five noise strengths, including both endpoints. Agreement is checked to absolute tolerance $10^{-12}$. It separately verifies normalization and the absence of odd-parity support.

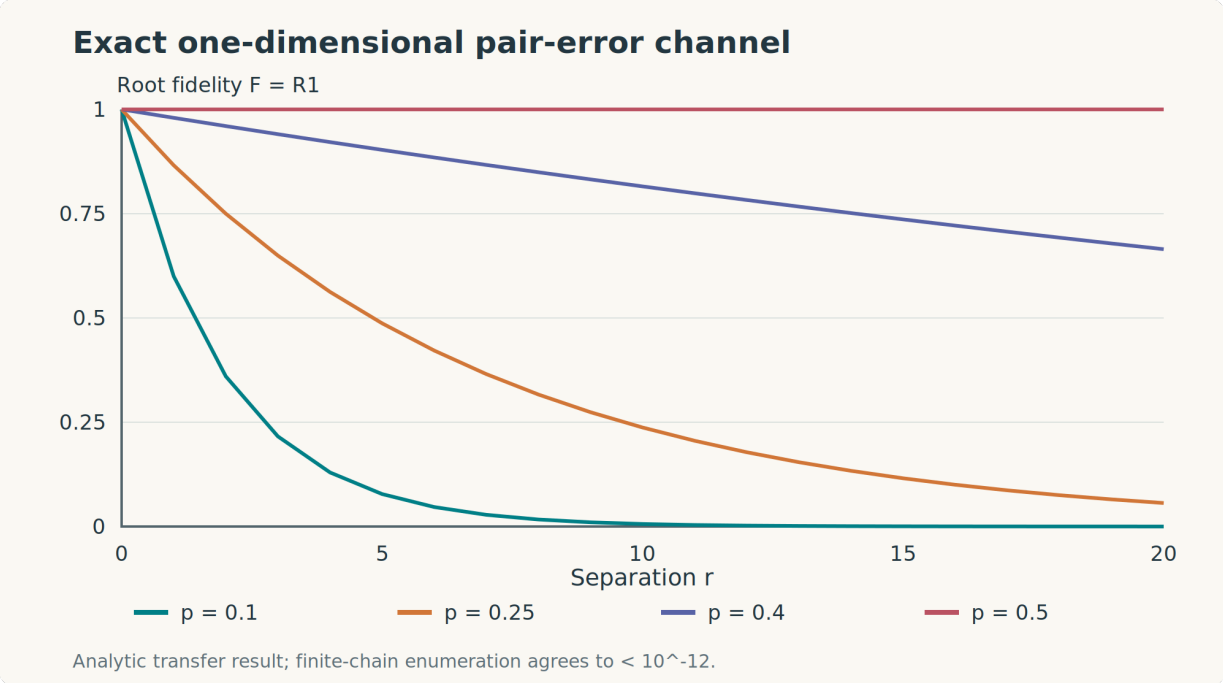



Figure 17.2: Exact one-dimensional fidelity correlators. A curve close to one over a short interval need not have nonzero long-distance order.

For $0 < p < 1/2$, $\xi_F^{-1} = -\ln[2\sqrt{p(1-p)}]$. Write $p = 1/2 - \delta$. Then $2\sqrt{p(1-p)} = \sqrt{1-4\delta^2}$, so $\xi_F \simeq 1/(2\delta^2)$. The divergence at complete dephasing is quadratic in the inverse distance to that endpoint. Translating $p$ to time via continuous pair dephasing produces an exponentially growing information length in this discrete-symmetry model. That is different from the continuous-charge models where other growth laws arise.

## 17.4 What changes on a square lattice?

The same update rule works on any graph. On a periodic square lattice, different bond configurations can have the same boundary because they differ by closed loops. The unique-pattern factorization used for the chain is lost. Summing over those loops is precisely the statistical-mechanical problem underlying the two-dimensional transition.

The data file `square-finite.csv` contains exact $L = 3, 4$ periodic-lattice results at selected $p$ values. For these small sizes, all probabilities are explicitly represented; there is no sampling error. There is substantial finite-size error if one tries to infer an infinite-lattice threshold. The numbers near $0.109$ and $0.178203$ are included to compare with the independently derived fidelity and Rényi-2 transition values discussed in the decoherence chapter, not to remeasure those thresholds.

A meaningful scaling study would require sizes larger than the critical correlation length, observables with a controlled scaling form, uncertainty from both statistics and finite-size corrections, and an explicit choice of diagnostic. Crossing curves of $F$ and $R_2$ on sixteen spins cannot establish a common phase boundary.

## 17.5 A noncommuting example: the correlators differ

Equality $F = R_1$ above follows from simultaneous diagonalization. It is not a universal identity. The script constructs a positive random matrix $MM^\dagger$, normalizes it, and embeds it in the even-$X$ sector of four spins. With $A = Z_1 Z_2$, it evaluates

$$F_A = \|\sqrt{\rho} A \sqrt{\rho}\|_1, \quad R_{1,A} = \mathrm{Tr}(\sqrt{\rho} A \sqrt{\rho} A^\dagger), \quad R_{2,A} = \frac{\mathrm{Tr}(\rho A \rho A^\dagger)}{\mathrm{Tr}\,\rho^2}.$$

Matrix square roots come from the Hermitian eigenvalue decomposition. Tiny negative eigenvalues from rounding are clipped only at zero after positivity is guaranteed by the construction. Trace norms come from singular values. The output is recorded in `noncommuting.csv`, including the seed. The checked bounds are

$$R_1 \le F \le \sqrt{R_1}.$$

These inequalities concern normalized states and unitary insertions. They do not provide a dimension-independent equivalence between $R_2$ and $F$. Nor does a four-site noncommuting example establish spontaneous order: there is no long-distance limit in that calculation. Its purpose is to test the definitions and implementation.

## 17.6 Diffusion: an exact conservation check

For a single particle hopping symmetrically on a periodic chain at rate $D$,

$$\dot{n}_x = D(n_{x+1} + n_{x-1} - 2n_x).$$

The Fourier mode $e^{ikx}$ has eigenvalue $-4D\sin^2(k/2)$. Starting from $n_x(0) = \delta_{x0}$, the exact solution is

$$n_x(t) = \frac{1}{N} \sum_{m=0}^{N-1} e^{-4D\sin^2(\pi m/N)t} e^{2\pi i m x/N}. \tag{17.6}$$

The zero mode stays constant, ensuring $\sum_x n_x(t) = 1$. For long wavelengths and times shorter than the finite-size equilibration time, the profile approaches a Gaussian of width $\sqrt{2Dt}$. The script checks normalization and positivity at all displayed times. For the symmetric exclusion process, the same linear equation holds exactly for the mean density even though the many-particle probability distribution has interactions through exclusion.

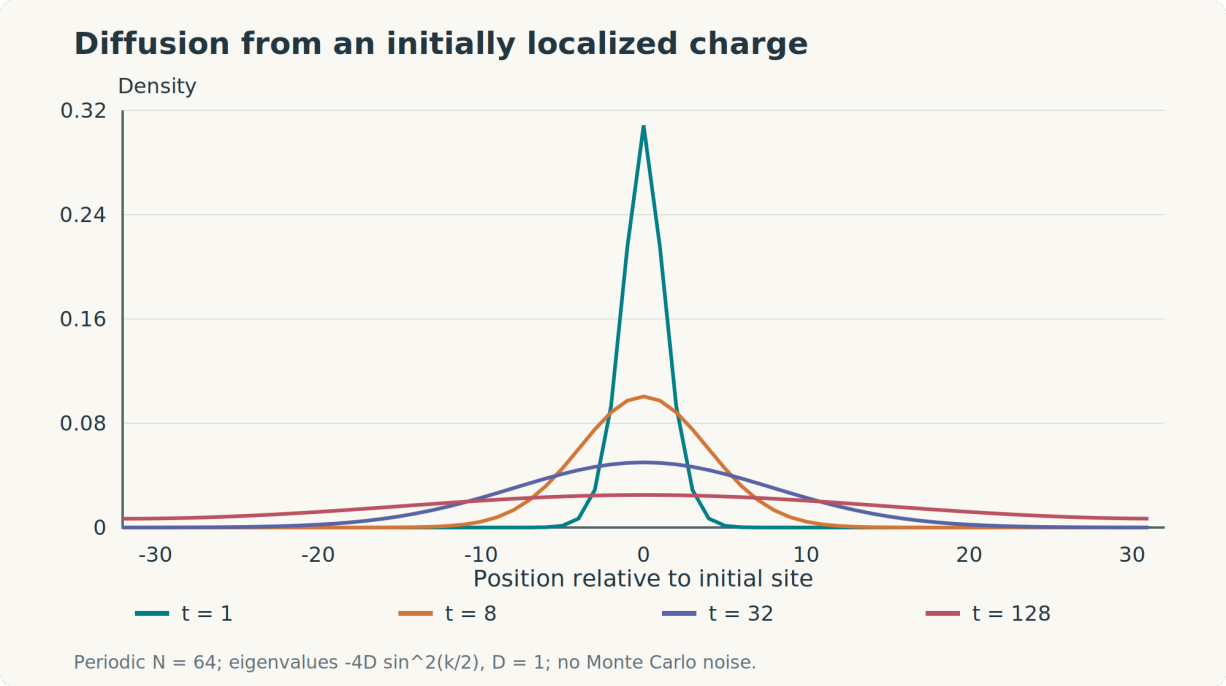



Figure 17.3: Exact diffusion profiles for a periodic 64-site lattice. Diffusion spreads density on a square-root-of-time scale. Its relation to nonlinear information lengths requires additional analysis, developed in the hydrodynamics chapter.

This plot demonstrates diffusion, not SW-SSB by itself. To diagnose SW-SSB in a stationary charge-conserving many-body ensemble, one must also evaluate the nonlinear charge-transfer correlators and their limits.

## 17.7 Canonical block fluctuations

The last figure compares the exact hypergeometric variance from Equation 2.4 with its independent-site grand-canonical counterpart. At fixed small block size they agree as $N$ grows. At the full system size the canonical variance vanishes.

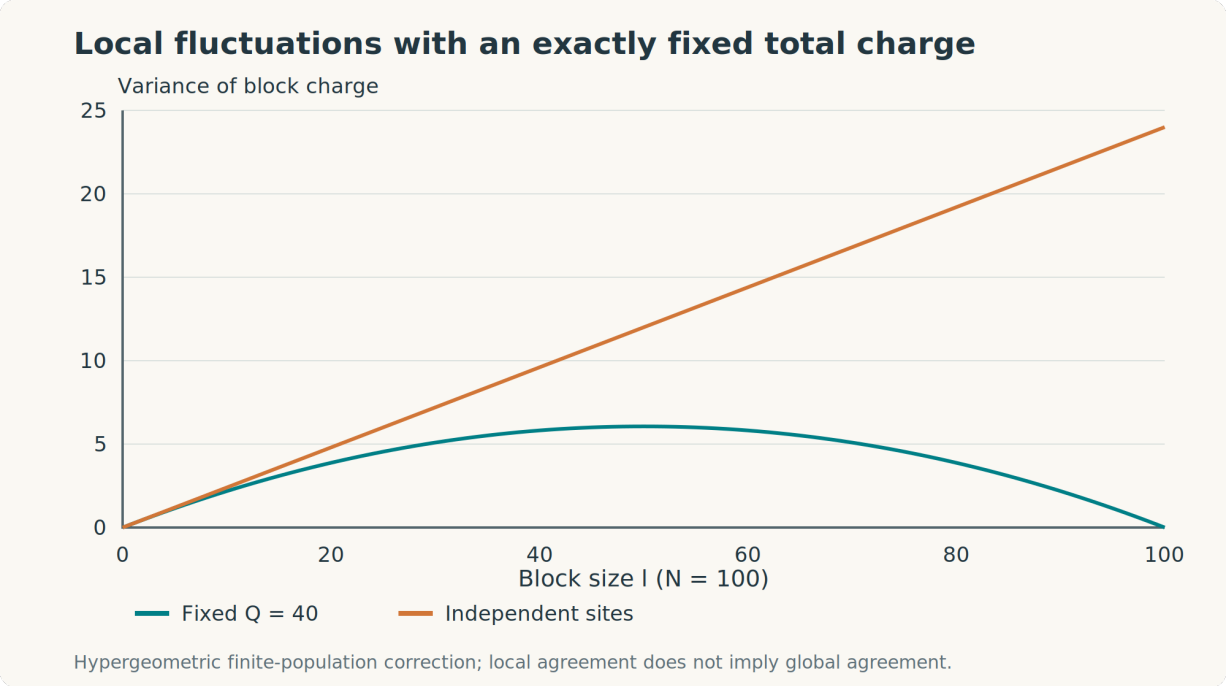



Figure 17.4: Block-charge fluctuations in a uniform fixed-charge ensemble and in the corresponding independent-site ensemble. The fixed total charge is visible in the finite-population correction.

Extensive variance on small fractions of the system is consistent with a globally sharp charge. Conversely, extensive block variance alone does not establish SW-SSB. The charge-scrambling chapter develops explicit reasons why both directions of an unconditional equivalence fail.

## 17.8 Exercises and worked solutions

### 17.8.1 1. Verify the strong-breaking response

Show that $\mathrm{Tr}(\rho(p)U) = \lambda$ and that the one-point fidelity in Equation 17.2 is maximal when this global parity expectation vanishes.

**Solution**

The identity part has zero trace against $U$, while $U^2 = I$ gives $\mathrm{Tr}(\lambda U^2/2^N) = \lambda$. The fidelity is $\sqrt{1-\lambda^2}$, maximal at $\lambda = 0$. This is a relation between a nonlocal linear observable and a nonlinear local-operation overlap for this special family, not a general identity for all mixed states.

### 17.8.2 2. Why is the open-chain map injective?

Given endpoint spin flips $s_i$, reconstruct the bond-error bits $e_i$ on an open chain. Explain the even-parity constraint.

**Solution**

The boundary equations are $s_1 = e_1$, $s_i = e_{i-1} \oplus e_i$ for interior sites, and $s_N = e_{N-1}$. Thus $e_i = s_1 \oplus \cdots \oplus s_i$ is unique. The last equation holds precisely when $\bigoplus_i s_i = 0$. Adding a cycle destroys uniqueness because toggling every edge of the cycle leaves the boundary unchanged.

### 17.8.3 3. Extract a length without fitting

Derive the asymptotic divergence of $\xi_F$ near $p = 1/2$ and compare it with $\xi_2$.

**Solution**

With $\delta = 1/2 - p$, $a_F = \sqrt{1 - 4\delta^2} = 1 - 2\delta^2 + O(\delta^4)$, so $\xi_F \simeq 1/(2\delta^2)$. The Rényi-2 base is $(1-4\delta^2)/(1+4\delta^2) = 1 - 8\delta^2 + O(\delta^4)$, so $\xi_2 \simeq 1/(8\delta^2)$. They diverge at the same endpoint in this chain but have different lengths. Their common endpoint is not evidence that all dimensions or models share the same transitions.

### 17.8.4 4. The diffusion equilibration time

Find the slowest nonzero Fourier decay rate for the periodic chain and its large-$N$ scaling.

**Solution**

The $m = 1, N - 1$ modes have rate $4D \sin^2(\pi/N) \simeq 4\pi^2 D/N^2$. Thus the inverse spectral rate scales as $N^2/(4\pi^2 D)$. A specified mixing error can introduce logarithmic factors. The conserved zero mode is excluded from the gap rather than treated as a slowly decaying fluctuation.

### 17.8.5 5. Design a useful numerical extension

How would you test the difference between a local SW diagnostic and a global finite-sector diagnostic on the parity family?

**Solution**

Compute the full one-point fidelity, then trace out one site and recompute it on the remaining state. For the exact fixed-parity input, the full one-point fidelity is zero because the charge sectors are orthogonal. The reduced state is maximally mixed, so its one-point fidelity is one. Repeat while varying the region and total size separately. The calculation reveals a difference in which information is retained, rather than a numerical inconsistency between two implementations.

# 18 Synthesis and research problems

## 18.1 A connected picture

The simplest SW-SSB state carries a sharp global charge without ordinary long-range order. A local charge insertion changes its sector and produces an orthogonal state; a distant neutral pair insertion can have finite overlap. The charge is therefore globally constrained and locally mobile in the information-theoretic sense. This picture connects the major themes of the book.

Density matrices separate ensemble invariance from support in a charge sector. Purification turns nonlinear mixed-state overlaps into linear correlations in an enlarged space. Stability explains why strongly symmetric shallow dynamics can enter an ordered regime but cannot freely leave it. Recovery translates the order into a restriction on repairing local damage. Gauging converts charge-transfer inference into topological error correction. Hydrodynamics connects delocalized continuous charge to slow density modes, provided the required static and dynamical assumptions hold.

None of these connections erases the distinctions developed along the way. A laboratory records particular observables with particular resources. A theorem quantifies over a specified class of states and maps. A phase equivalence retains a specified kind of information. Scientific progress comes from making those choices explicit enough that two apparently competing statements can be compared.

## 18.2 A ledger of what has been established

| Statement | Status and conditions |
|---|---|
| Strong state symmetry restricts support to one charge sector. | Exact finite-dimensional linear algebra; for non-Abelian groups a common eigenvalue for every group element is a one-dimensional representation. |
| $R_1 \le F \le \sqrt{R_1}$. | Exact for normalized states with unitary insertions in the conventions of Chapter 6. Nonunitary operators require the stated homogeneous or conditioned normalization. |
| Rényi-2 order is interchangeable with fidelity order. | False in general. Explicit spectral counterexamples and different transitions in the square-lattice decoherence model demonstrate the failure. |
| Strongly symmetric bounded-depth local channels preserve global fidelity order. | Theorem with bounded local Hilbert spaces, local gates, and appropriately dressed bounded-support charged operators; growing depth requires quantitative care. |
| SW-SSB produces a local recovery obstruction. | Established in the global fixed-sector setting with the relevant charged diagnostics. The retained charge constraint and allowed recovery region matter. |
| Long-range ordinary CMI characterizes every local infinite-volume SW state. | False without qualification. The complete-system finite-volume CMI and infinite-volume local CMI are different limits; symmetry-resolved constructions add assumptions. |
| The two-dimensional independent pair-error model has one intrinsic fidelity/recovery threshold near $p = 0.109$. | Established mapping with numerical Nishimori critical input; the distinct clean Rényi-2 threshold near $0.178203$ is analytic from the clean Ising critical coupling. |

| Statement | Status and conditions |
|---|---|
| Every state with extensive subsystem charge variance has SW-SSB. | False. Sparse fixed-charge distributions provide counterexamples. |
| Rényi-1 order forces extensive charge variance. | Conditional theorem; sufficiently rapid approach to long-distance order is essential. Dephased superfluids show why. |
| Conservation implies diffusion. | False. Diffusion follows with additional constitutive and equilibration assumptions; exact models and suitable effective theories realize it. |
| Information-critical topological phases can preserve a fraction of logical information. | Demonstrated in specified decohered $\mathbb{Z}_N$ models through mappings, continuum arguments, and numerical evidence; not a universal claim about every code or noise channel. |
| Small quantum processors establish an infinite-system singularity. | No. The experiments provide finite-size signatures and scaling evidence, with channel and inference assumptions described in Chapter 16. |

This ledger is a guide to inference rather than a hierarchy of value. An exact counterexample can be as illuminating as a universal theorem. A controlled experiment can reveal a phenomenon before its full classification exists.

## 18.3 How to formulate a precise project

A proposal involving SW-SSB should specify six pieces of data: the family of states; the symmetry and its representation; the diagnostic and normalization; the order of limits; the allowed operations or measurement access; and the criterion for deciding the question. Without them, a statement such as "noise destroys the phase" can refer to several inequivalent calculations.

A useful first project has a controlled limiting case. Start from an exactly solvable parity or fixed-number ensemble, a stabilizer state, or a classical diagonal channel. Verify the limit before introducing interactions, correlated noise, disorder, or non-Abelian fusion. Distinguish finite-size deviations from the failure of the proposed principle.

## 18.4 Project 1: local versus global information

**Question.** Which operational resources preserve the distinction between a fixed-parity sequence and the unrestricted maximally mixed sequence when their finite-region limits coincide?

**Starting calculation.** Reproduce their full and reduced one-point fidelities, full-system CMI, and exact parity discrimination probabilities. Then impose a measurement architecture: local measurements with a global classical parity computation, contiguous local tomography, or circuits of bounded depth with bounded classical communication.

**Deliverable.** A resource-dependent distinguishability statement. A nonlocal classical computation of all local $X_i$ outcomes accesses global parity, even though no single bounded region does. This is different from implementing a bounded-support channel.

**Open direction.** Compare noisy parity readout, finite sampling, and scaling of accessible regions with the extension-based infinite-volume symmetry framework. A precise result should say whether it concerns a finite-volume encoding, a local state, or an observer's inference task. The conceptual disagreement in Chapter 8 supplies the motivation, not a predetermined answer.

## 18.5 Project 2: when a nonlinear transition is intrinsic

**Question.** Which modifications of a noise model move a replica transition without changing the fidelity/recovery phase boundary?

**Starting calculation.** Use the spectral counterexamples of Chapter 6 and the exact chain of Chapter 9. Add a neutral local flag or a second eigenvalue scale. Track $F, R_1, R_2$, purity, and trace distance together.

**Deliverable.** An explicit family showing which singularity is a spectral reweighting effect. If studying a two-dimensional lattice numerically, derive the statistical model for each observable before fitting a crossing.

**Open direction.** Find physically natural, locally prepared families where higher-Rényi quantities supply controlled bounds on intrinsic diagnostics over a useful parameter range. Equality need not hold universally to make a proxy valuable experimentally.

## 18.6 Project 3: information length versus transport length

**Question.** How does the scale on which charge becomes locally indistinguishable compare with the scale of density transport?

**Starting calculation.** Reproduce the exact exclusion-process density modes, then evaluate overlap diagnostics of its probability distribution for several times and separations. Compare the transport scale $\sqrt{Dt}$ with the nonlinear length predicted by the finite-time analysis in Chapter 13.

**Deliverable.** A scaling plot in which both lengths are defined from explicit observables. State the initial condition, charge sector, discretization, and window before finite-size saturation.

**Open direction.** Introduce dipole conservation, kinetic constraints, or weak integrability breaking. A different transport exponent does not automatically determine the nonlinear information exponent. Deriving their relation is the research question.

## 18.7 Project 4: what non-Abelian noise leaves behind

**Question.** Which information about a total non-Abelian charge can be decoded from local records, and which requires access to fusion channels?

**Starting calculation.** Work through the fusion-tree example of Chapter 12 for a finite group or a small spin-singlet space. Separate center charge, irrep labels, and multiplicity labels. Construct two ensembles with the same coarse charge statistics but different retained fusion information.

**Deliverable.** A hierarchy of operational tasks and fidelities, with an explicit measurement algebra for each. An irrep label is not the same as a scalar charge, and a multiplicity space may hold information invisible to that label.

**Open direction.** Relate these decoding hierarchies to the entanglement forced by non-Abelian symmetry, and determine which features are stable under the chosen local-channel equivalence. The recent non-Abelian work provides model-based evidence rather than a final classification.

## 18.8 Project 5: an experimentally controlled estimator

**Question.** How much can a structured estimator certify when the preparation model is only approximately known?

**Starting calculation.** Implement the diagonal probability-overlap estimator or known-channel Gram-kernel calculation in Chapter 16 on the exact distributions supplied with this book. Perturb the reference model while retaining the same measured low-order moments.

**Deliverable.** Bias and variance estimates separated from finite-size extrapolation. Include purity dependence if using $R_2$. Report whether the estimator supplies an upper bound, a lower bound, or a model-dependent point estimate.

**Open direction.** Combine local tomography, certified tensor-network bounds, and physical noise calibration to obtain an error bar on an intrinsic diagnostic. General hardness results explain why a useful protocol must state its access assumptions; they do not preclude efficient measurements in structured families.

## 18.9 Several questions that remain connected

Theories of mixed-state phases now retain different amounts of local recovery and higher-form symmetry data. A useful classification must explain which experimental or dynamical questions each equivalence relation answers. Non-Abelian symmetry and anomalies can enforce entanglement even when Abelian SW order admits classical representatives. Their relation to hydrodynamics remains less complete than the U(1) picture. Information-critical phases raise the possibility that logical memory and decoding have structures richer than a single threshold. Learning-based detection makes the observer's prior information part of the operational problem.

These are not independent curiosities. They all ask which information is local, which is globally constrained, and what operations can move between those descriptions. The best next calculation is often a small one that isolates a disputed implication.

## 18.10 Exercises and worked solutions

### 18.10.1 1. A statement with missing hypotheses

Repair the sentence "A finite-depth channel cannot create or destroy SW-SSB."

**i** Solution

For global fidelity-based order, a strongly symmetric bounded-depth local channel preserves existing SW-SSB under the assumptions of the stability theorem, allowing a dressed order parameter. Such a channel can create order from an initially unordered state; the pair-error model is an example. A weak-only covariant channel can remove the exact strong-sector condition. Thus neither arbitrary channels nor two-way invariance are justified by the one-way theorem.

### 18.10.2 2. A proposed fluctuation diagnostic

An experiment measures $\mathrm{Var}(Q_A) \propto |A|$. Is that sufficient to claim SW-SSB?

**i** Solution

No. Extensive variance does not determine connectivity of the density matrix's support under local charge transfers. Sparse fixed-sector mixtures can reproduce the variance while lacking the relevant Rényi-1 order. The experiment needs a nonlinear diagnostic or additional verified assumptions linking its family to one. It should also distinguish coherent variance from classical mixing if that distinction is part of the claim.

### 18.10.3 3. Two ways to "undo" erasure

Why does preparing a fresh parity ensemble not contradict its local recovery obstruction?

**i** Solution

A global preparation can reset every site and build a new constraint. A local recovery must leave a distant region untouched while restoring its correlations with the damaged region. The former task can discard the old information; the latter must preserve it. A reverse channel connecting two selected states need not reverse each local part of the original deformation. That is why local reversibility is a stronger criterion

than two-way channel reachability.

# 19 A conceptual source atlas

SW-SSB emerged where several research programs met: symmetry sectors of dissipative generators, doubled descriptions of thermalization, decoherence and decoding thresholds, average-symmetry topology, and information-theoretic recovery. Their convergence explains both the framework's reach and the disagreements about its boundaries. This atlas organizes the sources by physical questions and links them to the textbook's derivations.

The cutoff is **22 October 2026**, rather than the unelapsed remainder of that month. Research began with Chong Wang's review, first posted on 1 June and revised on 8 June 2026; its primary references and subsequent submissions were then checked independently [8]. At the cutoff it is a review submitted to *Annual Review of Condensed Matter Physics*.

Dates below identify a verified publication year, first preprint submission, or explicitly named revision. **Derivation** means that the book reconstructs a representative result and its mechanism. **Framework** means that the definition, assumptions, and consequences are explained with examples. **Frontier** identifies a specialized extension whose evidential status and connection to the main theory are explained; it does not imply that an unresolved classification has become a theorem. These labels describe the textbook's treatment, not scientific importance.

## 19.1 From open-system symmetry to spontaneous order

Strong and weak symmetry first organized dissipative generators and invariant subspaces. The later state-based question is different: can a density matrix have an exact finite-size charge while becoming insensitive to distant charge transfers? Thermalization and decoherence supplied routes to that question before the fidelity phase theory was formulated.

| Physical question and primary source | Verified record | Role in this book |
|---|---|---|
| Completely positive continuous-time dynamics: Gorini–Kossakowski–Sudarshan and Lindblad [2,3] | Generator theorems, 1976 | Master equation and assumptions in Chapter 3 |
| Sector preservation versus covariance: Buča–Prosen; Albert–Jiang [5,6] | New Journal of Physics 2012; PRA 2014 | Algebraic derivation in Chapter 5 |
| Many-body delocalisation as symmetry breaking, Garratt–Chalker [9] | First 21 December 2020; PRL 2021 | Historical precursor: doubled-history/transfer-matrix order in Floquet dynamics |
| Quantum criticality under decoherence or weak measurement, Lee–Jian–Xu [12] | January 2023 preprint; PRX Quantum 2023 | Nonlinear decoherence transition and solvable model in Chapter 9 |
| Symmetry-enforced many-body separability transitions, Chen–Grover [15] | 2023 preprint; PRX Quantum 2024 | Ensemble structure and separability in Chapter 7 |
| Purification perspective, Sala–Gopalakrishnan–Oshikawa–You [38] | PRB 2024 | System order and a specified environment, Chapter 7 and Chapter 11 |
| SW-SSB in mixed quantum states, Lessa–Ma–Zhang–Bi–Cheng–Wang [7] | 2024 preprint; PRX Quantum 2025 | Fidelity definition, stability, CMI and exact noise models in Chapter 6 through Chapter 9 |

Garratt–Chalker's transfer-matrix symmetry is a precursor, not a reason to identify every localization transition with the later density-matrix definition. Similarly, an algebraic symmetry of a Lindbladian does not prove that every steady state exhibits SW-SSB. The textbook separates generator algebra, positive-matrix support, and thermodynamic order before relating them.

The main fidelity paper brings together mixed-state symmetry, topology, and quantum information. Its ideas enter the textbook where needed: the support theorem in Chapter 5, distinguishability in Chapter 6, and finite-depth locality in Chapter 8. The reader can follow that conceptual chain without first learning the authors' research histories.

## 19.2 Which diagnostic and which phase relation?

Canonical purification uses $\sqrt{\rho}$ and has marginal $\rho$. Normalized Choi vectorization uses $\rho$ and has marginal $\rho^2/\operatorname{Tr}\rho^2$. Their order parameters can become singular at different noise strengths. A second fork concerns whether one compares complete finite-system families, intrinsic local thermodynamic states, or states connected by particular channel classes.

| Primary route | Record and status | Coverage |
|---|---|---|
| Rényi-1/canonical purification, Weinstein [10] | 2410.23512; PRL 2025 | Affinity/fidelity bounds and conditional preparation resource |
| Wightman/thermofield diagnostic, Liu–Chen–Zhang–Zhou–Zhang [11] | 2410.09327; Communications Physics 2025 | Equivalent stable order diagnostic and thermal examples |
| Susceptibility/information geometry, Zhang–Xu–Xu [14] | 2409.18944, 27 September 2024 | Nonanalytic response metrics and correlator bounds; response convention must be specified |
| Approximate recovery, Fawzi–Renner [4] | 2014 preprint; mathematical theorem, 2015 | Fidelity/CMI relation in Chapter 4 |
| Fast dissipative reachability, Coser–Pérez-García [18] | 1810.05092; Quantum 3, 174 (2019) | Equivalence and partial order with stated polylogarithmic time/locality resources |
| Correlation-preserving RG and decoding, Sang–Zou–Hsieh [19] | 2310.08639; PRX 2024; v2 26 February 2026 | Constructive two-way channels; thermal versus dephased toric-code families |
| Dynamical uniformity, Rakovszky–Gopalakrishnan–von Keyserlingk [20] | PRX 14, 041031 (2024) | Stability in a neighborhood of generators; generic gap/relaxation links remain conjectural |
| Finite Markov length, Sang–Hsieh [17] | PRL 2025 | Recovery along dissipative paths in Chapter 8 |
| Local reversibility, Sang–Lessa–Mong–Grover–Wang–Hsieh [16] | 2507.02292, v2 25 July 2025 | Framework preserving locally recoverable quantum and classical information |
| Local fidelity, Divi–Lessa–Wang [21] | 2605.28967, 27 May 2026 | Local/global comparison and accessible region-size scaling |
| Marginal fidelity and dynamics, Carolyn Zhang [22] | 2605.29113, 27 May 2026 | Conditional local/global bounds and absorbing-state examples |
| Infinite-volume strong symmetry, Liu–Yi–Else [23] | 2605.28925v3, 12 August 2026 revision | Operator-algebraic framework and competing thermodynamic interpretation |
| CMI constraints on RG, Chen–Grover [78] | 2602.04961v2, 2 August 2026 revision | Frontier inequalities assuming UV-finiteness and scaling hypotheses |

These formulations retain different information:

| Criterion | Operational question |
|---|---|
| Global finite-family fidelity | How does the complete state respond to distant charged insertions, including its global sector constraint? |
| Intrinsic local thermodynamic state | Which properties survive in all fixed-region expectation values and consistent extensions? |
| Two-way shallow channel connectivity | Can each complete state be produced from the other? |
| Locally reversible channel circuits | Can restricted local deformations also be reversed nearby? |
| Stochastic local-channel entanglement frameworks | Which quantum long-range structures survive after specified classical correlations are discounted? |
| Choi/Rényi-2 construction | What order exists in the purity-reweighted doubled state? |

For example, fixed-parity and unconstrained maximally mixed families have identical proper local marginals, while complete-system fidelity diagnostics distinguish them. An intrinsic local limit cannot distinguish a bit that disappears from every fixed region. This is a substantive difference in the object and access being classified. Chapter 8 presents both perspectives rather than asserting their universal equivalence. Positive finite-region fidelity also bounds a limiting local order parameter from above; it does not certify a nonzero limit without a convergence argument.

## 19.3 Noise, inference, and quantum memory

The statistical-mechanical bridge asks which global error class occurred after a syndrome is observed. Nishimori matching ties the disorder distribution to the actual error probabilities. Gauging turns charge transfer into a logical/topological process; keeping or forgetting measurement records changes the inference task.

| Source and connection | Verified record | Coverage and qualification |
|---|---|---|
| Nishimori: matched disordered Ising statistics [24] | 1980; exact statistical identities | Matching derived in Chapter 9 |
| Dennis–Kitaev–Landahl–Preskill: topological quantum memory [25] | 2001 preprint; J. Math. Phys. 2002 | Error chains, homology and posterior sums in Chapter 10 |
| Wang–Harrington–Preskill: faulty-syndrome mapping [27] | 2002 preprint; Ann. Phys. 2003 | Distinguishes one-shot errors from spacetime decoding |
| Lee–Ji–Bi–Fisher: measurement-prepared phases [30] | v2 6 September 2022 | Records, decoding and Ising/gauge connections |
| Fan–Bao–Altman–Vishwanath: mixed topological diagnostics [28] | 2023 preprint; PRX Quantum 2024 | Coherent information, relative entropy and negativity for specified Pauli noise |
| Chen–Grover: topological separability transitions [52] | 2023 preprint; PRL 132, 170602 (2024) | Convex mixtures of short-range-entangled states, distinct from full on-site separability |
| Bao–Fan–Vishwanath–Altman: errorfield double [29] | v2 1 May 2026; PRL 2026 | Doubled topological sectors and temporal decoherence defects |
| Kim–Altman–Lee: SYK code threshold [77] | First 31 October 2024; PRB 2026 | Approximate-code/large-$N$ saddle mechanism, Chapter 15 |
| Temkin–Weinstein–Fan–Podolsky–Altman: charge-informed correction [73] | 26 December 2025 preprint | Integer loops, optimal BKT decoding theory and loop-glass numerical evidence |
| Vijay–Lee: information-critical phases [72] | First 26 December 2025; v2 10 August 2026 | Clock posterior/coherent-information derivation and partial-memory mechanism |

An optimal threshold belongs to a code/noise/information task; a practical decoder's threshold is algorithm dependent. Maximum likelihood sums histories in a class, whereas minimum weight selects one representative. Topological negativity can persist after useful quantum memory is lost. Nonmaximal coherent information does not mean that a particular integer number of logical qudits is perfectly protected in one finite block. These distinctions connect the stabilizer calculations of Chapter 10 with the critical-angle physics of Chapter 15.

## 19.4 Average symmetry and protected mixed-state structure

SPT order supplies symmetry actions on defects and boundaries without intrinsic bulk topological degeneracy. Forgetting a disorder or measurement record is a physical operation, so an average-symmetry ensemble and its density matrix must be compared carefully.

| Primary sources | Verified record | Coverage and limitation |
|---|---|---|
| Ma–Wang, average SPT [31] | PRX 2023 | Decorated disorder defects and average anomalies |
| Ma–Zhang–Bi–Cheng–Wang, decohered, disordered and intrinsic [32] | PRX 2025 | Different admissibility conditions for disorder and decoherence |
| Lee–You–Xu, SPT under decoherence [36] | Quantum 2025 | Type-I/type-II strange diagnostics and doubled response |
| Ma–Turzillo; Xue–Lee–Bao [33,34] | 2403.13280, PRX Quantum 2025; 2403.17069, 2024 preprint | Doubled/tensor classifications with positivity, Hermiticity and injectivity/locality hypotheses |
| Guo–Zhang–Zhang–Yang–Bi [35] | 2403.16978, v3 24 April 2025; PRX 2025 | Locally purified density operators and the role of a specified local purification |
| Zhang–Qi–Bi, fidelity strange correlators [37] | 2022 preprint; Science Bulletin 2026 | Fidelity-based ASPT diagnostic; publication year is not submission year |
| Guo–Yang; Shah et al. [40,44] | 2410.13734, PRB 2025; 2410.12900, Quantum 2025 | Double-ASPT mechanism versus instability of particular parent-Lindbladian steady states |
| Lu–Silveira–You, topological response [41] | 26 February 2026 preprint | Defect-decoration/dephasing mechanism and extensions |
| Kuno–Orito [42,43] | 2606.22507, 21 June; 2609.27530, 23 September 2026 | Rényi-2 subsystem hierarchy and filtering/postselection; retain those qualifiers |

Chapter 11 reconstructs cluster stabilizers, edge algebra, string attenuation, and strong/weak protection. Specialized classifications extend those mechanisms. A Choi-state cohomology result is not presented as a completed classification under every channel equivalence, and state-phase stability is distinguished from robustness as the stationary state of a perturbed generator.

## 19.5 Generalized symmetry and non-Abelian information

Higher-form symmetry moves the charged object from a point to a loop or surface. Non-Abelian charge adds fusion multiplicities. Both reveal information absent from a list of local densities.

| Primary route | Record and status | Coverage |
|---|---|---|
| Wang–Wu–Wang; Sohal–Prem; Ellison–Cheng [49–51] | 2307.13758, 2403.13879, 2405.02390; PRX Quantum 2025 | Intrinsically mixed orders, anyon proliferation, partial/proposed classifications |

| Primary route | Record and status | Coverage |
|---|---|---|
| Zhang–Xu–Zhang–Xu–Bi–Luo [48] | 2409.17530; PRB 2025; v3 19 February 2026 | One-form SW breaking and revised Rényi-2 Markov-length assumptions |
| Lessa–Cheng–Wang; Lessa et al. [39,53] | PRX 2025; 2503.12792 | Anomaly/entanglement obstructions with theorem-specific dimensions |
| Hsin–Kobayashi–Prem [54] | 2504.10569, v2 5 November 2025 | Higher-form obstruction and mixed order without complete remote detectability |
| Yang–Shi–Lee [56] | 2506.04221, v3 5 October 2026; PRX 2026 | Axiomatic mixed topological phases under recoverability/uniformity premises |
| Song–Zhang [55] | 2509.24179, v3 20 September 2026 | Noninvertible higher-form symmetry in noisy quantum doubles |
| Schafer-Nameki et al.; Qi et al. [45,46] | 2507.05350, 2507.05335 | SymTFT/folded-bilayer frameworks with stated short-range Choi premises |
| Moharramipour–Lessa–Wang–Hsieh–Sahu; Li–Pollmann–Read–Sala [57,58] | 2406.08542, PRX Quantum 2024; PRX 2025 | Symmetry-enforced entanglement in singlet-sector maximal mixtures and stationary states |
| Sala–Temkin–Xu–Podolsky–Altman [59] | 2610.06692, 5 October 2026 | Center ordering and nested fusion-decoder hierarchy |

The classical loop ensemble, crossing-string algebra, and four-spin singlet example in Chapter 12 make the distinction between classical constraints and quantum entanglement explicit. Non-Abelian singlet ensembles add entanglement even at maximal mixing within a symmetry sector. The center-order work is a major post-review development; its finite-hierarchy statements are not a solution of unrestricted optimal non-Abelian decoding. Pre-modular and categorical proposals remain proposals where no completeness theorem has been established.

Another bridge uses channel-state duality to turn repeated $d$-dimensional channels into boundary states of a $(d+1)$-dimensional sequential circuit. In the construction of Lu–Liu–Gopalakrishnan–You, boundary SW-SSB and its CMI inherit bulk topological structure [83] (2511.19597v2, 1 September 2026 revision). This is a concrete tensor-network duality, rather than an assertion that all mixed states have one universal holographic dual. Vijay–Lee similarly relates symmetric circuits to a dynamically generated noisy surface code, connecting charge sharpening with bulk logical information and a confinement description [84] (2511.21685v2, 8 May 2026 revision). Its $N > 4$ intermediate regime ties the circuit's inference times to the critical-memory mechanisms of Chapter 15.

## 19.6 Hydrodynamics, charge scrambling, and learning

The shared theme is a globally fixed charge that is locally indefinite. Whether that indefiniteness implies a slow mode, nonlinear order, or failed inference depends on additional conditions.

| Primary sources | Verified record | Coverage |
|---|---|---|
| Ogunnaike–Feldmeier–Lee; Moudgalya–Motrunich [60,61] | 2304.13028, PRL 2023; 2309.15167, PRX Quantum 2024 | Microscopic superoperator/single-mode hydrodynamic constructions |
| Huang–Qi–Zhang–Lucas [63] | 2407.08760, v2 20 March 2025; PRB 2025 | Response-field diffusion EFT in Chapter 13 |
| Gu–Wang–Wang; Watanabe–Katsura–Lee [62,64] | PRB 2025; PRL 2024 | Liouvillian models and frustration-free low-dimensional qualifications |

| Primary sources | Verified record | Coverage |
|---|---|---|
| Ziereis–Moudgalya–Knap [76] | 2509.09669, 11 September 2025 | Steady-state SW order and weak-postselection transitions; distinguish conditioned operations from CPTP channels |
| Hauser–Su–Ha–Lloyd–Kiely–Vasseur–Gopalakrishnan–Xu–Fisher [67] | 2602.16045, 17 February 2026 | Discrete/continuum distinction, ballistic information length, 2D BKT-like evidence |
| Lee [66] | 2605.05288, 6 May 2026 | Fluctuation bound, skew information and both directions of counterexample |
| Sarma [74] | 2609.37965, 29 September 2026 | Exact all-index infinite-dephasing XY/FCS mapping; finite-dephasing even-index analysis |
| Agrawal et al.; Singh et al. [68,69] | PRX 2022; 2503.10308, PRB 2026 | Charge sharpening, entanglement and noisy mixed-state inference |
| Zerba–Gopalakrishnan–Knap; Kumar et al. [70,71] | 2512.14830, 16 December 2025; 2602.11262v2, 18 March 2026 revision | Dipole-learning hierarchy and restricted-query hardness |

The exclusion-process and Gaussian calculations in Chapter 13 derive the transport/information separation before using a Keldysh field theory. Chapter 14 shows why a summable nonlinear tail can force extensive variance, and why neither direction of an unconditional variance/order equivalence holds. Chapter 15 derives the counting-statistics identity and its Rényi-index dependence. These bridges connect Lucas's effective theory, Lee's static qualifications, Fisher and Gopalakrishnan's microscopic/inference perspective, and the experimental estimators.

The relevant time limits must be stated. One-dimensional information lengths can grow without finite-time long-range order; an infinite-time fixed-sector stationary ensemble can nevertheless order. Two-dimensional algebraic order is different from a nonzero long-distance constant. A conserved density and SW-SSB do not alone determine a diffusion coefficient, and an equal-time structure factor is not automatically a response susceptibility outside equilibrium.

## 19.7 Detection and the experimental timeline

| Access or observation | Verified date/status | Interpretation |
|---|---|---|
| Canonical-purification route; randomized measurements [10,79] | PRL 2025 | Conditional resources and an implementable Rényi-2 protocol |
| Feng–Cheng–Ippoliti hardness [13] | First 16 April 2025; PRL 2025 | State-agnostic obstruction, consistent with structured protocols |
| Liu–Zou certification [80] | 19 January 2026 preprint | MPDO representation and variational fidelity certificates |
| Wang et al., dephased Fermi gas [81] | 2604.16137, 17 April 2026 | Snapshots plus learned Gaussian reference; finite-temperature/subsystem qualifications |
| Cao–Bi–Murch, superconducting array [47] | 2608.24993, 25 August 2026 | Five-qubit cluster, discarded records, finite-size fidelity signatures |
| Fechisin et al., long-range-channel theory [75] | 2610.00480, first 30 September 2026 | Rényi-2 MPS phase diagram and high-purity route to hardware |
| Wei et al., quantum processor [82] | 2610.26229, first **19 October 2026** | Known-channel reconstruction, finite-size crossings and collapse |

The processor paper is included because its primary submission history explicitly precedes the cutoff, despite the high sequence number in its identifier. Chapter 16 derives its characteristic-kernel/Gram estimator and the cold-gas probability-ratio estimator. Measurement settings, classical processing, and shot complexity are different resources. Few settings alone do not establish efficient precision for arbitrary highly mixed states. Finite-size evidence is described as such rather than represented as a literal measurement of an infinite-volume singularity.

## 19.8 Established core and open boundaries

The established core developed here includes finite-system support definitions, fidelity/Rényi-1 bounds under stated operator assumptions, finite-depth stability, solvable examples, and recoverability consequences with specified limits. Numerical thresholds, BKT evidence, large-$N$ saddle transitions, and decoder comparisons remain results for their model families. Universal mixed-state classifications, unrestricted non-Abelian optimal decoding, and a general equivalence of SW-SSB and hydrodynamics remain open.

Review references described only as "to appear" were treated as research leads rather than turned into invented papers. Revisions are included only when available by the cutoff. The explicit source maps make the scope inspectable and extendable; essential arguments along the book's pedagogical path have been reconstructed in the preceding chapters. Specialized extensions are connected to those arguments with their assumptions and unresolved questions retained.

## 19.9 Exercises in interpreting a frontier claim

### 19.9.1 A correlator, a threshold, and a slow mode

An abstract reports nonlinear order, an error threshold, and diffusion. What must be checked before calling these one transition?

> **i** Solution
>
> Specify the correlator and normalization; finite-system family versus local limit; code, noise and decoder or optimal-recovery task; large-size and time limits; and the generator, structure factor and constitutive assumptions behind diffusion. Then verify a theorem or mapping connecting those particular objects. Shared terminology alone supplies none of those links.

### 19.9.2 A revised phase criterion

A later revision adds a finite Rényi-2 Markov-length assumption to a phase statement. How should the result be quoted?

> **i** Solution
>
> Name the version and the operational equivalence accompanying the claim. Explain the changed assumption when it affects the interpretation. Combining conclusions from different versions into a stronger theorem would misrepresent both. This is especially consequential when comparing Choi-space order with locally reversible fidelity-based phases.

# A Mathematical toolkit

## A.1 Spectra, singular values, and matrix functions

A Hermitian matrix has an orthonormal eigenbasis, $H = V\,\mathrm{diag}(h_a)V^\dagger$. A positive matrix has $h_a \geq 0$. For any real function defined on its spectrum,

$$f(H) = V\,\mathrm{diag}(f(h_a))V^\dagger.$$

The square root in fidelity is the positive square root. The logarithm in entropy is applied only to positive eigenvalues, with zero contributions understood by continuity. An inverse such as $\rho^{-1/2}$ in a recovery map is restricted to the support unless a full-rank assumption is made.

For an arbitrary matrix $M$, its singular-value decomposition is $M = V\Sigma W^\dagger$, with nonnegative diagonal entries $s_a$ in $\Sigma$. Its singular values are the square roots of the eigenvalues of $M^\dagger M$. The polar decomposition is $M = U|M|$, with $|M| = \sqrt{M^\dagger M}$ and a partial isometry $U$ that can be extended to a unitary when the spaces permit.

For numerical work, do not diagonalize a non-Hermitian matrix and assume its eigenvalues are singular values. A trace norm is a sum of singular values, not the sum of absolute eigenvalues except in appropriate normal cases.

## A.2 Three norms and their roles

The operator norm is $\|M\|_\infty = \max_a s_a$. It bounds the largest amplification of a vector. The Hilbert-Schmidt norm is $\|M\|_2 = (\mathrm{Tr}\, M^\dagger M)^{1/2}$. The trace norm is $\|M\|_1 = \sum_a s_a$.

If the rank is $r$,

$$\|M\|_\infty \leq \|M\|_2 \leq \|M\|_1 \leq \sqrt{r}\,\|M\|_2.$$

The last factor can grow exponentially with volume. A finite-dimensional norm equivalence therefore need not establish a thermodynamically uniform relation between order parameters.

The trace pairing obeys Hölder's inequality,

$$|\,\mathrm{Tr}(M^\dagger N)| \leq \|M\|_1 \|N\|_\infty,$$

and Hilbert-Schmidt Cauchy-Schwarz,

$$|\,\mathrm{Tr}(M^\dagger N)|^2 \leq \mathrm{Tr}(M^\dagger M)\,\mathrm{Tr}(N^\dagger N).$$

The latter follows from positivity of $\|M - zN\|_2^2$ and minimization over complex $z$. It is the same geometry as the vector Cauchy-Schwarz inequality, applied to matrices as a vector space.

The variational formula

$$\|M\|_1 = \max_{V\ munitary} |\,\mathrm{Tr}(VM)|$$

follows immediately from polar decomposition. For Hermitian $M$, another form is $\|M\|_1 = \max_{\|O\|_\infty \leq 1} \mathrm{Tr}(OM)$. These identities turn a norm into an optimized overlap or measurement, and are repeatedly used in stability and irrecoverability proofs.

## A.3 Vectorization without ambiguity

Fix a basis and define

$$|M\rangle\!\rangle = \sum_{ij} M_{ij}|i\rangle_L|j\rangle_R.$$

Then

$$\langle\!\langle M|N\rangle\!\rangle = \mathrm{Tr}(M^\dagger N), \qquad (A\otimes B^T)|M\rangle\!\rangle = |AMB\rangle\!\rangle. \tag{A.1}$$

The transpose appears because of index placement. In particular $(U\otimes U^*)|M\rangle\!\rangle = |UMU^\dagger\rangle\!\rangle$. Thus weak symmetry becomes a diagonal doubled action.

The canonical purification $|\sqrt{\rho}\rangle\!\rangle$ has norm one, and its left reduced state is $\rho$. The raw vector $|\rho\rangle\!\rangle$ has norm squared $\mathrm{Tr}\,\rho^2$. Once normalized, its left reduced state is $\rho^2/\mathrm{Tr}\,\rho^2$. The two constructions have different eigenvalue weights and can have different phases in their doubled descriptions.

For any matrices $A, B$,

$$\langle\!\langle M|A\otimes B^T|M\rangle\!\rangle = \mathrm{Tr}(M^\dagger AMB).$$

Setting $M=\sqrt{\rho}$ and $B=A^\dagger$ gives the canonical overlap correlator. Setting $M=\rho$ and dividing by purity gives its Rényi-2 counterpart. This single index identity is safer than memorizing several copy-space formulas with inconsistent conjugations.

## A.4 Cumulants and small twists

For a Hermitian charge $Q_A$, let $Z_A(\theta)=\langle e^{i\theta Q_A}\rangle$. Near zero, provided the required moments exist,

$$\ln Z_A(\theta) = \sum_{n\geq 1}\frac{(i\theta)^n}{n!}\kappa_n,$$

where $\kappa_1=\langle Q_A\rangle$ and $\kappa_2=\mathrm{Var}(Q_A)$. Hence $\ln|Z_A(\theta)| = -\theta^2\,\mathrm{Var}(Q_A)/2 + O(\theta^4)$. Higher cumulants matter if the small-angle expansion is used outside its uniform range as the region grows. A Gaussian central-limit approximation does not follow from the variance alone.

For a classical mixture labeled by $a$, the law of total variance is

$$\mathrm{Var}(Q) = \sum_a p_a\,\mathrm{Var}(Q|a) + \sum_a p_a\big(\langle Q\rangle_a - \langle Q\rangle\big)^2.$$

It follows by adding and subtracting $\sum_a p_a\langle Q\rangle_a^2$. It is useful when passing from a symmetry-selected ensemble to a symmetric mixture. For noncommuting observables, the expression still holds for the ordinary variance of one Hermitian operator in a convex mixture; it should not be interpreted as a joint classical distribution for incompatible observables.

## A.5 Fourier transforms and zero modes

On a periodic chain of $N$ sites with unit spacing,

$$n_k = \frac{1}{\sqrt{N}} \sum_x e^{-ikx} n_x, \qquad n_x = \frac{1}{\sqrt{N}} \sum_k e^{ikx} n_k, \qquad k = \frac{2\pi m}{N}.$$

Orthogonality is $\sum_x e^{i(k-k')x} = N\delta_{kk'}$. The discrete Laplacian has eigenvalue $e^{ik} + e^{-ik} - 2 = -4\sin^2(k/2)$, approaching $-k^2$ at small $k$. In a fixed-charge sector the fluctuating zero mode is absent: $\delta n_{k=0} = 0$ exactly.

For translation-invariant covariance $C(x-y) = \langle \delta n_x \delta n_y \rangle$, define $S(k) = \langle \delta n_k \delta n_{-k} \rangle$. A block sum has

$$\mathrm{Var}(Q_A) = \frac{1}{N} \sum_k S(k) \left| \sum_{x \in A} e^{ikx} \right|^2 .$$

This formula makes the order of $k \to 0$ and $N \to \infty$ explicit. A function can satisfy $S(0) = 0$ at every finite $N$ yet have positive $S(k \to 0^+)$ after the thermodynamic limit. There is no inconsistency with a fixed total charge.

## A.6 Errors, loops, and boundaries over binary arithmetic

For $\mathbb{Z}_2$ noise on a graph, an edge configuration is a list $e_\ell \in \{0, 1\}$. Its boundary at vertex $v$ is the parity of incident occupied edges:

$$(\partial e)_v = \bigoplus_{\ell \ni v} e_\ell .$$

Two edge configurations give the same syndrome precisely when their difference has no boundary: $\partial(e \oplus e') = 0$. Such a difference is a cycle. On a simply connected region, contractible cycles can be built from plaquette boundaries. On a torus, noncontractible cycles carry homology labels and can act as logical operators.

For general $\mathbb{Z}_N$, replace exclusive-or by addition modulo $N$ and assign orientations. Charge conservation is the statement that the boundary of a boundary vanishes. This elementary algebra explains why decoding naturally sums over loop configurations and why different topological sectors can share the same local syndrome.

## A.7 Turning a posterior into a Boltzmann weight

For independent binary errors,

$$P(e) = \prod_\ell p^{e_\ell} (1-p)^{1-e_\ell}.$$

Set $\tau_\ell = (-1)^{e_\ell}$ and $K = \frac{1}{2} \ln[(1-p)/p]$. Then

$$P(e) = [p(1-p)]^{|E|/2} \exp\left( K \sum_\ell \tau_\ell \right).$$

Conditioning on a syndrome restricts the sum to compatible errors. Parameterizing compatible differences by loops or dual spins introduces a classical partition function. The equality $e^{-2K} = p/(1-p)$ is the Nishimori relation for this noise model. A different prior, correlated noise, measurement error, or replica weight gives a different effective statistical model. A clean Ising critical point cannot be inserted into a disordered inference problem merely because both contain spins.

## A.8 Scaling statements and their quantifiers

An exact statement holds at every specified finite size. A thermodynamic statement concerns a sequence as $N \to \infty$. A long-distance statement requires separation tending to infinity while staying within a controlled geometry. A hydrodynamic limit also scales time and wavelength. "Finite depth" normally means independent of $N$; "polylogarithmic depth" does not.

A numerical critical exponent requires a scaling window and corrections to scaling. A finite plateau can be preasymptotic. A proof using an inequality with a prefactor growing as $N$ must keep that prefactor when limits are taken. If the last step of an argument discards the only factor carrying system-size dependence, it may have discarded the physical question.

## A.9 Exercises and worked solutions

### A.9.1 1. Rank-dependent norm bounds

Take $M = I_d/d$. Compute its three norms and identify the size-dependent relation.

**Solution**

All $d$ singular values equal $1/d$. Thus $\|M\|_\infty = 1/d$, $\|M\|_2 = 1/\sqrt{d}$, and $\|M\|_1 = 1$. The bound $\|M\|_1 \leq \sqrt{d}\|M\|_2$ is saturated. A Hilbert-Schmidt quantity vanishing exponentially in volume can coexist with a trace norm of order one.

### A.9.2 2. Vectorizing a commutator

Write $|[H, \rho]\rangle\!\rangle$ as an operator acting on $|\rho\rangle\!\rangle$.

**Solution**

The first term is $(H \otimes I)|\rho\rangle\!\rangle$ and the second is $(I \otimes H^T)|\rho\rangle\!\rangle$. Therefore the commutator is generated by $H_L - H_R^T$. A missing transpose changes the formula for Hamiltonians with complex matrix elements.

### A.9.3 3. Why a loop changes no syndrome

Toggle every edge around one square plaquette. What happens to its four vertex boundary parities?

**Solution**

Each vertex touches exactly two toggled edges. Adding two modulo two gives zero, so every vertex syndrome is unchanged. On a torus a noncontractible closed path also has no endpoints but may change a logical winding label. Local syndrome agreement therefore does not imply logical equivalence.

# B Glossary and notation

## B.1 States and limits

| Term or symbol | Meaning in this book |
|---|---|
| $\mathcal{H}$, $\rho$ | Hilbert space and positive trace-one density matrix. |
| $P_\rho$ | Projector onto the support of $\rho$, the span of positive-eigenvalue eigenvectors. |
| Pure state | Rank-one density matrix; $\operatorname{Tr}\rho^2 = 1$. |
| Convex decomposition | A representation $\rho = \sum_a p_a \lvert\psi_a\rangle\langle\psi_a\rvert$; generally not unique. |
| Trajectory | A state conditioned on a particular measurement or environment record. |
| Separability | Expressibility as a convex mixture of product states for a specified partition. |
| Purification | A pure enlarged-system state reducing to $\rho$ after an auxiliary system is discarded. |
| Canonical purification | $\lvert\sqrt{\rho}\rangle\rangle$ in a fixed vectorization convention. |
| Choi double of a state | In this context, the normalized vectorization $\lvert\rho\rangle\rangle/\sqrt{\operatorname{Tr}\rho^2}$; distinct from a channel's Choi matrix. |
| Thermodynamic family | A specified sequence of finite systems, including boundary conditions and charge constraints. |
| Local infinite-volume state | Compatible expectation values or density matrices on every finite region; may forget global finite-size constraints. |
| Local operator | Supported in a bounded region as system size and separation grow. |
| Long-range order | A specified correlator has a nonzero asymptotic limit. Algebraic quasi-order is stated separately. |

## B.2 Symmetry and operations

| Term or symbol | Meaning in this book |
|---|---|
| $U_g$ | Unitary representation of a symmetry element. |
| Strong state symmetry | Finite-volume condition $U_g\rho = \chi(g)\rho$: the support has one common charge character. |
| Weak state symmetry | $U_g\rho U_g^\dagger = \rho$: the ensemble is invariant under conjugation. |
| Charged operator | $U_g O U_g^\dagger = \chi_O(g)O$ for a nontrivial character, or an explicitly stated higher-dimensional multiplet. |
| Strongly symmetric channel | A channel with charge-preserving Kraus operators, or an equivalent charge-preserving dilation in the convention used here. |
| Covariant channel | $\mathcal{E}(U_g\rho U_g^\dagger) = U_g\mathcal{E}(\rho)U_g^\dagger$; can exchange charge sectors. |
| CPTP | Completely positive and trace preserving. |
| Kraus operators | Matrices $K_a$ representing a channel through $\sum_a K_a\rho K_a^\dagger$, with $\sum_a K_a^\dagger K_a = I$. |

| Term or symbol | Meaning in this book |
|---|---|
| Stinespring dilation | A unitary or isometric enlarged-system realization of a channel. |
| Instrument | Outcome-resolved completely positive maps whose sum is a channel. |
| Lindbladian $\mathcal{L}$ | Generator of a Markovian quantum dynamical semigroup. |
| Jump operator $L_a$ | A dissipative transition in a Lindblad representation. |
| Unital channel | $\mathcal{E}(I) = I$; different from trace preservation. |
| Finite-depth circuit | A bounded number of bounded-range local gate layers independent of system size. |
| Two-way connectivity | Existence of shallow channels between two selected states in both directions. |
| Local reversibility | Recovery of local pieces of a deformation in slightly enlarged regions, with controlled error; stronger than two-way connectivity. |

## B.3 Diagnostics and information

| Term or symbol | Meaning in this book |
|---|---|
| $C_O(x, y)$ | Ordinary charged-pair correlator $\mathrm{Tr}(\rho O_x O_y^\dagger)$. |
| $F(\rho, \sigma)$ | Root fidelity $\mathrm{Tr}\,\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$. |
| $F_O(x, y)$ | Fidelity with the state altered by a neutral charged pair. |
| $R_1(A)$ | Canonical affinity $\mathrm{Tr}(\sqrt{\rho}A\sqrt{\rho}A^\dagger)$; unitary convention unless stated otherwise. |
| $R_2(A)$ | Purity-normalized overlap $\mathrm{Tr}(\rho A\rho A^\dagger)/\,\mathrm{Tr}\,\rho^2$. |
| Global SW-SSB | Strongly symmetric finite-volume family with nonlinear fidelity long-range order and no ordinary charged long-range order. |
| Local SW diagnostic | Limiting fidelity of growing reduced states with a local charged insertion; broader than global two-point SW order. |
| $S(\rho)$ | Von Neumann entropy, using natural logarithms. |
| $D(\rho\|\sigma)$ | Quantum relative entropy; unlike trace distance, not a metric. |
| $T(\rho, \sigma)$ | Trace distance $\frac{1}{2}\|\rho - \sigma\|_1$, distinguished from relative entropy $D(\rho\|\sigma)$. |
| MI | Mutual information $S_A + S_C - S_{AC}$. |
| CMI | Conditional mutual information $S_{AB} + S_{BC} - S_B - S_{ABC}$ in a stated geometry and limit. |
| Markov length $\xi_M$ | A decay scale of buffered CMI, when such exponential decay holds uniformly. |
| Petz recovery | A state-dependent recovery construction using reference density matrices and inverse square roots on their supports. |
| Skew information | $I_{\mathrm{WY}}(\rho, Q) = -\frac{1}{2}\,\mathrm{Tr}[\sqrt{\rho}, Q]^2$, a measure of charge coherence. |
| Edwards-Anderson order | Order within a specified ensemble of samples or trajectories; ensemble dependence must be distinguished from intrinsic mixed-state quantities. |

## B.4 Topology, dynamics, and evidence

| Term | Meaning in this book |
|---|---|
| Stabilizer | A commuting operator fixing a code or state; generators define a common eigenspace. |
| Syndrome | The measured local constraint violations produced by an error. |
| Logical operator | An operator acting on encoded information while preserving the syndrome. |
| Decoder | A rule inferring an error class or logical correction from available records. |
| Nishimori relation | The equality between a noise prior and a disordered statistical model's coupling, such as $e^{-2K} = p/(1-p)$. |
| Coherent information | $S(B) - S(RB)$ for a channel output $B$ and reference $R$ in the specified encoded-state experiment. |
| SPT phase | Symmetry-protected topological phase; trivialization is obstructed within a specified symmetry-preserving class of operations. |
| Average SPT | Mixed-state SPT structure involving a weak ensemble symmetry together with strong protection, with the precise framework stated. |
| Higher-form symmetry | Symmetry acting on extended charged objects and represented by operators on corresponding closed supports. |
| Anomaly | An obstruction to realizing a symmetry structure in a specified strictly local setting; not simply any unusual correlation. |
| Fusion channel | An allowed way non-Abelian charges combine; may contain information beyond a total irrep or center charge. |
| Diffusion | Relaxation with long-wavelength rate $Dk^2$ under the required constitutive assumptions. |
| Charge scrambling | Delocalization of charge information; a nonlinear static diagnostic, a fluctuation footprint, and a dynamical spreading process must be distinguished. |
| Information-critical phase | A model-defined phase with critical information or decoding structure, potentially retaining fractional logical information. |
| Established result | An exact derivation or theorem within stated assumptions. |
| Numerical evidence | A finite-size or controlled numerical finding with a stated extrapolation and model. |
| Conjecture | A proposed general statement whose proof or sufficient evidence is not supplied by the cited work. |

# References

Primary sources are cited at the point where their results enter the discussion. The literature map groups them by concept and distinguishes original arguments, numerical evidence, experimental reports, and unsettled extensions. Preprints are identified as such when a journal publication was not verified within the cutoff.


[1] N. D. Mermin and H. Wagner, Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic heisenberg models, Physical Review Letters **17**, 1133 (1966).

[2] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, Completely positive dynamical semigroups of n-level systems, Journal of Mathematical Physics **17**, 821 (1976).

[3] G. Lindblad, On the generators of quantum dynamical semigroups, Communications in Mathematical Physics **48**, 119 (1976).

[4] O. Fawzi and R. Renner, Quantum conditional mutual information and approximate markov chains, Communications in Mathematical Physics **340**, 575 (2015).

[5] B. Buča and T. Prosen, A note on symmetry reductions of the lindblad equation: Transport in constrained open spin chains, New Journal of Physics **14**, 073007 (2012).

[6] V. V. Albert and L. Jiang, Symmetries and conserved quantities in lindblad master equations, Physical Review A **89**, 022118 (2014).

[7] L. A. Lessa, R. Ma, J.-H. Zhang, Z. Bi, M. Cheng, and C. Wang, Strong-to-weak spontaneous symmetry breaking in mixed quantum states, PRX Quantum **6**, 010344 (2025).

[8] C. Wang, Strong-to-weak spontaneous symmetry breaking, (2026).

[9] S. J. Garratt and J. T. Chalker, Many-body delocalization as symmetry breaking, Physical Review Letters **127**, 026802 (2021).

[10] Z. Weinstein, Efficient detection of strong-to-weak spontaneous symmetry breaking via the Rényi-1 correlator, Physical Review Letters **134**, 150405 (2025).

[11] Z. Liu, L. Chen, Y. Zhang, S. Zhou, and P. Zhang, Diagnosing strong-to-weak symmetry breaking via wightman correlators, Communications Physics **8**, 274 (2025).

[12] J. Y. Lee, C.-M. Jian, and C. Xu, Quantum criticality under decoherence or weak measurement, PRX Quantum **4**, 030317 (2023).

[13] X. Feng, Z. Cheng, and M. Ippoliti, Hardness of observing strong-to-weak symmetry breaking, Physical Review Letters **135**, 200402 (2025).

[14] J.-H. Zhang, C. Xu, and Y. Xu, Fluctuation-dissipation theorem and information geometry in open quantum systems, (2024).

[15] Y.-H. Chen and T. Grover, Symmetry-enforced many-body separability transitions, PRX Quantum **5**, 030310 (2024).

[16] S. Sang, L. A. Lessa, R. S. K. Mong, T. Grover, C. Wang, and T. H. Hsieh, Mixed-state phases from local reversibility, (2025).

[17] S. Sang and T. H. Hsieh, Stability of mixed-state quantum phases via finite markov length, Physical Review Letters **134**, 070403 (2025).

[18] A. Coser and D. Pérez-García, Classification of phases for mixed states via fast dissipative evolution, Quantum **3**, 174 (2019).

[19] S. Sang, Y. Zou, and T. H. Hsieh, Mixed-state quantum phases: Renormalization and quantum error correction, Physical Review X **14**, 031044 (2024).

[20] T. Rakovszky, S. Gopalakrishnan, and C. von Keyserlingk, Defining stable phases of open quantum systems, Physical Review X **14**, 041031 (2024).

[21] F. Divi, L. A. Lessa, and C. Wang, Local strong-to-weak spontaneous symmetry breaking, (2026).

[22] C. Zhang, Local diagnostics for strong-to-weak spontaneous symmetry breaking and non-equilibrium phase transitions, (2026).

[23] R. Liu, J. Yi, and D. V. Else, A local description of strong symmetries and strong-to-weak symmetry breaking in quantum many-body systems, (2026).

[24] H. Nishimori, Exact results and critical properties of the ising model with competing interactions, Journal of Physics C: Solid State Physics **13**, 4071 (1980).

[25] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, Topological quantum memory, Journal of Mathematical Physics **43**, 4452 (2002).
[26] E. Knill and R. Laflamme, Theory of quantum error-correcting codes, Physical Review A **55**, 900 (1997).
[27] C. Wang, J. Harrington, and J. Preskill, Confinement-higgs transition in a disordered gauge theory and the accuracy threshold for quantum memory, Annals of Physics **303**, 31 (2003).
[28] R. Fan, Y. Bao, E. Altman, and A. Vishwanath, Diagnostics of mixed-state topological order and breakdown of quantum memory, PRX Quantum **5**, 020343 (2024).
[29] Y. Bao, R. Fan, A. Vishwanath, and E. Altman, Mixed-state topological order and the errorfield double formulation of decoherence-induced transitions, Physical Review Letters **136**, 220402 (2026).
[30] J. Y. Lee, W. Ji, Z. Bi, and M. P. A. Fisher, Decoding measurement-prepared quantum phases and transitions: From ising model to gauge theory, and beyond, arXiv Preprint (2022).
[31] R. Ma and C. Wang, Average symmetry-protected topological phases, Physical Review X **13**, 031016 (2023).
[32] R. Ma, J.-H. Zhang, Z. Bi, M. Cheng, and C. Wang, Topological phases with average symmetries: The decohered, the disordered, and the intrinsic, Physical Review X **15**, 021062 (2025).
[33] R. Ma and A. Turzillo, Symmetry-protected topological phases of mixed states in the doubled space, PRX Quantum **6**, 010348 (2025).
[34] H. Xue, J. Y. Lee, and Y. Bao, Tensor network formulation of symmetry protected topological phases in mixed states, arXiv Preprint (2024).
[35] Y. Guo, J.-H. Zhang, H.-R. Zhang, S. Yang, and Z. Bi, Locally purified density operators for symmetry-protected topological phases in mixed states, Physical Review X **15**, 021060 (2025).
[36] J. Y. Lee, Y.-Z. You, and C. Xu, Symmetry protected topological phases under decoherence, Quantum **9**, 1607 (2025).
[37] J.-H. Zhang, Y. Qi, and Z. Bi, Fidelity strange correlators for average symmetry-protected topological phases, Science Bulletin **71**, 688 (2026).
[38] P. Sala, S. Gopalakrishnan, M. Oshikawa, and Y. You, Spontaneous strong symmetry breaking in open systems: Purification perspective, Physical Review B **110**, 155150 (2024).
[39] L. A. Lessa, M. Cheng, and C. Wang, Mixed-state quantum anomaly and multipartite entanglement, Physical Review X **15**, 011069 (2025).
[40] J. Shah, C. Fechisin, Y.-X. Wang, J. T. Iosue, J. D. Watson, Y.-Q. Wang, B. Ware, A. V. Gorshkov, and C.-J. Lin, Instability of steady-state mixed-state symmetry-protected topological order to strong-to-weak spontaneous symmetry breaking, Quantum **9**, 1912 (2025).
[41] S. Lu, L. Q. Silveira, and Y. You, Nonequilibrium topological response under charge dephasing, arXiv Preprint (2026).
[42] Y. Kuno and T. Orito, Hierarchy of mixed symmetry protected topological states in extended cluster states under subsystem decoherence, arXiv Preprint (2026).
[43] Y. Kuno and T. Orito, Fate of average symmetry-protected topological states under symmetry-preserving quantum operations, arXiv Preprint (2026).
[44] Y. Guo and S. Yang, Strong-to-weak spontaneous symmetry breaking meets average symmetry-protected topological order, Physical Review B **111**, L201108 (2025).
[45] M. Qi, R. Sohal, X. Chen, D. T. Stephen, and A. Prem, The symmetry taco: Equivalences between gapped, gapless, and mixed-state SPTs, arXiv Preprint (2025).
[46] S. Schafer-Nameki, A. Tiwari, A. Warman, and C. Zhang, SymTFT approach for mixed states with non-invertible symmetries, arXiv Preprint (2025).
[47] Q. Cao, Z. Bi, and K. W. Murch, Mixed-state symmetry-protected topology and strong-to-weak spontaneous symmetry breaking in a superconducting qubit array, arXiv Preprint (2026).
[48] C. Zhang, Y. Xu, J.-H. Zhang, C. Xu, Z. Bi, and Z.-X. Luo, Strong-to-weak spontaneous breaking of 1-form symmetry and intrinsically mixed topological order, Physical Review B **111**, 115137 (2025).
[49] R. Sohal and A. Prem, Noisy approach to intrinsically mixed-state topological order, PRX Quantum **6**, 010313 (2025).
[50] T. D. Ellison and M. Cheng, Toward a classification of mixed-state topological orders in two dimensions, PRX Quantum **6**, 010315 (2025).
[51] Z. Wang, Z. Wu, and Z. Wang, Intrinsic mixed-state topological order, PRX Quantum **6**, 010314 (2025).
[52] Y.-H. Chen and T. Grover, Separability transitions in topological states induced by local decoherence, Physical Review Letters **132**, 170602 (2024).
[53] L. A. Lessa, S. Sang, T.-C. Lu, T. H. Hsieh, and C. Wang, Higher-form anomaly and long-range entanglement of mixed states, arXiv Preprint (2025).

[54] P.-S. Hsin, R. Kobayashi, and A. Prem, Higher-form anomalies imply intrinsic long-range entanglement, arXiv Preprint (2025).
[55] Z. Song and J.-H. Zhang, Strong-to-weak spontaneous symmetry breaking of higher-form non-invertible symmetries in kitaev's quantum double model, arXiv Preprint (2025).
[56] T.-H. Yang, B. Shi, and J. Y. Lee, Topological mixed states: Phases of matter from axiomatic approaches, Physical Review X **16**, 041002 (2026).
[57] A. Moharramipour, L. A. Lessa, C. Wang, T. H. Hsieh, and S. Sahu, Symmetry enforced entanglement in maximally mixed states, PRX Quantum **5**, 040336 (2024).
[58] Y. Li, F. Pollmann, N. Read, and P. Sala, Highly entangled stationary states from strong symmetries, Physical Review X **15**, 011068 (2025).
[59] P. Sala, V. Temkin, C. Xu, D. Podolsky, and E. Altman, Climbing the fusion tree: Center ordering and decoding hierarchy in non-abelian mixed states, arXiv Preprint (2026).
[60] O. Ogunnaike, J. Feldmeier, and J. Y. Lee, Unifying emergent hydrodynamics and lindbladian low-energy spectra across symmetries, constraints, and long-range interactions, Physical Review Letters **131**, 220403 (2023).
[61] S. Moudgalya and O. I. Motrunich, Symmetries as ground states of local superoperators: Hydrodynamic implications, PRX Quantum **5**, 040330 (2024).
[62] H. Watanabe, H. Katsura, and J. Y. Lee, Critical spontaneous breaking of U(1) symmetry at zero temperature in one dimension, Physical Review Letters **133**, 176001 (2024).
[63] X. Huang, M. Qi, J.-H. Zhang, and A. Lucas, Hydrodynamics as the effective field theory of strong-to-weak spontaneous symmetry breaking, Physical Review B **111**, 125147 (2025).
[64] D. Gu, Z. Wang, and Z. Wang, Spontaneous symmetry breaking in open quantum systems: Strong, weak, and strong-to-weak, Physical Review B **112**, 245123 (2025).
[65] L. V. Delacrétaz, Boulder lectures on thermal dynamics and hydrodynamic EFTs, (2026).
[66] J. Y. Lee, Charge scrambling in strong-to-weak spontaneous symmetry breaking, (2026).
[67] J. Hauser, K. Su, H. Ha, J. Lloyd, T. G. Kiely, R. Vasseur, S. Gopalakrishnan, C. Xu, and M. P. A. Fisher, Strong-to-weak symmetry breaking in open quantum systems: From discrete particles to continuum hydrodynamics, (2026).
[68] U. Agrawal, A. Zabalo, K. Chen, J. H. Wilson, A. C. Potter, J. H. Pixley, S. Gopalakrishnan, and R. Vasseur, Entanglement and charge-sharpening transitions in U(1) symmetric monitored quantum circuits, Physical Review X **12**, 041002 (2022).
[69] H. Singh, R. Vasseur, A. C. Potter, and S. Gopalakrishnan, Mixed-state learnability transitions in monitored noisy quantum dynamics, Physical Review B **113**, 054305 (2026).
[70] C. Zerba, S. Gopalakrishnan, and M. Knap, Strong-to-weak symmetry breaking in monitored dipole conserving quantum circuits, (2025).
[71] T. A. Kumar, Y. Zou, A.-R. Negari, R. G. Melko, and T. H. Hsieh, Unlearnable phases of matter, (2026).
[72] A. Vijay and J. Y. Lee, Information critical phases under decoherence, (2026).
[73] V. Temkin, Z. Weinstein, R. Fan, D. Podolsky, and E. Altman, Charge-informed quantum error correction, (2025).
[74] A. Sarma, Strong-to-weak spontaneous symmetry breaking of dephased fermions, (2026).
[75] C. Fechisin, T.-C. Lu, Z.-Y. Wei, J. Shah, Y.-X. Wang, A. V. Gorshkov, and C.-J. Lin, Mixed-state phases induced by power-law quantum channels, (2026).
[76] N. Ziereis, S. Moudgalya, and M. Knap, Strong-to-weak symmetry breaking phases in steady states of quantum operations, (2025).
[77] J. Kim, E. Altman, and J. Y. Lee, Error threshold of SYK codes from strong-to-weak parity symmetry breaking, Physical Review B **114**, L051101 (2026).
[78] Y.-H. Chen and T. Grover, Non-perturbative constraints on stability and renormalization group flows in nonequilibrium matter, (2026).
[79] N. Sun, P. Zhang, and L. Feng, Scheme to detect the strong-to-weak symmetry breaking via randomized measurements, Physical Review Letters **135**, 090403 (2025).
[80] Y. Liu and Y. Zou, Polynomial-time certification of fidelity for many-body mixed states and mixed-state universality classes, (2026).
[81] S. Wang et al., Observation of strong-to-weak spontaneous symmetry breaking in a dephased fermi gas, (2026).
[82] Z.-Y. Wei, C. Fechisin, Y.-X. Wang, J. Shah, T.-C. Lu, C.-J. Lin, and A. V. Gorshkov, Strong-to-weak spontaneous symmetry breaking transitions on a quantum processor, (2026).

[83] T.-C. Lu, Y.-J. Liu, S. Gopalakrishnan, and Y. You, Holographic duality between bulk topological order and boundary mixed-state order, PRX Quantum 7, 033041 (2026).
[84] A. Vijay and J. Y. Lee, Holographically emergent gauge theory in symmetric quantum circuits, (2026).